FACULDADE DE ENGENHARIA DA UNIVERSIDADE DO PORTO

Low-Velocity Impact Damage Assessment and Numerical Modelling of Glass/PPS Laminated Composites Using Commercial Finite Element Tools

António José Freitas Machado Dórdio Gomes

Master in Mechanical Engineering - Aeronautical and Vehicles Structures

Supervisor FEUP: Prof. Dr. Volnei Tita

Host Institution Supervisor: Prof. Dr. Antonio Caporale

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Approved in oral examination by the committee:

President: Prof. Dr. Rui Miranda Guedes
Referee: Dr. Gabriel Sales Candido Souza
Referee: Prof. Dr. Volnei Tita

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Resumo

O impacto de baixa velocidade (LVI) constitui um cenário crítico de dano em estruturas compósitas termoplásticas, uma vez que ativa múltiplos mecanismos de falha interdependentes difíceis de capturar através de modelos numéricos convencionais. Este trabalho investiga o comportamento ao impacto de laminados compósitos termoplásticos Glass/PPS com arquitetura tecida, através de uma abordagem combinada experimental e numérica. Ensaios de impacto realizados a diferentes níveis de energia revelaram uma clara correlação entre a energia de impacto e a severidade do dano, observando-se um aumento da profundidade de indentação, do comprimento da fissura na face posterior e da delaminação interna. Inspeções por ultrassons através de mapeamento C-scan permitiram quantificar a evolução do dano interlaminar, evidenciando um aumento da área de delaminação de aproximadamente 20 mm^2 a 7.5 J para 143 mm^2 a 27 J. Foi desenvolvido um modelo numérico progressivo em Abaqus/Explicit para reproduzir a resposta ao impacto observada experimentalmente. O dano intralaminar foi representado através do critério de falha de Hashin 2D implementado com elementos continuum shell, enquanto a delaminação interlaminar foi modelada através de elementos coesivos de interface. Para capturar a indentação permanente sob o impactor, foi implementada uma estratégia de modelação bifásica incorporando elementos sólidos localizados com plasticidade dependente da pressão. A análise combinada dos resultados experimentais e numéricos mostra que a resposta ao impacto do laminado Glass/PPS investigado é governada pela interação entre fratura intralaminar dominada pelas fibras, delaminação interlaminar e deformação localizada da matriz. A modelação do dano intralaminar reproduziu a morfologia da fissura observada na face posterior e a resposta dinâmica global em termos da história temporal da aceleração, indicando que a fratura dominada pelas fibras constitui o principal mecanismo responsável pela formação da fissura neste laminado. As inspeções por ultrassons confirmaram igualmente que a delaminação contribui de forma significativa para a evolução global do dano, enquanto a plasticidade da matriz governa a indentação localizada sob o impactor. Os valores de energia de fratura intralaminar inicialmente calibrados em configurações de modelação simplificadas (aproximadamente 9% dos valores de referência reportados na literatura) devem ser interpretados como parâmetros efetivos, representando implicitamente mecanismos de dissipação de energia não explicitamente modelados nas fases iniciais da simulação. Uma validação adicional baseada em medições C-scan mostrou que a utilização de valores de energia de fratura mais elevados, próximos dos reportados na literatura para compósitos tecidos de fibra de vidro, melhora a concordância entre as áreas de delaminação previstas numericamente e as observadas experimentalmente.

Palavras-chave: Impacto de baixa velocidade, compósitos termoplásticos, laminados Glass/PPS, dano intralaminar, delaminação, modelação por elementos finitos.

Abstract

Low-velocity impact (LVI) is a critical damage scenario for thermoplastic composite structures, as it activates multiple interacting failure mechanisms that are difficult to capture using conventional numerical models. This work investigates the impact behaviour of woven Glass/PPS thermoplastic composite laminates through a combined experimental and numerical study. Impact tests at different energy levels revealed a clear correlation between impact energy and damage severity, with increasing indentation depth, back-face crack length, and internal delamination. Ultrasonic C-scan inspections quantified the evolution of interlaminar damage, showing delamination footprints increasing from approximately 20 mm^2 at 7.5 J to 143 mm^2 at 27 J. A progressive finite element modelling framework was developed in Abaqus/Explicit to reproduce the observed impact response. Intralaminar damage was represented using the Hashin 2D failure criterion with continuum shell elements, while interlaminar delamination was modelled through cohesive interface elements. To capture permanent indentation beneath the impactor, a bi-phasic modelling strategy incorporating localized solid elements with pressure-dependent plasticity was implemented. The combined numerical and experimental analysis shows that the impact response of the investigated Glass/PPS laminate is governed by the interaction between intralaminar fibre fracture, interlaminar delamination, and localized matrix deformation. Intralaminar damage modelling reproduced the experimentally observed back-face crack morphology and global acceleration–time response, indicating that fibre-dominated fracture is the primary mechanism controlling crack formation in the investigated laminate. Ultrasonic inspections also confirmed that delamination contributes significantly to the overall damage evolution, while matrix plasticity governs localized indentation beneath the impactor. The intralaminar fracture energy values initially calibrated in simplified modelling configurations (approximately 9% of literature benchmarks) should therefore be interpreted as effective parameters, implicitly accounting for energy dissipation mechanisms not explicitly represented in the early modelling stages. Additional validation against ultrasonic C-scan measurements showed that adopting higher fracture energy values, closer to literature values for woven glass composites, improves the agreement between predicted and experimentally observed delamination footprints.

Keywords: Low-velocity impact, thermoplastic composites, Glass/PPS laminates, intralaminar damage, delamination, finite element modelling.

UN Sustainable Development Goals

This dissertation aligns its contributions in advanced structural modelling, sustainable materials engineering, and improved predictive design methodologies with several United Nations Sustainable Development Goals (SDGs). The selected goals are drawn directly from the official UN SDG descriptions to ensure fidelity to their original framing.[1]

The research focuses on the experimental characterization and numerical modelling of impact damage in Glass/PPS thermoplastic composite laminates. These materials are widely used in aerospace structures due to their high strength-to-weight ratio and potential for lightweight structural design. Improving the understanding and prediction of damage mechanisms contributes to safer, more efficient structures with optimized material usage and extended service life.

The SDGs most closely aligned with this research are the following:

- **SDG 9 – Industry, Innovation and Infrastructure:** Build resilient infrastructure, promote inclusive and sustainable industrialization and foster innovation.
- **SDG 12 – Responsible Consumption and Production:** Ensure sustainable consumption and production patterns.
- **SDG 13 – Climate Action:** Take urgent action to combat climate change and its impacts.

SDG	Goal	Relevance to this dissertation
9	Industry, innovation and infrastructure	Development of numerical modelling methodologies for impact damage in thermoplastic composite laminates contributes to innovation in aerospace structural design and improves the reliability of advanced engineering infrastructure.
12	Responsible consumption and production	Improved prediction of damage mechanisms enables more efficient structural design and optimized material use, supporting reduced waste and more sustainable utilization of composite materials.
13	Climate action	Advances in modelling and design of lightweight composite structures support more energy-efficient aerospace systems, contributing to reduced operational emissions and supporting climate mitigation efforts.

[1] https://sdgs.un.org/goals

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António Gomes

"[Em] Itália, o presidente tem oito anos (...); vamos fazer os melhores gelados e vamos ter os carros mais rápidos."

António A. Coutinho, Humorista

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List of Acronyms

FRP Fibre-Reinforced Polymer
LVI Low-Velocity Impact
NDT Non-Destructive Technique
BVID Barely Visible Impact Damage
ASTM American Society for Testing and Materials
PPS Polyphenylene Sulphide
PEEK Polyetheretherketone
CFRTP Carbon Fiber Reinforced Thermoplastic
FE Finite Element
CDM Continuum Damage Mechanics
CZM Cohesive Zone Model
VUMAT Vectorized User Material Subroutine
UMAT User-Defined Material Subroutine
FAA Federal Aviation Administration
EASA European Union Aviation Safety Agency
CbA Certification by Analysis
CS Certification Specification
AC Advisory Circular
AMC Acceptable Means of Compliance
SEA Specific Energy Absorption
EA Absorbed Energy
PMCs Polymer Matrix Composites
MMCs Metal Matrix Composites
CMCs Ceramic Matrix Composites
AFP Automated Fiber Placement
PEKK Polyetherketoneketone
UD Unidirectional
GFRPs Glass Fibre-reinforced Polymers
CFRPs Carbon Fibre-reinforced Polymers
AFRPs Aramid Fibre-reinforced Polymers

PDM Progressive Damage Modelling

FT Fiber Tension

FC Fiber Compression

MT Matrix Tension

MC Matrix Compression

BK Benzeggagh-Kenane

DIC Digital Image Correlation

ENF End-Notched Flexure

Chapter 1

Introduction

This chapter introduces the scope and motivation of the present thesis. It first contextualizes the engineering relevance of low-velocity impact in thermoplastic composite laminates and defines the main objectives of the work. The adopted methodology is then described, clarifying the use of previously obtained experimental impact data, the post-impact non-destructive testing performed in this study, and the development of the numerical modelling framework using commercially available tools. Finally, the structure of the thesis is outlined to guide the reader through the organization of the document.

1.1 Objectives and Contextualization

Fibre-Reinforced Polymer (FRP) composites have become fundamental materials in modern aerospace structures due to their high specific strength, stiffness-to-weight ratio, corrosion resistance, and fatigue performance [1, 2]. The progressive increase in composite content in commercial aircraft structures reflects the continuous pursuit of lightweight airframes and improved fuel efficiency [3, 4]. As composite materials are increasingly employed in primary load-bearing components, ensuring their structural integrity under realistic service loading conditions becomes a critical engineering requirement.

Among the different in-service loading scenarios, Low-Velocity Impact (LVI) represents one of the most critical threats to laminated composite structures. Such impacts may occur during maintenance operations, tool drops, debris strikes, or handling incidents, and are characterized by relatively low impact velocities but significant local damage [5]. Unlike high-velocity ballistic events, low-velocity impacts frequently induce complex internal damage mechanisms without obvious external visibility. These mechanisms typically include matrix cracking, fibre fracture, fibre/matrix debonding, and interlaminar delamination, which interact and evolve during the impact event [6, 7]. The presence of barely visible impact damage may significantly reduce stiffness and residual strength, making reliable prediction of impact behaviour a key design challenge.

Recent investigations have highlighted differences in impact response between thermoset and thermoplastic composite systems. Comparative studies on woven carbon fibre laminates have shown distinct damage patterns and energy absorption characteristics depending on matrix type [8, 9]. In particular, thermoplastic matrices such as Polyphenylene Sulphide (PPS) have attracted attention in aerospace applications due to their processing characteristics, thermal stability, and resistance to environmental degradation [10, 11]. Experimental studies on PPS-based laminates subjected to impact loading have reported characteristic delamination propagation and damage mechanisms that influence the residual mechanical response of the laminate [12, 13]. These aspects justify further investigation of the low-velocity impact behaviour of Glass fibre-reinforced PPS laminates from an engineering modelling perspective.

From a numerical standpoint, accurate simulation of low-velocity impact remains challenging due to the strong coupling between contact dynamics, material nonlinearity, progressive intralaminar failure, and interlaminar delamination growth. Progressive damage modelling strategies based on failure criteria such as Hashin's formulation [14] and continuum damage mechanics approaches have been widely adopted to represent intralaminar failure in composite laminates [15, 16]. Interlaminar delamination is commonly described using cohesive zone models governed by traction–separation laws and mixed-mode fracture criteria [17, 18]. In addition to conventional laminate modelling strategies, a bi-phasic numerical approach is adopted in this work to better capture matrix-dominated and through-thickness deformation effects occurring under low-velocity impact. In this strategy, the composite laminate is primarily represented using continuum-shell elements to model the fibre-reinforced plies, while solid elements are introduced to explicitly represent the matrix-dominated response. Two configurations are investigated: a localized insertion of solid elements in the impact zone and a fully solid laminate representation. This comparative approach enables evaluation of the influence of through-thickness discretization on indentation behaviour, damage prediction, and global impact response while remaining within the modelling capabilities available in Abaqus/Explicit [19, 20]. Although advanced user-defined implementations exist in the literature, there remains significant engineering interest in assessing the predictive capability of the constitutive and damage formulations natively available in commercial finite element software.

Within this context, the present thesis focuses on the numerical investigation and post-impact damage assessment of woven Glass/PPS laminated composite plates subjected to low-velocity impact loading. The experimental impact campaign was conducted during a preceding research stage, and the resulting acceleration-time histories, back face crack length, and indentation diameter data are used in this work as a reference for numerical correlation.

A particular focus of this work is the investigation of the relative contribution of different damage mechanisms to the global impact response of the laminate. Low-velocity impact in composite materials involves the simultaneous activation of intralaminar fibre fracture, interlaminar

delamination, and matrix-dominated deformation mechanisms. Accurately capturing the interaction between these mechanisms remains a major challenge for numerical modelling approaches. By progressively introducing intralaminar damage, interlaminar cohesive interfaces, and localized matrix plasticity within the modelling framework, this study aims to assess how each mechanism contributes to the observed impact response and to identify the dominant processes governing crack formation, delamination development, and indentation behaviour in Glass/PPS laminates.

The primary contribution of this thesis lies in the experimental post-impact Non-Destructive Technique (NDT) characterization performed to evaluate internal damage distribution, as well as in the development of a finite element modelling framework capable of reproducing the observed impact response using exclusively the constitutive models available in Abaqus/Explicit [20, 21]. The work aims to identify the dominant damage mechanisms, implement progressive intralaminar and interlaminar damage formulations within a commercial finite element environment, and critically assess the predictive capability of this approach through systematic comparison with experimental measurements. By restricting the modelling strategy to commercially available formulations, the study emphasizes engineering applicability and reproducibility.

1.2 Methodology

The methodology adopted in this work combines experimental post-impact characterization and numerical modelling to investigate the low-velocity impact response of Glass/PPS laminated composite plates.

The low-velocity impact experimental campaign was conducted during a preceding research stage. The available dataset includes acceleration-time histories, back-face crack length, and indentation diameter. These results are used in the present work as a reference for model validation.

The research activities performed in this thesis comprise the following main stages:

- Processing and critical evaluation of the available experimental impact data;
- Post-impact NDT performed in this work to characterize internal damage distribution and extent;
- Development of a baseline finite element model in Abaqus/Explicit using continuum-shell elements to represent the composite laminate [20, 21];
- Implementation of intralaminar progressive damage modelling using built-in failure criteria and interlaminar cohesive zone modelling for delamination prediction [14, 17, 18];
- Development of a bi-phasic modelling strategy introducing solid elements to represent matrix-dominated behaviour, including:

  - A localized solid-element insertion in the impact zone;
  - A fully solid laminate model for the entire plate;

- Comparative assessment of the different modelling strategies in terms of global impact response, indentation prediction, and damage evolution;
- Calibration and sensitivity assessment of selected modelling parameters within the adopted commercial-model framework;
- Quantitative correlation between numerical predictions and experimental measurements, including additional validation against post-impact ultrasonic delamination maps.

This structured approach enables systematic evaluation of the influence of modelling assumptions and discretization choices on the predictive capability of commercially available constitutive formulations under low-velocity impact conditions.

1.3 Thesis Structure

Chapter 1 introduces the engineering motivation for studying low-velocity impact in fibre-reinforced thermoplastic composites. The scope of the work is defined, clarifying the use of previously obtained impact test data, the post-impact non-destructive testing performed in this thesis, and the development of a numerical modelling framework using commercially available tools.

Chapter 2 presents the state of the art on composite materials in aerospace applications, low-velocity impact damage mechanisms, and thermoplastic laminate behaviour. Numerical approaches for modelling intralaminar and interlaminar damage are also reviewed.

Chapter 3 describes the theoretical background underlying the numerical modelling strategy, including intralaminar failure criteria, progressive damage modelling concepts, and cohesive zone formulations for delamination modelling.

Chapter 4 summarizes the available low-velocity impact experimental campaign used as a reference in this work. The impact setup, specimen configuration, and recorded acceleration-time histories, back-face crack length, and indentation diameter are presented for completeness.

Chapter 5 details the finite element modelling framework developed in Abaqus/Explicit. This includes the baseline continuum-shell model, the bi-phasic modelling strategies (localized solid

insertion and fully solid laminate representation), and the implementation of progressive intralaminar damage and interlaminar cohesive formulations. Numerical results are systematically compared with the available experimental impact measurements in order to assess predictive capability and to investigate the interaction between intralaminar fracture, interlaminar delamination, and matrix-dominated deformation mechanisms governing the impact response.

Chapter 6 presents the post-impact NDT work performed in this thesis. The inspection methodology and experimental damage characterization are described, with particular emphasis on the identification and quantification of internal delamination damage through ultrasonic inspection techniques, and on the additional validation of the cohesive numerical model through comparison with the experimentally observed damage patterns.

Chapter 7 summarizes the main conclusions of the thesis, highlighting the key findings regarding the interaction of damage mechanisms governing low-velocity impact in Glass/PPS laminates. The chapter also discusses the limitations of the present modelling framework and outlines possible future developments related to experimental characterization and numerical modelling strategies.

The overall organization of the thesis and the main methodological stages adopted in this work are illustrated in Figure 1.1, with particular emphasis on the numerical modelling strategy developed in Chapter 5.

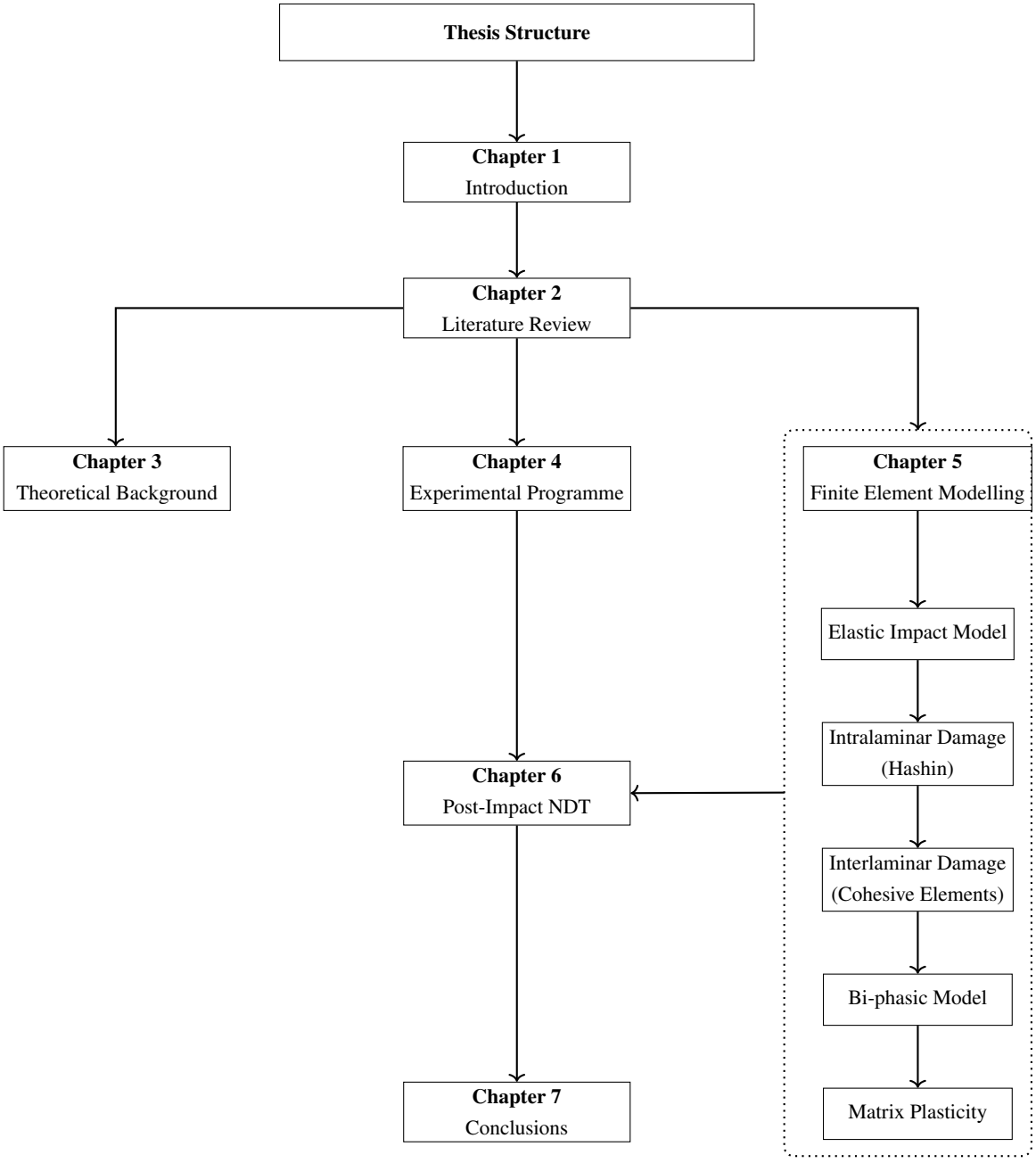


Figure 1.1: Structure of the thesis, highlighting the overall chapter organisation and the progressive development of the numerical modelling framework within Chapter 5.

Chapter 2

Literature Review

The increasing adoption of composite materials in aerospace structures has been driven by the demand for lightweight, high-performance systems capable of meeting stringent mechanical, environmental, and safety requirements. Fibre-reinforced polymer composites offer superior specific stiffness and strength, enhanced fatigue resistance, and improved corrosion behaviour when compared to conventional metallic materials, making them particularly attractive for both primary and secondary aircraft structures. However, the heterogeneous and anisotropic nature of composite materials introduces complex damage mechanisms and modelling challenges, especially under dynamic loading conditions such as impact.

This chapter presents a comprehensive review of the state of the art in composite materials for aerospace applications, with particular emphasis on aspects relevant to impact behaviour and numerical modelling. The discussion begins with an overview of composite material fundamentals and their historical evolution in the aerospace sector, followed by a detailed examination of matrix systems, fibre reinforcements, and laminate architectures commonly employed in aircraft structures. Special attention is given to thermoplastic composite systems due to their growing industrial relevance and distinctive damage and energy absorption characteristics under impact loading.

The subsequent sections focus on the impact response of composite laminates, highlighting experimental observations, dominant damage mechanisms, and material-dependent behaviours reported in the literature. Particular attention is given to intralaminar fibre fracture, interlaminar delamination, and matrix-dominated deformation under low-velocity impact conditions, as well as to numerical modelling strategies used to reproduce these phenomena. Non-destructive testing techniques for post-impact damage characterization are also discussed, providing context for the ultrasonic inspections performed in this work. This review establishes the scientific background necessary to interpret the experimental impact tests and to support the numerical modelling strategies developed in the following chapters.

2.1 Composite Materials for Aerospace Structures

The adoption of composite materials has profoundly transformed the design of modern aircraft structures, enabling significant weight reduction while maintaining high levels of structural performance and durability. By combining high-strength reinforcements with lightweight matrix materials, composites offer superior specific stiffness and strength compared to conventional metallic alloys, together with enhanced fatigue and corrosion resistance. These attributes have motivated the progressive integration of composites into both primary and secondary aircraft structures over the past decades [1, 2].

The increasing structural relevance of composites has been accompanied by new design, manufacturing, and certification challenges. Unlike metallic materials, composite laminates exhibit anisotropic behaviour and complex damage mechanisms, which must be carefully understood and controlled to ensure structural integrity and operational safety. Lessons learned from early composite aircraft programmes have highlighted the importance of material selection, damage tolerance, and reliable modelling approaches in achieving the anticipated performance benefits [22].

This section presents an overview of composite materials as applied to aerospace structures, with emphasis on their fundamental characteristics, historical evolution, and structural behaviour. Particular attention is given to polymer matrix composites and, subsequently, to thermoplastic composite systems, which are increasingly adopted in impact- and damage-critical aircraft components [11, 23]. The concepts introduced in this section provide the material background required for the discussion of impact behaviour, crashworthiness, and numerical modelling strategies developed in the subsequent chapters.

2.1.1 Introduction to Composite Materials

Composite materials are engineered material systems formed by combining two or more distinct constituents at the macroscopic scale in order to obtain a set of properties that cannot be achieved by any constituent alone [24, 25]. In structural applications, the most common class is fibre-reinforced composites, in which strong and stiff fibres are embedded within a continuous matrix phase to produce lightweight structures with high specific mechanical performance [26, 27].

2.1.1.1 Constituents and roles: matrix and reinforcement

In fibre-reinforced composites, the reinforcement is typically responsible for most of the load-carrying capability and provides high stiffness and strength along the fibre direction(s). The matrix binds the fibres, transfers shear stresses between them, stabilises the reinforcement against buckling, and provides environmental protection (e.g., against moisture and chemical exposure) [24, 26]. The overall response is therefore governed by the properties of each phase, their volume fraction, and the architecture of the reinforcement (e.g., fibre orientation and textile form), resulting in

a strongly directional (anisotropic) behaviour that can be tailored to the load paths of the structure [25, 27].

The fibre/matrix interface plays a critical role in determining how efficiently loads are transferred between constituents and how damage initiates and propagates under service loading. Interfacial debonding or degradation can reduce stiffness, promote delamination-type damage in laminates, and accelerate strength loss, particularly in environments involving combined mechanical and thermal or chemical exposure [26, 28].

2.1.1.2 Classification of composites

A common classification is based on the matrix material, which largely determines processing routes, operating temperature limits, and environmental resistance. Polymer Matrix Composites (PMCs) are the most widely used in aerospace due to their low density and excellent manufacturability, while Metal Matrix Composites (MMCs) and Ceramic Matrix Composites (CMCs) are employed when higher temperature capability or specific functional requirements are needed [2, 29]. Within PMCs, the matrix may be thermosetting or thermoplastic, leading to important differences in processing, damage tolerance, repairability, and end-of-life options (e.g., recycling), which are addressed in detail later in this chapter [10, 30].

Composites can also be classified by reinforcement form, such as continuous fibres, discontinuous fibres, particles, or textile reinforcements [27, 31]. For high-performance aerospace structures, continuous fibre reinforcement is most common because it enables efficient load transfer and superior specific stiffness and strength. Textile reinforcements (woven, braided, or stitched fabrics) are widely used to provide multi-directional reinforcement, improved drapeability, and manufacturing robustness in complex geometries [32, 33].

2.1.1.3 Advantages and limitations in aerospace structures

The main advantages of structural composites include high specific stiffness and strength, design flexibility through property tailoring, and enhanced fatigue and corrosion performance compared with conventional metallic alloys [1, 27]. These benefits have driven the progressive integration of composites into aircraft structures, where mass reduction directly contributes to improved fuel efficiency and operational performance [1, 34].

At the same time, composites present important limitations and challenges. Their anisotropy and layered architecture introduce complex failure and damage mechanisms, and their structural performance can be sensitive to manufacturing quality (e.g., fibre waviness, porosity, and interface integrity) and to environmental effects [26, 31]. Furthermore, design and certification require reliable methods to address damage tolerance and inspection, since certain damage modes may be difficult to detect without dedicated non-destructive evaluation procedures [35].

Overall, the combination of tailored performance potential and structural complexity motivates a careful discussion of composite architectures, material systems, and application-driven requirements. The following subsections therefore focus on the evolution of composites in aerospace and on polymer matrix composites, with particular emphasis on thermoplastic composite systems relevant to the present work.

2.1.2 Evolution of Composite Materials in Aerospace

The use of composite materials in aerospace structures has evolved progressively over several decades, driven by the continuous demand for weight reduction, improved structural efficiency, and enhanced durability. Early applications of composites in aircraft date back to the mid-20th century, initially limited to secondary and non-load-bearing components due to limited material maturity, manufacturing capabilities, and insufficient understanding of long-term behaviour [22, 36].

During the 1970s and 1980s, advances in fibre production and polymer matrix systems, together with the development of improved analytical and experimental tools, enabled the gradual introduction of fibre-reinforced polymer composites into more demanding structural roles. Glass fibre composites were among the first to be adopted due to their cost-effectiveness and favourable damage tolerance, while carbon fibre reinforced polymers offered superior specific stiffness and strength for weight-critical applications [1, 31]. Nevertheless, expectations of very large weight savings were often moderated in practice by design conservatism, certification requirements, and the need to ensure adequate damage tolerance and inspectability [22].

From the 1990s onwards, composite materials became increasingly integrated into primary aircraft structures, supported by extensive experimental databases and accumulated service experience. This period marked a transition from component-level applications towards large-scale structural assemblies, including wings, fuselage sections, and control surfaces. The adoption of composites in these roles required the development of new design philosophies accounting for anisotropy, layered construction, and progressive damage mechanisms, as well as the establishment of reliable inspection and maintenance strategies [33, 35].

A major milestone in the evolution of aerospace composites was reached with the introduction of composite-intensive commercial aircraft, in which polymer matrix composites constitute a substantial fraction of the primary structure. These developments demonstrated that composites could meet stringent airworthiness and durability requirements when supported by appropriate design allowables, manufacturing control, and certification procedures [1, 37]. At the same time, the growing structural relevance of composites highlighted the need for accurate modelling tools capable of capturing complex damage and failure processes under operational and accidental loading scenarios [34].

More recently, attention has increasingly shifted towards advanced composite systems and manufacturing technologies aimed at improving production rates, sustainability, and structural performance. Thermoplastic matrix composites, in particular, have emerged as promising alternatives to traditional thermoset systems due to their potential for improved damage tolerance, rapid processing, and recyclability [10, 23]. The progressive integration of these materials into aerospace structures reflects both technological maturation and evolving design priorities, especially in applications where impact resistance, reparability, and lifecycle considerations are critical.

Overall, the historical evolution of composite materials in aerospace illustrates a transition from cautious adoption in secondary components to widespread use in primary load-bearing structures. This evolution has been enabled by advances in materials, manufacturing, and analysis methods, and it continues to shape current research and development efforts focused on next-generation composite systems. The following subsection therefore concentrates on polymer matrix composites and their specific relevance to modern aerospace structures.

The progressive adoption of composite materials in civil aviation has been driven by the need for significant weight reduction, improved fatigue resistance, and enhanced corrosion performance compared to conventional metallic airframes. Over the past decades, these advantages have enabled a gradual transition from secondary composite components to primary load-bearing structures, culminating in modern aircraft designs that rely extensively on fibre-reinforced polymer composites for fuselage and wing structures. This evolution is illustrated in Fig. 2.1, where aircraft such as the Boeing 787 and Airbus A350 exemplify the shift towards composite-intensive airframe architectures.

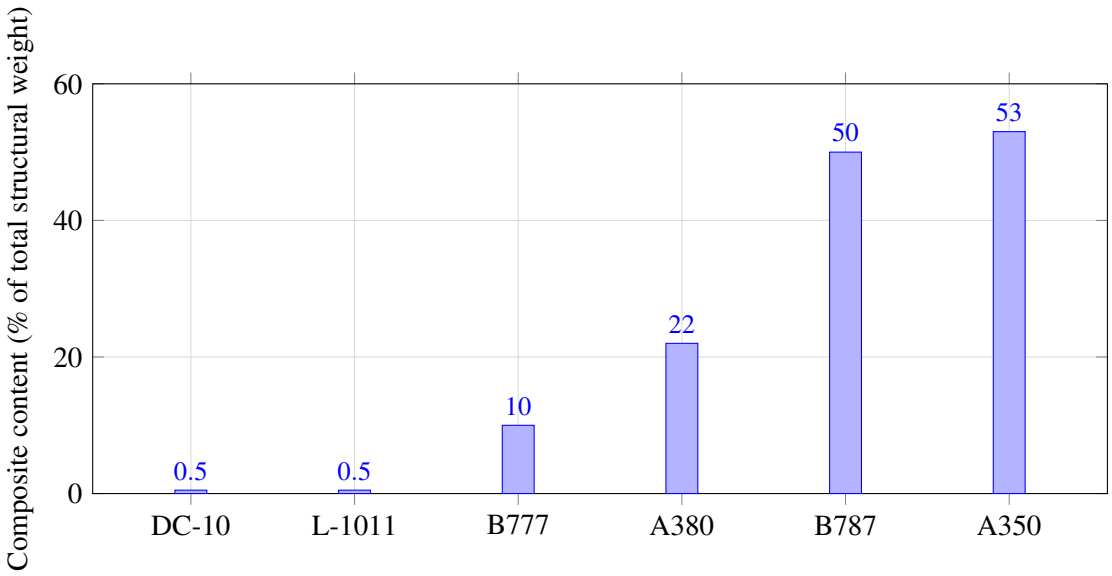


Figure 2.1: Approximate evolution of composite material usage in selected commercial aircraft, expressed as percentage of total structural weight attributed to composites. Values are taken from the trend reported by GAO (note: percentages are approximate) and complemented with Airbus public technical facts for the A350. [3, 4]

2.1.3 Thermoset and Thermoplastic Matrix Systems in Aerospace Composites

Polymer matrix composites employed in aerospace structures are commonly classified according to the nature of the polymer matrix into thermoset and thermoplastic systems. This distinction has a decisive influence on manufacturing routes, damage tolerance, reparability, and sustainability considerations, and therefore plays a central role in the design and certification of composite aircraft components.

Thermoset composites, typically based on epoxy, polyester, or phenolic resins, have historically dominated aerospace applications due to their relatively low processing temperatures, good fibre wet-out, and mature manufacturing technologies such as autoclave curing and resin transfer moulding [30, 33]. In these systems, curing leads to the formation of an irreversible cross-linked molecular network, resulting in good dimensional stability and thermal resistance. However, this cross-linked structure also contributes to limited strain-to-failure, increased brittleness, and reduced resistance to impact-induced damage, which are critical limitations for structures subjected to accidental loading scenarios [1, 27].

Thermoplastic composites, by contrast, employ matrices such as PPS, Polyetheretherketone (PEEK), or Polyetherketoneketone (PEKK), which soften upon heating and solidify upon cooling without undergoing chemical cross-linking [11, 23]. This reversible behaviour enables manufacturing processes based on melting and consolidation, including thermoforming and welding, and results in higher fracture toughness, improved damage tolerance, and enhanced resistance to

delamination growth compared to thermoset-based systems [38, 39]. These characteristics are particularly advantageous for aerospace structures exposed to impact events such as tool drops, runway debris, or bird strikes.

Automated Fiber Placement (AFP) represents a key manufacturing technology for large composite aerospace structures and can be applied to both thermoset and thermoplastic matrix systems. In thermoset AFP, prepreg tows are deposited using localized heating to promote tack and placement stability, while final consolidation and curing are performed in an autoclave or oven [1, 33]. This approach is widely adopted for primary aircraft structures, including fuselage and wing components of modern commercial aircraft. In thermoplastic AFP, the material must be heated above its melting temperature to achieve in-situ consolidation during placement, typically requiring high-energy heat sources such as laser or flashlamp systems [10, 40]. While thermoplastic AFP enables reduced production cycles and eliminates post-deposition curing, it also introduces additional challenges related to thermal management, crystallinity control, and residual stress development.

Beyond mechanical performance, thermoplastic composites offer significant advantages in terms of storage stability and sustainability. Unlike thermoset prepregs, thermoplastic materials do not require refrigerated storage and exhibit essentially unlimited shelf life, reducing logistical constraints and material waste [10]. Furthermore, the ability to re-melt and reprocess thermoplastic matrices provides opportunities for recycling and repair, aligning with the increasing emphasis on circular economy principles within the aerospace sector [2, 41].

Despite these advantages, the broader adoption of thermoplastic composites in primary aircraft structures remains constrained by higher processing temperatures, tooling requirements, and process-induced residual stresses [11, 27]. Consequently, thermoplastic systems are often introduced progressively, initially in secondary or semi-primary structures, before being considered for more critical load-bearing applications.

Within this framework, glass fibre-reinforced thermoplastic composites, and in particular Glass/PPS systems, represent a balanced compromise between mechanical performance, impact resistance, and manufacturability. Their favourable damage tolerance, thermal stability, and processing flexibility make them especially attractive for aerospace structures subjected to dynamic loading and impact conditions, providing strong motivation for the experimental and numerical investigations presented in this work.

A concise comparison between thermoset and thermoplastic matrix systems, highlighting their respective advantages and limitations in aerospace applications, is provided in Table 2.1.

Table 2.1: Comparison between thermoset and thermoplastic matrix systems used in aerospace composite structures.

Aspect	**Thermoset Composites**	**Thermoplastic Composites**
Matrix structure	Cross-linked polymer network formed by irreversible curing [30, 33]	Linear or semi-crystalline polymer structure, re-meltable upon heating [11, 23]
Typical matrices	Epoxy, phenolic resins, polyester [30]	PPS, PEEK, PEKK [10, 23]
Processing routes	Autoclave curing, resin transfer moulding, thermoset AFP [1, 33]	Thermoforming, welding, thermoplastic AFP [10, 40]
Consolidation mechanism	Post-placement chemical curing [30]	In-situ or post-deposition melting and consolidation [10]
Impact resistance	Moderate, with brittle matrix behaviour [1]	High, due to increased matrix ductility [38, 39]
Fracture toughness	Lower fracture toughness [27]	Higher fracture toughness [38]
Damage tolerance	Limited tolerance to delamination growth [1]	Improved resistance to delamination propagation [39]
Repairability	Limited reparability once cured [27]	Good reparability through localized reheating and welding [11]
Storage requirements	Refrigerated storage required for prepregs [33]	Essentially unlimited shelf life at room temperature [10]
Recyclability	Poor recyclability due to cross-linked structure [27]	Good recyclability and reprocessing potential [2, 41]
Aerospace maturity	Very high, extensively certified and industrialized [1]	Increasing, with growing adoption in secondary and semi-primary structures [10]

2.1.4 Fibre Reinforcement Systems for Aerospace Composites

The impact and crashworthiness performance of fibre-reinforced polymer composites is strongly governed by the mechanical characteristics and failure behaviour of the reinforcing fibres. In aerospace structures subjected to impact loading, the selection of fibre type directly influences energy absorption mechanisms, damage propagation, and residual load-carrying capacity. Among the most relevant reinforcement systems for impact-critical applications are glass, carbon, aramid, and natural fibres, each exhibiting distinct mechanical responses and damage tolerance characteristics.

Glass fibres represent one of the most widely used reinforcements in aerospace composites, particularly in structures where impact resistance, damage tolerance, and cost efficiency are critical design drivers. Glass Fibre-reinforced Polymers (GFRPs) exhibit moderate stiffness and strength

compared to carbon fibres, but significantly higher strain-to-failure and fracture toughness, leading to more progressive and stable damage mechanisms under impact loading [31, 41]. Under transverse and low-velocity impact, GFRPs dissipate energy through a combination of matrix cracking, fibre debonding, delamination, and fibre fracture, resulting in distributed damage zones rather than localized brittle failure [42]. These characteristics make glass fibre composites particularly suitable for impact-prone aerospace components, including fuselage skins, fairings, leading edges, and secondary structural panels.

In contrast, carbon fibres are primarily selected for their exceptional specific stiffness and strength, enabling lightweight and highly efficient load-bearing structures. However, Carbon Fibre-reinforced Polymers (CFRPs) exhibit relatively brittle behaviour under impact, with limited plastic deformation and a tendency toward abrupt fibre fracture and stiffness degradation [1, 36]. While CFRPs can dissipate energy through delamination and matrix damage, their lower damage tolerance often leads to reduced residual strength following impact, necessitating careful damage-tolerant design and inspection strategies in aerospace applications.

Aramid fibres, such as Kevlar®, provide a markedly different impact response due to their high toughness, low density, and excellent tensile strength. Aramid Fibre-reinforced Polymers (AFRPs) are characterized by large deformation capability and progressive failure modes, including fibre stretching, fibrillation, and pull-out, which promote efficient kinetic energy dissipation [43, 44]. These properties make aramid fibres particularly effective in applications requiring high impact and ballistic resistance, such as aircraft leading edges, containment structures, and protective panels. Nevertheless, their relatively low compressive strength and susceptibility to moisture absorption limit their use as primary load-bearing reinforcements.

Increasing attention has also been directed toward natural fibre-reinforced composites as sustainable alternatives for non-primary aerospace structures. Natural fibres such as flax, hemp, and jute offer advantages including low density, renewability, and reduced environmental impact [45]. From an impact perspective, natural fibre composites demonstrate favourable energy absorption at low-to-moderate impact energies due to extensive fibre pull-out and interfacial debonding [46]. Although their mechanical performance remains inferior to glass and aramid systems, natural fibres are increasingly considered for secondary structures and hybrid configurations where sustainability considerations are prioritized.

To exploit the complementary advantages of different reinforcements, hybrid fibre composites have been widely investigated for impact-resistant aerospace applications. Hybrid systems combining glass, carbon, aramid, and/or natural fibres enable tailored mechanical responses, balancing stiffness, strength, and energy absorption [47]. In particular, glass-based hybrids have demonstrated improved damage tolerance and controlled failure progression compared to pure CFRPs systems, making them attractive candidates for crashworthy and impact-critical components.

Beyond structural performance, glass fibre-reinforced composites also play an important role in functional aerospace applications, particularly where electromagnetic transparency or shielding requirements are present. Unlike carbon fibres, which are electrically conductive and can interfere with electromagnetic wave transmission, GFRPs exhibit favourable dielectric properties, making them well suited for radomes, antenna covers, and electromagnetic shielding or isolation components [41, 48]. The combination of impact resistance, corrosion durability, and electromagnetic compatibility further reinforces the widespread adoption of glass fibre composites in aerospace structures subjected to complex multi-physics requirements.

In addition to fibre type, the architectural form of the reinforcement significantly influences impact behaviour. Unidirectional (UD) laminates provide high stiffness and strength along the fibre direction but exhibit pronounced anisotropy and limited damage tolerance under off-axis or transverse impact loading. As a result, impact damage in UD composites often localizes along preferred crack paths, leading to rapid stiffness degradation and reduced residual strength. Conversely, woven fabric reinforcements promote more isotropic in-plane properties and enhanced damage distribution due to fibre interlacing and load sharing between warp and weft directions [32]. Under impact loading, woven glass fibre composites typically exhibit more diffuse damage zones, increased delamination resistance, and improved energy absorption compared to UD configurations, making them particularly attractive for impact-critical aerospace structures.

In summary, glass fibre composites occupy a central position in impact-resistant aerospace structures due to their balanced mechanical performance, damage tolerance, and economic viability. When combined with carbon, aramid, or natural fibres in hybrid architectures, GFRPs provide a versatile design platform capable of meeting stringent crashworthiness and impact requirements. These characteristics are directly relevant to the material system investigated in the present work, where the impact behaviour and damage mechanisms of glass fibre-reinforced thermoplastic composites are of primary interest.

To clarify the distinct roles of the main fibre reinforcement systems employed in aerospace composites, Table 2.2 summarizes the key characteristics, impact behaviour, and typical applications of glass, carbon, aramid, and natural fibres. This comparison highlights the trade-offs between stiffness, damage tolerance, energy absorption, and functional performance that govern fibre selection for impact-resistant aerospace structures.

Table 2.2: Comparison of fibre reinforcement systems for impact-resistant aerospace composites.

Fibre type	**Key advantages**	**Impact behaviour**	**Typical aerospace applications**
GFRPs	Moderate stiffness, high strain-to-failure, good fracture toughness, low cost, electromagnetic transparency	Progressive damage with matrix cracking, fibre debonding, delamination and fibre fracture; good energy absorption under low- and intermediate-velocity impact [31, 41]	Fuselage skins, fairings, radomes, leading edges, secondary structures [41, 48]
CFRPs	Very high specific stiffness and strength; excellent load-carrying efficiency	Brittle fibre fracture and localized damage; limited plastic deformation; reduced residual strength after impact [1, 36]	Primary load-bearing structures such as wings, fuselage barrels, frames and spars [1]
AFRPs	Low density, high tensile strength, exceptional toughness and energy absorption	Progressive failure through fibre stretching, fibrillation and pull-out; excellent impact and ballistic resistance [43, 44]	Leading edges, containment systems, helicopter floors, ballistic and protective panels [44]
Natural fibres	Low density, renewability, reduced environmental impact, low cost	Energy dissipation through fibre pull-out, interfacial debonding and matrix cracking; effective at low-to-moderate impact energies [45, 46]	Secondary and non-primary structures; hybrid and sustainability-driven applications [47]

In addition to fibre type, the architectural configuration of the reinforcement has a significant influence on the impact response of composite laminates. Table 2.3 provides a comparative overview of unidirectional, woven, and hybrid reinforcement architectures, emphasizing their mechanical characteristics, dominant impact damage mechanisms, and typical aerospace applications.

Table 2.3: Comparison of reinforcement architectures and their influence on impact response of composite laminates.

Architecture	**Mechanical characteristics**	**Impact damage mechanisms**	**Typical use cases**
UD	High stiffness and strength along fibre direction; strong anisotropy	Localized matrix cracking, fibre fracture and delamination along preferred orientations; limited damage spreading under transverse impact [32, 34]	Primary load paths where stiffness efficiency dominates (e.g., spars, caps)
Woven fabrics	More isotropic in-plane properties; lower peak stiffness due to crimp	Distributed damage with enhanced delamination resistance; improved energy absorption through fibre interlacing and load redistribution [32, 49]	Impact-prone and damage-tolerant structures such as skins, panels and secondary components
Hybrid architectures	Tailored balance of stiffness, strength and toughness through multi-fibre stacking	Controlled and progressive failure; enhanced energy absorption and delayed perforation due to synergistic fibre interaction [39, 47]	Crashworthy and impact-critical aerospace components requiring optimized performance

2.1.5 Impact Behaviour of Thermoplastic Composite Materials

Thermoplastic composite materials have attracted increasing attention in aerospace applications subjected to impact loading due to their enhanced damage tolerance, fracture toughness, and strain-rate-dependent mechanical response compared to conventional thermoset systems. Under impact conditions, the matrix plays a critical role in governing damage initiation and propagation, particularly in terms of delamination resistance, crack arrest capability, and energy absorption mechanisms. As a result, thermoplastic composites exhibit a distinct impact response that warrants dedicated consideration at the material behaviour level.

One of the defining characteristics of thermoplastic matrices is their intrinsic ductility and capacity for plastic deformation. Unlike thermoset resins, which tend to fail in a brittle manner once the elastic limit is exceeded, thermoplastics can undergo significant plastic flow prior to fracture. Under impact loading, this behaviour promotes progressive damage development through matrix yielding, shear band formation, and crack blunting, which delays catastrophic failure and enhances energy dissipation [10, 11]. Consequently, thermoplastic composites often display reduced

delamination growth and improved post-impact residual strength when compared to equivalent thermoset-based laminates.

The strain-rate sensitivity of thermoplastic matrices further influences their impact response. At elevated loading rates, commonly associated with impact events, thermoplastics typically exhibit increased stiffness and strength, while maintaining a relatively high failure strain. This rate-dependent behaviour contributes to improved resistance against crack initiation and propagation, particularly in interlaminar regions [1, 2]. In contrast, thermoset matrices show limited rate sensitivity and tend to experience abrupt matrix cracking and delamination under similar loading conditions.

From a damage mechanics perspective, impact loading in thermoplastic composites activates a combination of intralaminar and interlaminar failure mechanisms. Intralaminar damage is primarily associated with matrix plastic deformation, fibre–matrix debonding, and fibre fracture, while interlaminar damage manifests as delamination driven by through-thickness stresses. Due to the higher fracture toughness of thermoplastic matrices, delamination growth is often more stable and distributed, leading to larger but less severe damaged areas [38, 39]. This behaviour is particularly advantageous for aerospace structures, where damage containment and residual load-bearing capability are critical design requirements.

The synergistic interaction between fibre reinforcement and thermoplastic matrix is especially relevant in glass fibre-reinforced thermoplastic composites. Glass fibres, characterized by relatively high strain-to-failure and good compatibility with thermoplastic matrices, promote progressive damage evolution under impact loading. Experimental studies have shown that glass fibre thermoplastic laminates dissipate impact energy through a combination of matrix plasticity, fibre breakage, and controlled delamination, resulting in enhanced damage tolerance compared to both carbon fibre thermoplastics and glass fibre thermosets [41, 42]. These characteristics make glass fibre thermoplastic composites particularly suitable for impact-prone aerospace components such as skins, panels, and leading-edge structures.

In addition to mechanical performance, thermoplastic composites offer manufacturing and operational advantages that indirectly influence impact behaviour. The absence of chemical curing reactions reduces residual stresses and manufacturing-induced defects, which can otherwise act as crack initiation sites under impact loading. Furthermore, the potential for reheating and reshaping enables improved repairability following impact events, contributing to extended service life and reduced maintenance costs [10, 23].

In summary, the impact behaviour of thermoplastic composites is characterized by enhanced damage tolerance, progressive failure mechanisms, and strong strain-rate sensitivity, distinguishing them from traditional thermoset-based systems. When combined with glass fibre reinforcements, thermoplastic matrices provide a balanced structural response that couples energy absorption with damage containment, aligning closely with the performance requirements of aerospace structures subjected to impact loading. The understanding of these material-level mechanisms provides a necessary foundation for the numerical damage modelling strategies introduced in subsequent chapters.

2.2 Crashworthiness

Crashworthiness plays a critical role in aircraft structural safety, as it addresses the response of the airframe under emergency and crash landing scenarios where severe transient loading and structural damage may occur. Unlike conventional strength and fatigue considerations, crashworthiness is primarily concerned with occupant protection, survivable structural behaviour, and post-impact evacuation capability under dynamic loading conditions. The increasing use of composite materials further complicates crashworthiness assessment due to their distinct damage mechanisms and energy dissipation behaviour compared to metallic structures.

This section presents the regulatory framework and certification philosophy governing aircraft crashworthiness, with particular emphasis on the requirements applicable to composite aircraft structures. The emergency landing conditions defined in Certification Specification (CS)-25, composite-specific certification guidance, and the role of experimental and numerical methods in crashworthiness assessment are discussed. The concepts introduced in this section provide the regulatory and conceptual basis for the interpretation of impact response and crashworthy behaviour examined in the subsequent chapters.

2.2.1 Crashworthiness in Aircraft Certification

In aircraft certification, crashworthiness refers to the capability of the aircraft structure and associated systems to protect occupants and maintain survivable conditions during emergency and crash landing events. Unlike conventional static strength or fatigue requirements, crashworthiness addresses the structural response under severe transient loading, where structural damage is anticipated but must not compromise occupant protection, evacuation capability, or post-impact survivability.

For transport-category aeroplanes, crashworthiness requirements are established within the certification framework defined by CS-25. In particular, CS 25 introduces emergency landing conditions that explicitly recognise the possibility of structural damage, provided that the aircraft is designed to protect occupants. These requirements emphasise the retention and positioning of

items of mass, the prevention of injury or fire hazards, and the avoidance of structural deformation that could impede rapid evacuation following an emergency landing [50].

Crashworthiness within CS-25 is therefore framed through a performance-based philosophy. Rather than prescribing specific structural concepts or energy-absorbing design solutions, the regulation defines loading environments and safety objectives that must be satisfied. Compliance may be demonstrated through an appropriate combination of testing, analysis, and validated numerical simulation, particularly where full-scale experimental demonstration is impractical. This approach underpins the increasing relevance of Certification by Analysis (CbA) for complex aircraft structures and advanced materials systems [51].

The growing adoption of composite materials in modern aircraft structures introduces additional challenges in the interpretation of crashworthiness requirements. Unlike metallic structures, which primarily dissipate energy through plastic deformation, composite structures absorb energy through progressive damage mechanisms such as matrix cracking, fibre fracture, delamination, and fragmentation. While these mechanisms can provide high specific energy absorption, their effectiveness depends strongly on material system and structural configuration. Consequently, aviation authorities require explicit consideration of impact behaviour and damage tolerance when demonstrating equivalent safety for composite aircraft structures, as reflected in composite certification guidance and supported by extensive crashworthiness research [52–54].

In this context, crashworthiness assessment forms a critical link between regulatory requirements and the experimental and numerical evaluation of structural impact response. The certification framework established in CS-25 provides the basis for interpreting crashworthy behaviour, while research on composite crashworthiness supports the development and validation of analysis methods suitable for advanced composite aircraft structures.

2.2.2 Emergency Landing Conditions and Dynamic Crashworthiness Requirements

Emergency landing conditions constitute the primary regulatory basis for crashworthiness assessment in transport-category aircraft. Within CS-25, these requirements define the loading environments and safety objectives associated with emergency and crash landing scenarios, explicitly recognising that structural damage may occur provided that occupant protection and survivable conditions are maintained [50].

CS-25.561 establishes the general emergency landing conditions and associated inertia load cases acting on the aircraft structure. These requirements address the structural response under prescribed accelerations and place particular emphasis on occupant protection and post-impact safety. Specifically, CS-25.561 requires that items of mass be restrained and positioned so as to avoid injury to occupants, creation of fire hazards, or obstruction of escape routes, and that seats and their supporting structures must not deform in a manner that would impede rapid evacuation

following an emergency landing [50]. As such, CS-25.561 links structural integrity directly to survivability and evacuation capability.

While CS-25.561 addresses emergency landing conditions primarily through inertial loading, CS-25.562 introduces explicit *dynamic* crashworthiness requirements related to occupant protection. This regulation defines dynamic test conditions for seats and restraint systems using prescribed changes in velocity and floor deceleration pulse characteristics. CS-25.562 specifies two principal dynamic impact conditions: a vertical condition, involving a change in downward velocity of not less than 10.7 m/s (35 ft/s) with peak floor deceleration occurring within 0.08 s and reaching at least 14 g; and a longitudinal condition, involving a change in forward velocity of not less than 13.4 m/s (44 ft/s) with peak floor deceleration occurring within 0.09 s and reaching at least 16 g [50]. These requirements demonstrate that crashworthiness in certification is governed not only by peak acceleration levels, but also by the time-dependent characteristics of the acceleration pulse.

Although CS-25.562 focuses on seats and restraint systems, the defined dynamic loading conditions have direct implications for the surrounding aircraft structure. The structural layout, stiffness distribution, and local deformation behaviour influence the acceleration environment transmitted to the cabin floor and occupant systems. Consequently, the assessment of structural response under impact loading is closely linked to crashworthiness performance, motivating the evaluation of transient acceleration response and deformation mechanisms in representative structural components [55].

For composite aircraft structures, the interpretation of emergency landing and dynamic requirements requires particular care. Composite materials dissipate energy through progressive damage mechanisms rather than through ductile yielding, and their dynamic response can be sensitive to laminate architecture and damage evolution. Regulatory guidance therefore requires explicit consideration of impact behaviour and damage tolerance when demonstrating compliance for composite structures, often supported by a combination of testing and validated numerical analysis [52, 53]. In this context, experimental impact tests and correlated numerical simulations provide a practical means of assessing crashworthy structural response in a manner consistent with the certification framework [51, 54].

In summary, CS-25 emergency landing and dynamic conditions establish the regulatory foundation for crashworthiness assessment in transport-category aircraft, defining performance objectives centred on occupant protection and survivability under transient loading. These requirements motivate the evaluation of structural impact response and acceleration characteristics, particularly for composite structures, and provide the basis for the crashworthiness metrics and assessment methods discussed in the following subsection.

2.2.3 Certification Philosophy for Composite Aircraft Structures

The certification of composite aircraft structures is governed by a dedicated philosophy that reflects the fundamental differences between composite and metallic materials. While metallic structures rely primarily on plastic deformation and ductility to accommodate overload and impact conditions, composite structures exhibit anisotropic behaviour and dissipate energy through progressive damage mechanisms such as matrix cracking, fibre fracture, delamination, and fragmentation. As a result, the certification of composite structures requires explicit consideration of damage initiation, damage progression, and residual structural capability.

In the Federal Aviation Administration (FAA) certification framework, this philosophy is formalised through Advisory Circular (AC)-20-107B, which provides guidance for the substantiation of composite aircraft structures. AC-20-107B establishes that composite structures must demonstrate a level of safety equivalent to that of conventional metallic designs, despite the differences in material behaviour. To achieve this, the guidance emphasises the need for damage tolerance, accounting for manufacturing defects, service-induced damage, and impact events, as well as the validation of analysis methods through appropriate testing [52].

Similarly, within the European Union Aviation Safety Agency (EASA) framework, Acceptable Means of Compliance (AMC)-20-29 defines acceptable means of compliance for composite aircraft structures. This guidance adopts a performance-based approach, allowing compliance to be demonstrated through a combination of tests, analysis, and validated numerical simulations. AMC-20-29 recognises that composite structures may experience local damage under impact or emergency landing conditions, provided that such damage does not lead to catastrophic failure and that the overall safety objectives defined in CS-25 are satisfied [53].

A central concept in composite certification is the *building-block approach*, which is adopted by both FAA and EASA guidance. This methodology involves the progressive validation of material behaviour and structural response through a hierarchy of tests, ranging from coupon-level characterisation to element, subcomponent, and full-scale structural evaluations. Numerical analysis plays an increasingly important role within this framework, enabling the interpretation of test results and the extension of validated behaviour to configurations that are not directly tested [51].

Impact behaviour occupies a particularly important position within the composite certification philosophy. Both FAA and EASA guidance require that the effects of impact damage be explicitly considered when demonstrating compliance, especially for structures exposed to operational threats or emergency landing conditions. Unlike metallic structures, where impact damage may be readily detectable and associated with local plastic deformation, composite impact damage can be highly localised and subsurface, necessitating dedicated assessment of damage tolerance and residual strength [52, 53].

From a crashworthiness perspective, the composite certification philosophy does not mandate specific structural energy-absorption concepts, but it requires that the dynamic response and damage behaviour of composite structures be understood and appropriately substantiated. This requirement motivates the use of experimental impact testing and correlated numerical simulations to characterise progressive damage mechanisms and transient structural response under crash-relevant loading conditions. Such approaches are widely adopted in crashworthiness research on composite aircraft structures and are consistent with the certification-by-analysis paradigm [54].

In summary, the certification philosophy for composite aircraft structures is centred on demonstrating equivalent safety through damage tolerance, validated analysis, and progressive substantiation across multiple structural scales. This philosophy provides the regulatory context for evaluating the crashworthiness and impact behaviour of composite structures and forms the basis for the crashworthiness metrics and assessment methodologies discussed in the following subsection.

2.2.4 Crashworthiness Metrics and Assessment Criteria

The assessment of crashworthiness requires quantitative metrics capable of characterising the dynamic structural response under emergency and crash landing conditions. Within the aircraft certification framework, crashworthiness metrics are not prescribed explicitly as design targets; instead, they are employed as assessment tools to evaluate whether structural response and occupant-related systems satisfy the safety objectives defined in CS-25. Consequently, crashworthiness metrics are used to interpret acceleration environments, energy dissipation behaviour, and damage progression in a manner consistent with regulatory intent.

From a certification perspective, the most critical crashworthiness indicators are those related to occupant protection and survivable conditions. CS-25 emphasises the importance of limiting hazardous accelerations transmitted to occupants, preserving evacuation capability, and preventing the release of injurious items of mass under emergency landing conditions [50]. While CS-25.562 defines explicit dynamic pulses for seat and restraint certification, the structural response of surrounding components influences the acceleration environment experienced by the occupant, motivating the use of structural-level crashworthiness metrics.

In crashworthiness research and structural assessment, several quantitative metrics are commonly employed to characterise impact and crash response. These metrics enable comparison between different structural concepts and materials and provide insight into the mechanisms governing energy dissipation and transient response. Among the most widely used metrics are absorbed energy, specific energy absorption, peak acceleration, mean acceleration over the impact duration, and contact or pulse duration [54, 56].

Absorbed energy represents the portion of the initial kinetic energy dissipated through structural deformation and damage during impact. For composite structures, this energy is dissipated

through progressive damage mechanisms such as matrix cracking, fibre fracture, delamination, and frictional interactions between damaged plies. Specific Energy Absorption (SEA), defined as the absorbed energy normalised by structural mass, is frequently used to evaluate the crashworthiness efficiency of lightweight structures, particularly in aerospace applications where mass penalties are critical [57, 58].

Acceleration-based metrics play a central role in crashworthiness assessment due to their direct connection with occupant loading and injury risk. Peak acceleration provides an indication of the maximum severity of the impact event, while mean acceleration and pulse duration describe the overall shape of the acceleration-time history. It is well established that crashworthiness performance cannot be assessed solely on the basis of peak values; the duration and temporal distribution of the acceleration pulse strongly influence the dynamic response of both the structure and occupant systems [55, 59].

In numerical and experimental investigations, these metrics are typically evaluated in combination to provide a comprehensive description of crashworthy behaviour. Energy-based metrics quantify the efficiency of energy dissipation, while acceleration-based metrics characterise the severity and temporal evolution of the dynamic response. The joint interpretation of these indicators enables assessment of structural crashworthiness in a manner consistent with certification objectives, without imposing artificial design targets not specified by the regulations.

Table 2.4 summarises the principal crashworthiness metrics used in aircraft structural assessment, together with their physical meaning and relevance to certification-oriented crashworthiness evaluation.

Table 2.4: Principal crashworthiness metrics used in aircraft structural assessment and their relevance to certification-oriented evaluation.

Metric	**Definition**	**Relevance to crashworthiness**
Absorbed Energy (EA)	Energy dissipated through structural deformation and damage during impact	Quantifies the capability of the structure to dissipate kinetic energy under crash loading conditions
Specific Energy Absorption (SEA)	Absorbed energy normalised by structural mass	Measures crashworthiness efficiency of lightweight structures, relevant for aerospace applications
Peak acceleration ($a_{\max}$)	Maximum acceleration recorded during the impact event	Indicator of impact severity; related to occupant loading and injury risk
Mean acceleration ($\bar{a}$)	Average acceleration over the effective impact or pulse duration	Represents overall severity of the acceleration environment transmitted to occupants
Pulse or contact duration (t_c)	Time interval over which significant acceleration or contact force is applied	Characterises the temporal distribution of the impact event; longer pulses generally reduce injury risk
Impact force or impulse	Resultant force or integral of force over time during impact	Provides insight into load transfer mechanisms and structural engagement during crash

2.2.5 Role of Impact Testing and Numerical Simulation in Crashworthiness Assessment

Within the aircraft certification framework, crashworthiness is demonstrated through a combination of experimental evidence and validated analytical or numerical methods. While CS-25 defines the safety objectives associated with emergency and crash landing conditions, it does not prescribe detailed methodologies for structural-level crashworthiness testing. Instead, compliance is typically established using representative tests supported by analysis, particularly when full-scale demonstrations are impractical [50]. Detailed descriptions of impact test methodologies and experimental procedures are presented in a dedicated literature review section; here, the focus is limited to the role of impact testing and numerical simulation within the crashworthiness and certification context.

Impact testing provides a practical means of reproducing transient loading conditions and assessing crashworthy structural response in a controlled manner. In aircraft structures, such tests are commonly performed at the panel, element, or subcomponent level to evaluate damage development, energy dissipation, and acceleration response under crash-relevant conditions. Within

the composite certification building-block approach, these tests contribute to the progressive substantiation of material and structural behaviour, while remaining compatible with certification requirements [51].

For composite aircraft structures, the role of impact testing is particularly important due to the complex and often non-visible nature of damage. Impact events may produce subsurface delamination, matrix cracking, and local fibre failure, which can significantly influence crashworthy response and residual structural capability. Experimental impact data therefore serve not only to characterise damage mechanisms, but also to provide reference benchmarks for the validation of numerical models used in crashworthiness assessment [54, 56].

Numerical simulation has become an essential component of crashworthiness assessment, supporting both design development and certification activities. Explicit finite element formulations are widely adopted for crashworthiness and impact analyses due to their ability to handle highly transient loading, complex contact interactions, and progressive material damage. When appropriately validated against experimental data, numerical models enable the evaluation of acceleration histories, energy dissipation, and damage evolution under crash-relevant loading scenarios, extending the applicability of experimental results to configurations that are not directly tested [51].

In the context of composite aircraft structures, the fidelity of numerical crashworthiness assessment depends on the accurate representation of intralaminar and interlaminar damage mechanisms, as well as on the correct modelling of contact and structural deformation. Studies on composite aircraft fuselage sections and energy-absorbing structures demonstrate that, when properly correlated, numerical simulations can reproduce key crashworthiness indicators such as acceleration-time histories, absorbed energy, and deformation patterns [55, 59].

Overall, impact testing and numerical simulation play complementary roles in the assessment of crashworthiness within the certification framework. Experimental tests provide physical evidence of crashworthy behaviour, while validated numerical models support interpretation, sensitivity studies, and extension to additional scenarios. This integrated approach is consistent with the performance-based philosophy of CS-25 and with composite-specific certification guidance, and it forms the basis for the experimental and numerical methodology adopted in the present work.

2.3 Impact Tests

LVI testing is widely employed to characterise the behaviour of composite laminates subjected to out-of-plane dynamic loads. Because such impacts can trigger complex internal damage while leaving only limited surface evidence, they represent a critical scenario for structural integrity assessment in aerospace and automotive applications. This section reviews the fundamental concepts associated with LVI, including damage mechanisms, characteristic response features, and

the standardised methodologies used to generate reproducible impact conditions. The discussion provides the scientific context needed to interpret the experimental results presented in Chapter 4 and to support the development and validation of the numerical models introduced in Chapter 5.

2.3.1 Importance of Impact Testing

LVI represents one of the most critical loading scenarios for composite structures because it can trigger complex internal damage while leaving only limited visible evidence on the impacted surface. This phenomenon, commonly referred to as Barely Visible Impact Damage (BVID), is widely reported in the literature and is well known to compromise the stiffness and residual strength of laminated composites even at relatively low impact energies [60]. Such internal damage may propagate through the thickness in the form of matrix cracking, fibre–matrix debonding, and delamination, making LVI a key concern for the aerospace, automotive, and wind-energy sectors.

The need for controlled and reproducible LVI tests is widely emphasised in experimental and numerical studies. These tests provide the force–time or acceleration–time histories necessary to characterise the structural response and identify damage events, including sudden load drops and stiffness degradation during impact [61]. LVI experiments also provide quantitative metrics—such as peak force, contact duration, and absorbed energy—that are essential for validating modelling strategies and calibrating material constitutive laws. Recent work involving both experimental testing and numerical simulation highlights the importance of accurate LVI characterisation to support predictive modelling of composite impact behaviour [62].

The relevance of LVI testing becomes even more pronounced for thermoplastic composite systems, which exhibit fundamentally different damage mechanisms and energy-absorption characteristics compared with conventional thermoset laminates. Thermoplastic matrices, such as PPS, demonstrate enhanced toughness and extensive matrix plasticity, which can delay delamination onset and redistribute impact loads [12]. These material-specific behaviours require dedicated experimental characterisation to ensure reliable modelling of impact events. As a result, LVI testing has become a central tool in the development and validation of numerical models for thermoplastic composites, forming the basis for the simulation work carried out in Chapter 5.

2.3.2 Typical Damage Mechanisms in Composite Laminates

LVI initiates a sequence of interacting damage mechanisms within composite laminates due to their layered architecture and anisotropic behaviour. The first damage events typically arise in the matrix, where transverse tension and shear stresses generate microcracks in the resin-rich regions or at fibre–matrix interfaces. These microcracks may form early during the rising portion of the impact response and can be observed experimentally as the initial visible damage on the impact surface, often preceding more severe internal degradation [60]. In woven systems, the interlaced yarn paths promote local stress concentrations that further facilitate matrix cracking and

interface separation, consistent with observations reported for various woven composite systems under impact [63].

As impact energy increases, matrix cracks may link and promote delamination between adjacent plies. Delamination is widely recognised as one of the dominant damage modes under LVI because it significantly alters the bending stiffness of the laminate and enables more extensive deformation. Experimental studies demonstrate that load drops or slope changes in the force–time curve often coincide with delamination onset or growth, reflecting sudden reductions in structural stiffness during impact [61]. Delamination propagation may continue during the unloading phase, especially under repeated or progressively higher impact energies, where additional interface separation becomes more pronounced [64].

At higher energy levels, intralaminar fibre damage becomes increasingly relevant. Depending on the local stress state, fibres may fail through tensile breakage, compressive kinking, or shear-driven rupture, contributing to a marked reduction in impact resistance. In thermoplastic composites such as PPS, matrix plasticity plays a central role in redistributing stresses prior to fibre failure, resulting in damage morphologies that combine extensive matrix cracking, stable delamination, and localised fibre breakage. Recent characterisation of PPS-based laminates under impact loading confirms the coexistence of these mechanisms and highlights the importance of capturing both intra- and interlaminar failure modes when modelling thermoplastic composite impacts [12]. Figure 2.2 illustrates the typical sequence of intralaminar and interlaminar damage modes observed during low-velocity impact, complementing the mechanisms described above.

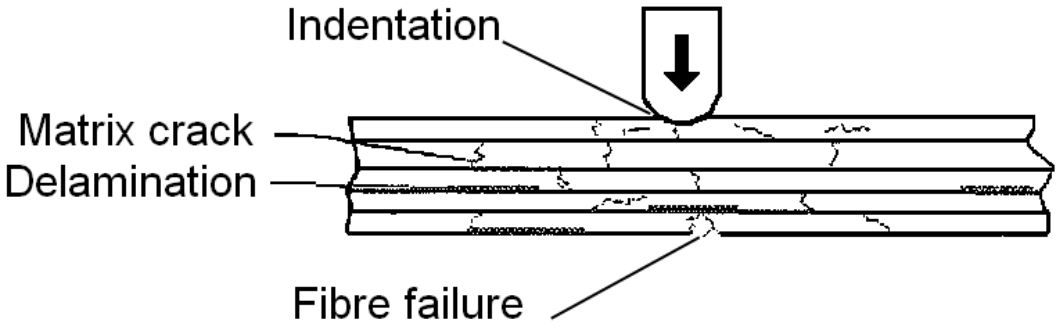


Figure 2.2: Damage modes observed during LVI [65].

2.3.3 Characteristics of Low-Velocity Impact Response

The structural response of composite laminates under low-velocity impact is commonly interpreted through characteristic features of the force–time, force–displacement, and energy–time histories. These curves reveal the sequence of physical events occurring during impact, including elastic deformation, damage initiation, damage propagation, and potential perforation. A typical force–time response begins with a nearly linear region representing the elastic bending of the

laminate, where the slope reflects the undamaged structural stiffness. Multiple studies report this initial linear segment followed by distinct nonlinear deviations as damage develops, often marked by the first load drop that signals matrix cracking or the onset of delamination [61]. Similar observations are reported for thermoplastic systems, including PPS laminates, where damage initiation manifests as a sudden reduction in load after the initial stiffness-controlled rise [12]. These load drops are widely used as indicators of the transition from elastic behaviour to progressive failure.

The force–displacement response provides additional insight into stiffness degradation, failure modes, and structural integrity. After the initial linear regime, the curves typically exhibit a peak load followed by a post-peak decline associated with through-thickness cracking, delamination growth, or fibre failure. For both thermoset and thermoplastic laminates, the enclosed area under the force–displacement curve corresponds to absorbed energy, while changes in curvature or slope between the displacement at peak load (PD_m) and the maximum displacement (PD_u) reflect the extent of damage within the laminate [61]. Thermoplastic composites often display smoother post-peak behaviour due to matrix plasticity, whereas more brittle thermoset laminates show sharper drops and higher stiffness slopes. In PPS laminates, distinct phases—elastic response, nonlinear damage growth, peak load, and unloading—are clearly observed and linked to the evolution of matrix cracking, delamination, and fibre breakage [12].

Energy–time curves provide a complementary perspective by decomposing the impact energy into elastic, absorbed, and dissipated components. The peak energy corresponds to the incident kinetic energy transferred to the plate, while the absorbed energy reflects the portion used to create damage, including matrix cracking, delamination, and fibre failure [61]. Experimental characterisation of PPS laminates shows that absorbed energy increases consistently with impact energy, accompanied by increases in damaged area and indentation depth [12]. Studies on energy-based damage metrics further highlight that different layups dissipate energy at characteristic rates, with the crack-failure energy curve stabilising after the onset of fibre breakage [66]. Together, the evolution of load, displacement, and energy reveals critical transitions in structural behaviour, enabling the identification of key phenomena such as barely visible impact damage, supercritical failure, and perforation. Figure 2.3 provides a schematic representation of the characteristic load–time, energy–time, and load–displacement responses typically observed in low-velocity impact events, illustrating the main features referenced throughout this section.

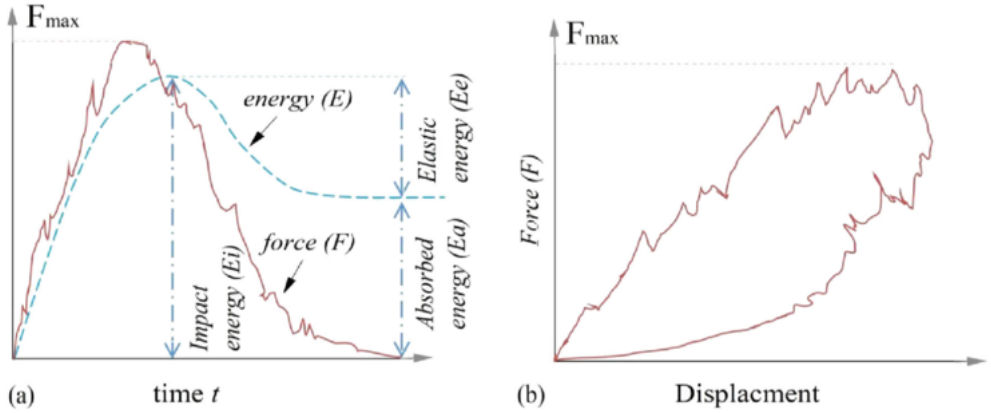


Figure 2.3: Schematic illustration of typical low-velocity impact responses. (a) Force–time and energy–time curves showing elastic loading, damage initiation, peak load, and energy partition. (b) Force–displacement curve highlighting stiffness, maximum displacement, and post-peak behaviour [67].

2.3.4 Standardised low-velocity impact tests: ASTM D7136/D7136M

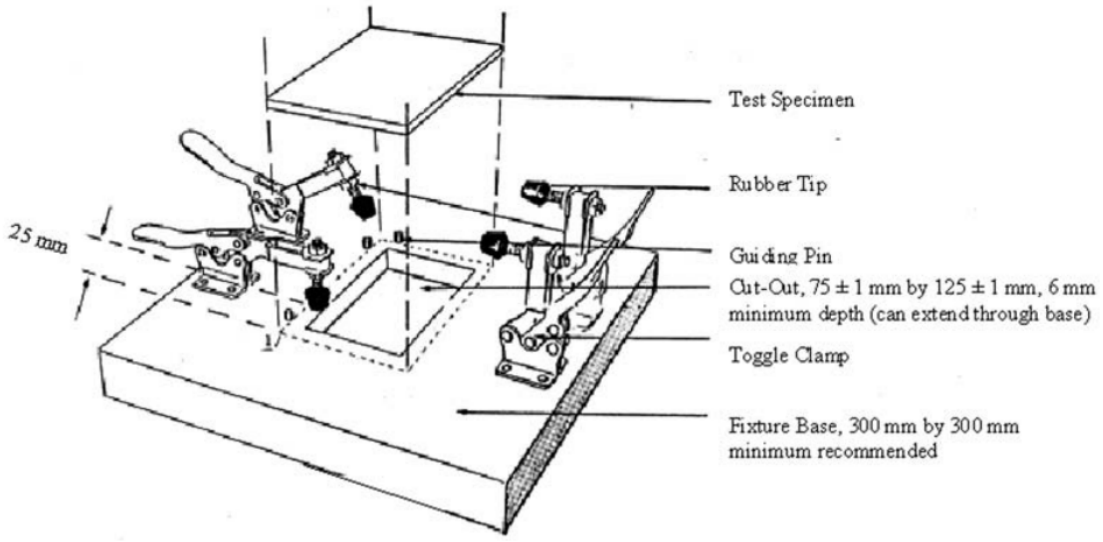
The LVI tests considered in this work are framed within the American Society for Testing and Materials (ASTM) D7136/D7136M standard for drop-weight impact on fibre-reinforced polymer-matrix laminates. This method defines a reference configuration in which a flat, balanced and symmetric laminate plate is impacted in the normal direction by a guided drop weight with a hemispherical striker. The objective is to characterise the damage resistance of the laminate in terms of dent depth, damage size and internal damage state, typically as a precursor to compression-after-impact tests performed in accordance with ASTM D7137/D7137M [68].

ASTM D7136/D7136M specifies both the laminate construction and the specimen geometry. For comparative material screening, the laminate thickness is restricted to a narrow band (4–6 mm, with a nominal target of 5 mm) and the stacking sequence is prescribed as a quasi-isotropic lay-up, either based on unidirectional plies $[45/0/-45/90]_{N\mathrm{S}}$ or on woven fabrics $[(\pm 45)/(0/90)]_{N\mathrm{S}}$, with N chosen to achieve the required thickness. The plate specimen is rectangular and centrally clamped over a rectangular cut-out in a rigid support fixture, whose opening and tolerances are also standardised to control boundary conditions and minimise the influence of edge effects on the damage pattern [68].

The apparatus section of the standard defines the key elements of the impact rig. The support fixture consists of a steel or aluminium plate at least 20 mm thick, with a central opening of 75 mm $\times$ 125 mm and four edge clamps located a fixed distance from the cut-out. The clamps, fitted with neoprene pads, must provide a minimum normal load to prevent slip while still allowing the plate to bend freely under impact. The impact device includes a rigid frame, a vertical guide, a rebound catcher and a drop weight. The reference impactor has a total mass of 5.5 kg and a 16 mm

diameter hardened hemispherical tup; alternative impactors may be used, but must be explicitly reported as deviations from the standard configuration [68].

In addition to geometry and hardware, ASTM D7136/D7136M provides detailed prescriptions for specimen preparation, conditioning, measurement accuracy and data acquisition. Thickness tolerances and machining requirements are given to avoid artificial notches or tapers that would increase scatter in the results. A minimum number of specimens per configuration is recommended, and the use of calibrated micrometers, depth gauges and balances with defined accuracy is required. Optional instrumentation is described for recording impact force and velocity with sufficient sampling rate and frequency response, enabling the reconstruction of force–time and force–displacement histories and the determination of characteristic quantities such as maximum force, absorbed energy and dent depth [68]. Figure 2.4 illustrates the reference ASTM D7136/D7136M support fixture.



NOTE—Clamp tip centered 6 mm from edge of cut-out.

Figure 2.4: Impact Support Fixture (SI Version) [68].

Table 2.5 summarises the principal geometric, mechanical, and procedural parameters defined by ASTM D7136/D7136M, complementing the configuration illustrated in Figure 2.4.

Table 2.5: Key configuration parameters of ASTM D7136/D7136M.

Item	ASTM D7136/D7136M specification
Specimen type	Balanced, symmetric laminate plate
Nominal thickness	4–6 mm (target 5 mm)
Recommended lay-up	$[45/0/-45/90]_{NS}$ (UD) or $[(\pm 45)/(0/90)]_{NS}$ (woven)
Plate geometry	Rectangular plate, centrally clamped over opening
Fixture opening	75 ± 1 mm $\times$ 125 ± 1 mm
Number of clamps	4 rubber-tipped edge clamps ($\geq$ 1100 N capacity)
Impactor mass	5.5 ± 0.25 kg
Striker geometry	Hemispherical tip, 16 ± 0.1 mm diameter, hardened steel
Primary outputs	Dent depth, damage size, force/energy metrics

2.3.5 Low-Velocity Impact Behaviour of Thermoplastic Composite Laminates

Thermoplastic composites exhibit distinct LVI behaviour compared with traditional thermoset systems due to their higher matrix toughness, better fibre–matrix adhesion and greater capacity for plastic deformation. Experimental work on PPS- and PEEK-based laminates consistently shows that thermoplastic matrices accommodate a larger portion of the impact-induced strain through viscoplastic deformation, resulting in more gradual stiffness degradation and delayed onset of catastrophic failure [12, 69]. This behaviour is reflected in the smoother load drops and extended non-linear regions observed in force–time and force–displacement curves, indicating that thermoplastic systems can redistribute stresses and resist crack propagation more effectively than their thermoset counterparts. The enhanced ductility of the matrix also provides increased tolerance to out-of-plane loading, improving damage containment especially in woven configurations.

The morphology of impact damage in thermoplastic laminates is strongly influenced by matrix plasticity and the more robust nature of fibre–matrix interfaces. Studies on PPS laminates under LVI report a combination of distributed matrix cracking, stable delamination growth and localised fibre rupture, with damage progressing through multiple stages rather than abrupt brittle failure [12]. In acrylic thermoplastic systems, such as Elium-based laminates, impact tests reveal a characteristic separation between elastic energy and absorbed energy components, with absorbed energy increasing more steadily as the matrix undergoes ductile deformation [61]. These mechanisms result in wider damage zones with smoother crack paths and reduced fragmentation compared with thermoset laminates, and are consistent with the energy–time signatures reported across several thermoplastic impact studies.

Beyond PPS and Elium systems, hybrid and high-performance thermoplastic composites such as PEEK also demonstrate sensitivity to matrix properties and operating conditions. Degradation

of PEEK due to thermal exposure has been shown to significantly reduce impact resistance, primarily by limiting the matrix's ability to undergo plastic deformation and support interlaminar stress redistribution [69]. Likewise, experimental and numerical analyses of non-crimp thermoplastic fibre architectures highlight the importance of fibre alignment and local strain concentrations, which influence both the initial stiffness response and subsequent delamination development during impact [62]. Collectively, these studies show that thermoplastic composites do not follow a single failure pattern under impact; instead, their behaviour is governed by a balance between matrix ductility, interlaminar strength, fibre architecture and environmental conditioning. Understanding these influences is essential for developing accurate constitutive models and for ensuring meaningful comparison with the experimental dataset used in Chapter 4. Table 2.6 summarises the qualitative differences between thermoset and thermoplastic laminates reported in the literature, emphasising the role of matrix ductility and interface behaviour in shaping the low-velocity impact response.

Table 2.6: Qualitative comparison between thermoset and thermoplastic composite behaviour under low-velocity impact.

Characteristic	**Thermoset Laminates**	**Thermoplastic Laminates**
Matrix behaviour	Brittle response with limited plasticity [60]	Ductile matrix with pronounced plastic deformation [12, 62]
Force–displacement curve	Sharp peak followed by abrupt load drop [60]	Gradual peak with smoother post-peak response [61, 62]
Energy absorption	Lower; damage tends to be localised [60]	Higher; plasticity and stable delamination increase dissipation [12, 61]
Damage morphology	Fragmented cracking and brittle delamination [60]	Stable delamination, matrix shear yielding, distributed cracks [12, 70]
Sensitivity to temperature	Generally low to moderate	Significant, particularly in PEEK systems [69]
Impact tolerance	Lower impact tolerance and sudden stiffness loss [60]	Higher tolerance with progressive, ductile damage accumulation [12, 62]

2.3.6 Literature Using Low-Velocity Impact for FE Model Validation

LVI tests are widely employed in the literature as a benchmark for the calibration and validation of Finite Element (FE) models of composite laminates. Typical validation quantities include the force-time, displacement-time, and energy-time responses, as well as permanent indentation, delamination area and damage morphology obtained through visual inspection or non-destructive evaluation. These experimental outputs are then used to tune intralaminar damage

models, cohesive-zone parameters, and contact formulations in explicit solvers such as Abaqus/-Explicit [61, 71].

A significant body of research uses LVI experiments to calibrate intralaminar damage laws. For example, Zhao et al. developed a detailed numerical model for woven Carbon Fiber Reinforced Thermoplastic (CFRTP) laminates in Abaqus/Explicit, employing a user-defined Vectorized User Material Subroutine (VUMAT) incorporating visco-plasticity and multiple damage variables; the model parameters were calibrated to reproduce the experimentally measured force-time history, peak load and damage progression [70]. Similar approaches are reported for other thermoplastic and thermoset systems, where Continuum Damage Mechanics (CDM) or Hashin-type criteria are adjusted to match the experimentally observed stiffness degradation, load drops and residual deformation [62, 69].

LVI tests also form the basis for identifying Cohesive Zone Model (CZM) parameters governing interlaminar behaviour. Shang et al. used a drop-weight impact campaign to calibrate a layered CZM model, tuning the mixed-mode fracture energy and interfacial strengths until the simulated delamination area and absorbed energy matched the experiments [66]. Similar cohesive interface strategies have been used to predict delamination initiation and propagation in woven and unidirectional laminates under impact loading [12, 63]. These studies demonstrate that LVI experiments provide essential reference data for establishing accurate interlaminar constitutive behaviour.

Overall, the studies reviewed here demonstrate that LVI data provide one of the most sensitive and informative experimental bases for evaluating numerical formulations that combine intralaminar damage models with cohesive-zone representations of delamination. In this context, low-velocity impact tests are frequently used as a validation strategy for assessing the predictive capability of material models and failure criteria under dynamic loading conditions. Once the modelling framework has been validated at the laminate level, the calibrated constitutive formulations can subsequently be applied to the analysis of more complex composite structures subjected to similar impact loading scenarios.

2.3.7 Reference Experimental Dataset from Martina D'Aiello

A comprehensive experimental campaign following the ASTM D7136 standard was conducted by Martina D'Aiello at Politecnico di Milano and is documented in a dedicated Technical Report [72]. This campaign generated a consistent set of LVI tests on Glass/PPS laminates, including the measured initial velocities, processed acceleration–time histories, and post-impact assessments. The standardised drop-weight configuration and controlled boundary conditions ensure the reliability and repeatability of the resulting dataset.

These experiments constitute the primary empirical reference for the present dissertation. To maintain a clear distinction between prior work and the contributions of this thesis, the detailed

experimental configurations and results are reported in Chapter 4. The role of this section is therefore limited to positioning the dataset within the broader context of LVI research and highlighting its relevance as a consistent benchmark for later numerical developments.

2.3.8 Summary of Impact Tests

In summary, the literature establishes LVI testing as a rigorous framework for characterising the dynamic response and failure mechanisms of composite laminates. The collective insights regarding matrix-dominated cracking, delamination processes, energy absorption behaviour and the influence of test configuration provide a comprehensive foundation for understanding the mechanical phenomena explored throughout this thesis. The standardised methodology defined by ASTM D7136 and the extensive body of work employing LVI tests for constitutive model calibration guide the modelling strategies adopted in the numerical sections that follow. Chapter 4 builds on this foundation by presenting the specific experimental dataset used in this thesis, while Chapter 5 develops the corresponding explicit FE models and validation procedures.

2.4 Dynamic Explicit Finite Element Modeling

The numerical simulation of impact events on composite structures presents significant challenges due to the highly transient nature of the loading, the presence of complex contact interactions, and the progressive development of material damage and failure. Accurate prediction of impact response therefore requires a numerical framework capable of handling strong nonlinearity, high-frequency wave propagation, and rapid stiffness degradation without loss of numerical stability. For these reasons, explicit finite element formulations are widely adopted for the analysis of impact and crash-related problems in aerospace structures.

This section presents the theoretical and numerical background of the explicit dynamic formulation implemented in ABAQUS/EXPLICIT, with particular emphasis on its suitability for composite impact simulations. The fundamental principles of explicit time integration, stability constraints, contact modelling, and progressive damage representation are discussed, together with their implications for numerical efficiency and modelling strategy. The concepts introduced in this section provide the basis for the numerical implementation and simulation methodology adopted in the present work.

2.4.1 Motivation for Using the Explicit Method in Composite Impact Simulations

The numerical simulation of impact events on composite laminates requires a computational framework capable of capturing highly transient dynamic behaviour, strongly nonlinear material degradation, complex contact interactions, and the propagation of failure across multiple length scales. The ABAQUS/EXPLICIT solver is widely established in the literature as an efficient and robust strategy for analysing both low-velocity and high-velocity impacts in composite structures

[73, 74]. Its suitability derives from the characteristics of explicit time integration, the nature of wave propagation in laminated media, and the severe numerical challenges posed by progressive damage and delamination.

In an impact event, the structural response occurs over extremely short durations, typically on the order of 10^{-4}–10^{-3} s. During this interval, the response is dominated by inertial forces rather than stiffness-controlled behaviour. Explicit solvers integrate the equations of motion using a central-difference scheme, advancing the solution without iterative equilibrium corrections and without forming or factorising a global stiffness matrix, as documented in the Abaqus Theory Guide [20] As a result, explicit dynamics avoids the repeated formation and inversion of the global stiffness matrix that characterises implicit schemes. This feature makes the explicit formulation inherently stable and efficient for problems involving rapidly changing contact conditions and high strain-rate deformation.

A fundamental requirement of explicit dynamics is that the time increment remains below a critical threshold governed by the smallest characteristic length of the mesh and the material wave speed. The stable time increment can be approximated by Equation 2.1,

$$\Delta t_{\text{stable}} \leq \frac{\ell_{\min}}{c} = \frac{\ell_{\min}}{\sqrt{E/\rho}}, \tag{2.1}$$

where $\ell_{\min}$ is the characteristic element size and c is the dilatational wave speed. This expression is discussed explicitly in [73], where the authors show that for composite laminates with small ply thicknesses—and especially when cohesive interfaces are introduced—the element size in the thickness direction severely reduces the stability limit if solid elements are used. By employing shell-based or surface-based formulations, the detrimental effect of the ply thickness on Δt is eliminated, yielding significantly more efficient simulations. This relationship between element discretisation and time increment explains why explicit analysis is the de facto choice in impact studies involving fine through-thickness resolution.

Besides its temporal characteristics, explicit dynamics is also well suited for *progressive damage modelling*. Composite failure involves a combination of intralaminar mechanisms (matrix cracking, fibre breakage, in-plane shear degradation) and interlaminar mechanisms (delamination). These mechanisms induce abrupt stiffness changes that frequently cause convergence difficulties in implicit solvers, especially when stiffness degradation drives the tangent matrix toward singularity. In explicit dynamics, stiffness degradation does not compromise numerical stability, since accelerations are computed directly from nodal forces at each time increment [20]. This feature is critical when modelling failure criteria such as Hashin-based intralaminar degradation or cohesive-zone-driven delamination, which produce strong local softening and element deletion. Numerical studies on progressive damage, such as the progressive damage formulation proposed by Lapczyk and Hurtado [15] for the 2D Hashin criterion implemented in Abaqus, demonstrate

that the finite element response under strong softening remains numerically well behaved when implemented in an explicit formulation.

Delamination modelling in particular benefits from the explicit approach. Cohesive elements or cohesive surfaces introduce additional stiffness and damage evolution laws at the interfaces between plies. These interfaces often possess very small characteristic dimensions and exhibit highly nonlinear traction–separation responses. Studies such as [73] show that delamination initiation and growth can cause rapid oscillations in the internal force field and sharp stiffness gradients, which explicit dynamics handles robustly [20]. Moreover, contact between the interfaces after decohesion may control the stable time increment, as Abaqus assigns surface density to cohesive surfaces and adjusts nodal masses accordingly [73]. Therefore, explicit solvers provide both robustness and accuracy when modelling interlaminar fracture.

Another advantage of explicit dynamics is its ability to handle complex contact conditions. During impact, the striker undergoes rapid changes in contact state, ranging from initial indentation to separation and rebound. Explicit penalty-based contact algorithms are inherently stable under such conditions and do not require iterative enforcement [21]. The review of impact simulations in [73] highlights the importance of reliable contact computation in capturing force history, peak impact load, and energy transfer.

From a computational-efficiency perspective, explicit dynamics scales exceptionally well on modern parallel architectures. The NASA technical report [75] shows that explicit solvers exhibit near-linear scaling when distributed across a large number of processors due to the absence of global matrix factorisation. Although the small stable time increment demands many time steps, the high scalability of explicit models—combined with the short total physical duration of impact events—makes the method computationally feasible.

However, the efficiency of explicit modelling is strongly affected by mesh design. High spatial resolution near the impact region improves the prediction of delamination patterns and local failure but increases computational cost by reducing Δt_{stable}. This trade-off is well documented in low-velocity impact studies and motivates local refinements rather than uniform fine meshes.

Mass scaling is another mechanism often considered to increase efficiency by artificially increasing Δt_{stable}. The study in [76] shows that, when applied judiciously, mass scaling can reduce simulation time by more than 90% without significantly altering force-displacement or damage responses. Nevertheless, the authors emphasise that excessive mass scaling may distort inertia-controlled behaviour and is unsuitable for high-velocity impact, where wave propagation accuracy is essential.

In summary, the explicit integration scheme in ABAQUS/EXPLICIT is ideally suited for composite impact simulations due to:

- its ability to resolve short-duration, inertia-driven events without iterative equilibrium;
- its robustness in the presence of progressive damage, element deletion, and strong stiffness degradation;
- its compatibility with cohesive-zone and intralaminar damage models;
- its stable and efficient treatment of evolving contact conditions;
- and its favourable computational scaling on modern high-performance systems.

Given that the present work involves high-frequency wave propagation, delamination via cohesive interfaces, severe contact changes, and progressive Hashin-type intralaminar degradation, the explicit formulation provides the numerical robustness and accuracy required for reliable simulation of impact on glass/PPS composite laminates.

2.4.2 Central-Difference Time Integration Scheme

The explicit formulation implemented in ABAQUS/EXPLICIT is based on a central-difference time integration scheme, as described in the Abaqus Theory Guide [20], which advances the solution of the equations of motion explicitly in time. The semi-discrete form of the dynamic equilibrium equation can be written as

$$\mathbf{M}\,\ddot{\mathbf{u}}(t) = \mathbf{F}_{\text{ext}}(t) - \mathbf{F}_{\text{int}}(t), \tag{2.2}$$

where $\mathbf{M}$ is the (lumped) mass matrix, $\ddot{\mathbf{u}}$ is the nodal acceleration vector, and $\mathbf{F}_{\text{ext}}$ and $\mathbf{F}_{\text{int}}$ are the external and internal force vectors, respectively. In explicit dynamics, the mass matrix is diagonal, allowing the accelerations to be computed directly without solving a coupled system of equations [20].

The central-difference scheme approximates the acceleration and velocity at time t as

$$\ddot{\mathbf{u}}^{t} = \frac{\mathbf{u}^{t+\Delta t} - 2\mathbf{u}^{t} + \mathbf{u}^{t-\Delta t}}{\Delta t^{2}}, \tag{2.3}$$

$$\dot{\mathbf{u}}^{t+\frac{1}{2}} = \frac{\mathbf{u}^{t+\Delta t} - \mathbf{u}^{t}}{\Delta t}. \tag{2.4}$$

Using these expressions, nodal accelerations are first evaluated from the force imbalance, after which velocities and displacements are updated sequentially. This staggered update procedure eliminates the need for iterations and ensures computational efficiency for highly nonlinear transient problems.

A defining characteristic of the central-difference method is that it is *conditionally stable*. Stability is guaranteed only if the time increment Δt is smaller than a critical value governed by the smallest element in the mesh and the wave propagation speed of the material, as defined in the Abaqus Theory Guide [20]. A commonly adopted estimate for the stable time increment is

$$\Delta t_{\text{stable}} \leq \frac{\ell_{\text{min}}}{c}, \tag{2.5}$$

where ℓ_{min} is the characteristic length of the smallest finite element and c is the dilatational wave speed, defined as

$$c = \sqrt{\frac{E}{\rho}}, \tag{2.6}$$

with E the elastic modulus and ρ the material density. This stability criterion is explicitly discussed in composite impact studies using ABAQUS/EXPLICIT, where it is shown that the smallest element dimension dominates the computational cost of the simulation [73].

In composite laminates, the stable time increment is particularly sensitive to the discretisation strategy adopted through the thickness. When solid elements or continuum shell elements are used, the small ply thickness directly reduces ℓ_{min}, resulting in extremely small stable time increments. González et al. [73] demonstrate that this effect can dominate the computational cost when modelling delamination using three-dimensional elements. As a consequence, shell-based formulations combined with cohesive surfaces or interface elements are frequently preferred, as they decouple the stability limit from the ply thickness while still allowing accurate representation of interlaminar failure.

The influence of the time increment extends beyond computational efficiency. If Δt approaches the stability limit, numerical dispersion and high-frequency noise may contaminate the solution, particularly in the presence of wave propagation and impact-induced oscillations. For this reason, ABAQUS automatically selects a conservative fraction of Δt_{stable} during the analysis. This choice improves numerical robustness but further increases the total number of time increments required.

In practice, ABAQUS automatically selects a conservative fraction of the estimated stable time increment to ensure numerical robustness, particularly in the presence of contact and material nonlinearity [20]. Contact interactions can further restrict the stable time increment. In explicit analyses involving penalty-based contact formulations, the effective contact stiffness and the mass associated with contact surfaces influence the local wave speed. González et al. [73] report that contact definitions—especially when cohesive surfaces and surface-based interactions are employed—may control the stable time increment rather than the bulk material properties. This observation is particularly relevant for impact simulations, where repeated opening and closing of contact interfaces occurs.

The central-difference scheme also interacts strongly with damage and stiffness degradation. As intralaminar damage variables evolve or cohesive tractions soften, the local stiffness decreases, modifying internal force transmission and wave propagation characteristics. Unlike implicit solvers, the explicit formulation remains stable under these conditions, as no global equilibrium iterations are required [20]. Enhanced progressive damage models implemented in explicit frameworks have demonstrated stable behaviour even under severe softening and element deletion [77].

From a computational perspective, the requirement of very small time increments is mitigated by the excellent parallel scalability of explicit solvers. The NASA technical report in [75] shows that explicit finite element analyses scale efficiently on modern high-performance computing architectures, as the absence of global matrix factorisation allows near-linear speedup with increasing processor count. This characteristic makes explicit dynamics particularly suitable for large-scale impact models with local mesh refinement.

In summary, the central-difference integration scheme provides a numerically robust and efficient framework for simulating impact on composite laminates. Its conditional stability directly links mesh design, material properties, contact definitions, and computational cost, making careful control of element size and interface modelling essential. These considerations strongly influence the modelling strategy adopted in the present work and are revisited in detail in the numerical implementation chapter.

2.4.3 Stability, Mass Scaling, and Numerical Efficiency in Explicit Impact Simulations

The explicit time integration scheme employed in ABAQUS/EXPLICIT is conditionally stable, meaning that numerical stability is guaranteed only if the time increment remains below the critical limit dictated by the smallest element dimension and the material wave speed. While this constraint ensures numerical robustness, it also represents the primary source of computational cost in explicit impact simulations. As discussed in the Abaqus Theory Guide, the stable time increment is evaluated automatically throughout the analysis and updated as material properties, contact conditions, or element dimensions evolve [20].

In composite impact simulations, stability constraints are particularly severe due to the presence of thin plies, refined meshes in the impact region, and interface formulations such as cohesive elements or cohesive surfaces. As shown by González et al. [73], the smallest characteristic length often corresponds either to the through-thickness dimension of continuum elements or to the effective contact stiffness associated with interface definitions. This coupling between mesh resolution, interface modelling, and stable time increment means that explicit impact analyses are inherently sensitive to discretisation choices.

From a numerical perspective, reduced-integration elements are commonly adopted in explicit dynamics to mitigate computational cost. However, reduced integration introduces the possibility of hourglass modes, which represent non-physical zero- energy deformation patterns. ABAQUS/-EXPLICIT employs several hourglass control strategies to suppress these modes, and the energy associated with hourglass deformation is explicitly tracked during the analysis [20]. It is generally recommended that the hourglass energy remains a small fraction of the total internal energy in order to ensure the physical reliability of the results.

The influence of hourglass control on impact simulations has been investigated systematically by Bartan Kumbasar and Gider [76]. Their study demonstrates that excessive hourglass energy can significantly alter force histories and damage evolution, particularly at higher impact velocities. The authors report that while hourglass effects are less pronounced in low-velocity impact scenarios, they become increasingly important as impact speed and mesh distortion increase. These findings underline the necessity of monitoring hourglass energy throughout explicit composite impact simulations.

Given the restrictive stability limit imposed by fine meshes, mass scaling is often considered as a strategy to improve computational efficiency. Mass scaling artificially increases nodal masses, thereby reducing the effective wave speed and increasing the stable time increment. ABAQUS supports both fixed and automatic mass scaling techniques, which may be applied globally or selectively to specific regions of the model [20]. However, the use of mass scaling must be carefully justified in dynamic problems dominated by inertia.

The effects of mass scaling on composite impact simulations are examined in detail in [76]. The authors show that moderate mass scaling can reduce computational time by more than 90% while preserving global response metrics such as impact force and displacement histories in low-velocity impact cases. Nevertheless, they emphasise that excessive mass scaling distorts inertial effects and alters damage evolution, particularly when wave propagation plays a dominant role. For high-velocity impact problems, where accurate representation of stress waves and dynamic interactions is essential, mass scaling is therefore generally discouraged.

In addition to stability and mass scaling considerations, the overall numerical efficiency of explicit impact simulations is strongly influenced by parallel scalability. Unlike implicit solvers, explicit dynamics does not require global matrix assembly or factorisation, resulting in highly favourable scaling on modern parallel computing architectures. The NASA technical report [75] demonstrates that explicit finite element analyses exhibit near- linear speedup with increasing processor count, provided that the workload is sufficiently large. This property makes explicit dynamics particularly suitable for high-resolution impact models involving local mesh refinement and complex contact definitions.

Despite its favourable scalability, explicit dynamics remains computationally demanding when extremely small time increments are required. As a result, a balanced modelling strategy is essential. Local mesh refinement should be restricted to regions where high stress gradients and damage evolution are expected, such as the impact and crack propagation zones, while coarser meshes may be adopted elsewhere. This approach minimises the reduction in stable time increment while preserving the accuracy of the predicted damage mechanisms [73].

In summary, numerical stability and efficiency in explicit composite impact simulations are governed by a combination of mesh design, element formulation, contact definitions, and solver parameters. While mass scaling and reduced- integration elements can significantly improve computational performance, their use must be carefully controlled to avoid compromising physical fidelity. Explicit monitoring of stability limits, hourglass energy, and global energy balance is therefore essential to ensure the reliability of explicit impact analyses. These considerations directly inform the modelling strategy adopted in the present work and are revisited in detail in the numerical implementation chapter.

2.4.4 Contact Modelling in Explicit Composite Impact Simulations

Contact modelling plays a central role in the numerical simulation of impact events, as the interaction between the impactor and the composite structure governs the load transfer, force history, and energy dissipation mechanisms. In explicit dynamics, contact conditions evolve rapidly, often involving repeated opening, closing, sliding, and rebound. The robustness of the contact algorithm is therefore a critical requirement for reliable impact simulations. The contact formulations implemented in ABAQUS/EXPLICIT are specifically designed to address such highly transient interactions and are documented in detail in the Abaqus Analysis User's Guide [21].

In ABAQUS/EXPLICIT, contact interactions are typically enforced using penalty-based formulations. In this approach, contact constraints are satisfied approximately by introducing penalty forces proportional to the penetration between contacting surfaces. The magnitude of the contact force is governed by an effective contact stiffness, which is automatically determined based on material properties, element dimensions, and surface definitions [21]. Unlike constraint- based formulations, penalty methods do not require iterative enforcement of contact conditions and are therefore well suited for explicit time integration.

The penalty-based contact algorithm offers several advantages for impact simulations. First, it remains stable under large relative velocities and rapidly changing contact states, which are characteristic of both low- and high-velocity impact scenarios. Second, it allows for efficient treatment of separation and rebound without numerical instabilities. These features make penalty contact the preferred option for composite impact analyses, as confirmed by numerous numerical studies employing ABAQUS/EXPLICIT [73, 74].

In composite impact problems, contact modelling is required not only between the impactor and the laminate but also between internal interfaces following delamination. When cohesive elements or cohesive surfaces are employed, complete decohesion may lead to local separation between plies, followed by re-contact under compressive loading. Explicit contact algorithms can naturally handle this behaviour without additional constraints or convergence difficulties [20]. This capability is essential for accurately capturing post-impact behaviour and internal damage interactions.

The definition of contact properties directly influences numerical stability and computational efficiency. In penalty-based contact, the effective contact stiffness affects the local wave speed and may therefore restrict the stable time increment. González et al. [73] report that in explicit composite impact simulations, contact definitions—particularly when cohesive surfaces are present—can control the global stability limit rather than the bulk material properties. This highlights the strong coupling between contact modelling and time integration in explicit dynamics.

Normal and tangential contact behaviours must be defined carefully to ensure physical accuracy. In impact simulations, normal contact is typically modelled using a hard contact formulation, preventing penetration while allowing separation once contact forces vanish. Tangential behaviour is often represented using frictionless or low- friction assumptions to avoid excessive constraint of in-plane motion during impact [21]. The choice of tangential behaviour influences the force history and energy dissipation, particularly in oblique impact scenarios.

The treatment of contact is further complicated by the presence of reduced-integration elements and local mesh refinement. Contact algorithms in ABAQUS/EXPLICIT are fully compatible with reduced-integration shell and continuum shell elements, but the accuracy of contact force transmission depends on mesh density and surface discretisation. Poorly resolved contact surfaces may lead to spurious oscillations or numerical noise, especially in high-frequency impact events [20].

From a numerical perspective, the explicit contact formulation interacts strongly with damage evolution and stiffness degradation. As material damage progresses or cohesive interfaces soften, local changes in contact conditions may occur, including the activation of new contact pairs between previously bonded surfaces. The explicit solver accommodates these transitions seamlessly, as contact constraints are enforced locally at each time increment without reliance on global equilibrium iterations [21]. This feature is particularly advantageous for composite impact simulations involving extensive delamination and fragmentation.

In summary, contact modelling in ABAQUS/EXPLICIT relies on robust penalty-based algorithms that are well suited for highly transient impact phenomena. The interaction between contact definitions, stability limits, and damage evolution plays a decisive role in both the accuracy and efficiency of composite impact simulations. Careful selection of contact properties and surface

discretisation is therefore essential, and these aspects are addressed explicitly in the numerical modelling strategy adopted in the present work.

2.4.5 Progressive Damage and Element Deletion in Explicit Composite Impact Analysis

The accurate numerical representation of composite impact phenomena requires a modelling framework capable of capturing the progressive initiation and evolution of damage mechanisms at both intralaminar and interlaminar levels. Impact loading induces highly localised stress concentrations, rapid stiffness degradation, and complex interactions between multiple failure modes, including matrix cracking, fibre breakage, shear-driven damage, and delamination. These characteristics pose significant numerical challenges, particularly when stiffness softening and material separation occur. In this context, the explicit formulation implemented in ABAQUS/EXPLICIT provides substantial advantages for modelling progressive damage and failure in composite structures.

Intralaminar damage models describe failure mechanisms occurring within individual plies. Commonly employed criteria, such as Hashin-type formulations, distinguish between fibre and matrix failure under tensile and compressive loading conditions. Once a damage initiation criterion is met, material stiffness is progressively degraded according to an evolution law, often driven by fracture energy dissipation. This progressive stiffness reduction introduces strong nonlinearity into the constitutive response. In implicit solvers, such softening behaviour frequently leads to convergence difficulties due to loss of positive definiteness of the tangent stiffness matrix. In contrast, explicit solvers update nodal accelerations directly from the internal force imbalance, allowing the simulation to remain numerically stable even as local stiffness approaches zero [20, 77].

Interlaminar damage is commonly modelled using CZM, which represent the traction–separation behaviour of ply interfaces. Cohesive elements or cohesive surfaces enable both initiation and propagation of delamination without the need for predefined cracks. The traction–separation law introduces an initial elastic response followed by progressive softening governed by fracture energy parameters. During impact, delamination growth is often driven by mixed-mode loading and may evolve rapidly. The explicit formulation is particularly well suited for cohesive modelling, as it can robustly handle abrupt changes in interface stiffness and localcontact conditions following complete decohesion [20, 73].

Element deletion is frequently employed to represent complete material failure once a critical damage state is reached. In intralaminar damage models, element deletion may be triggered when damage variables exceed a prescribed threshold, preventing non-physical stress transfer through fully failed material. Similarly, cohesive elements may be deleted once the traction capacity is exhausted and the fracture energy fully dissipated. In implicit formulations, element removal can

introduce severe convergence issues due to sudden topology changes. Explicit dynamics avoids this limitation, as element deletion does not affect the stability of the integration scheme [20]. This capability is essential for simulating perforation, fragmentation, and extensive delamination in impact scenarios.

The interaction between intralaminar and interlaminar damage mechanisms further highlights the suitability of explicit dynamics. Impact-induced delamination often alters the stress distribution in adjacent plies, promoting sublaminate buckling, matrix cracking, and fibre failure. Conversely, intralaminar damage can modify load transfer paths and influence delamination growth. Explicit solvers can naturally capture these interactions without the need for special numerical treatments, as damage evolution and contact activation are resolved locally at each time increment. Numerical studies of composite impact consistently rely on explicit formulations to model such coupled damage processes [73, 74].

Mesh dependency is an inherent challenge in progressive damage modelling, particularly when strain-softening laws are employed. Fracture-energy-based damage evolution reduces, but does not eliminate, sensitivity to element size. Recent developments in enhanced progressive damage models emphasise the importance of characteristic length definitions and mesh objectivity to obtain physically meaningful results [77]. In explicit impact analyses, mesh refinement is therefore typically localised to regions of expected damage initiation and propagation, while coarser meshes are adopted elsewhere to maintain computational efficiency.

Finally, the combination of progressive damage modelling, cohesive interfaces, and element deletion imposes stringent requirements on energy balance monitoring. In explicit dynamics, it is essential to ensure that artificial energy contributions, such as hourglass energy or excessive numerical damping, remain small compared to the physical internal and kinetic energies of the system [20]. Proper energy tracking provides confidence that damage evolution and failure processes are governed by physical mechanisms rather than numerical artefacts.

In summary, the explicit formulation provides a robust numerical framework for modelling progressive intralaminar and interlaminar damage in composite impact simulations. Its ability to accommodate stiffness degradation, cohesive softening, and element deletion without convergence loss makes it particularly suitable for capturing the complex failure mechanisms observed in impacted composite laminates. These characteristics form the basis for the damage modelling strategy adopted in the present work.

2.4.6 Synthesis and Modelling Implications

The explicit dynamic formulation implemented in ABAQUS/EXPLICIT provides a numerically robust and computationally efficient framework for simulating impact phenomena in composite laminates. As discussed in the preceding subsections, the explicit approach is particularly well

suited for problems characterised by short time scales, inertia-dominated response, complex contact interactions, and severe material nonlinearity.

The central-difference time integration scheme ensures stable and efficient resolution of transient dynamics, provided that the stability condition imposed by the smallest element dimension and material wave speed is respected. While this constraint leads to small time increments and increased computational cost, it also enables accurate capture of wave propagation, force histories, and damage initiation under impact loading. The strong coupling between mesh design, contact definitions, and numerical stability necessitates careful discretisation strategies, especially in regions of expected damage and delamination.

Explicit dynamics also offers decisive advantages for modelling progressive damage and failure in composite structures. The ability to accommodate intralaminar stiffness degradation, cohesive softening, and element deletion without convergence loss makes the explicit formulation particularly suitable for impact simulations involving complex failure mechanisms. Furthermore, the robustness of penalty-based contact algorithms ensures stable treatment of rapidly evolving contact conditions, including impactor rebound and post-delamination ply interaction.

From a computational standpoint, the excellent parallel scalability of explicit solvers mitigates the cost associated with small stable time increments, enabling large-scale high-resolution simulations on modern high-performance computing architectures. However, numerical efficiency remains strongly dependent on appropriate mesh refinement, controlled use of reduced-integration elements, careful monitoring of hourglass energy, and judicious consideration of mass scaling techniques.

The key numerical features of the explicit formulation and their implications for composite impact modelling are summarised in Table 2.7. These considerations directly inform the modelling strategy adopted in the present work and provide the theoretical foundation for the numerical implementation described in Chapter 5.

Table 2.7: Key numerical characteristics of the explicit formulation in ABAQUS/EXPLICIT and their implications for composite impact modelling.

Aspect	**Numerical characteristic**	**Implication for modelling**
Time integration	Central-difference, conditionally stable explicit scheme	Very small time increments required; suitable for short-duration impact events
Stability limit	Controlled by smallest element dimension and wave speed	Local mesh refinement must be carefully managed to limit computational cost
Mass matrix	Diagonal (lumped) mass matrix	No global stiffness matrix inversion; efficient for highly nonlinear dynamics
Progressive damage	Stable under stiffness degradation and element deletion	Suitable for Hashin intralaminar damage and cohesive delamination modelling
Cohesive interfaces	Traction–separation laws with progressive softening	Robust simulation of delamination initiation and propagation
Contact formulation	Penalty-based, non-iterative enforcement	Stable treatment of impact, separation, and rebound phenomena
Hourglass control	Energy-based suppression of zero-energy modes	Hourglass energy must remain small relative to internal energy
Mass scaling	Artificial increase of stable time increment	May improve efficiency for low-velocity impact; generally avoided for high-velocity impact
Parallel scalability	Excellent scaling on multi-core architectures	Enables high-resolution impact simulations with local mesh refinement

2.5 Sensitivity and Uncertainty Considerations in Material Property Identification

Numerical simulations of composite structures subjected to impact loading rely on a large set of material parameters whose values are often affected by intrinsic variability, experimental scatter, and modelling assumptions. In composite materials, uncertainties arise from multiple sources, including fibre and matrix properties, ply-level elastic constants, damage-related parameters, and manufacturing-induced variations. As a consequence, the structural response predicted by finite

element models may exhibit significant sensitivity to relatively small changes in selected input parameters, particularly in nonlinear damage-driven analyses [78, 79].

In this context, sensitivity analysis provides a valuable framework to assess the influence of uncertain material properties on the numerical response, enabling the identification of dominant parameters and guiding model calibration efforts. Rather than performing a full probabilistic uncertainty quantification, which typically requires extensive statistical data and high computational cost, many studies adopt deterministic sensitivity-based approaches in which selected parameters are systematically varied within predefined bounds to evaluate their relative influence on key response metrics [80, 81]. Such approaches are particularly suitable for impact simulations of composite laminates, where the response is highly nonlinear and experimental reference data are often limited.

This section reviews the fundamental concepts of uncertainty and sensitivity analysis as applied to composite structural modelling, with emphasis on deterministic parameter influence studies used to support material property identification and numerical–experimental correlation. The objective is to establish the theoretical background that justifies the sensitivity-based calibration strategy adopted in the present work, while clearly defining its scope and limitations prior to the detailed numerical modelling presented in subsequent chapters.

2.5.1 Motivation for Sensitivity Analysis in Numerical Impact Modelling

Numerical modelling of composite structures subjected to impact loading involves a large number of material and modelling parameters whose values are often affected by uncertainty. In composite materials, this uncertainty originates from intrinsic variability in fibre and matrix properties, manufacturing-induced scatter, imperfect control of ply orientation and thickness, and simplifications introduced in constitutive damage models. As a consequence, even when the same modelling framework is employed, relatively small variations in selected input parameters may lead to significant differences in the predicted structural response, particularly in highly nonlinear impact simulations [78, 79].

This sensitivity is especially pronounced in damage-driven analyses, where the onset and evolution of matrix cracking, fibre failure, and delamination depend strongly on material parameters that are difficult to determine experimentally with high accuracy. Reported values of elastic constants, strength limits, and fracture-related properties in the literature often exhibit considerable scatter, reflecting differences in material systems, processing routes, testing methodologies, and environmental conditions. Consequently, the direct adoption of nominal material properties may not guarantee an accurate correlation between numerical predictions and experimental observations [81].

Sensitivity analysis provides a structured approach to address this challenge by quantifying the influence of uncertain input parameters on the numerical response. Rather than treating all parameters as equally relevant, sensitivity-based methods enable the identification of those properties that exert a dominant effect on key response metrics, such as acceleration history, contact force, or energy absorption. This information is essential for guiding model refinement and focusing calibration efforts on the most influential parameters, thereby improving both numerical efficiency and physical consistency [78, 80].

In the context of impact modelling of composite laminates, sensitivity analysis is particularly valuable because the response is governed by coupled material and geometric nonlinearities, making intuitive assessment of parameter influence unreliable. Furthermore, the high computational cost associated with explicit finite element simulations often limits the feasibility of extensive probabilistic studies. For these reasons, deterministic sensitivity-based approaches have been widely adopted in the literature as a practical and effective means of supporting material property identification and numerical–experimental correlation [80, 82].

2.5.2 Sources of Uncertainty in Composite Impact Simulations

Uncertainty in numerical simulations of composite structures arises from multiple sources that span different modelling levels, ranging from material characterization to structural idealization. In composite impact analyses, these uncertainties are particularly relevant due to the heterogeneous nature of the material system and the strong coupling between material behaviour and damage evolution. Identifying and classifying the main sources of uncertainty is therefore a necessary step prior to any sensitivity-based assessment of model response [78, 79].

Material-related uncertainties represent one of the most significant contributors to variability in composite impact simulations. Elastic properties, strength limits, and damage-related parameters are often derived from experimental tests that are sensitive to specimen preparation, testing configuration, strain rate, and environmental conditions. As a result, reported material properties in the literature frequently exhibit noticeable scatter, even for nominally identical material systems. This variability is further amplified in thermoplastic composites, where matrix ductility and strain-rate sensitivity can influence the effective stiffness and failure response under impact loading [81, 82].

Additional uncertainty arises from laminate configuration and manufacturing-related factors. Variations in ply thickness, fibre volume fraction, local fibre waviness, and ply orientation deviations can alter the stress distribution within the laminate and affect damage initiation and propagation. Although such effects are often neglected or idealized in numerical models, their influence may become significant in impact simulations, where local stress concentrations govern failure mechanisms [78].

Modelling assumptions and simplifications constitute another important source of uncertainty. The choice of constitutive laws, damage initiation criteria, and degradation models introduces model-form uncertainty, as different formulations may predict distinct damage patterns and response levels for the same input parameters. Moreover, assumptions related to boundary conditions, contact definitions, and numerical discretization can further influence the predicted impact response, particularly in explicit finite element analyses [79].

Given the multiplicity of uncertainty sources and their potential interactions, it is often impractical to address all uncertainties simultaneously within a single modelling framework. Consequently, many studies focus on the most influential parameters, typically material properties directly governing stiffness and damage evolution, while treating other sources in a simplified or nominal manner. This approach enables a targeted sensitivity analysis that balances modelling fidelity with computational feasibility, and forms the basis for the deterministic parameter influence studies discussed in the following subsection [80].

2.5.3 Deterministic Sensitivity Analysis in Composite Impact Simulations

Given the multiple sources of uncertainty inherent to composite impact simulations discussed in the previous subsection, it becomes necessary to adopt a structured strategy to assess their influence on the numerical response. Among the available approaches, deterministic sensitivity analysis has been widely employed in composite structural modelling as a practical means of evaluating parameter influence while maintaining manageable computational cost.

Deterministic sensitivity analysis focuses on the controlled variation of selected input parameters around a nominal baseline, with all remaining parameters held constant. The resulting changes in the numerical response provide direct insight into the relative importance of each parameter and its contribution to global metrics such as stiffness degradation, peak acceleration, contact duration, and damage evolution. This approach is particularly suitable when the objective is to identify dominant parameters or support model calibration, rather than to quantify full probabilistic bounds [78, 79].

In the context of laminated composite structures, sensitivity-based methodologies have demonstrated that variations in a limited subset of material and damage-related parameters can lead to significant changes in structural response. Studies on angle-ply and multidirectional laminates have shown that elastic constants, strength values, and fracture-related parameters often dominate the sensitivity landscape, justifying their prioritization in deterministic analyses [81]. By introducing bounded perturbations, the influence of each parameter can be evaluated in a transparent and physically interpretable manner, without introducing probabilistic assumptions.

More recent investigations have reinforced the effectiveness of deterministic sensitivity analysis for identifying critical material properties in composite simulations. Systematic parameter

variation has been successfully used to rank the influence of elastic, strength, and damage parameters, allowing modelling efforts to be focused on the properties that govern the dominant response mechanisms [80]. This aspect is particularly relevant for impact simulations, where nonlinear effects associated with contact, progressive damage, and strain-rate sensitivity can amplify the impact of parameter uncertainty.

Furthermore, the interaction between deterministic parameter variations and progressive damage modelling has been shown to play a central role in the predictive capability of composite simulations. When combined with intralaminar and interlaminar damage formulations, deterministic sensitivity approaches allow controlled exploration of degradation-related parameters while preserving numerical robustness and model interpretability [82].

In the present work, uncertainty is therefore addressed through a deterministic sensitivity analysis in which selected material and damage parameters are varied independently within bounded ranges. This strategy follows established practices in composite structural analysis, where controlled parameter perturbations are employed to identify dominant sensitivities and support correlation with experimental results, rather than to perform a full stochastic uncertainty quantification [78, 79, 81]. The adopted approach enables direct comparison between numerical predictions and experimental acceleration–time responses, while maintaining numerical stability and computational efficiency.

Chapter 3

Theoretical Modelling of Damage Mechanisms in Composite Laminates

The accurate numerical simulation of impact and damage phenomena in composite laminates requires modelling strategies capable of representing a wide range of interacting physical mechanisms occurring at different length scales. These mechanisms include matrix cracking and fibre failure within the plies, delamination between adjacent layers, and highly localised matrix-dominated deformation under complex three-dimensional stress states. To address these phenomena, a variety of theoretical and numerical formulations have been developed in the literature, each relying on distinct assumptions regarding material behaviour, damage evolution, and mechanical interaction.

This chapter presents a theoretical framework for damage modelling in composite laminates, with emphasis on numerical approaches commonly adopted in dynamic impact analyses. The discussion focuses on intralaminar damage modelling based on continuum damage mechanics, interlaminar damage modelling using cohesive-zone formulations, and advanced bi-phasic modelling strategies aimed at representing matrix-dominated nonlinear behaviour. Rather than describing a specific numerical implementation, the objective of this chapter is to review the underlying theoretical concepts, constitutive assumptions, and modelling philosophies associated with each approach, thereby providing the conceptual basis for the numerical models developed and assessed in the subsequent chapters.

3.1 Intralaminar Damage Modelling in Composite Laminates

Intralaminar damage refers to failure mechanisms that develop within individual plies of a composite laminate, such as fiber breakage, fiber kinking, matrix cracking, and matrix crushing. Under impact loading conditions, these mechanisms play a fundamental role in governing the local stiffness degradation, stress redistribution, and energy absorption capacity of the laminate.

An accurate numerical representation of intralaminar damage is therefore essential to capture the overall structural response of composite materials subjected to dynamic loading.

In the present work, intralaminar damage is modelled using a progressive damage modelling framework based on continuum damage mechanics. Damage initiation is described through two-dimensional Hashin failure criteria, which allow the distinction between fiber-dominated and matrix-dominated failure modes under tensile and compressive loading. Following damage initiation, the evolution of damage is governed by an energy-based softening law that progressively degrades the material stiffness as a function of an equivalent displacement measure. This section presents the theoretical formulation of the intralaminar damage model adopted in this work, including the damage initiation criteria, the damage evolution law, and the associated modelling assumptions and numerical implementation aspects.

3.1.1 Intralaminar Damage Mechanisms in Composite Laminates

Intralaminar damage refers to damage mechanisms that develop within the plies of a composite laminate, as opposed to interlaminar damage, which occurs at the interfaces between adjacent plies. Under mechanical and impact loading, intralaminar damage is primarily associated with fiber failure and matrix-dominated cracking processes, which progressively reduce the stiffness and load-carrying capability of the laminate [14, 83].

Following the classification originally proposed by Hashin, intralaminar failure mechanisms are commonly divided into four distinct modes: fiber tensile failure, fiber compressive failure, matrix tensile failure, and matrix compressive failure [14]. In this context, fiber tensile and compressive failures correspond to failure of the ply under tension and compression in the fibre direction (direction 1), respectively, while matrix tensile and compressive failures correspond to failure of the ply under tension and compression in the transverse direction (direction 2). This separation reflects the fundamentally different physical origins of fiber- and matrix-dominated damage and forms the basis for mode-dependent failure criteria used in progressive damage modelling. While fiber-dominated damage is often associated with brittle fracture under high longitudinal stresses, matrix-dominated damage typically initiates at lower stress levels and manifests through transverse cracking, shear-driven degradation, and localized crushing under compressive loading conditions.

In the context of impact loading, matrix-dominated intralaminar damage is frequently the first damage mechanism to occur and plays a key role in the early stages of energy dissipation and stiffness degradation [84]. Experimental and numerical studies have shown that such damage mechanisms significantly influence the global structural response prior to the onset of extensive delamination or fiber fracture, particularly in low- to medium-energy impact scenarios.

From a numerical modelling perspective, intralaminar damage is commonly represented using homogenised continuum approaches, in which each composite ply is treated as an equivalent

anisotropic material. Within this framework, damage is not modelled as discrete cracks but is instead represented through a progressive degradation of material stiffness at the material point level, enabling efficient simulation of damage initiation and evolution within individual plies [83, 85]. Although this smeared damage representation does not capture discrete crack paths, it provides a computationally efficient and robust means of accounting for the effects of intralaminar damage in dynamic analyses such as impact simulations.

Despite its limitations, an accurate representation of intralaminar damage constitutes a fundamental component of composite impact modelling, as it governs the initial stiffness reduction and energy absorption mechanisms that precede the development of interlaminar damage and global structural failure [84].

For clarity, the main intralaminar damage mechanisms commonly considered in continuum damage modelling of composite laminates are summarised in Table 3.1. This classification follows the mode-dependent failure concept originally proposed by Hashin and subsequently adopted in progressive damage formulations for composite materials [14, 83].

Table 3.1: Classification of intralaminar damage mechanisms in composite laminates.

Damage mode	**Dominant stress state**	**Physical interpretation**
Fiber tensile failure	Longitudinal tension	Fiber rupture
Fiber compressive failure	Longitudinal compression	Fiber microbuckling or kinking
Matrix tensile failure	Transverse tension	Matrix cracking
Matrix compressive failure	Transverse compression / shear	Matrix crushing or shear-driven damage

3.1.2 Progressive Damage Modelling Framework

Progressive Damage Modelling (PDM) provides a numerical framework for simulating the initiation and evolution of damage in composite materials through a gradual degradation of material properties. Within this framework, damage is represented using continuum damage mechanics concepts, in which the onset of failure is identified through stress- or strain-based criteria and the subsequent material degradation is governed by damage evolution laws [14, 83].

The general workflow of progressive damage modelling adopted in this work is schematically illustrated in Figure 3.1. The flowchart highlights the sequential nature of the constitutive update process typically used in explicit dynamic simulations. At each time increment, the strain increment provided by the explicit solver is used to compute a trial elastic stress state. The damage initiation criteria are then evaluated, and if the activation conditions are satisfied, internal damage variables are updated and the corresponding material stiffness is degraded. The updated stresses and state variables are subsequently returned to the solver to advance the dynamic solution. In

explicit formulations such as Abaqus/Explicit, this update procedure is performed directly at the material point level and does not require the computation of a consistent Jacobian matrix, which is normally necessary in implicit solution schemes [86].

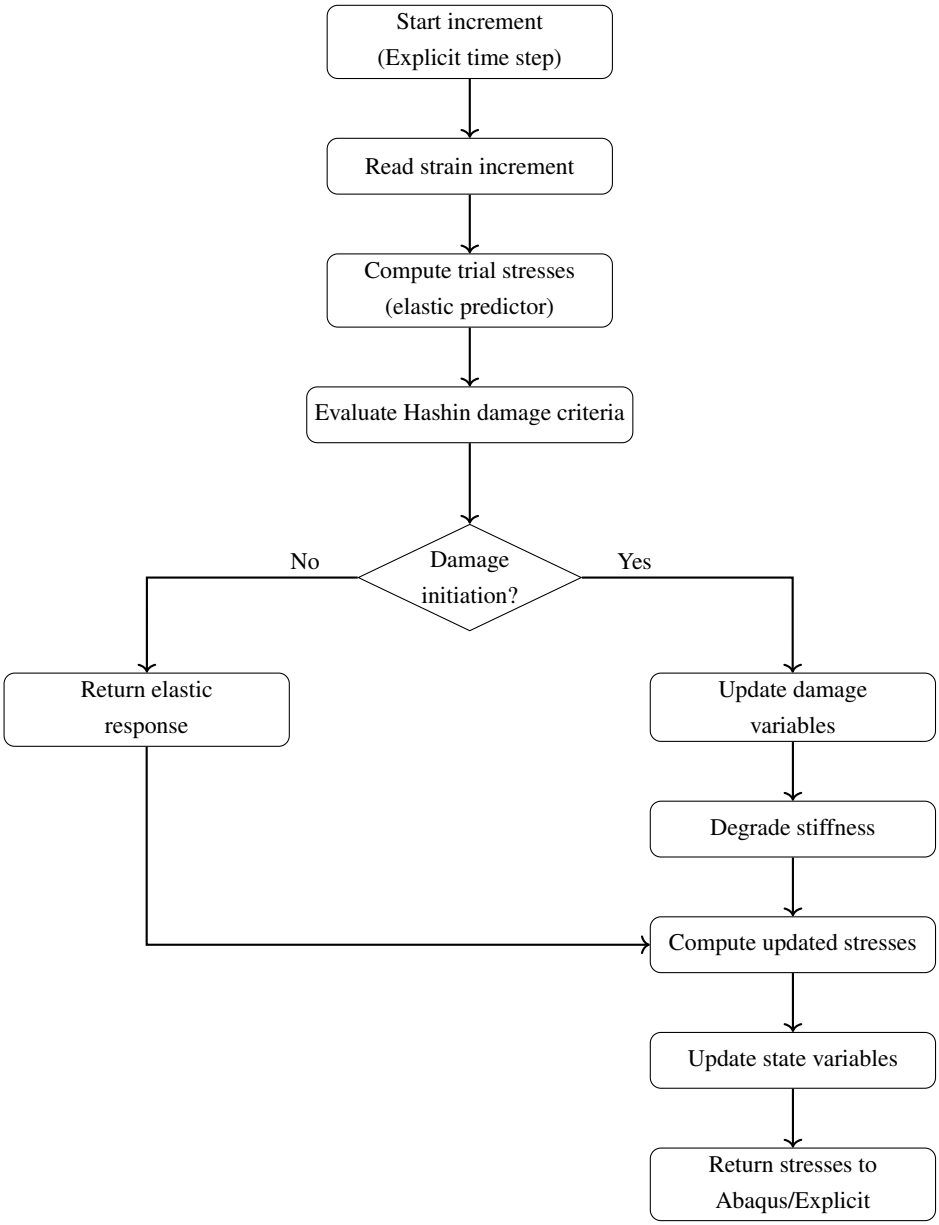


Figure 3.1: Flowchart of the constitutive update procedure adopted for the explicit intralaminar damage model implementation. In Abaqus/Explicit, the material update is performed through stress and state-variable evolution at each increment, without requiring the consistent Jacobian matrix needed in implicit formulations.

A key feature of progressive damage models is the explicit separation between damage initiation and damage evolution. Damage initiation criteria define the conditions under which a specific damage mode is activated, while damage evolution laws control the post-initiation response of the material, typically through a progressive reduction of stiffness. This distinction is essential for

modelling composite laminates under dynamic loading, as it enables the representation of gradual stiffness degradation and energy dissipation rather than abrupt, brittle failure [83].

In continuum-based implementations, intralaminar damage is treated as a smeared phenomenon at the material point level. Once a damage initiation criterion is satisfied, internal damage variables are introduced to degrade the elastic stiffness associated with the corresponding failure mode. The evolution of these damage variables is commonly driven by energy-based formulations, in which fracture energy parameters define the amount of energy dissipated during damage growth and help mitigate mesh dependency effects inherent to softening constitutive laws [83, 85].

Progressive damage modelling has been widely adopted for the simulation of impact loading in composite laminates, as it allows the interaction between multiple damage mechanisms to be captured within a unified numerical framework. Studies focused on impact scenarios have shown that the progressive development of intralaminar damage significantly influences the global force and acceleration response of the structure prior to the onset of extensive delamination or fiber fracture [84]. Consequently, PDM constitutes a fundamental modelling approach for accurately representing the mechanical response of composite laminates subjected to impact loading.

3.1.3 Hashin 2D Failure Criterion for Intralaminar Damage Initiation and Evolution

The initiation of intralaminar damage in the present work is described using the two-dimensional plane-stress Hashin failure criteria, which distinguish between fiber-dominated and matrix-dominated damage mechanisms under tensile and compressive loading conditions [14]. Within the PDM framework, these criteria are employed to detect the onset of damage, while the post-initiation response is governed by a damage evolution law coupled to the degradation of the ply stiffness [83].

3.1.3.1 Damage Initiation

The initiation of intralaminar damage in the present work is described using the two-dimensional plane-stress Hashin failure criteria, which distinguish between fiber-dominated and matrix-dominated damage mechanisms under tensile and compressive loading conditions [14]. Within the framework of progressive damage modelling, these criteria are employed exclusively to detect the onset of damage, while the subsequent material degradation is governed by a continuum damage mechanics formulation based on stiffness degradation and strain-softening behaviour [15, 16].

In the Hashin two-dimensional formulation, four independent intralaminar failure modes are considered: Fiber Tension (FT), Fiber Compression (FC), Matrix Tension (MT), and Matrix Compression (MC). Damage initiation is evaluated through scalar failure indices F_I, which reach a critical value equal to unity at the onset of damage for the corresponding mode.

Fiber tensile failure ($\sigma_{11} \geq 0$)

Fiber tensile damage initiation is assumed to occur when the longitudinal normal stress is tensile and the following interaction criterion is satisfied:

$$F_{ft} = \left(\frac{\sigma_{11}}{X_T}\right)^2 + \left(\frac{\tau_{12}}{S_L}\right)^2 = 1, \tag{3.1}$$

where σ_{11} is the longitudinal normal stress, τ_{12} is the in-plane shear stress, X_T is the longitudinal tensile strength, and S_L is the longitudinal shear strength.

Fiber compressive failure ($\sigma_{11} < 0$)

Fiber compressive damage initiation is governed by the longitudinal compressive stress and is expressed as:

$$F_{fc} = \frac{-\sigma_{11}}{X_C} = 1, \tag{3.2}$$

where X_C is the longitudinal compressive strength. In this formulation, shear stresses are neglected in the fiber compression mode, reflecting the dominant role of fiber microbuckling and kinking mechanisms [14].

Matrix tensile failure ($\sigma_{22} \geq 0$)

Matrix tensile damage initiation is associated with transverse tensile stresses and shear interaction and is defined by:

$$F_{mt} = \left(\frac{\sigma_{22}}{Y_T}\right)^2 + \left(\frac{\tau_{12}}{S_T}\right)^2 = 1, \tag{3.3}$$

where σ_{22} is the transverse normal stress, Y_T is the transverse tensile strength, and S_T is the transverse shear strength.

Matrix compressive failure ($\sigma_{22} < 0$)

Matrix compressive damage initiation accounts for transverse compression and shear interaction through the following criterion:

$$F_{mc} = \left(\frac{\sigma_{22}}{2S_T}\right)^2 + \left[\left(\frac{Y_C}{2S_T}\right)^2 - 1\right]\frac{\sigma_{22}}{Y_C} + \left(\frac{\tau_{12}}{S_T}\right)^2 = 1, \tag{3.4}$$

where Y_C is the transverse compressive strength. This formulation captures the nonlinear interaction between transverse compressive and shear stresses that characterises matrix crushing and shear-driven failure mechanisms [14].

For clarity, the intralaminar failure modes considered in the Hashin two-dimensional formulation are summarised in Table 3.2. The table highlights the stress conditions associated with each

failure mode, the dominant stress components involved in damage initiation, and the corresponding physical interpretation.

Table 3.2: Intralaminar failure modes considered in the Hashin two-dimensional damage initiation criterion.

Failure mode	**Stress condition**	**Stress components**	**Physical interpretation**
Fiber tension (FT)	$\sigma_{11} \geq 0$	$\sigma_{11},\ \tau_{12}$	Fiber rupture under longitudinal tension
Fiber compression (FC)	$\sigma_{11} < 0$	σ_{11}	Fiber microbuckling or kinking
Matrix tension (MT)	$\sigma_{22} \geq 0$	$\sigma_{22},\ \tau_{12}$	Matrix cracking and shear-driven damage
Matrix compression (MC)	$\sigma_{22} < 0$	$\sigma_{22},\ \tau_{12}$	Matrix crushing and shear-induced failure

3.1.3.2 Damage Evolution and Stiffness Degradation

Once intralaminar damage initiation has been detected according to the two-dimensional Hashin failure criteria, it is necessary to define how damage evolves and how it affects the mechanical response of the material. In composite laminates, this post-initiation behaviour is commonly described using CDM formulations, in which damage is represented by internal variables that progressively degrade the material stiffness and govern energy dissipation during failure [15, 16].

In the present work, damage evolution is formulated within a progressive damage modelling framework. Hashin's criteria are employed exclusively to identify damage initiation, while the subsequent evolution is controlled through an energy-based softening law expressed in terms of equivalent displacement, fracture energy, and damage variables.

Figure 3.2 schematically illustrates the progressive damage mechanism adopted in this work. The figure simultaneously represents the elastic response up to damage initiation, the subsequent linear softening governed by fracture energy, and the evolution of the damage variable as a function of the equivalent displacement. This combined representation highlights the coupling between stress degradation and damage accumulation inherent to CDM-based formulations.

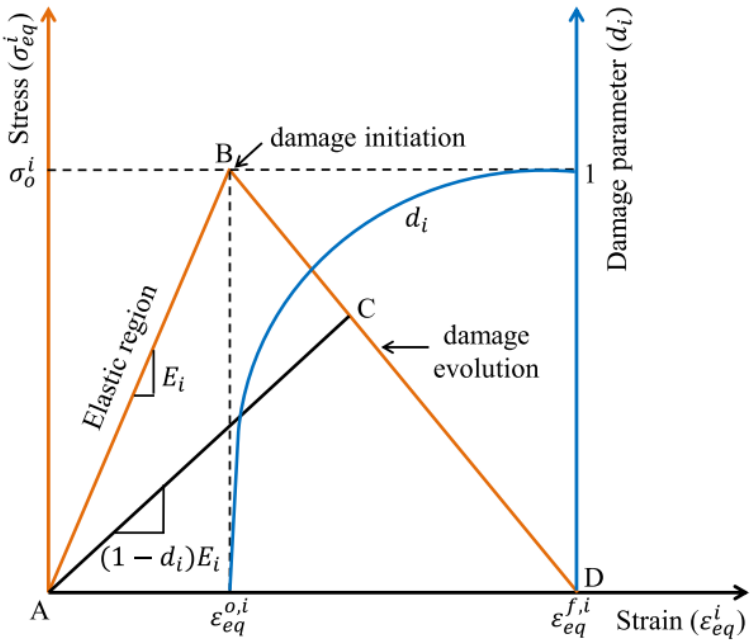


Figure 3.2: Schematic representation of the progressive damage model, illustrating the elastic response up to damage initiation, the linear softening regime governed by fracture energy, and the evolution of the damage variable with increasing equivalent displacement [87].

Equivalent displacement

The evolution of damage is driven by an equivalent displacement measure $\delta_{I,eq}$, defined for each intralaminar failure mode $I \in \{ft, fc, mt, mc\}$. This quantity relates strain measures to a displacement-like variable through a characteristic length l_c, ensuring regularisation of the softening response and reducing mesh dependency effects. For the two-dimensional plane-stress formulation, the equivalent displacements are defined as:

$$\delta_{ft,eq} = l_c \sqrt{\langle \varepsilon_{11} \rangle^2 + \varepsilon_{12}^2}, \tag{3.5}$$

$$\delta_{fc,eq} = l_c \langle -\varepsilon_{11} \rangle, \tag{3.6}$$

$$\delta_{mt,eq} = l_c \sqrt{\langle \varepsilon_{22} \rangle^2 + \varepsilon_{12}^2}, \tag{3.7}$$

$$\delta_{mc,eq} = l_c \sqrt{\langle -\varepsilon_{22} \rangle^2 + \varepsilon_{12}^2}. \tag{3.8}$$

Here, ε_{ij} are the in-plane strain components and $\langle \cdot \rangle$ denotes the Macauley operator, defined as:

$$\langle x \rangle = \frac{x + |x|}{2}, \quad \forall x \in \mathbb{R}. \tag{3.9}$$

Equivalent stress and fracture energy

Once the equivalent displacement has been determined, the corresponding equivalent stress $\sigma_{I,eq}$ is computed in order to relate the stress state to the softening law:

$$\sigma_{ft,eq} = \frac{l_c\left(\langle\sigma_{11}\rangle\langle\varepsilon_{11}\rangle + \sigma_{12}\varepsilon_{12}\right)}{\delta_{ft,eq}}, \tag{3.10}$$

$$\sigma_{fc,eq} = \frac{l_c\langle-\sigma_{11}\rangle\langle-\varepsilon_{11}\rangle}{\delta_{fc,eq}}, \tag{3.11}$$

$$\sigma_{mt,eq} = \frac{l_c\left(\langle\sigma_{22}\rangle\langle\varepsilon_{22}\rangle + \sigma_{12}\varepsilon_{12}\right)}{\delta_{mt,eq}}, \tag{3.12}$$

$$\sigma_{mc,eq} = \frac{l_c\left(\langle-\sigma_{22}\rangle\langle-\varepsilon_{22}\rangle + \sigma_{12}\varepsilon_{12}\right)}{\delta_{mc,eq}}. \tag{3.13}$$

The equivalent displacement at damage onset is defined as:

$$\delta^0_{I,eq} = \frac{\delta_{I,eq}}{\sqrt{F_I}}, \tag{3.14}$$

where F_I is the failure index obtained from the corresponding Hashin damage initiation criterion. The equivalent stress at damage onset is given by:

$$\sigma^0_{I,eq} = \frac{\sigma_{I,eq}}{\sqrt{F_I}}. \tag{3.15}$$

The final equivalent displacement corresponding to complete damage is obtained from the fracture energy $G_{I,c}$ associated with the respective failure mode:

$$\delta^f_{I,eq} = \frac{2G_{I,c}}{\sigma^0_{I,eq}}. \tag{3.16}$$

Damage variable evolution

The evolution of damage is described through scalar damage variables $d_I \in [0,1]$, defined as:

$$d_I = \frac{\delta^f_{I,eq}\left(\delta_{I,eq} - \delta^0_{I,eq}\right)}{\delta_{I,eq}\left(\delta^f_{I,eq} - \delta^0_{I,eq}\right)}, \qquad \delta^0_{I,eq} \leq \delta_{I,eq} \leq \delta^f_{I,eq}. \tag{3.17}$$

This definition ensures that $d_I = 0$ at damage onset and that d_I approaches unity as the equivalent displacement reaches its final value, consistently with the softening behaviour illustrated in Figure 3.2.

Damaged constitutive response

The effect of damage on the mechanical response of the ply is introduced through the degradation of the elastic stiffness. The constitutive relation in the presence of damage is written as:

$$\sigma = \mathbf{C}_d \varepsilon, \tag{3.18}$$

where $\mathbf{C}_d$ is the damaged elasticity matrix. For the two-dimensional plane-stress case, $\mathbf{C}_d$ is expressed as:

$$\mathbf{C}_d = \frac{1}{D}\begin{bmatrix} C_{11} & C_{12} & 0 \\ C_{21} & C_{22} & 0 \\ 0 & 0 & C_{66} \end{bmatrix}, \tag{3.19}$$

with

$$\begin{aligned} C_{11} &= d_f E_{11}, \\ C_{22} &= d_m E_{22}, \\ C_{12} &= C_{21} = d_f d_m E_{11} \nu_{21}, \\ C_{66} &= d_s G_{12} D, \end{aligned} \tag{3.20}$$

and

$$\begin{aligned} d_f &= (1-d_{ft})(1-d_{fc}), \\ d_m &= (1-d_{mt})(1-d_{mc}), \\ d_s &= 1-d_f d_m, \\ D &= 1-d_f d_m \nu_{12} \nu_{21}. \end{aligned} \tag{3.21}$$

Here, E_{ii} are the elastic moduli, G_{12} is the in-plane shear modulus, and ν_{ij} are Poisson coefficients. The variable d_s accounts for the degradation of shear stiffness induced by the combined effect of fiber- and matrix-dominated damage mechanisms.

3.1.3.3 Intralaminar Damage Modelling Assumptions and Numerical Implementation

The intralaminar damage formulation presented in the previous subsections provides the theoretical framework used in this work to model fiber- and matrix-dominated damage mechanisms within composite plies. In order to apply this formulation within a finite element environment, a set of modelling assumptions and implementation choices is required. These are outlined in the following to clarify the scope and limitations of the adopted approach.

Scope of the intralaminar damage formulation

In the present work, intralaminar damage is modelled using the plane-stress form of the Hashin failure criteria combined with a continuum damage mechanics formulation for damage evolution. Although the finite element discretisation employs three-dimensional elements, the adopted formulation corresponds to the two-dimensional Hashin criterion, which is derived from laminate theory under a plane-stress assumption.

This modelling choice is justified by the thin laminate configuration considered in this study, where the laminate thickness is small compared to its in-plane dimensions. Under these conditions, through-thickness stresses are typically much smaller than the in-plane stress components, and the dominant intralaminar damage mechanisms can therefore be described using in-plane normal and shear stresses.

An additional practical consideration concerns the numerical implementation available within commercial finite element software. In Abaqus, the built-in Hashin composite damage formulation is based on a plane-stress representation of ply behaviour, even when applied within three-dimensional finite element discretisations. The use of this formulation enables progressive intralaminar damage to be simulated using standard modelling tools without requiring custom user-defined material subroutines such as User-Defined Material Subroutine (UMAT) or VUMAT. One of the objectives of the present work is to investigate the impact response of composite laminates using modelling strategies that rely primarily on commercially available constitutive models.

Within this framework, intralaminar damage is therefore described using the in-plane stress components associated with the Hashin failure criteria, while out-of-plane failure mechanisms are addressed separately through interlaminar modelling approaches. This separation between intra- and interlaminar damage is commonly adopted in CDM-based formulations for laminated composite structures [16].

Numerical implementation

The intralaminar damage model is implemented using the built-in composite damage modelling framework available in the finite element solver. Hashin's failure criteria are employed to detect damage initiation within the composite plies, while damage evolution is governed by an energy-based softening law defined through fracture energy parameters. This modelling strategy follows standard progressive damage modelling approaches for fiber-reinforced composites, allowing the initiation and propagation of intralaminar damage to be captured in a numerically stable manner within an explicit dynamic analysis framework [15].

Representation of woven composite plies

Although the Hashin failure criteria were originally developed for unidirectional composite plies, they are commonly applied to woven laminates by treating each ply as an equivalent orthotropic material. In this work, the woven architecture is homogenised at the ply level, with the warp and weft directions aligned with the material principal directions. Fiber- and matrix-dominated damage modes are therefore interpreted in terms of damage along the warp and weft directions of the woven fabric. This homogenised representation is appropriate at the mesoscale level considered in the present simulations and is widely adopted in continuum-based damage modelling of woven composite laminates [16].

Model limitations and interaction with interlaminar damage

The intralaminar damage model adopted in this work represents damage in a smeared manner through stiffness degradation and does not explicitly capture discrete cracking or fiber kinking mechanisms. Furthermore, delamination and other interlaminar damage phenomena are not represented within the intralaminar formulation. These mechanisms are instead modelled separately using cohesive zone approaches, which allows a clearer distinction between intra- and interlaminar damage processes and avoids artificial coupling between the two within a single constitutive law [15].

3.2 Interlaminar Damage Modelling using Cohesive Elements

Interlaminar damage in composite laminates is primarily associated with delamination, a failure mechanism characterised by the progressive separation of adjacent plies along their interfaces. Owing to the layered nature of laminated composites, delamination may develop under relatively low through-thickness and interfacial shear stresses, significantly affecting the stiffness, load transfer capability, and damage tolerance of the structure. Accurate numerical modelling of interlaminar damage therefore requires a dedicated interface-level description capable of capturing both the initiation and propagation of delamination under general loading conditions.

In the present work, interlaminar damage is modelled using cohesive zone elements, which provide a physically motivated framework for representing interface fracture through a traction-separation relationship. This section introduces the physical origin of delamination and the general concepts underlying cohesive zone modelling, followed by a detailed description of the traction-separation formulation implemented in Abaqus and the corresponding damage initiation and evolution criteria adopted in the numerical model.

3.2.1 Physical Origin of Delamination and Modelling Motivation

Interlaminar damage in composite laminates is primarily associated with delamination, which corresponds to the progressive separation of adjacent plies along their interfaces. Unlike intralaminar damage mechanisms, which develop within the plies themselves, delamination originates at the ply interfaces and is driven by through-thickness stresses and interfacial shear stresses. Under transverse impact and indentation loading, such stresses are often significant, making delamination one of the dominant damage mechanisms governing the structural integrity and residual performance of laminated composite structures [88, 89].

CZM has become a standard framework to represent interfacial damage, as it enables both the initiation and propagation of delamination to be captured through a constitutive relationship between interfacial tractions and relative displacements [90]. In contrast to approaches requiring the explicit definition of initial cracks and subsequent crack-front tracking, CZM allows delamination

to develop naturally once a critical traction- or energy-based condition is satisfied, which is particularly advantageous for composite laminates where crack initiation sites are not known a priori [90].

Furthermore, delamination growth in composites is strongly influenced by mixed-mode loading conditions, where normal opening and shear sliding mechanisms act simultaneously at the interface. This mixed-mode nature motivates the adoption of cohesive formulations capable of combining Mode I and Mode II contributions within a unified damage evolution framework [88, 91]. Based on these considerations, cohesive zone modelling is adopted in the present work to describe interlaminar damage. The general concepts underlying CZM are introduced in the following subsection, before detailing the specific traction-separation formulation implemented in Abaqus and the corresponding modelling assumptions.

3.2.2 General Cohesive Zone Modelling Concepts

In its general form, a cohesive zone model is defined by a traction-separation law that governs the interfacial response from the undamaged state to complete decohesion. The traction-separation relationship typically exhibits an initial elastic regime, followed by a softening branch in which the interfacial tractions progressively decrease as the relative separation increases. Complete separation occurs when the traction reduces to zero, corresponding to full loss of load transfer across the interface. The area under the traction-separation curve represents the fracture energy dissipated during delamination, providing an energy-based description of interlaminar failure [88].

To aid the interpretation of the cohesive zone modelling concepts introduced above, schematic representations of the fundamental delamination modes and of a typical mixed-mode traction-separation response are presented. These figures are intended to provide a physical and conceptual understanding of the interfacial separation mechanisms and the associated energy dissipation processes, without reference to any specific numerical implementation.

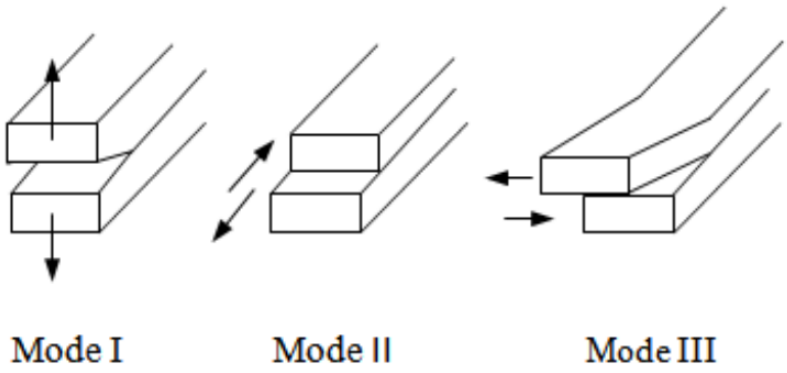


Figure 3.3: Schematic representation of the fundamental delamination modes in composite laminates: Mode I (opening), Mode II (in-plane shear), and Mode III (out-of-plane shear) [92].

Delamination in composite laminates may occur under normal opening (Mode I), in-plane shear sliding (Mode II), or out-of-plane shear sliding (Mode III), as illustrated in Fig. 3.3. In practical loading scenarios, particularly under impact or indentation, these separation modes often act simultaneously, leading to mixed-mode delamination behaviour. Accurately capturing the interaction between normal and shear separation components is therefore essential for realistic modelling of interlaminar damage.

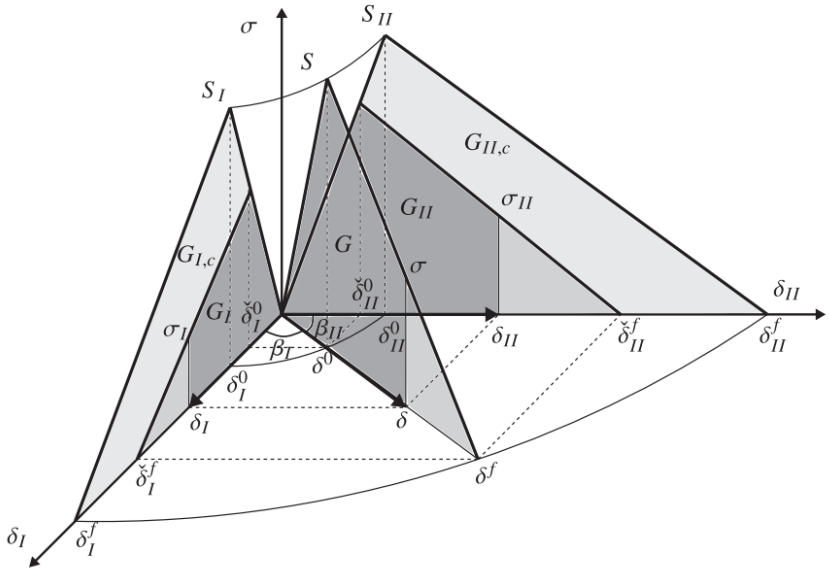


Figure 3.4: Schematic representation of a mixed-mode traction-separation law for cohesive zone modelling, illustrating damage initiation, softening behaviour, and fracture energy dissipation under combined normal and shear separation [17].

To account for mixed-mode loading, cohesive zone models can be extended through interaction criteria that combine the contributions of the different separation modes within a single damage evolution framework [91]. While CZM offers significant advantages in terms of modelling flexibility and physical interpretation, the predicted interlaminar response remains sensitive to the choice of cohesive parameters, such as interfacial strength and fracture toughness, as well as to numerical aspects including mesh discretisation and the assumed shape of the traction-separation law [90].

Although the concepts presented in this subsection describe cohesive zone modelling in a general and solver-independent manner, practical implementations may differ in terms of kinematic definitions, constitutive assumptions, and numerical treatment. In the present work, interlaminar damage is modelled using the traction-separation formulation implemented in Abaqus, which is described in detail in the following subsection.

3.2.3 Abaqus Traction-Separation Formulation for Cohesive Elements

In Abaqus, interlaminar damage is modelled using cohesive elements based on a traction-separation formulation, in which the mechanical response of the interface is defined directly in terms of nominal tractions and nominal separations. This formulation differs from strain-based cohesive approaches commonly found in the literature, as the constitutive behaviour is expressed with respect to relative displacements across the interface rather than continuum strains [93].

The nominal traction vector $\mathbf{t}$ is composed of one normal and two shear components,

$$\mathbf{t} = \{t_n,\, t_s,\, t_t\}, \tag{3.22}$$

which are energetically conjugate to the corresponding nominal separations,

$$\delta = \{\delta_n,\, \delta_s,\, \delta_t\}. \tag{3.23}$$

These separations represent the relative displacements between the opposing surfaces of the cohesive interface in the normal and tangential directions.

Prior to damage initiation, the elastic response of the cohesive element is governed by a linear traction-separation relationship,

$$\mathbf{t} = \mathbf{K}\delta, \tag{3.24}$$

where $\mathbf{K}$ is the cohesive stiffness matrix. In Abaqus, this matrix is typically defined as diagonal, such that independent penalty stiffness values are prescribed for the normal and shear directions. The stiffness parameters do not represent physical material properties, but rather numerical quantities that enforce displacement compatibility across the interface before damage initiation [88].

A key aspect of the Abaqus cohesive formulation is the use of a unit constitutive thickness by default. As a result, the traction-separation law is defined directly in terms of tractions and separations, without requiring the specification of an explicit interface thickness. This assumption simplifies the definition of cohesive properties and avoids ambiguities associated with thickness-dependent stiffness calibration [93].

Damage initiation and damage evolution are treated as distinct processes in Abaqus. Prior to damage initiation, the interface response remains fully elastic, regardless of the magnitudede of the separations. Once a damage initiation criterion is satisfied, the traction-separation response transitions to a softening regime governed by a damage evolution law, ultimately leading to complete decohesion. The specific criteria adopted for damage initiation and evolution are described in the following subsection.

3.2.4 Damage Initiation and Evolution in Abaqus Cohesive Elements

In Abaqus, damage initiation and damage evolution in cohesive elements are treated as distinct processes. Prior to damage initiation, the cohesive interface exhibits a linear elastic response governed by the traction-separation relationship described in the previous subsection. Damage initiation marks the onset of stiffness degradation, while damage evolution governs the progressive loss of load-carrying capacity leading to complete interfacial decohesion [93].

Damage initiation criterion

Damage initiation in Abaqus cohesive elements is commonly defined using a quadratic nominal stress criterion, expressed as

$$\left(\frac{\langle t_n \rangle}{t_n^0}\right)^2 + \left(\frac{t_s}{t_s^0}\right)^2 + \left(\frac{t_t}{t_t^0}\right)^2 = 1, \tag{3.25}$$

where t_n, t_s, and t_t are the nominal normal and shear tractions, respectively, and t_n^0, t_s^0, and t_t^0 denote the corresponding interfacial strength parameters. The Macaulay brackets $\langle \cdot \rangle$ ensure that compressive normal tractions do not contribute to damage initiation, reflecting the assumption that delamination cannot initiate under pure compression [93].

Once the damage initiation criterion is satisfied, a scalar damage variable is activated and the cohesive response transitions from the elastic regime to a softening regime governed by a damage evolution law.

Damage evolution law

Damage evolution in Abaqus cohesive elements is formulated in an energy-based manner, such that the degradation of the cohesive tractions is governed by the fracture energy dissipated during decohesion. The evolution of damage is controlled by a scalar damage variable d, which progressively reduces the effective tractions transmitted across the interface. For a linear softening law, the nominal traction vector may be written as

$$\mathbf{t} = (1-d)\,\mathbf{K}\boldsymbol{\delta}, \tag{3.26}$$

where $\mathbf{K}$ is the cohesive stiffness matrix and $\boldsymbol{\delta}$ is the nominal separation vector.

Complete decohesion occurs when the energy dissipated by the cohesive interface equals the critical fracture energy associated with the prevailing separation mode. The fracture energy corresponds to the area under the traction-separation curve and provides a physically meaningful measure of the resistance of the interface to delamination [17].

Mixed-mode interaction

Under general loading conditions, delamination is typically governed by mixed-mode separation involving both normal and shear components. To account for this behaviour, Abaqus allows mixed-mode damage evolution to be described through interaction criteria that combine the individual mode contributions to the fracture energy. A widely adopted formulation is the Benzeggagh-Kenane (BK) criterion, given by

$$G_c = G_{Ic} + (G_{IIc} - G_{Ic}) \left(\frac{G_{II}}{G_I + G_{II}} \right)^{\eta}, \tag{3.27}$$

where G_I and G_{II} are the energy release rates associated with Mode I and Mode II separation, respectively, G_{Ic} and G_{IIc} are the corresponding critical fracture energies, and η is a material-dependent interaction parameter.

The Benzeggagh-Kenane criterion was originally proposed on the basis of experimental measurements of mixed-mode delamination fracture toughness using a mixed-mode bending apparatus [18]. Although initially formulated within a fracture mechanics framework, this criterion has been widely adopted in cohesive zone modelling to describe mixed-mode damage evolution due to its simplicity and strong experimental foundation.

For clarity, the main quantities governing damage initiation and evolution in the Abaqus cohesive formulation are summarised in Table 3.3. The table highlights the parameters controlling elastic behaviour, damage onset, and mixed-mode damage evolution, together with their physical interpretation and role in the numerical model.

Table 3.3: Summary of main parameters governing damage initiation and evolution in Abaqus cohesive elements.

Quantity	**Description**	**Role in model**
t_n^0	Normal interfacial strength	Controls damage initiation under Mode I loading
t_s^0, t_t^0	Shear interfacial strengths	Controls damage initiation under shear loading
K_n, K_s, K_t	Elastic cohesive stiffnesses	Define the initial elastic traction—separation response
G_{Ic}	Mode I fracture energy	Controls the energy dissipated during normal separation
G_{IIc}	Mode II fracture energy	Controls the energy dissipated during shear separation
η	Benzeggagh-Kenane interaction exponent	Defines mixed-mode fracture energy evolution
d	Damage variable	Controls stiffness degradation after damage initiation

Unloading and irreversibility

Once damage initiation has occurred, the cohesive response in Abaqus is irreversible. Unloading from a damaged state follows a linear path towards the origin with a reduced stiffness proportional to the current value of the damage variable. Healing of the interface is not permitted, and the damage variable is constrained to be non-decreasing throughout the analysis. Upon complete decohesion, the cohesive element is unable to transmit tractions, and the interface surfaces may subsequently interact through contact if contact conditions are defined [93].

3.3 Mesoscale Bi-phasic Modelling Strategies for Matrix-Dominated Damage in Composite Laminates

Bi-phasic modelling strategies represent an alternative numerical formulation for the analysis of composite laminates, developed to address limitations inherent to classical single-phase damage approaches when dealing with matrix-dominated nonlinear behaviour. Rather than describing the composite response through a single constitutive law that implicitly couples fibre- and matrix-related mechanisms, bi-phasic approaches are based on the separation of mechanical roles between an idealised fibre-dominated contribution and an idealised matrix-dominated contribution. This section reviews the theoretical foundations of bi-phasic modelling, its constitutive assumptions, and its adaptation to composite laminates, with particular emphasis on the representation of matrix-driven deformation mechanisms under localised loading conditions. The

discussion builds upon the binary modelling framework originally proposed for textile composites and its subsequent extension to laminate configurations, providing the conceptual basis for the modelling strategy adopted in the present work.

3.3.1 Motivation and Limitations of Single-Phase Composite Damage Formulations

Numerical models for the impact response of composite laminates are commonly formulated within a single-phase continuum description, in which the macroscopic stress-strain response is governed by a unique constitutive framework that implicitly combines the mechanical contributions of fibres and matrix. Within this context, intralaminar damage models based on continuum damage mechanics CDM and interlaminar cohesive-zone formulations are typically employed to represent stiffness degradation, energy dissipation, and failure progression [19]. Although such approaches have proven effective for predicting global damage patterns and delamination growth, they are often associated with a very high computational cost, particularly when detailed intralaminar and interlaminar damage mechanisms are modelled simultaneously. This computational burden, rather than limitations in predictive capability alone, represents a key motivation for the development of bi-phasic modelling strategies.

This assumption becomes restrictive when addressing matrix-dominated deformation mechanisms. In particular, under highly localised loading conditions such as impact-induced contact, the composite response is governed by complex three-dimensional stress states characterised by transverse compression and shear, for which the polymer matrix plays a dominant role. Experimental evidence and numerical studies indicate that, in these conditions, the polymer matrix plays a key role in governing local deformation and indentation, while its relative contribution to global energy dissipation depends on the activation of fibre-dominated fracture mechanisms [19]. When interpreted within a single-phase damage framework, such matrix-driven nonlinearities are often forced to manifest as premature damage growth, leading to excessive stiffness degradation and an inaccurate representation of local permanent deformation.

To overcome these limitations, alternative modelling strategies have been proposed in which the composite response is not described by a single degrading constitutive law, but rather by the superposition of distinct mechanical contributions associated with different material roles. This conceptual shift forms the basis of binary or bi-phasic modelling approaches originally developed for textile composites, in which the reinforcement architecture and the surrounding matrix-dominated medium are represented through separate but interacting mechanical descriptions [94, 95]. In these formulations, nonlinear matrix behaviour can be introduced explicitly within a dedicated matrix-dominated phase, without directly compromising the fibre-dominated load-carrying capability of the composite.

Building on these ideas, bi-phasic approaches have subsequently been adapted to composite laminates to provide a physically motivated framework for representing matrix-driven intralaminar and interlaminar damage mechanisms within a unified formulation [19]. By distinguishing between an idealised fibre-continuity phase and an effective matrix phase, these models enable the explicit treatment of matrix nonlinearity while preserving the structural role of the fibres. This separation constitutes the fundamental motivation for the bi-phasic modelling strategies discussed in the following subsections.

3.3.2 Fundamental Assumptions of Bi-Phasic Modelling and Stress Decomposition

The defining feature of bi-phasic modelling strategies is the assumption that the macroscopic mechanical response of a composite can be expressed as the superposition of distinct contributions associated with different material roles. Rather than embedding fibre- and matrix-related mechanisms within a single constitutive law, bi-phasic approaches postulate that these mechanisms can be represented independently while remaining kinematically compatible and in equilibrium. This conceptual framework was originally introduced in the context of binary models for textile composites, where it was shown that the reinforcement architecture and the surrounding matrix-dominated medium may be described through separate mechanical representations without violating the principles of continuum mechanics [94, 95].

At the constitutive level, this assumption is commonly expressed through an additive decomposition of the macroscopic stress tensor,

$$\boldsymbol{\sigma} = \boldsymbol{\sigma}^{(f)} + \boldsymbol{\sigma}^{(m)}, \tag{3.28}$$

where $\boldsymbol{\sigma}^{(f)}$ denotes the stress contribution associated with the fibre-dominated phase, and $\boldsymbol{\sigma}^{(m)}$ represents the stress contribution associated with the matrix-dominated phase. This decomposition reflects the hypothesis that fibres and matrix contribute additively to the overall stress response while sharing the same deformation field. In the binary model formulation, such a superposition is justified by enforcing kinematic compatibility and global equilibrium, allowing the two phases to interact mechanically without the introduction of explicit interfaces or additional coupling laws [94].

To facilitate the interpretation of the bi-phasic modelling assumptions introduced above, Figure 3.5 presents a schematic representation of the hybrid numerical discretisation typically adopted in bi-phasic approaches for composite laminates. Rather than depicting a separation into distinct physical materials, the figure illustrates a superimposed mesh strategy in which different element formulations are used to represent complementary mechanical contributions within the same laminate volume. In this framework, a continuum shell discretisation is used to represent the laminate-level fibre-dominated in-plane response, while solid elements may be introduced locally to enhance the representation of matrix-dominated three-dimensional behaviour associated

with localized out-of-plane mechanisms such as indentation beneath the impactor. This hybrid discretisation reflects the distinct mechanical roles assigned to each phase in the bi-phasic framework: continuum shell elements efficiently capture the laminate stiffness associated with fibre continuity, whereas three-dimensional solid elements enable the accurate representation of matrix-dominated stress states and deformation mechanisms arising under localized contact loading. The two discretisations are kinematically coupled, ensuring strain compatibility and force equilibrium, and enabling the activation of matrix-driven nonlinear mechanisms while preserving fibre continuity within a unified numerical framework. The specific discretisation strategy adopted in the present work, together with the rationale for the use of continuum shell elements for the composite panel, is discussed in Chapter 5 (Section 5.1.7.2).

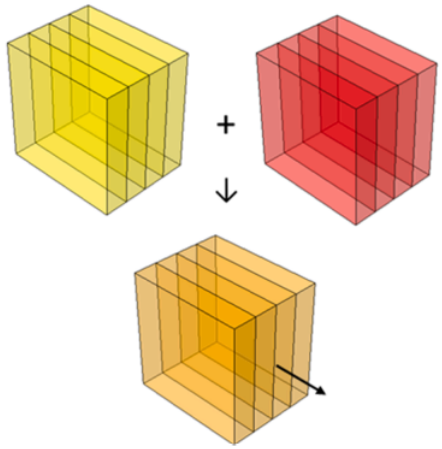


Figure 3.5: Schematic representation of a hybrid bi-phasic numerical discretisation for composite laminates based on superimposed continuum shell and solid elements, used to represent fibre-dominated and matrix-dominated mechanical contributions, respectively. [96].

A direct consequence of the stress decomposition in Eq. (3.28) is that distinct constitutive behaviours may be assigned to each phase. In the original binary models, the fibre-dominated phase is introduced to represent the continuity of the reinforcement architecture and is typically assumed to respond elastically, whereas the matrix-dominated phase embodies the nonlinear and inelastic response of the surrounding medium [95]. This separation enables matrix-driven nonlinearity to be introduced explicitly, without directly compromising the load-carrying capability associated with fibre continuity.

When adapted to composite laminates, the physical interpretation of the two phases differs from that of textile architectures but preserves the same underlying assumptions. In this context, the fibre-dominated phase does not represent the physical fibres alone, but rather an idealised phase responsible for the laminate stiffness associated with fibre continuity across plies. Conversely, the matrix-dominated phase is interpreted as an effective continuum medium that represents composite behaviour in the absence of fibre continuity, thereby governing matrix-driven intralaminar

cracking, interlaminar separation, and other damage mechanisms [19]. Despite this reinterpretation, the additive stress decomposition in Eq. (3.28) remains valid and constitutes the cornerstone of the bi-phasic modelling strategy.

The adoption of a stress-based decomposition is particularly advantageous for problems involving highly localised loading conditions. By allowing the matrix-dominated phase to develop nonlinear or inelastic behaviour independently of the fibre-dominated phase, bi-phasic models provide a physically consistent framework to represent localised matrix deformation while preserving the global structural response of the laminate. This feature is central to the application of bi-phasic approaches to impact and indentation problems, where matrix-dominated mechanisms play a primary role in the early and intermediate stages of the response [19].

3.3.3 Constitutive Description of the Fibre-Dominated Phase

Within the bi-phasic modelling framework, the fibre-dominated phase is introduced to represent the mechanical contribution associated with fibre continuity and load transfer in the composite. In contrast to single-phase damage formulations, where fibre- and matrix-related mechanisms are intrinsically coupled, the bi-phasic approach assigns a distinct constitutive role to the fibre-dominated phase, thereby allowing its response to be defined independently of matrix-driven nonlinearities [94, 95].

In its most common formulation, the fibre-dominated phase is assumed to behave elastically and is characterised by an orthotropic constitutive relation of the form

$$\boldsymbol{\sigma}^{(f)} = \mathbf{C}^{(f)} : \boldsymbol{\varepsilon}, \tag{3.29}$$

where $\mathbf{C}^{(f)}$ denotes the stiffness tensor associated with fibre continuity. This assumption reflects the observation that, under highly localised loading conditions such as impact or indentation, fibres may remain largely elastic while the surrounding matrix undergoes significant nonlinear deformation [19]. By prescribing an elastic response for the fibre-dominated phase, the bi-phasic formulation preserves the load-carrying capability and in-plane stiffness of the laminate even when matrix-dominated mechanisms become active.

It is important to emphasise that the fibre-dominated phase does not represent the physical fibres alone, but rather an idealised mechanical contribution that accounts for the stiffness associated with fibre continuity across the laminate [19]. As such, the constitutive properties assigned to this phase may be interpreted as effective quantities that capture the reinforcement-dominated response at the laminate scale. This interpretation is consistent with the original binary model philosophy, in which the reinforcement architecture is represented through a simplified mechanical description that excludes matrix-driven nonlinearity [94].

The elastic character of the fibre-dominated phase also plays a critical role in ensuring numerical robustness. By preventing matrix-induced nonlinearities from directly degrading the fibre-associated stiffness, the bi-phasic approach mitigates the risk of premature global stiffness loss and enables a clear separation between localised matrix deformation and global structural response [19, 95]. This feature is particularly advantageous in impact analyses, where the preservation of fibre continuity is essential for reproducing the overall dynamic response of the laminate.

In the context of laminate bi-phasic formulations, damage mechanisms associated with fibre failure may be incorporated within the fibre-dominated phase if required; however, such extensions are not intrinsic to the bi-phasic concept itself and depend on the specific modelling objectives [19]. At the level of the present numerical literature review, the fibre-dominated phase is therefore considered as an elastic contribution whose primary function is to represent reinforcement-driven stiffness and load transfer within the bi-phasic framework.

3.3.4 Constitutive Description of the Matrix-Dominated Phase

The matrix-dominated phase constitutes the central component of bi-phasic modelling strategies, as it is responsible for representing nonlinear and inelastic mechanisms that cannot be adequately captured within a purely elastic or damage-only formulation. In contrast to the fibre-dominated phase, which is introduced to preserve stiffness associated with fibre continuity, the matrix-dominated phase is designed to govern deformation modes driven by the polymer matrix, including transverse compression, shear localisation, and irreversible deformation under complex three-dimensional stress states [19].

Within the bi-phasic framework, the constitutive response of the matrix-dominated phase may be formulated by introducing an inelastic strain component into the stress-strain relation. In the present work, this formulation is adopted as a modelling choice to represent the thermoplastic behaviour of the PPS matrix, which is more appropriately described using an elastoplastic constitutive law based on flow plasticity theory. Accordingly, the stress response of the matrix-dominated phase is written as

$$\boldsymbol{\sigma}^{(m)} = \mathbf{C}^{(m)} : (\boldsymbol{\varepsilon} - \boldsymbol{\varepsilon}^{p}), \tag{3.30}$$

where $\mathbf{C}^{(m)}$ denotes the stiffness tensor associated with the matrix-dominated phase and $\boldsymbol{\varepsilon}^{p}$ represents the inelastic (e.g., plastic) strain tensor. This formulation reflects the assumption that matrix-driven nonlinearities are primarily associated with irreversible deformation rather than with immediate stiffness degradation [19, 95]. By separating inelastic deformation from elastic stiffness, the matrix-dominated phase allows permanent deformation to develop without directly impairing the load-carrying contribution of the fibre-dominated phase.

The evolution of the inelastic strain tensor is typically governed by a yield condition and an associated flow rule,

$$f\left(\boldsymbol{\sigma}^{(m)}\right) \leq 0, \tag{3.31}$$

where f denotes a generic yield function defined in terms of the stress state of the matrix-dominated phase. Although different yield criteria and hardening laws may be adopted depending on the material system and modelling objectives, the introduction of a yield surface serves a common purpose: it enables the representation of matrix-driven energy dissipation through inelastic processes rather than through damage-induced stiffness loss [19, 97]. At the level of the present literature review, the specific choice of yield function is of secondary importance compared to the underlying assumption that matrix nonlinearity is fundamentally inelastic in nature.

When applied to composite laminates, the matrix-dominated phase does not represent the polymer matrix alone, but rather an effective continuum medium introduced to describe matrix-driven deformation mechanisms within the laminate plies [19]. In the present modelling context, this effective interpretation is restricted to intralaminar matrix behaviour, including nonlinear deformation and damage processes developing under complex three-dimensional stress states. Interlaminar damage mechanisms, such as delamination between adjacent plies, are instead represented separately through dedicated cohesive-zone formulations. This separation of roles enables matrix-driven intralaminar behaviour to be modelled within the matrix-dominated phase while preserving a clear and consistent treatment of interlaminar damage at ply interfaces.

The use of an inelastic constitutive law for the matrix-dominated phase is particularly relevant for impact and indentation problems. Under localised contact loading, the stress state in the vicinity of the impactor is dominated by transverse compression and shear, conditions under which the polymer matrix may yield and undergo permanent deformation while fibres remain largely elastic [19]. By explicitly allowing inelastic deformation to develop within the matrix-dominated phase, bi-phasic models provide a physically consistent mechanism to reproduce local indentation and energy dissipation while preserving the global stiffness response of the laminate.

In summary, the constitutive description of the matrix-dominated phase represents a key departure from classical single-phase damage formulations. By interpreting matrix nonlinearity as an inelastic process rather than as immediate damage, the bi-phasic approach enables a clear separation between irreversible local deformation and global structural response, thereby offering a robust framework for modelling matrix-driven phenomena in composite laminates [19, 95].

3.3.5 Kinematic Compatibility and Interaction between Phases

A fundamental requirement of bi-phasic modelling strategies is that the fibre-dominated and matrix-dominated phases interact mechanically in a manner that is consistent with the principles of continuum mechanics. Although the two phases are associated with distinct constitutive

behaviours and mechanical roles, they are not intended to represent separate physical bodies. Instead, they are assumed to coexist at the same material point and to undergo identical kinematic fields, ensuring compatibility of deformation and equilibrium of internal forces [94, 95].

This requirement is commonly enforced by assuming that both phases share the same displacement field,

$$u^{(f)} = u^{(m)} = u, \tag{3.32}$$

where $u^{(f)}$ and $u^{(m)}$ denote the displacement vectors associated with the fibre-dominated and matrix-dominated phases, respectively. As a consequence, the strain tensor is identical in both phases, while the stress response differs according to the constitutive relations assigned to each contribution. This kinematic assumption ensures that the additive stress decomposition introduced in Eq. (3.28) remains mechanically admissible and that no artificial separation or sliding occurs between phases [94].

The enforcement of kinematic compatibility allows stress redistribution to occur naturally between the two phases as deformation progresses. When nonlinear or inelastic behaviour develops within the matrix-dominated phase, the fibre-dominated phase may continue to carry load elastically, thereby preserving the global stiffness of the composite. Conversely, under loading conditions where fibre-dominated mechanisms are activated, the matrix-dominated phase may contribute only marginally to the overall response. This interaction mechanism is a defining feature of bi-phasic models and distinguishes them from approaches that rely on explicit interface laws or coupling elements to transfer forces between distinct material regions [95].

In the context of composite laminates, this interaction framework provides a physically consistent means of representing the coexistence of localised matrix-driven deformation and global laminate response. As discussed in [19], the shared kinematic field ensures that matrix-dominated intralaminar and interlaminar mechanisms may develop within the effective matrix phase without inducing spurious discontinuities or compromising fibre continuity. The resulting stress redistribution between phases enables the model to accommodate localised damage or inelasticity while maintaining overall structural integrity.

From a modelling standpoint, the assumption of kinematic compatibility also contributes to numerical robustness. By avoiding the introduction of additional kinematic constraints or interface constitutive laws between phases, the bi-phasic framework limits the number of parameters required for calibration and reduces the risk of numerical instabilities associated with over-constrained formulations [94, 95]. This feature is particularly advantageous for dynamic impact simulations, where stable stress transfer and consistent energy dissipation are essential for accurate response prediction.

In summary, the enforcement of a common displacement field represents a key assumption underlying bi-phasic modelling strategies. It provides the mechanical foundation for stress superposition, enables natural interaction between fibre- and matrix-dominated contributions, and ensures that the resulting formulation remains both physically meaningful and computationally efficient when applied to composite laminates [19].

3.3.6 Relevance of Bi-Phasic Approaches for Impact-Induced Indentation

The relevance of bi-phasic modelling strategies becomes particularly evident in impact scenarios characterised by localised contact and indentation. In such configurations, the composite laminate is subjected to complex three-dimensional stress states dominated by transverse compression and shear in the vicinity of the impactor, while the global structural response remains largely governed by fibre continuity. Experimental observations and numerical investigations consistently indicate that, under these conditions, the polymer matrix plays a dominant role in governing local deformation and energy dissipation, whereas fibre failure is typically a secondary or delayed mechanism [19].

Classical single-phase damage formulations may encounter difficulties in reproducing this behaviour, as matrix-driven nonlinearities are often introduced through stiffness degradation mechanisms that simultaneously affect the global laminate response. As a result, permanent indentation and localised deformation may be accompanied by an unrealistically rapid loss of stiffness or by damage patterns that are not fully consistent with experimental evidence [19]. These limitations motivate the adoption of modelling strategies in which matrix-dominated deformation can be represented independently of fibre-dominated load transfer.

Within the bi-phasic framework, impact-induced indentation is naturally associated with the activation of inelastic mechanisms in the matrix-dominated phase, while the fibre-dominated phase remains predominantly elastic and continues to sustain the global load-carrying function of the laminate. This separation of mechanical roles enables the representation of permanent local deformation through matrix inelasticity, without enforcing premature or excessive degradation of fibre-associated stiffness [19, 95]. Consequently, the global response of the laminate may be preserved even in the presence of significant local indentation.

Furthermore, the shared kinematic field between phases ensures that local matrix-driven deformation is consistently embedded within the overall laminate response. Stress redistribution between the fibre- and matrix-dominated phases allows the model to accommodate highly localised inelastic zones while maintaining equilibrium and compatibility throughout the structure. This feature is particularly advantageous for impact analyses, where accurate representation of contact forces, energy absorption, and post-impact deformation patterns requires a balanced description of both local and global mechanisms [19].

From a numerical standpoint, bi-phasic approaches offer a robust alternative to purely damage-based formulations for the simulation of impact and indentation in composite laminates. By interpreting matrix nonlinearity as an inelastic process rather than as immediate damage, these models provide a physically grounded framework to capture indentation phenomena, progressive energy dissipation, and stiffness retention in a unified manner. This makes bi-phasic strategies especially well suited for the analysis of thermoplastic composite systems, where matrix yielding and plastic flow play a central role in the response to impact loading [19, 97].

Chapter 4

Experimental Tests

This chapter summarizes the essential experimental data used to calibrate and validate the numerical model developed in this work. All results presented here originate from the low-velocity impact tests performed by Martina D'Aiello at Politecnico di Milano, conducted according to ASTM D7136 on Glass/PPS composite laminates. More detailed information regarding the experimental setup and testing procedures can be found in the corresponding Technical Report [72]. Only the information strictly necessary for the finite element modelling—namely the test configuration, specimen geometry, impact energies, initial velocities, and key acceleration and damage-related metrics—is reported. The purpose is to establish the experimental conditions that define the boundary parameters for the simulations discussed in Chapter 5.

4.1 Test Standard and Setup

The experimental campaign followed the procedures of ASTM D7136/D7136M for low-velocity impact testing of composite laminates, as described in Chapter 2. In accordance with this standard, each specimen was positioned over a rectangular cut-out support fixture and clamped at four locations, 6 mm from the edge, to provide simply supported boundary conditions. A hemispherical steel striker with a diameter of 16 mm was mounted on a drop-weight assembly with a total mass of 5.538 kg, ensuring consistent loading conditions across all tests.

A schematic representation of the impact setup used in the campaign is shown in Fig. 4.1, highlighting the support fixture, striker alignment, and overall test configuration reproduced in the finite element model of Chapter 5.

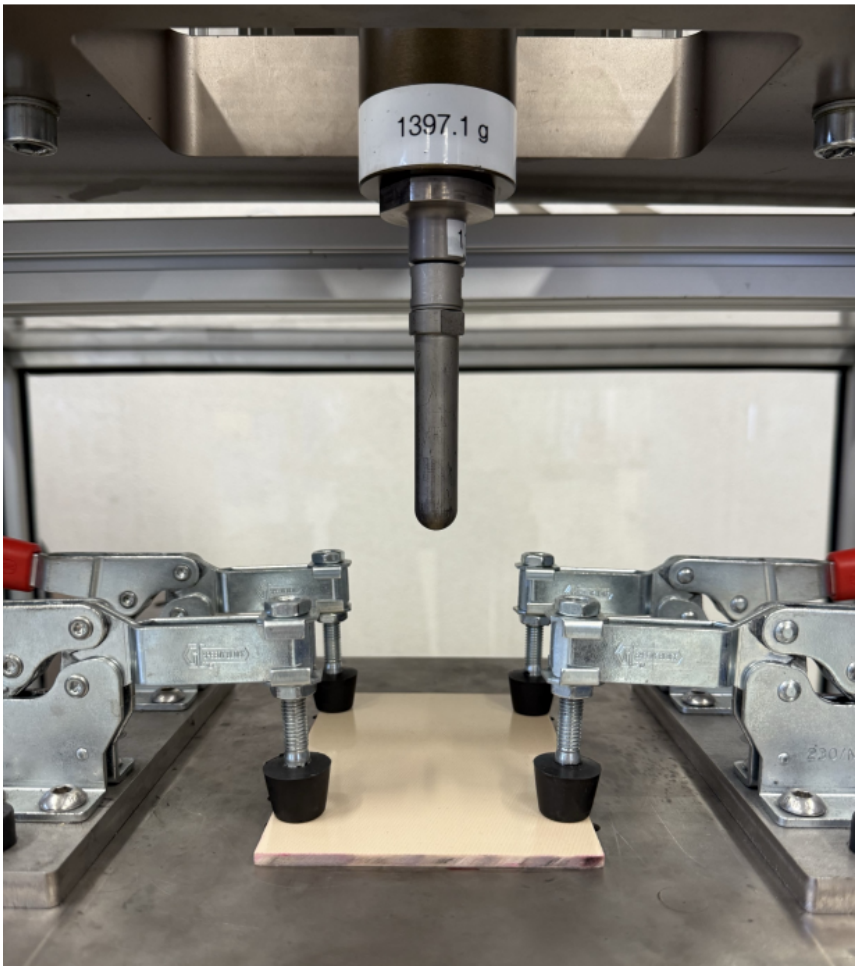


Figure 4.1: Experimental setup of the drop-weight impact device used in the campaign [72].

4.2 Specimen Geometry and Material Description

The specimens tested in the experimental campaign consisted of flat laminates manufactured from a Glass/PPS thermoplastic composite, representative of aerospace-grade impact-resistant materials. The material system corresponds to a commercial Glass/PPS laminate manufactured by TenCate Advanced Composites (currently Toray Advanced Composites). Each plate was cut to nominal in-plane dimensions of 148 mm by 100 mm, while the measured thickness across the ten samples varied within 3.7 ± 0.15 mm. The laminate was fabricated using a 5HS satin weave reinforcement with a symmetric $[0/90]_{8s}$ stacking sequence, consistent with the requirements of ASTM D7136/D7136M for woven composite laminates.

A schematic representation of the specimen geometry is shown in Fig. 4.2, illustrating the plate dimensions and the global reference directions adopted in the numerical model. The corresponding geometric parameters used in the simulations are summarized in Table 4.1.

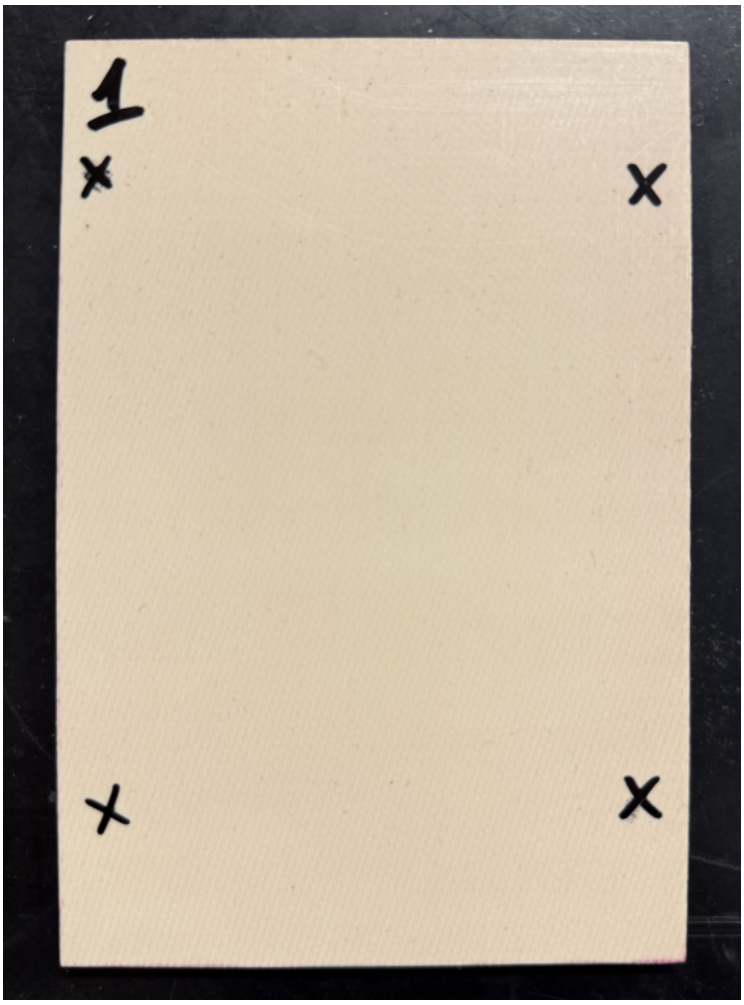


Figure 4.2: Specimen geometry used in the impact tests [72].

Table 4.1: Summary of specimen geometry and laminate configuration used in the experimental tests.

Parameter	**Value**
Plate length	148 mm
Plate width	100 mm
Measured thickness	3.7 ± 0.15 mm
Reinforcement type	5HS satin weave
Stacking sequence	$[0/90]_{8s}$
Matrix material	PPS (Polyphenylene Sulfide)
Fiber type	E-glass

4.3 Impact Energies and Initial Velocities

Each specimen was impacted at a prescribed energy level by releasing the 5.538 kg instrumented impactor from a calculated height, as required by ASTM D7136/D7136M. All impacts were conducted in the low-velocity regime typically used for assessing the damage resistance of composite laminates.

The initial velocity was obtained theoretically from the drop height and experimentally through the integrated velocity measurement system of the impact tower. Since the measured velocities account for frictional losses, the numerical model in Chapter 5 adopts these measured velocities as input conditions. The nominal impact energies and corresponding velocities are summarised in Table 4.2.

Table 4.2: Impact energy levels and corresponding initial velocities used in the experimental campaign. Mean values are computed from three specimens per case, except for Case 4.

Case	**Nominal Energy [J]**	**Theoretical v_0 [m/s]**	**Measured v_0 [m/s]**
1	≈ 4.2	~ 1.28	~ 1.31
2	≈ 7.5	~ 1.66	~ 1.63
3	≈ 14.2	~ 2.28	~ 2.29
4	≈ 27	~ 3.17	~ 3.22

4.4 Experimental Results Relevant for Model Validation

The experimental campaign provides several quantities required for the calibration and validation of the finite element model, namely the measured initial velocity, the front-surface indentation diameter, the rear-surface crack length, and the characteristic acceleration–time response for each impact case. These quantities evolve systematically with increasing impact energy and are used as target parameters in Chapter 5 to assess the predictive capability of the numerical model.

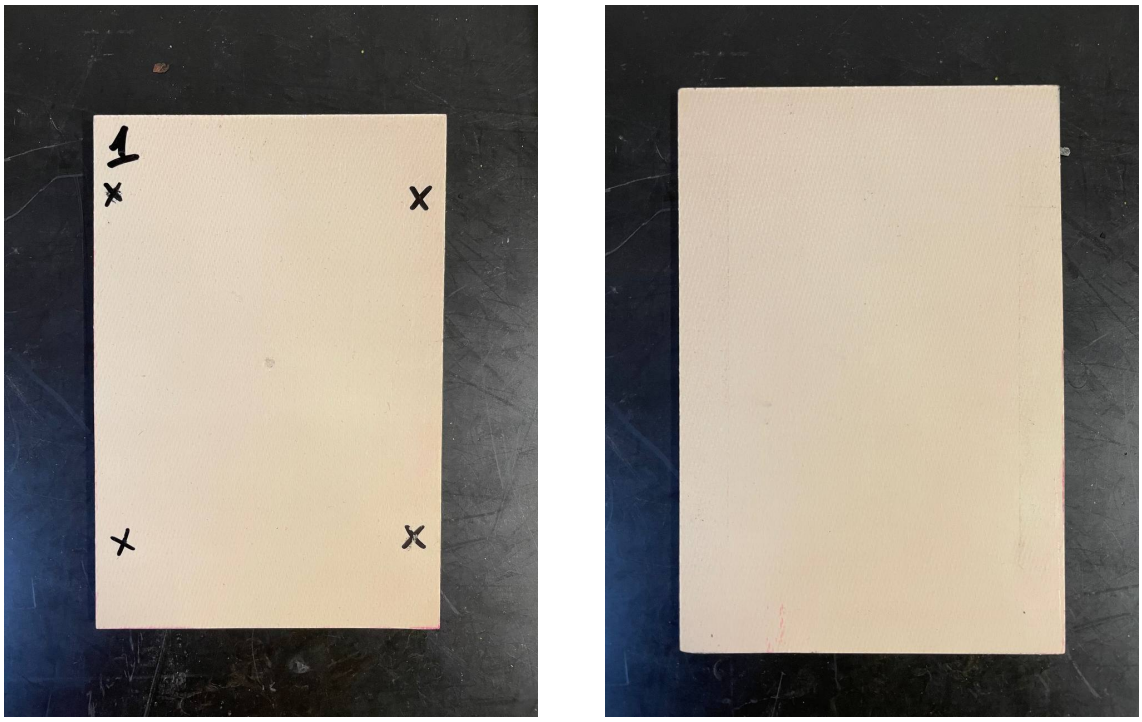
For each nominal impact energy, three specimens were tested to ensure repeatability, with the exception of the highest energy level, for which only one specimen was available. Mean values obtained from the tested specimens are summarised in Table 4.3. These represent the reference quantities used for numerical correlation.

Table 4.3: Summary of experimental quantities relevant for model validation. Mean values are computed from three specimens per case, except Case 4.

Case	**Measured v_0 [m/s]**	**Mean indentation [mm]**	**Mean crack length [mm]**	**No. specimens**
1	1.31	3.9	0.0	3
2	1.63	4.8	3.2	3
3	2.29	7.4	10.3	3
4	3.22	9.0	19.0	1

Case 1 (Nominal Energy $\approx$ 4.2 J)

Representative front- and rear-surface damage for Case 1 are shown in Fig. 4.3. As expected for this low-energy regime, surface indentation is shallow and no visible cracking is observed on the rear surface. The corresponding acceleration–time responses of the three specimens are presented in Fig. 4.4, illustrating the consistency and low scatter across the sample group.



(a) Front surface (indentation). (b) Rear surface (no visible cracks).

Figure 4.3: Representative damage observed for Case 1 specimens (nominal energy $\approx$ 4.2 J) [72].

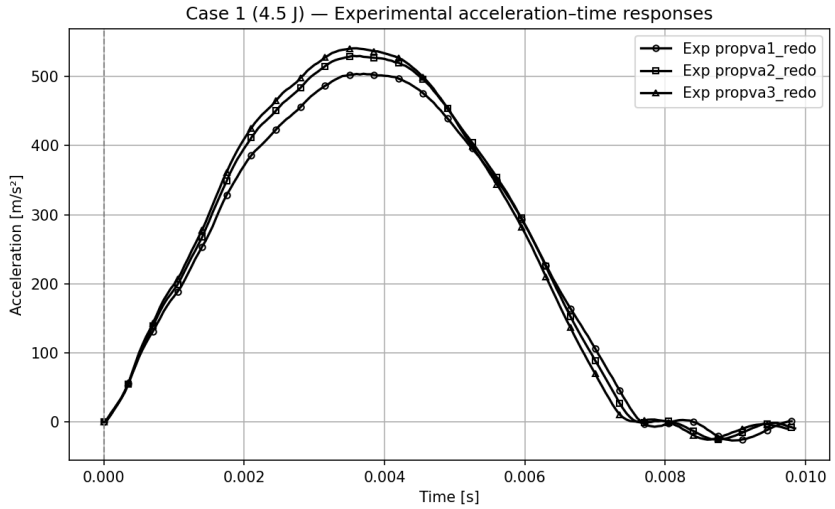
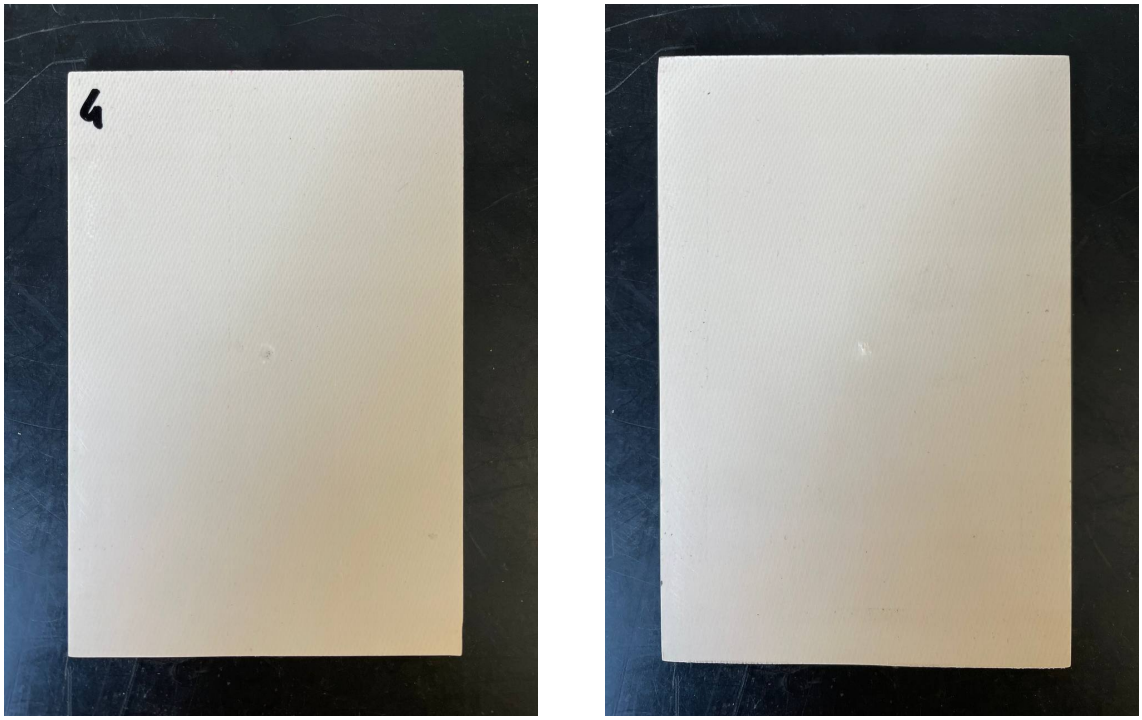


Figure 4.4: Acceleration–time responses of the three Case 1 specimens.

Case 2 (Nominal Energy ≈ 7.5 J)

Front and rear views of a representative Case 2 specimen are shown in Fig. 4.5. Compared to Case 1, a modest increase in indentation depth is observed, accompanied by the onset of rear-surface cracking. The three corresponding acceleration–time curves are displayed in Fig. 4.6, highlighting the expected rise in peak acceleration with impact velocity.



(a) Front surface (indentation). (b) Rear surface (crack formation).

Figure 4.5: Representative damage observed for Case 2 specimens (nominal energy ≈ 7.5 J) [72].

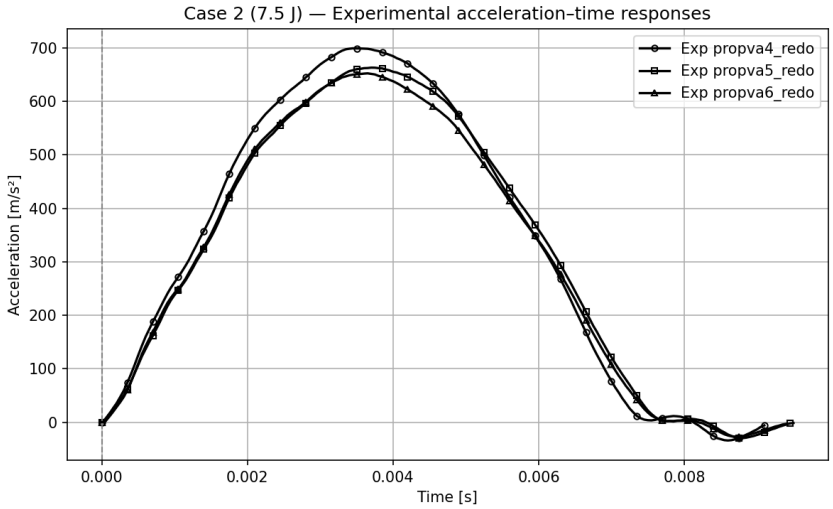


Figure 4.6: Acceleration–time responses of the three Case 2 specimens.

Case 3 (Nominal Energy ≈ 14.2 J)

The damage pattern for Case 3, shown in Fig. 4.7, exhibits more pronounced indentation and clearly identifiable rear-surface cracking, consistent with the mean values reported in Table 4.3. Acceleration–time curves in Fig. 4.8 reflect a further increase in peak response and contact duration.

(a) Front surface (indentation).

(b) Rear surface (crack propagation).

Figure 4.7: Representative damage observed for Case 3 specimens (nominal energy ≈ 14.2 J) [72].

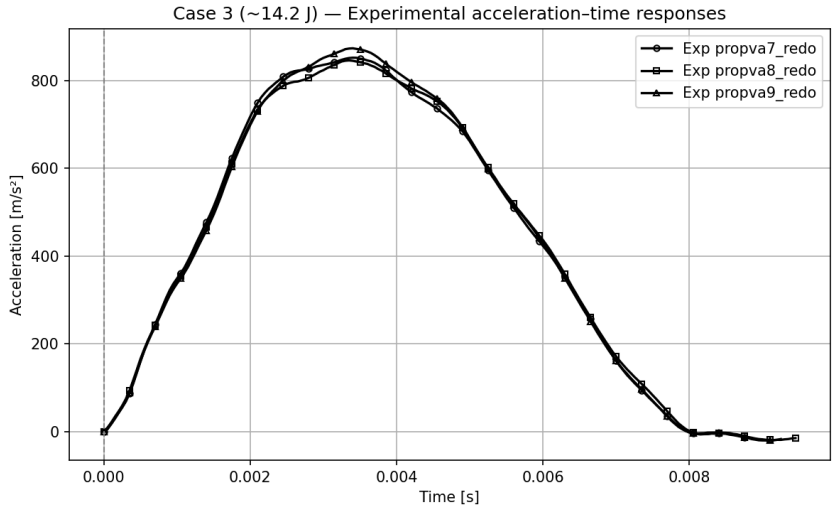
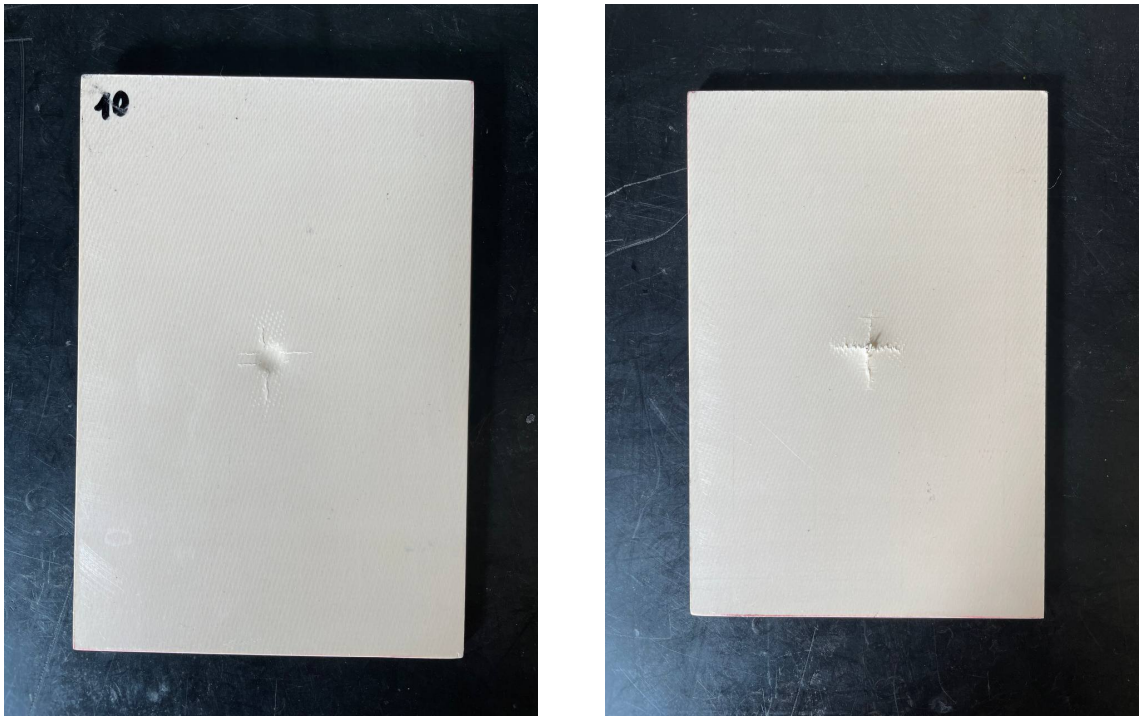


Figure 4.8: Acceleration–time responses of the three Case 3 specimens.

Case 4 (Nominal Energy ≈ 27 J)

For the highest energy level, only one specimen was available. The corresponding damage morphology is shown in Fig. 4.9, displaying the deepest indentation and the longest crack length among all tested conditions. The acceleration–time response for this specimen is shown in Fig. 4.10.



(a) Front surface (indentation). (b) Rear surface (crack propagation).

Figure 4.9: Representative damage observed for Case 4 specimen (nominal energy ≈ 27 J) [72].

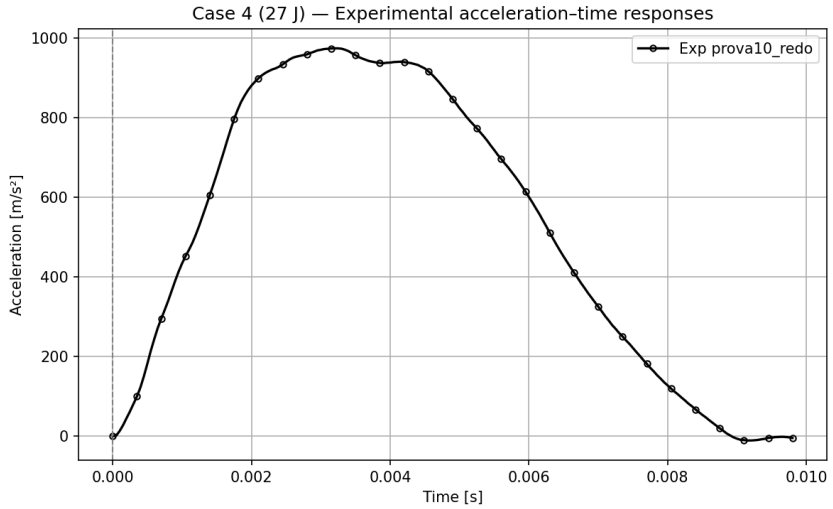


Figure 4.10: Acceleration–time response of the Case 4 specimen.

In addition to the case-specific acceleration responses presented previously, a consolidated comparison of the mean experimental curves for all impact cases is shown in Fig. 4.11. This plot provides a compact representation of the progressive change in the acceleration–time response with increasing impact energy, highlighting the systematic increase in peak acceleration and contact duration across the tested conditions. The use of mean curves reduces specimen-to-specimen variability and facilitates a clearer identification of the underlying trends, thereby offering a convenient reference for the numerical validation presented in Chapter 5.

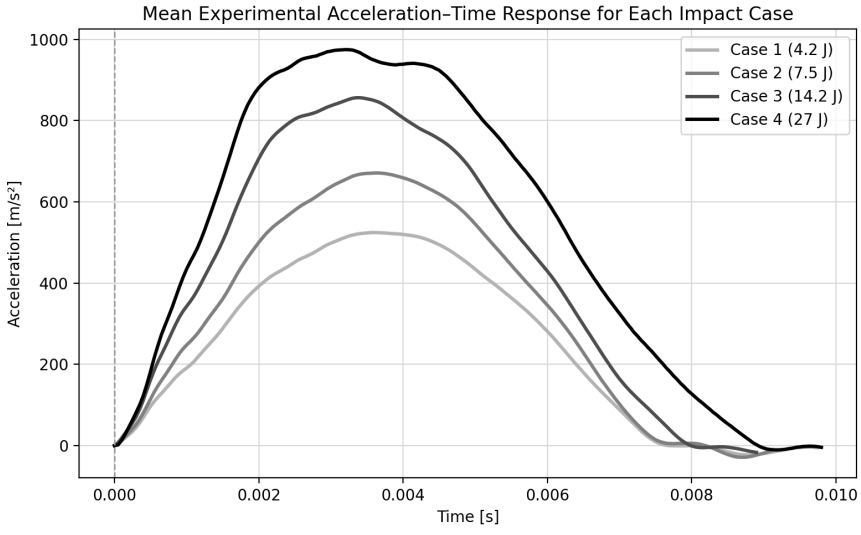


Figure 4.11: Mean experimental acceleration-time responses for all impact cases, obtained by averaging the processed signals within each energy level.

4.5 Summary of Experimental Observations

The experimental campaign set up a consistent set of damage metrics and impact conditions that are used in the numerical modeling in Chapter 5. The initial velocities, indentation diameters, and rear-surface crack lengths all showed clear patterns as the impact energy increased. For example, Case 1 (4.2 J) only caused shallow surface indentation and no visible cracking, while Case 4 (27 J) caused the biggest indentation and the longest visible rear crack. The acceleration–time responses showed that the peak response and contact time got longer over time. This is due to the higher energy levels, which caused more load transfer and structural deformation.

Using three samples for the first three energy levels made sure that the results could be repeated and made it possible to make averaged reference curves, which lessened the effect of differences between samples. These average responses serve as strong experimental benchmarks for confirming the finite element predictions of acceleration history, peak acceleration, indentation depth, and the extent of rear-side damage. In general, the experimental dataset provides a thorough and dependable basis for evaluating the predictive power of the numerical models that will be discussed in the next chapter.

Chapter 5

Development and Validation of the Numerical Impact Model

In order to complement the experimental investigation and provide deeper insight into the mechanisms governing impact damage in Glass/PPS laminates, a numerical modelling framework was developed using the commercial finite element software Abaqus/Explicit. The objective of this chapter is to describe the modelling strategy adopted to simulate the low-velocity impact tests and to progressively introduce the different physical mechanisms considered in the numerical model. The development of the model follows an incremental approach, beginning with a baseline elastic representation of the impact event and gradually incorporating additional features such as intralaminar damage, interlaminar delamination, and matrix plasticity. This step-by-step methodology allows the influence of each modelling assumption on the predicted structural response and damage evolution to be systematically evaluated, ultimately leading to a more comprehensive representation of the impact behaviour of the composite laminate. All numerical simulations were performed using the N–mm–ms–tonne unit system.

5.1 Numerical Model Overview and Modelling Strategy

This section presents the numerical model developed in Abaqus to reproduce the experimental impact tests described in Chapter 4. The modelling strategy was defined to ensure a faithful representation of the test configuration while maintaining numerical robustness and computational efficiency within an explicit dynamic framework. The geometry definition, assembly, interactions, boundary conditions, loading, analysis steps, and mesh strategy and discretization are described sequentially to provide a clear and reproducible description of the adopted approach.

5.1.1 Geometry Definition

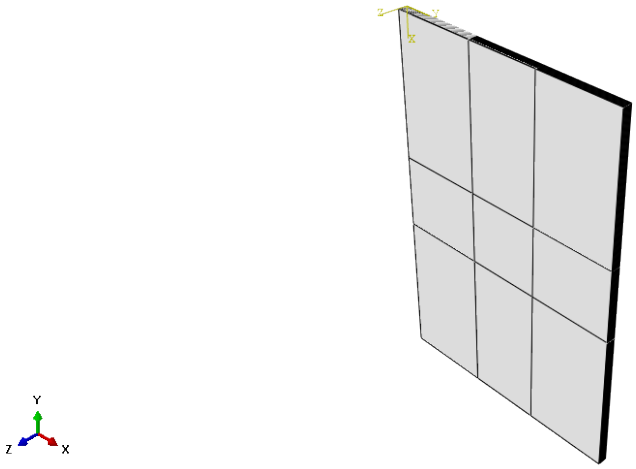
The numerical model geometry was defined to reproduce the experimental impact configuration described in Chapter 4, while maintaining a balance between geometrical fidelity and modelling efficiency. The model comprises the composite laminate plate, the impactor, and the

supporting fixture system, including the base and rubber clamping elements, which were idealized in accordance with the ASTM standard adopted in the experimental campaign.

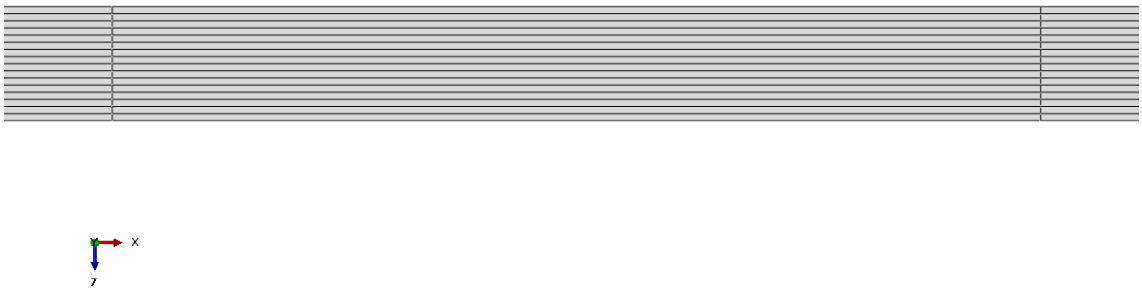
Composite Plate

The composite plate was modelled using its full geometry, without the application of symmetry conditions. This choice was motivated by the need to capture potential asymmetric damage patterns and stress-wave propagation effects arising from impact loading, particularly when interlaminar damage mechanisms are activated.

The panel dimensions were defined as $148 \times 100 \times 3.7$ mm³, corresponding directly to the dimensions measured for the tested specimens. A ply-wise partition strategy was adopted, whereby the laminate thickness was subdivided to explicitly represent each individual ply. This geometrical discretization enables a direct association between damage variables and specific laminate plies, which is essential for the analysis of intralaminar failure mechanisms and for the subsequent introduction of interlaminar damage models. Figure 5.1 represents this part modelled and ply partition strategy.



(a) Part representation.



(b) Ply partition strategy.

Figure 5.1: Composite Panel.

Impactor

The impactor was represented as a rigid body with a hemispherical geometry and a diameter of 16 mm, consistent with the experimental impactor used in the test campaign. The rigid body assumption is justified by the substantially higher stiffness of the impactor compared to the composite laminate, ensuring that all deformation and damage occur within the specimen.

The impactor was associated with a reference point, which allows the application of the prescribed initial velocity and the extraction of acceleration-time histories used for numerical–experimental correlation. Figure 5.2 represents this part modelled.

Figure 5.2: Impactor.

Supporting Base

The supporting base was modelled as a rigid frame with simplified geometry representative of the experimental fixture specified by the ASTM standard. The base consists of a rectangular outer frame with dimensions of 170 × 120 mm² and a uniform thickness of 20 mm, within which a rectangular inner opening of 125 × 75 mm² is defined. This inner opening establishes the effective unsupported area of the composite panel during impact and therefore plays a key role in determining the global stiffness and dynamic response of the specimen.

The base geometry was simplified to retain its mechanical function while avoiding unnecessary geometric detail not relevant to the impact response. Figure 5.3 presents the modelled support base part.

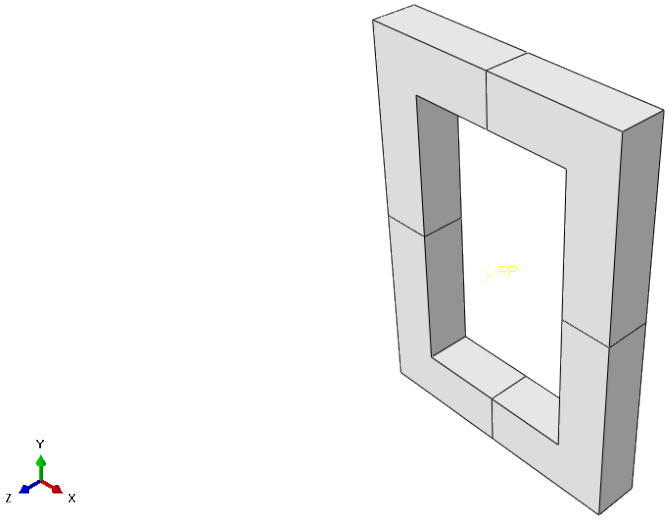


Figure 5.3: Supporting Base.

Rubber Clamping Elements

The rubber clamping elements were modelled with simplified cylindrical geometry consistent with the experimental fixture. Each clamp was defined with a diameter of 10 mm and a height of 5 mm, parameters that primarily govern the contact area with the laminate and the compliance introduced at the plate boundaries. These dimensions were retained explicitly in the numerical model in order to reproduce the boundary conditions imposed during the experimental impact tests, while maintaining a reduced geometrical complexity. Figure 5.4 presents the rubber clamp modelled part.

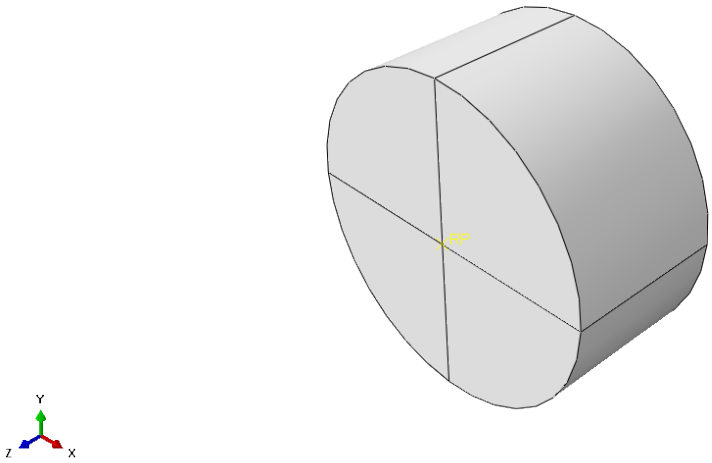


Figure 5.4: Rubber clamp.

Summary of Geometrical Modelling Parameters

For clarity and completeness, the main geometrical parameters adopted in the numerical model are summarized in Table 5.1 These dimensions directly reflect the experimental configuration and define the effective unsupported area, boundary conditions, and contact interfaces governing the impact response.

Table 5.1: Summary of geometrical parameters adopted in the numerical model

Component	Geometrical description
Composite panel	Rectangular plate, $148 \times 100 \times 3.7$ mm^3
Impactor	Hemispherical rigid body, 16 mm diameter
Supporting base	Outer frame 170×120 mm^2, inner opening 125×75 mm^2, thickness 20 mm
Rubber clamps	Cylindrical geometry, 10 mm diameter, 5 mm height

5.1.2 Assembly and Interaction Definition

The numerical assembly was defined to accurately reproduce the experimental impact configuration described in Chapter 4, ensuring consistent positioning of all components and a physically representative definition of kinematic interactions. The assembly comprises the composite panel, the supporting base, the rubber clamping elements, and the impactor, arranged to replicate the effective support conditions and impact alignment of the experimental fixture.

The composite panel was positioned centrally within the supporting base, such that the rectangular inner opening of the base defined the effective unsupported region of the laminate during impact. In the implemented configuration, the panel was supported by contact bands extending 11.5 mm from the top and bottom edges and 12.5 mm from each lateral edge. Accounting for the overlap at the four panel corners, this configuration resulted in a total support contact area of 5329 mm², corresponding to approximately 36.0% of the total panel planform area. This supported region governs the effective boundary stiffness of the specimen and plays a significant role in the global dynamic response during impact.

The rubber clamping elements were positioned between the composite panel and the supporting base to reproduce the compliant boundary conditions observed in the experimental setup. Each clamp was located with an offset of 6 mm from the lateral edges of the panel and 25 mm from the top and bottom edges, ensuring a consistent clamping configuration and a well-defined load transfer path at the panel boundaries.

The supporting base and the impactor were defined as rigid bodies and associated with dedicated reference points. Rigid body constraints were used to couple the motion of each rigid component

to its corresponding reference point, ensuring that their kinematic behaviour was fully defined by the reference point motion.

To reproduce the compliant behaviour of the rubber clamping system, the clamping elements were modelled as deformable bodies. A rigid body constraint was applied only to the top surface of each rubber clamp, coupling this surface to a reference point while leaving the remaining clamp volume free to deform. This modelling strategy allows the clamps to introduce boundary compliance while maintaining a controlled kinematic interface with the supporting base.

The impactor inertia was defined by assigning a lumped mass of 5.537 kg to the impactor reference point. This approach ensures a correct representation of the impactor inertial response while preserving the rigid body assumption adopted for its geometry.

The impactor was positioned normal to the laminate surface and aligned with the geometric centre of the composite panel, corresponding to a centred impact condition. An initial clearance was defined between the impactor and the panel to avoid initial contact at the start of the analysis. Figure 5.5 presents the final assembly.

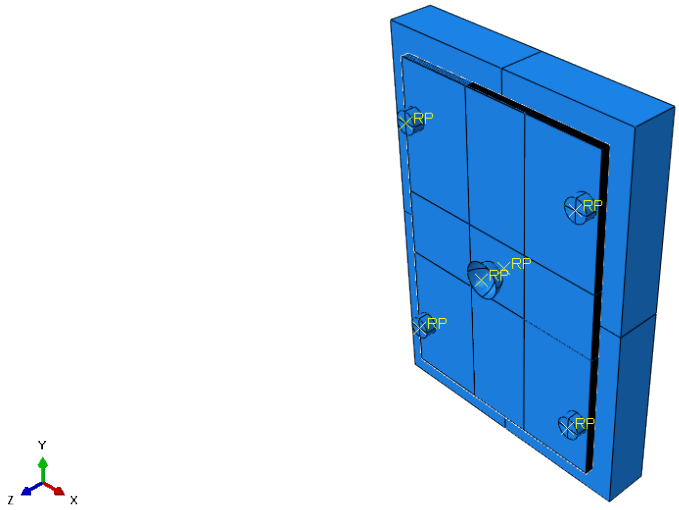


Figure 5.5: Assembly.

5.1.3 Contact and Interaction Modelling

Following the definition of the numerical assembly, contact and interaction formulations were introduced to govern force transmission between the impactor, composite panel, rubber clamping elements, and supporting base. A unified interaction framework was adopted to ensure robust and consistent treatment of all contact interfaces throughout the simulation.

Contact Formulation

All contact interactions in the model were defined using the General Contact formulation. This approach allows for the flexible and automatic detection of contact between multiple interacting surfaces, which is particularly suitable for impact simulations involving complex interaction states and evolving contact conditions.

The General Contact formulation was applied globally to account for interactions between:

- the impactor and the composite panel;
- the composite panel and the rubber clamping elements;
- the composite panel and the supporting base.

This strategy avoids the need for predefined contact pairs and ensures consistent contact handling during large deformations and dynamic interaction events.

Normal Contact Behaviour

The normal contact behaviour was defined using a hard contact formulation. This assumption enforces a non-penetration condition between interacting surfaces while allowing separation once contact forces are no longer transmitted. The hard contact definition is appropriate for impact problems, where normal force transmission dominates the interaction response and physical interpenetration between components must be avoided.

Tangential Contact Behaviour

Tangential contact behaviour was modelled using a penalty friction formulation with an arbitrary constant friction coefficient of 0.2. This value was selected to represent moderate frictional interaction between contacting surfaces while avoiding excessive numerical constraint that could artificially increase energy dissipation.

The penalty-based friction formulation provides a stable and computationally efficient representation of tangential force transfer during contact, which is particularly suitable for explicit dynamic analyses.

Modelling Considerations

The contact modelling strategy adopted in this work prioritizes numerical robustness and physical representativeness while maintaining a reduced level of modelling complexity. By combining a general contact formulation with hard normal behaviour and moderate penalty-based friction, the model ensures consistent force transmission across all interfaces without introducing unnecessary numerical constraints.

Detailed contact stiffness parameters were handled internally by the solver and were not explicitly modified, allowing the contact response to be governed primarily by the defined interaction laws and the evolving kinematics of the system.

5.1.4 Boundary Conditions

Boundary conditions were applied after the definition of the numerical assembly and contact interactions, ensuring that all constraints acted on physically meaningful reference points and reproduced the experimental fixture conditions without introducing artificial stiffness into the numerical model.

The reference points associated with the supporting base and with each rubber clamp were assigned an *encastre* boundary condition, enforcing

$$U_x = U_y = U_z = 0, \qquad R_x = R_y = R_z = 0.$$

These constraints prevent rigid body motion of the support and clamping system while maintaining a fixed reference for the numerical assembly. The boundary conditions are applied at the reference points that control the rigid-body motion of the base and clamps, as illustrated in Figure 5.6. This configuration reproduces the fixed support conditions of the experimental fixture without constraining the composite panel directly.

The rubber clamping elements were constrained through their kinematic coupling to the supporting base. As described in Section 5.1.2, a rigid body constraint was applied only to the top surface of each rubber clamp, coupling this surface to a reference point while leaving the remaining clamp volume free to deform. An encastre boundary condition was applied at each clamp reference point. This strategy prevents rigid body motion of the clamps while preserving their ability to deform and introduce compliance at the panel boundaries, reproducing the experimental clamping behaviour.

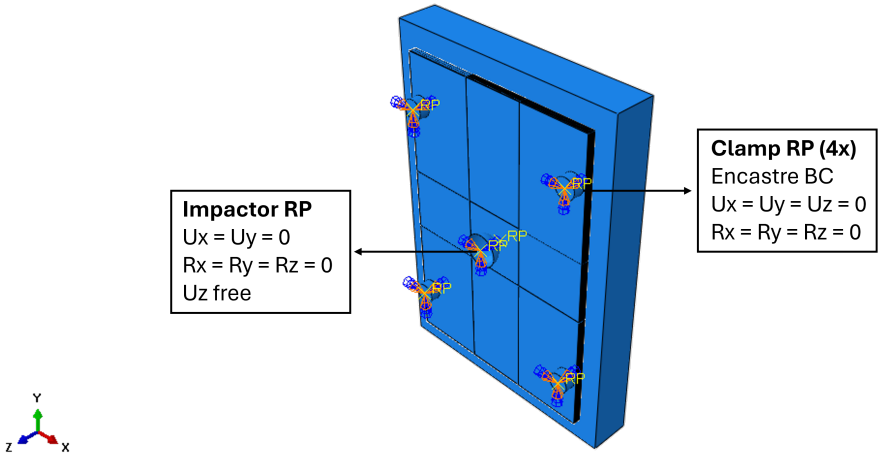
The motion of the impactor was controlled through its associated reference point. The impactor reference point was constrained such that motion was permitted only along the axis perpendicular to the composite panel, corresponding to the z-axis in the adopted coordinate system. Accordingly, the boundary conditions imposed were

$$U_x = U_y = 0, \qquad R_x = R_y = R_z = 0,$$

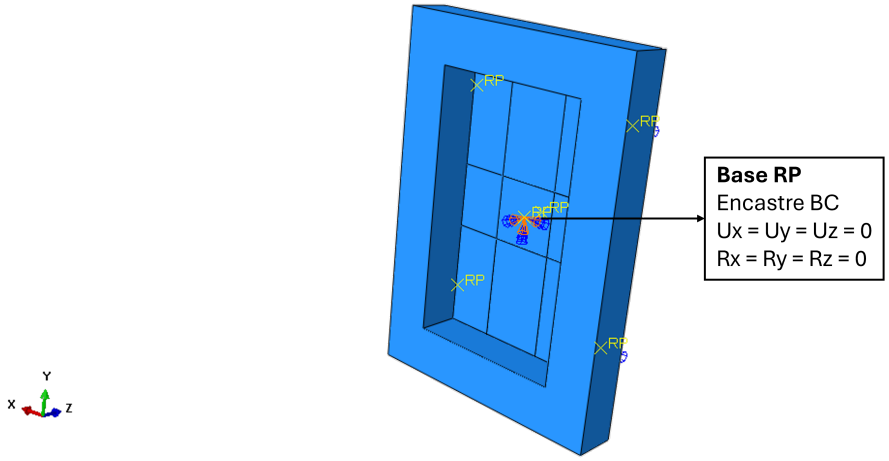
while the displacement along the z direction remained free to evolve according to the imposed inicial velocity. These constraints enforce a normal, centred impact condition consistent with the experimental configuration.

No boundary conditions were applied directly to the composite panel surfaces involved in contact with the impactor, rubber clamps, or supporting base. Deformation and force transmission within the laminate were therefore governed exclusively by the defined contact interactions and material response.

Figure 5.6 illustrates the implementation of the boundary conditions in the numerical model, highlighting the reference points associated with the supporting base and rubber clamps where the encastre constraints are applied.



(a) Front view of the numerical assembly showing the applied boundary conditions. The reference points associated with the four rubber clamps are assigned encastre boundary conditions ($U_x = U_y = U_z = 0$, $R_x = R_y = R_z = 0$). The impactor reference point is constrained such that $U_x = U_y = 0$ and $R_x = R_y = R_z = 0$, allowing motion only along the global z-axis.



(b) Rear view of the numerical assembly illustrating the encastre boundary condition applied to the supporting base reference point ($U_x = U_y = U_z = 0$, $R_x = R_y = R_z = 0$).

Figure 5.6: Implementation of the boundary conditions in the finite element model. The clamping system and supporting base are fully constrained through encastre boundary conditions applied at their respective reference points, while the impactor motion is restricted to the global z-direction in order to reproduce a normal impact configuration.

5.1.5 Loading Conditions

Impact loading was applied through a predefined velocity field assigned to the reference point associated with the impactor. The velocity was prescribed along the axis normal to the composite panel, corresponding to the z-axis in the adopted coordinate system, ensuring a normal impact condition consistent with the experimental configuration.

The magnitude of the prescribed initial velocity was selected to reproduce the experimental impact energies investigated in this work. For each simulated case, the initial kinetic energy of the impactor was defined through the combination of the imposed velocity and the impactor mass, allowing direct correspondence between numerical simulations and experimental tests.

The predefined velocity was applied at the beginning of the analysis step, after which no additional external loads were introduced. The subsequent response of the system was therefore governed exclusively by inertial effects, contact interactions, boundary conditions, and material behaviour.

5.1.6 Analysis Step and Solution Procedure

The impact event was simulated using a Dynamic, Explicit analysis step, which is well suited for short-duration transient problems involving complex contact interactions and nonlinear response. A single explicit step was employed to resolve the complete impact process.

The total step time was set to 0.01 s (10 ms). This duration was selected based on the experimental observations and was found to be sufficient to reproduce all relevant phases of the impact event recorded during testing, while avoiding unnecessary extension of the simulation beyond the physically meaningful response.

Geometric nonlinearity was enabled for all simulations, allowing for large displacements and rotations to be accurately captured during impact. No artificial mass scaling was applied, ensuring that the dynamic response of the system was governed solely by the physical mass distribution defined in the model.

Time integration was performed using the central difference scheme inherent to the explicit formulation. The stable time increment was automatically determined by the solver based on the smallest characteristic element dimension and the material wave speed, ensuring numerical stability throughout the analysis.

Output Requests

Field output variables were defined to capture the global and local response of the system during impact. The following field variables were recorded:

- A: acceleration;
- S: stress;
- E: strain;
- U: displacement;
- V: velocity;
- RF: reaction force.

Field outputs were requested at evenly spaced time intervals, with a total of 20 intervals over the analysis step. This choice was made to ensure temporal consistency with the experimental data acquisition and to facilitate direct numerical–experimental comparison.

Two distinct history output requests were defined to monitor the system response and assess numerical stability:

- Impactor reference point:
  - Acceleration (A3), reaction force (RF3), displacement (U3), and velocity (V3) were recorded along the z-direction, enabling direct comparison with experimentally measured impactor kinematics.
- Whole model (global energies):
  - Global energy quantities were monitored, including ALLAE, ALLFD, ALLIE, ALLKE, ALLVD, and ETOTAL. These outputs were used to verify the overall energy balance of the system and to assess the influence of numerical artefacts, particularly hourglass energy, during the simulation.

The definition of these output requests provides a comprehensive basis for evaluating both the physical response of the system and the numerical quality of the explicit solution. Detailed analysis of energy contributions and hourglass effects is presented in Section 5.1.7 and Appendix A.

5.1.7 Mesh Strategy and Discretization

The finite element discretization plays a central role in the accuracy and numerical stability of explicit dynamic simulations, particularly for impact problems involving thin composite laminates. The mesh strategy adopted in this work was therefore designed to ensure adequate spatial resolution in physically critical regions, while maintaining numerical robustness and reasonable computational cost. This section presents the element selection for each model component, the refinement strategy adopted for the composite panel, the verification of numerical stability through energy monitoring, and a summary of the final mesh configuration retained for the subsequent analyses.

5.1.7.1 Discretization strategy and element selection per component

The discretization strategy was defined to ensure numerical robustness, accurate representation of the physical response, and computational efficiency within the explicit dynamic framework. Different element formulations were adopted for each component of the numerical model based on their mechanical role, expected deformation modes, and interaction characteristics during impact.

The selection of element types followed a component-specific approach, guided by the following general criteria:

- expected level of deformation and damage;
- relevance of through-thickness stress representation;
- role in contact interactions;
- influence on numerical stability and computational cost.

Based on these considerations, the discretization choices for each component are described below.

Composite Panel

- Element type: three-dimensional continuum shell elements (SC8R);
- Rationale:
  - suitable for thin laminated structures subjected to impact loading;
  - allow representation of through-thickness stresses while maintaining computational efficiency;
  - compatible with ply-wise discretization strategies and damage modelling approaches.

The composite panel represents the primary load-bearing and energy-absorbing component of the system and therefore requires the highest level of numerical fidelity. A detailed discussion of the panel discretization strategy, the justification for the selection of SC8R elements, and the associated intralaminar damage formulation adopted for the laminate are provided in Subsection 5.1.7.2.

Rubber clamping elements

- Element type: three-dimensional reduced-integration solid elements (C3D8R);
- Rationale:
  - required to capture compressive deformation of the rubber material;

  - allow realistic representation of clamp compliance;
  - ensure consistent force transmission between the composite panel and the supporting base.

Supporting base

- Element type: three-dimensional reduced-integration solid elements (C3D8R);
- Rationale:
  - appropriate for a component primarily acting as a rigid support;
  - ensure stable contact interactions;
  - minimize computational cost while preserving numerical robustness.

Impactor

- Element type: rigid surface elements (R3D4);
- Rationale:
  - enable modelling of the impactor as a non-deformable body;
  - reduce the number of degrees of freedom;
  - allow accurate definition of kinematic and inertial properties through a reference point.

Overall, the adopted discretization strategy concentrates numerical resolution in the composite panel—where deformation, damage initiation, and energy absorption are critical—while employing simpler and computationally efficient formulations for components that do not govern the structural response.

5.1.7.2 Composite panel discretization and continuum shell formulation

The composite panel is the most critical component of the numerical model, as it governs the global structural response, damage initiation, and energy absorption during impact. Its discretization therefore required particular attention to ensure an accurate representation of the laminate behaviour, while maintaining numerical stability and computational efficiency within the explicit dynamic framework.

Discretization requirements for the composite panel

From both physical and numerical perspectives, the discretization of the composite panel must satisfy a set of requirements arising from its geometry, layered architecture, and role in the impact event:

- Ply-level resolution

  - The laminate is composed of multiple plies with distinct fibre orientations;
  - The discretization must allow stresses, strains, and damage variables to be associated with individual plies.

- Three-dimensional stress representation
  - Impact loading induces significant through-thickness stresses, particularly beneath the impactor;
  - The element formulation must therefore be capable of representing a three-dimensional stress state, including thickness-direction strains;
  - intralaminar failure initiation is evaluated using the two-dimensional plane-stress Hashin criteria described in Chapter 3, consistent with the progressive damage formulation implemented in Abaqus for laminated composites. Out-of-plane damage mechanisms are captured separately through the interlaminar cohesive elements introduced between plies.

- Robust contact behaviour
  - The panel is in direct contact with the impactor and the supporting fixture;
  - The discretization must remain stable under high localized contact pressures and indentation.

- Compatibility with explicit dynamic analysis
  - The stable time increment is governed by the smallest characteristic element dimension and the material wave speed;
  - Excessive through-thickness refinement using conventional solid elements would lead to prohibitively small time increments and increased computational cost.

- Numerical robustness under stiffness degradation
  - Progressive damage and stiffness reduction may exacerbate numerical instabilities;
  - The chosen formulation must remain stable in the presence of material softening.

Rationale for the use of continuum shell elements (SC8R)

Based on the above requirements, the composite panel was discretized using three-dimensional continuum shell elements (SC8R), as implemented in Abaqus/Explicit. This element formulation combines features of both solid and shell elements and has been widely adopted for the numerical modelling of thin composite laminates subjected to impact loading [7, 21].

The main reasons for adopting SC8R elements in the present work are summarised below:

- Three-dimensional formulation
  - Continuum shell elements possess distinct top and bottom faces and retain a full three-dimensional stress state;
  - This allows the computation of thickness-direction strains and stresses, which are relevant for impact loading and interlaminar damage mechanisms.
- Ply-wise discretization capability
  - The laminate thickness was partitioned such that one continuum shell element is used per ply;
  - This strategy provides ply-level resolution of the structural response; however, it is noted that SC8R elements are displacement-based, and the accuracy of bending and transverse shear stresses depends on the numerical integration scheme and on the transverse shear stiffness derived from the laminate stacking sequence.
- Improved numerical efficiency in explicit dynamics
  - Although SC8R elements have the same number of nodes and translational degrees of freedom as equivalent solid elements, their formulation alleviates locking phenomena that may affect thin solid discretizations;
  - more importantly, the use of continuum shell elements avoids extremely small element thicknesses, allowing significantly larger stable time increments in explicit analyses compared to fully solid discretizations of thin plies.
- Robust contact performance
  - Owing to their finite thickness and two-sided geometry, continuum shell elements are treated by contact algorithms in a manner analogous to solid elements;
  - this results in improved numerical robustness in contact-dominated problems, particularly under localized indentation and impact loading, compared to conventional shell formulations [8].

Final element formulation choice for the composite panel

Alternative modelling approaches, such as fully solid elements or conventional shell elements, can in principle be employed to represent composite laminates. However:

- fully solid discretizations of thin laminates impose severe restrictions on the stable time increment and lead to excessive computational cost in explicit dynamic analyses;
- conventional shell elements may exhibit limitations in representing through-thickness stresses and in handling localized contact effects under impact loading.

Considering these aspects, SC8R continuum shell elements were selected as the final discretization choice for the composite panel. This formulation provides an effective balance between physical representativeness, numerical stability, and computational efficiency and was retained consistently across all simulations presented in this work. The suitability of continuum shell elements for composite impact modelling is well documented in the literature and in the Abaqus Analysis User's Guide [21].

5.1.7.3 Mesh refinement strategy and local resolution

A non-uniform mesh refinement strategy was adopted to accurately resolve the highly localized response of the composite panel during impact while maintaining reasonable computational efficiency in the explicit dynamic analysis. Mesh refinement was therefore applied only in regions where high stress gradients and damage initiation were expected, while a coarser discretization was employed elsewhere.

Refinement strategy

The mesh design was guided by physical considerations associated with the impact event rather than by purely geometric criteria. In particular, localized refinement was introduced in the vicinity of the impact zone, where experimental observations revealed concentrated damage and a characteristic crack footprint.

The adopted strategy can be summarized as follows:

- a locally refined region was defined around the impact location to capture indentation and damage initiation;
- outside this region, the mesh density was controlled by a global seeding strategy;
- the resulting discretization represents a compromise between local resolution in the impact zone and overall computational efficiency.

Impact-zone refinement

A square refinement region with a side length of 30 mm was introduced at the centre of the composite panel, corresponding to the characteristic damage extension observed experimentally. Within this region, the in-plane mesh density was selected to be consistent with the through-thickness discretization adopted for the laminate.

Since the composite panel has a total thickness of 3.7 mm and is composed of 16 plies, a one-element-per-ply discretization results in a characteristic through-thickness element dimension of

$$t_{\text{ply}} = \frac{3.7}{16} \approx 0.231\,\text{mm}. \tag{5.1}$$

To achieve a near-unity in-plane to through-thickness element aspect ratio within the refined region, the 30 mm partition was discretized using 130 elements per side. This yields an in-plane element size of

$$\Delta_{\mathrm{IP}} = \frac{30}{130} \approx 0.231\,\mathrm{mm}, \tag{5.2}$$

resulting in elements with approximately equal in-plane and through-thickness dimensions at the ply level.

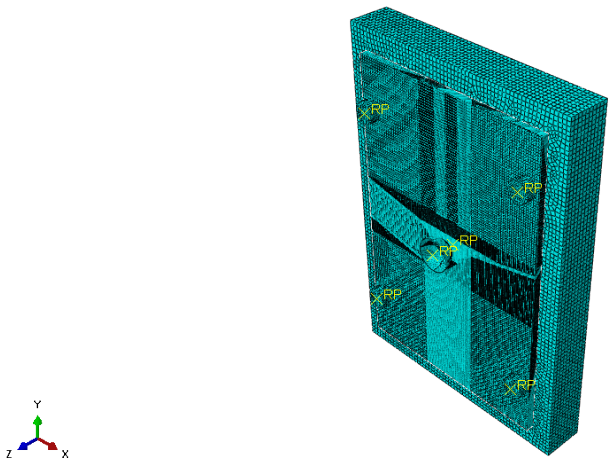
Mesh outside the impact region

Outside the refined impact zone, the mesh was generated using a global seeding strategy applied to the remaining area of the laminate. In these regions, a higher in-plane to through-thickness element aspect ratio was permitted, with target values of approximately AS ≈ 5, reflecting the lower stress gradients and absence of damage initiation away from the impact location.

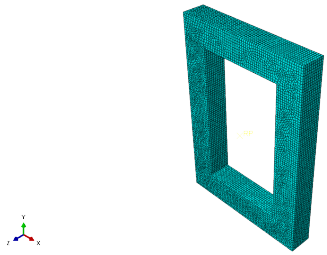
Auxiliary components

The supporting base and rubber clamping elements were discretized using a comparatively coarser mesh, as these components do not control damage initiation and undergo limited deformation relative to the composite panel. The adopted discretization was sufficient to ensure stable contact interactions and reliable transmission of reaction forces to the laminate boundaries. The impactor was modelled as a rigid body and therefore did not require local mesh refinement.

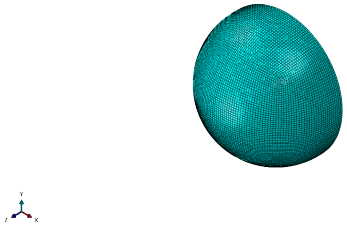
A representative view of the final mesh configuration of the numerical assembly is shown in Figure 5.7. The global view illustrates the discretization of the different model components, while the enlarged inset highlights the locally refined mesh adopted in the impact region of the composite panel. This refinement strategy enables an accurate representation of the contact interaction, stress gradients, and damage initiation mechanisms occurring beneath the impactor, while maintaining a coarser discretization in the surrounding regions to preserve computational efficiency.



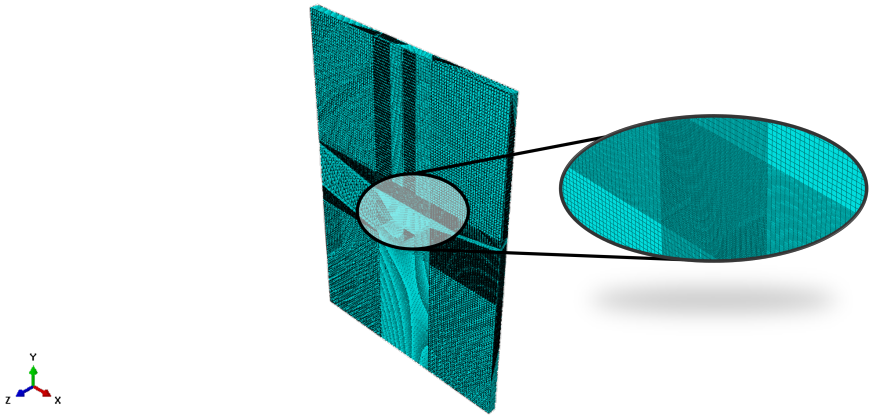
(a) Global mesh of the numerical assembly.



(b) Mesh of the supporting frame.



(c) Impactor discretization.



(d) Local mesh refinement in the impact region.

Figure 5.7: Finite element discretization of the numerical model and mesh refinement strategy adopted for the impact simulations.

5.1.7.4 Hourglass control and numerical stability

The use of reduced-integration elements in explicit dynamic analyses requires particular attention to numerical stability, as such formulations may exhibit non-physical zero-energy deformation modes commonly referred to as hourglassing. In the present model, numerical stability was ensured through a combination of mesh design choices and systematic monitoring of the global energy balance throughout the simulation.

Aspect-ratio considerations

Hourglass behaviour is strongly influenced by element geometry, particularly by the ratio between in-plane dimensions and through-thickness size. Elements with large aspect ratios are more prone to developing spurious deformation modes, which may artificially contribute to the internal energy and compromise solution accuracy.

For this reason, different aspect-ratio targets were adopted depending on the expected mechanical response:

- in the impact region, where high stress gradients, localized contact pressures, and damage initiation occur, a near-unity in-plane to through-thickness aspect ratio (AS ≈ 1) was enforced at the ply level;
- outside the impact region, where the structural response is smoother and no damage initiation is expected, a higher aspect ratio (AS ≈ 5) was permitted in order to reduce computational cost.

This strategy ensures balanced spatial resolution in the most critical region of the model while maintaining numerical efficiency elsewhere.

Energy-based verification of numerical stability

Numerical stability was assessed by monitoring the global energy components computed by Abaqus/Explicit throughout the simulation. In particular, the artificial strain energy was used as the primary indicator of hourglass activity, as it directly reflects the contribution of non-physical deformation modes associated with reduced-integration elements.

The following global energy quantities were extracted as history outputs for the whole model:

- total kinetic energy (ALLKE),
- total internal energy (ALLIE),
- total artificial strain energy (ALLAE),
- total viscous dissipation energy (ALLVD),
- total energy balance (ETOTAL).

Figure 5.8 presents the evolution of these energy components for a representative simulation. The results show a smooth and physically consistent exchange between kinetic and internal energy during the impacts. The artificial strain energy remains negligible throughout the simulation when compared to the physically meaningful energy components, indicating that hourglass effects

do not significantly influence the numerical response. In addition, the total energy remains effectively conserved, confirming the absence of spurious numerical dissipation and the stability of the explicit time integration.

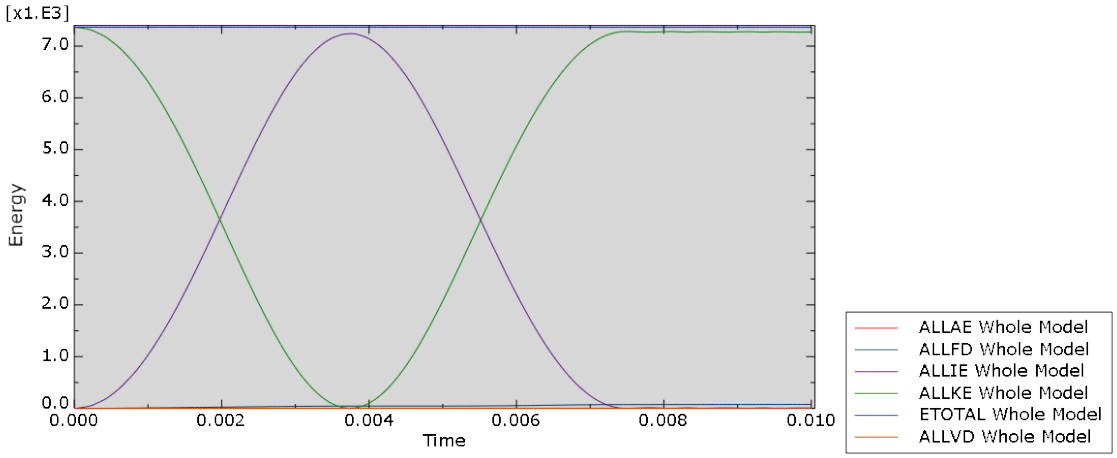


Figure 5.8: Global energy evolution of the explicit dynamic simulation, showing the exchange between kinetic and internal energy during impact and rebound. The artificial strain energy remains negligible relative to the physically meaningful energy components, confirming the absence of significant hourglass effects and the numerical stability of the model.

Verification and mesh sensitivity

To ensure that the observed numerical stability was not specific to a single discretization, additional simulations were performed using alternative mesh partitions and refinement strategies. These auxiliary models were employed to assess the sensitivity of the global energy balance and hourglass contribution to the adopted discretization choices.

The results of this mesh sensitivity study are reported in Appendix A. Based on this verification, the mesh configuration described in Section 5.1.7.3 was retained as the final discretization, as it provides a satisfactory compromise between numerical stability, physical fidelity, and computational efficiency.

5.1.7.5 Summary of the final mesh configuration

Based on the discretization strategy, refinement criteria, and numerical stability verification discussed in Sections 5.1.7.3 and 5.1.7.4, a reference mesh configuration was selected and adopted as the baseline discretization for the numerical analyses presented in this work. This mesh was used unchanged for the linear elastic and intralaminar damage simulations and serves as the geometric and discretization foundation for the subsequent modelling approaches.

The baseline mesh configuration is characterised by:

- ply-wise discretization of the composite laminate using one continuum shell element per ply;
- a locally refined impact region designed to achieve a near-unity in-plane to through-thickness element aspect ratio (AS ≈ 1);
- a coarser discretization outside the impact zone generated using a global seeding strategy, with an aspect ratio of approximately AS ≈ 5;
- negligible artificial strain energy relative to the physically meaningful energy components, confirming the absence of significant hourglass effects.

The baseline numerical model comprises a total of 808568 nodes and 750094 finite elements. The element types and their distribution among the different model components are summarised in Table 5.2.

Table 5.2: Summary of the baseline finite element mesh configuration.

Component	**Element type**	**Discretization strategy**	**Number of elements**
Composite panel	SC8R	Local refinement + global seed	705280
Rubber clamps	C3D8R	Global seed (4 clamps)	3360
Supporting base	C3D8R	Global seed	32270
Impactor	R3D4	Rigid surface	9184
Total	–	–	750094

For the interlaminar damage simulations, additional cohesive elements were introduced at selected ply interfaces, requiring local partitioning of the composite laminate. This modification results in a moderate increase in the total element count, while preserving the in-plane discretization and refinement strategy of the baseline mesh. The details of the cohesive discretization and the resulting mesh characteristics are presented in Section 5.4.

In the bi-phasic modelling approach, solid elements were locally introduced in the impact region to represent the matrix-dominated phase explicitly. This leads to a further local modification of the discretization and an associated increase in the number of elements. These changes are confined to the impact zone and are described in detail in Section 5.5.

Despite these model-specific modifications, the underlying mesh topology, refinement strategy, and global discretization philosophy remain consistent across all analyses, ensuring that differences in numerical response can be attributed to the adopted constitutive modelling approaches rather than to changes in spatial discretization.

5.1.8 Computational resources

All numerical simulations presented in this work were performed using Abaqus/Explicit on a dedicated Linux-based workstation. The computational resources employed are summarised in Table 5.3. The available hardware allowed the execution of large-scale explicit dynamic simulations without the use of mass scaling, ensuring that the physical time scales of the impact events were preserved.

The workstation is equipped with a dual-socket CPU configuration, providing a total of 64 physical cores (128 logical threads). Simulations were executed using up to 64 CPU cores, depending on the analysis case and model complexity. The adopted hardware configuration proved adequate for the mesh sizes and time integration requirements associated with the explicit impact analyses conducted in this work.

Table 5.3: Computational resources used for the numerical simulations.

Architecture	x86_64
Operating system	Linux (64-bit)
Processor	AMD EPYC 7513 32-Core Processor
Number of sockets	2
Physical CPU cores	64
Logical CPU threads	128
Threads per core	2
NUMA nodes	2
Solver	Abaqus/Explicit

5.2 Baseline Elastic Impact Model

This section presents the numerical results obtained from the finite element simulations and their correlation with the experimental impact data. The analysis is structured around a progressive assessment of the elastic behaviour of the laminate, starting from a baseline linear elastic model and followed by a deterministic sensitivity study of the in-plane elastic properties. The objective is to establish a physically consistent elastic reference configuration capable of reproducing the experimental response in the elastic-dominated impact regime, which will subsequently serve as the foundation for the damage-inclusive numerical models introduced in the following sections.

5.2.1 Elastic Material Properties and Model Definition

In order to establish a reference numerical response, the laminate was first modelled assuming a linear elastic orthotropic constitutive behaviour. This baseline model is intended to quantify the influence of the elastic constants on the global impact response, prior to the introduction of damage mechanisms.

The elastic properties adopted in the numerical model were obtained from an independent experimental characterisation campaign performed on specimens extracted from the same Glass/PPS laminate used in the present study. The experimental investigation reported by Calefe *et al.* [13] included quasi-static tensile, compressive, and in-plane shear tests, with strain fields measured through Digital Image Correlation (DIC).

The material consists of a woven Glass/PPS laminate with a 5-harness satin (5HS) architecture. The 0° direction corresponds to the warp direction (material direction 1), while the 90° direction corresponds to the weft direction (material direction 2). The resulting elastic constants reflect the orthotropic in-plane response of the laminate. Although the experimental results indicated a non-linear in-plane shear response associated with fibre reorientation effects typical of woven fabrics, a linearised shear modulus was adopted for the present elastic simulations, as reported in Table 5.4.

Table 5.4: Elastic properties adopted for the baseline linear elastic model, obtained from the experimental characterisation of the same Glass/PPS laminate (5HS) reported by Calefe *et al.* [13].

Property	**Direction 1 (warp)**	**Direction 2 (weft)**
Tensile Young's modulus, E [GPa]	22.49	19.29
In-plane shear modulus, G_{12} [GPa]	3.76	3.51
Poisson's ratio, ν_{12} [-]	0.113	–

A single representative in-plane shear modulus $G_{12} = 3.6$ GPa was adopted for the numerical model, corresponding to the average of the experimentally measured warp and weft values reported in Table 5.4.

The remaining engineering constants required to fully define the orthotropic elastic model were completed using micromechanical considerations and effective-property assumptions appropriate for woven laminates.

Constituent properties and fibre volume fraction

A fibre volume fraction of $V_f = 0.50$ was assumed. The constituent elastic properties were defined as

$$E_m = 4.0 \text{ GPa}, \qquad \nu_m = 0.36, \tag{5.3}$$

for the PPS matrix, and

$$E_f = 72 \text{ GPa}, \qquad \nu_f = 0.22, \tag{5.4}$$

for the E-glass fibres.

These values are consistent with typical properties reported for PPS engineering thermoplastics [98] and E-glass fibres commonly used in structural composite applications [99, 100].

Assuming isotropic behaviour for the constituents, the corresponding shear moduli were obtained from

$$G = \frac{E}{2(1+\nu)}, \tag{5.5}$$

yielding

$$G_m = 1.47 \text{ GPa}, \qquad G_f = 29.51 \text{ GPa}. \tag{5.6}$$

Transverse shear moduli

The transverse shear moduli G_{13} and G_{23} were initially estimated using the Halpin–Tsai formulation for shear-dominated behaviour,

$$\eta = \frac{G_f/G_m - 1}{G_f/G_m + \xi}, \tag{5.7}$$

$$G_{\text{HT}} = G_m \frac{1+\xi\eta V_f}{1-\eta V_f}, \tag{5.8}$$

where a shape parameter $\xi = 1$ was adopted. Substitution of the constituent properties and fibre volume fraction resulted in

$$G_{\text{HT}} \approx 3.9 \text{ GPa}. \tag{5.9}$$

To account for fibre waviness, crimp effects, and reduced load transfer efficiency inherent to woven laminate architectures, the Halpin–Tsai estimate was reduced by an effectiveness factor k, defined as

$$G_{13} = G_{23} = k\, G_{\text{HT}}. \tag{5.10}$$

A nominal value of $k = 0.8$ was adopted, leading to

$$G_{13} = G_{23} = 3.0 \text{ GPa}. \tag{5.11}$$

Through-thickness Young's modulus

The through-thickness Young's modulus E_3 was estimated using an inverse rule-of-mixtures (Reuss) formulation, appropriate for transverse loading,

$$\frac{1}{E_3^R} = \frac{V_f}{E_f} + \frac{1-V_f}{E_m}. \tag{5.12}$$

For the adopted constituent properties, this resulted in

$$E_3^R \approx 7.6 \text{ GPa}. \tag{5.13}$$

Consistent with the treatment of the transverse shear moduli, the same effectiveness factor k was applied to account for laminate architecture effects,

$$E_3 = k E_3^R, \tag{5.14}$$

yielding the final value

$$E_3 = 6.1 \text{ GPa}. \tag{5.15}$$

Transverse Poisson's ratios

The transverse Poisson's ratios ν_{13} and ν_{23} were assumed to be matrix-dominated and were therefore defined as effective values close to the Poisson's ratio of the PPS matrix. A nominal value of

$$\nu_{13} = \nu_{23} = 0.32 \tag{5.16}$$

was adopted, lying between the matrix-dominated and rule-of-mixtures bounds.

Modelling considerations

The present numerical model employs continuum-shell elements. For this formulation, the global response under low-velocity impact loading is primarily governed by the in-plane elastic constants and the transverse shear stiffness, while the sensitivity to the through-thickness normal modulus is reduced relative to fully three-dimensional solid discretisations. Consequently, E_3 is treated as a secondary parameter within the elastic baseline model, and its approximate definition is considered adequate for the purposes of the present elastic analysis and deterministic sensitivity study.

Density

The laminate density was defined using a rule-of-mixtures formulation,

$$\rho = V_f \rho_f + (1 - V_f) \rho_m, \tag{5.17}$$

where ρ_f and ρ_m denote the fibre and matrix densities, respectively. Assuming $\rho_f = 2.55$ g/cm^3 for E-glass fibres, $\rho_m = 1.35$ g/cm^3 for the PPS matrix, and a fibre volume fraction $V_f = 0.50$, the resulting laminate density is

$$\rho = 0.5 \times 2.55 + 0.5 \times 1.35 = 1.95 \text{ g/cm}^3. \tag{5.18}$$

This value was directly adopted in the numerical model.

Table 5.5 summarises the complete set of elastic material properties adopted for the Glass/PPS laminate in the baseline linear elastic numerical model. The values correspond to the engineering constants required by Abaqus for continuum-shell elements and combine experimentally measured in-plane properties with micromechanically estimated transverse constants and density. This set of properties defines the reference elastic configuration used for the deterministic sensitivity analysis and for comparison with the subsequent damage modelling approaches.

Table 5.5: Summary of elastic material properties adopted for the Glass/PPS laminate in the baseline linear elastic model. Values correspond to the engineering constants required by Abaqus for continuum-shell elements.

Property	**Value**
Young's modulus (warp)	$E_1 = 22.49$ GPa
Young's modulus (weft)	$E_2 = 19.29$ GPa
Through-thickness modulus	$E_3 = 6.10$ GPa
Poisson's ratio	$\nu_{12} = 0.113$
Poisson's ratio	$\nu_{13} = 0.32$
Poisson's ratio	$\nu_{23} = 0.32$
In-plane shear modulus	$G_{12} = 3.60$ GPa
Transverse shear modulus	$G_{13} = 3.00$ GPa
Transverse shear modulus	$G_{23} = 3.00$ GPa
Density	$\rho = 1.95$ g/cm^3

Auxiliary material properties

In addition to the composite laminate, auxiliary components were included in the numerical model to reproduce the experimental boundary conditions, namely the rubber clamps used to constrain the specimen and the steel base supporting the laminate during impact. These components were modelled using linear elastic isotropic material behaviour, as their role in the simulations is limited to load transfer, geometric constraint, and inertial effects rather than detailed material response.

The rubber clamps were assumed to be made of styrene-butadiene rubber (SBR) and were defined using representative elastic and density properties. The material was assigned a Young's modulus of $E = 6.00$ GPa and a Poisson's ratio of $\nu = 0.10$, together with a density of $\rho = 0.94$ g/cm^3. These values are consistent with typical engineering data reported for SBR elastomers and were considered sufficient to reproduce the global compliance and mass contribution of the clamping system without introducing additional constitutive complexity [101].

The steel base was modelled as a linear elastic isotropic material with standard structural steel properties, namely a Young's modulus of $E = 210.00$ GPa, a Poisson's ratio of $\nu = 0.30$, and a density of $\rho = 7.8$ g/cm^3. Given the large stiffness contrast between the steel base and the composite laminate, the deformation of this component was negligible and did not influence the global impact response [102].

Table 5.6 summarises the elastic and density properties adopted for the auxiliary components in the baseline elastic numerical model.

Table 5.6: Elastic material properties adopted for the auxiliary components in the numerical model. Representative values for styrene-butadiene rubber (SBR) and structural steel were selected based on typical engineering data reported in the literature [101, 102].

Property	**Rubber clamps (SBR)**	**Steel base**
Young's modulus, E [GPa]	6	210
Poisson's ratio, ν [-]	0.10	0.30
Density, ρ [g/cm^3]	0.94	7.80

5.2.2 Baseline Elastic Impact Response

Response metrics and correlation strategy

The baseline elastic impact response was analysed using the acceleration–time history as the primary response quantity. This choice was motivated by the fact that acceleration is the quantity directly measured during the experimental impact tests, while other kinematic and dynamic variables, such as velocity, displacement, contact force, and absorbed energy, are obtained through numerical integration of the acceleration signal.

Adopting the acceleration–time response as the reference quantity avoids uncertainties associated with signal integration, such as baseline drift and sensitivity to contact alignment, and enables a more direct and robust comparison between numerical and experimental results. For this reason, all elastic model correlations presented in this section are primarily based on the acceleration response.

To quantify the elastic stiffness of the system and establish a consistent correlation metric, the initial slope of the acceleration–time curve was selected as the primary comparison parameter. This quantity reflects the effective contact stiffness governing the early stages of the impact event, during which the structural response remains purely elastic and is dominated by the elastic properties of the laminate and the contact configuration.

The initial slope was evaluated from the acceleration–time response during the early stage of the impact event, within the approximately linear region preceding the peak acceleration. This metric

therefore represents the initial rate of increase of the acceleration signal during contact and was found to be particularly sensitive to variations in the elastic material parameters. It is therefore well suited for assessing the baseline elastic model and supporting the subsequent deterministic sensitivity analysis.

In order to assess the robustness of the correlation metric, the initial slope was evaluated over multiple portions of the acceleration–time history. Instead of using a fixed time interval, the evaluation window was defined as a fraction of the time required to reach the peak acceleration. Denoting the time at which the maximum acceleration occurs as t_{peak}, the slope was calculated using data from the beginning of contact up to $t_f = f\,t_{peak}$, where f represents the selected fraction. Consequently, the horizontal axis in the slope-correlation plots represents the percentage of the peak acceleration time, expressed as $100f$. For example, a value of 30% corresponds to evaluating the slope from $t = 0$ up to $0.30\,t_{peak}$. This normalization allows the slope evolution to be analysed in a consistent manner across different impact responses while remaining independent of small variations in the absolute duration of the impact event.

Baseline elastic response at 4.5 J

The baseline elastic numerical response was first evaluated for the lowest impact energy level ($E = 4.5$ J), corresponding to Group 1 of the experimental campaign. At this energy level, the structural response is predominantly elastic, making it particularly suitable for assessing the adequacy of the linear elastic material model and for defining a reference response prior to the introduction of damage mechanisms.

Figure 5.9 compares the experimental acceleration-time histories with the corresponding response predicted by the linear elastic numerical model. The numerical response reproduces the overall shape of the experimental curves, including the initial acceleration rise, the peak acceleration region, and the subsequent unloading phase associated with impactor rebound. Minor oscillations observed in the numerical signal are attributed to elastic structural vibrations inherent to the undamaged formulation.

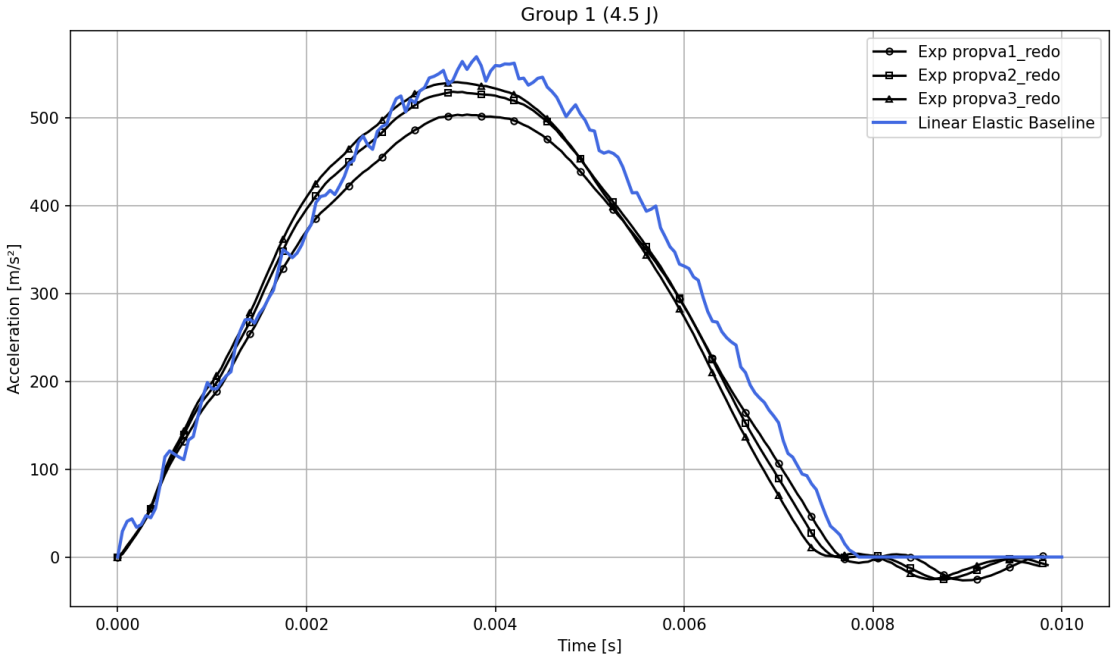


Figure 5.9: Acceleration-time response for Group 1 (4.5 J): experimental results and linear elastic baseline numerical prediction.

Quantitative correlation was assessed using the initial slope of the acceleration-time response, as defined in Section 5.2.2.1. The comparison between the numerical prediction and the experimental mean initial slope is reported in Figure 5.10. A deviation of approximately 5.6% was obtained, indicating good agreement in terms of effective elastic stiffness at this energy level.

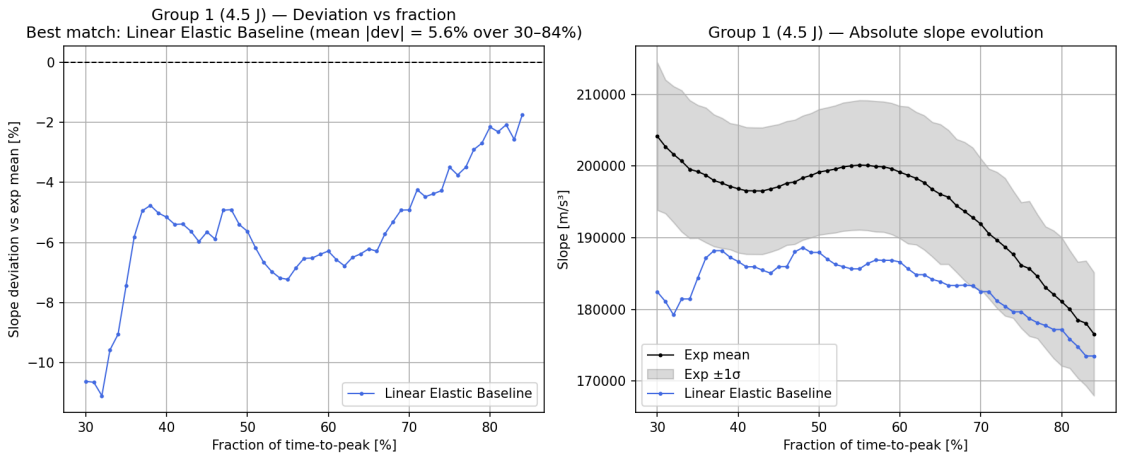


Figure 5.10: Comparison of the initial acceleration-time slope between the experimental mean response and the linear elastic baseline model for Group 1 (4.5 J).

Overall, the results obtained at 4.5 J confirm that the baseline linear elastic model is capable of accurately reproducing the early-stage impact response and effective stiffness of the Glass/PPS

laminate.

Baseline elastic response at 7.5 J

The baseline elastic model was subsequently evaluated for an intermediate impact energy of $E = 7.5$ J (Group 2). At this energy level, the experimental response remains largely elastic, although the influence of geometric nonlinearity and localised deformation becomes more pronounced.

The acceleration-time histories for the experimental tests and the numerical prediction are shown in Figure 5.11. The linear elastic model continues to capture the global shape of the response and the timing of the peak acceleration. However, a slight overprediction of the peak acceleration magnitude is observed, particularly during the unloading phase.

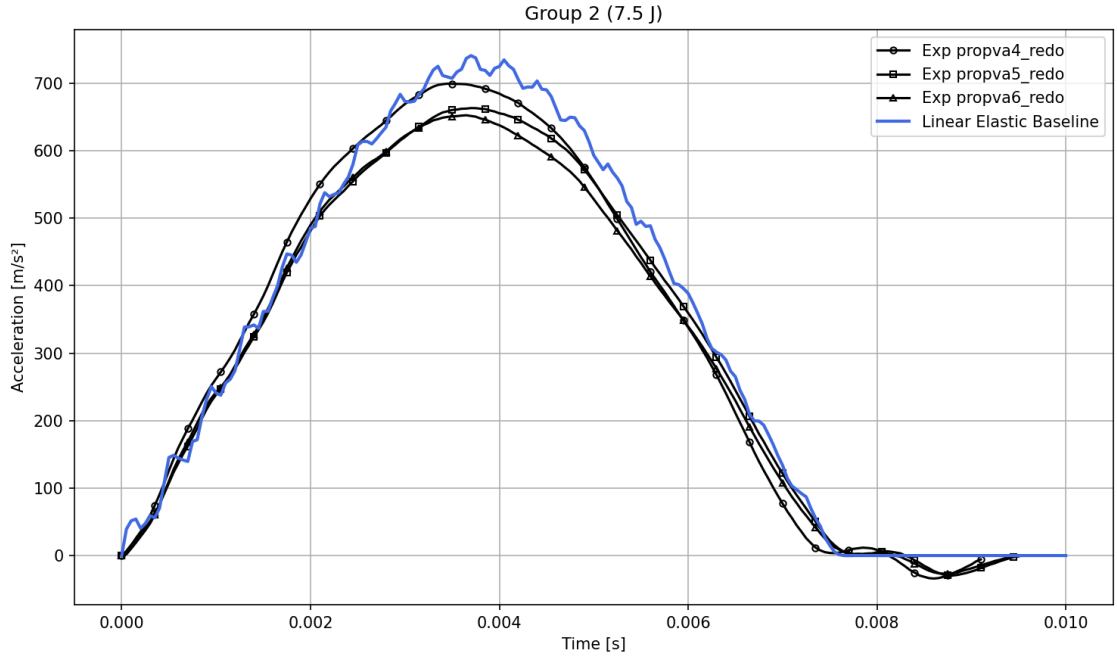


Figure 5.11: Acceleration-time response for Group 1 (7.5 J): experimental results and linear elastic baseline numerical prediction.

The initial slope correlation, presented in Figure 5.12, indicates a deviation of approximately 4.7% with respect to the experimental mean value. This result confirms that the elastic stiffness of the numerical model remains consistent with the experimental response for this energy level, despite the increasing role of nonlinear effects in the later stages of the impact event.

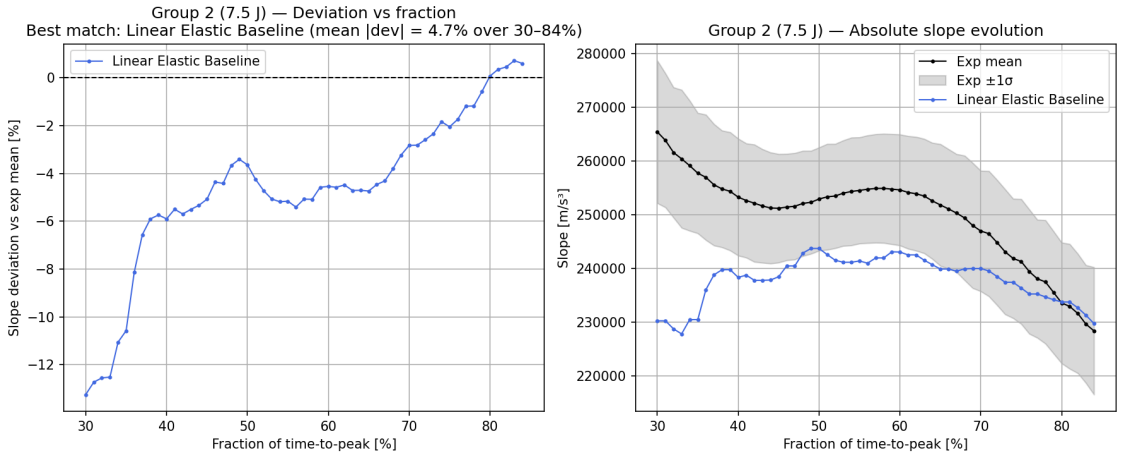


Figure 5.12: Comparison of the initial acceleration-time slope between the experimental mean response and the linear elastic baseline model for Group 1 (4.5 J).

Baseline elastic response at approximately 14.2 J

For the higher intermediate impact energy of approximately $E \approx 14.2$ J (Group 3), the limitations of the purely elastic formulation become more evident. Experimental observations at this energy level indicate the onset of damage mechanisms, which are not accounted for in the baseline elastic model.

As shown in Figure 5.13, the numerical acceleration–time response systematically overpredicts the peak acceleration compared to the experimental curves. This behaviour reflects the absence of stiffness degradation and energy dissipation mechanisms in the elastic formulation.

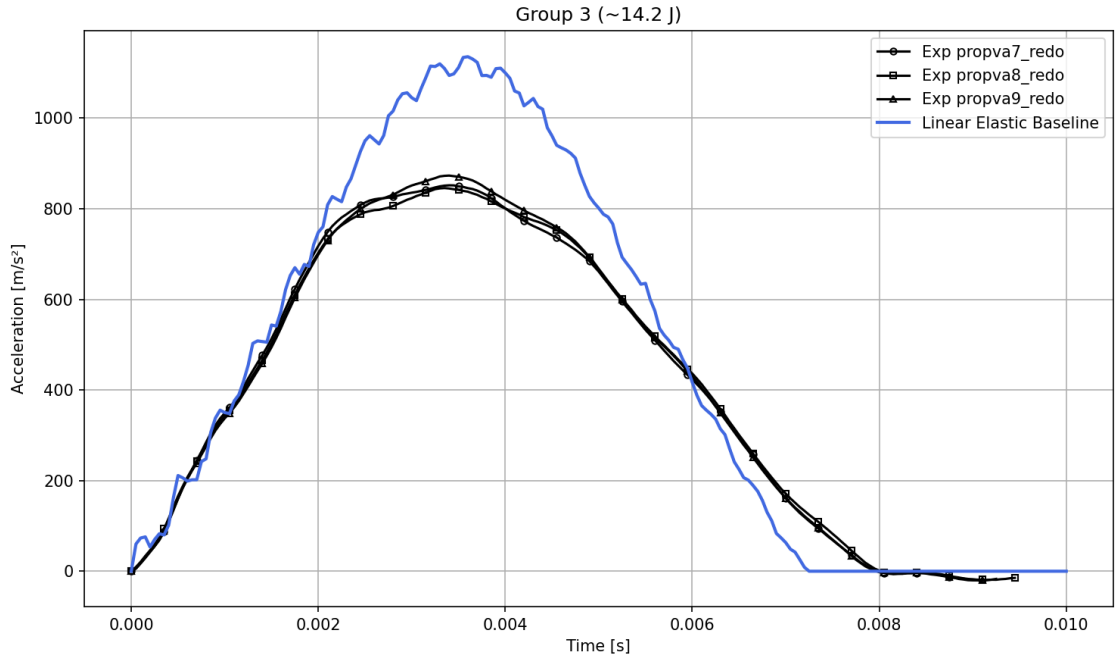


Figure 5.13: Acceleration-time response for Group 1 (14.2 J): experimental results and linear elastic baseline numerical prediction.

The initial slope comparison, reported in Figure 5.14, yields a deviation of approximately 6.1% relative to the experimental mean value. While the early-stage stiffness is still reasonably captured, the divergence observed in the later stages of the response highlightsthe inability of the elastic model to reproduce damage-driven compliance effects.

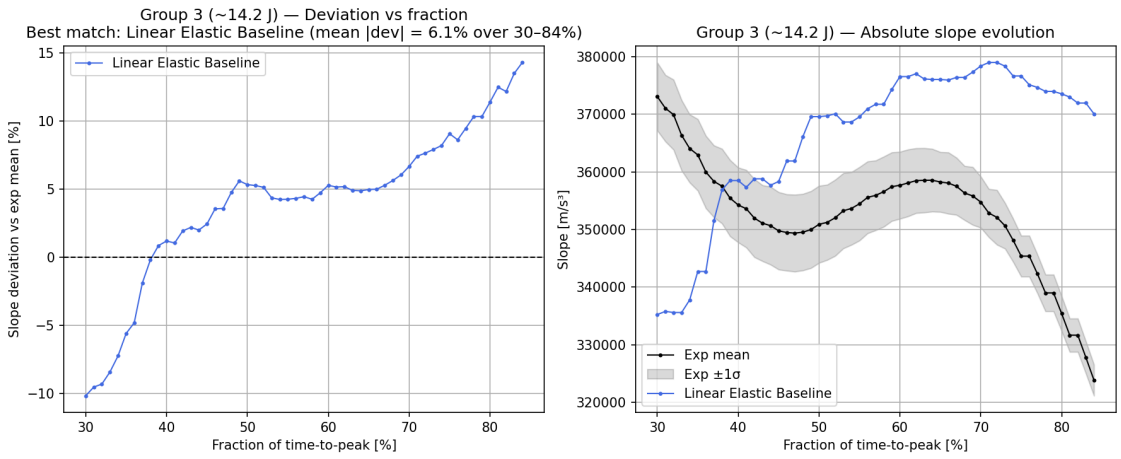


Figure 5.14: Comparison of the initial acceleration-time slope between the experimental mean response and the linear elastic baseline model for Group 1 (14.2 J).

Baseline elastic response at 27 J

The highest impact energy considered in the present study, $E = 27$ J (Group 4), represents a loading regime where extensive damage is experimentally observed. Under these conditions, the assumptions underlying the linear elastic formulation are no longer valid.

Figure 5.15 shows a pronounced discrepancy between the numerical and experimenacceleration-timetime responses. The elastic model significantly overpredicts the peak acceleration and exhibits a shorter contact duration, reflecting an unrealistically stiff response due to the absence of damage and energy dissipation mechanisms.

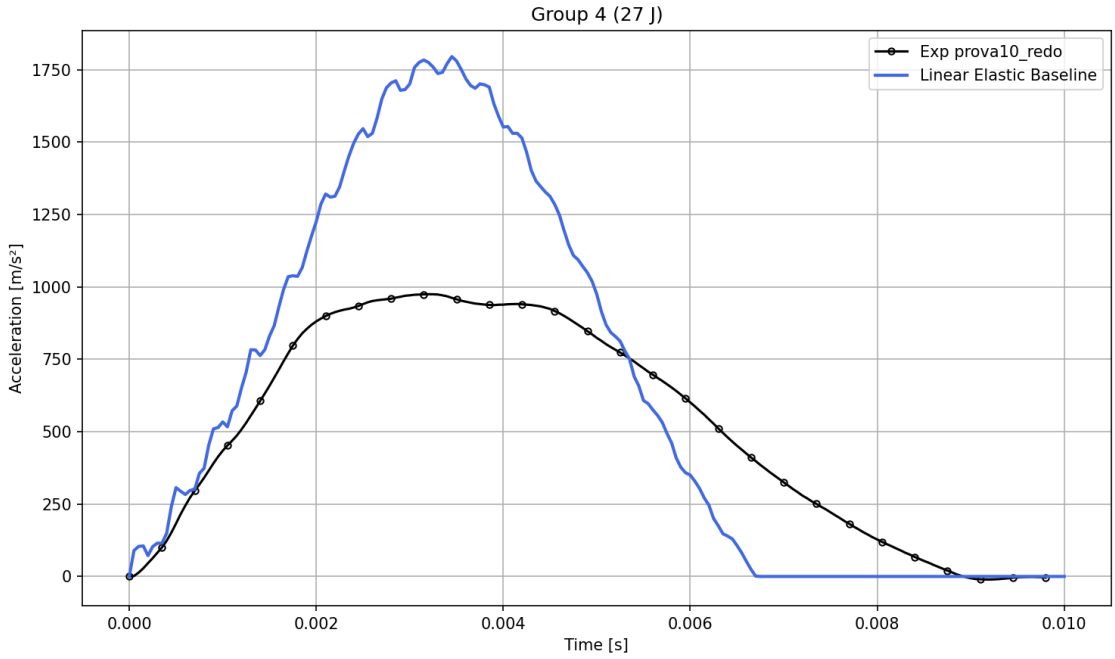


Figure 5.15: Acceleration-time response for Group 1 (14.2 J): experimental results and linear elastic baseline numerical prediction.

The initial slope comparison, presented in Figure 5.16, indicates a deviation of approximately 28.9% with respect to the experimental mean value. This large discrepancy confirms that, at high impact energies, the effective stiffness of the laminate is strongly governed by damage mechanisms that cannot be captured by a purely elastic model.

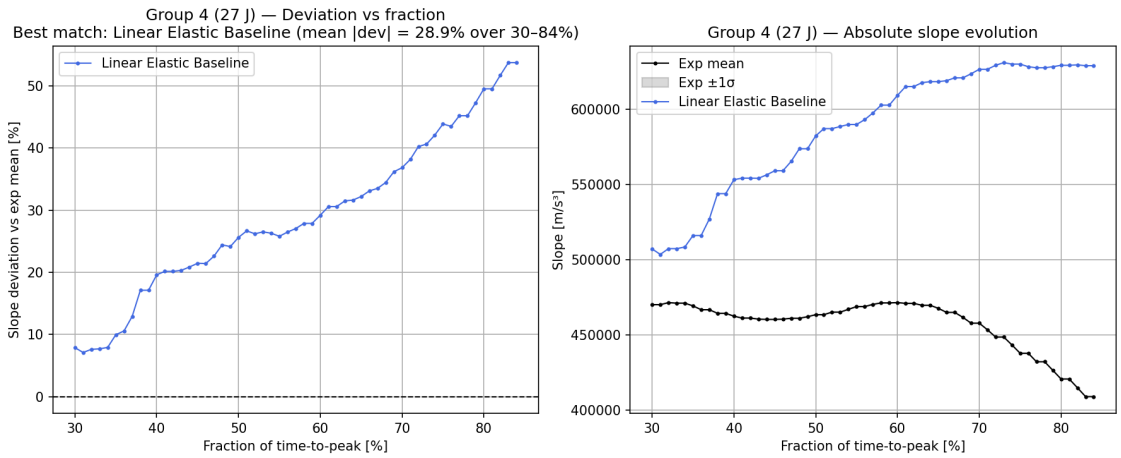


Figure 5.16: Comparison of the initial acceleration-time slope between the experimental mean response and the linear elastic baseline model for Group 1 (14.2 J).

The results obtained at 27 J clearly demonstrate the limitations of the baseline elastic formulation and motivate the introduction of progressive damage modelling approaches in the subsequent sections.

In summary, the baseline linear elastic model provides an accurate representation of the early-stage impact response and effective stiffness of the Glass/PPS laminate at low and moderate impact energies. However, as the impact energy increases and damage mechanisms become dominant, the elastic formulation progressively loses predictive capability. These observations establish a clear reference framework for the deterministic sensitivity analysis and for the evaluation of damage-enabled numerical models presented in the following sections.

5.2.3 Deterministic Sensitivity Analysis of In-Plane Elastic Properties

The correlation between numerical and experimental responses for the elastic model was performed using the acceleration-time history, which corresponds to the directly measured quantity in the experimental impact tests. Displacement, velocity, and energy histories were obtained by time-integration of the acceleration signal and are therefore considered derived quantities. Although energy plots provide useful insight into global energy absorption, the acceleration-time response was selected as the primary metric for elastic correlation due to its direct experimental relevance and its strong sensitivity to the initial structural stiffness.

For the linear elastic simulations, the comparison focused on the early portion of the acceleration-time curve. The slope of this initial segment is primarily governed by the effective in-plane stiffness of the laminate and is therefore well suited for assessing the adequacy of the adopted elastic constants prior to the introduction of damage mechanisms.

The deterministic sensitivity analysis investigated the influence of the in-plane Young's moduli E_1 (warp) and E_2 (weft), which were varied simultaneously. This choice reflects the balanced woven architecture of the laminate and avoids introducing artificial anisotropy through independent modification of the two in-plane directions. Based on the slightly compliant response observed in the baseline elastic simulations, only positive deviations were considered. Two stiffness increments were investigated: $+5\%$ and $+10\%$ relative to the baseline elastic properties.

4.5 J impact energy

Figure 5.17 presents the acceleration-time response for the 4.5 J impact case, comparing the experimental measurements with the baseline elastic simulation and the two stiffness-increased variants. The baseline model exhibits a slightly lower initial slope than the experimental response, indicating a marginally compliant behaviour.

Increasing E_1 and E_2 by 5% improves the agreement in the initial acceleration slope, while the 10% increase leads to a slightly stiffer response that more closely matches the experimental mean for this specific energy level. This behaviour is confirmed by the slope deviation analysis shown in Figure 5.18, where the $+10\%$ configuration minimises the deviation with respect to the experimental reference.

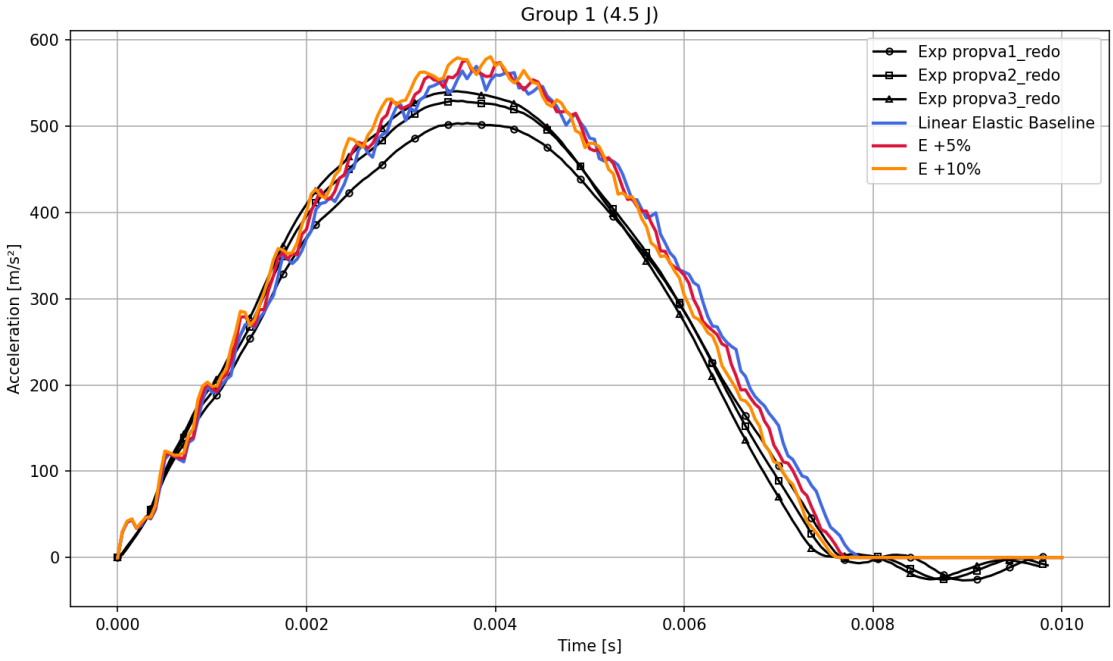


Figure 5.17: Acceleration-time response for the 4.5 J impact case: experimental data compared with the baseline elastic simulation and stiffness-increased variants.

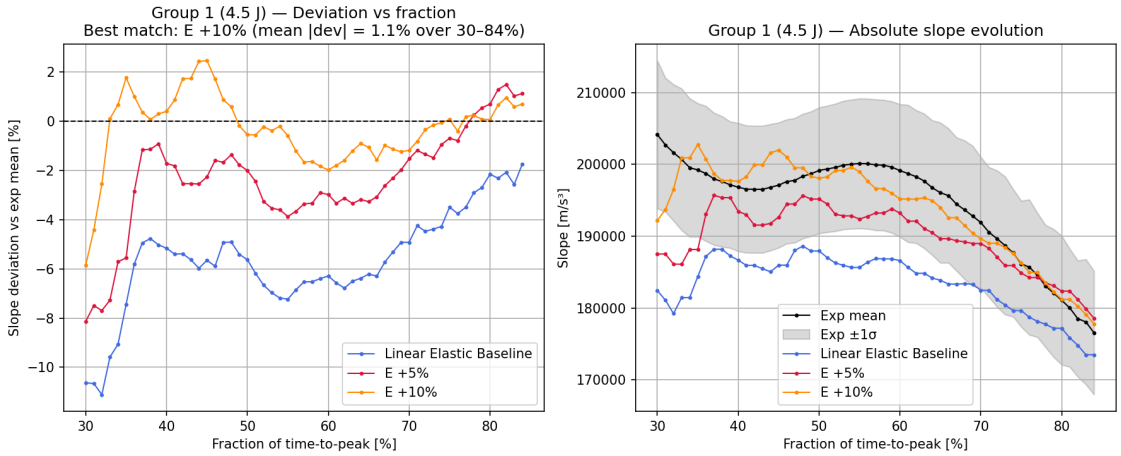


Figure 5.18: Initial slope comparison for the 4.5 J impact case: deviation of elastic simulations with respect to the experimental mean.

7.5 J impact energy

The acceleration-time response for the 7.5 J impact case is shown in Figure 5.19. As observed for the 4.5 J case, the baseline elastic model slightly underestimates the initial stiffness relative to the experimental response.

For this energy level, the 5% increase in E_1 and E_2 provides the best overall agreement in terms of initial slope, while the 10% increase results in a consistently stiffer response. The slope deviation analysis in Figure 5.20 confirms that the +5% configuration minimises the deviation from the experimental mean across the considered time window.

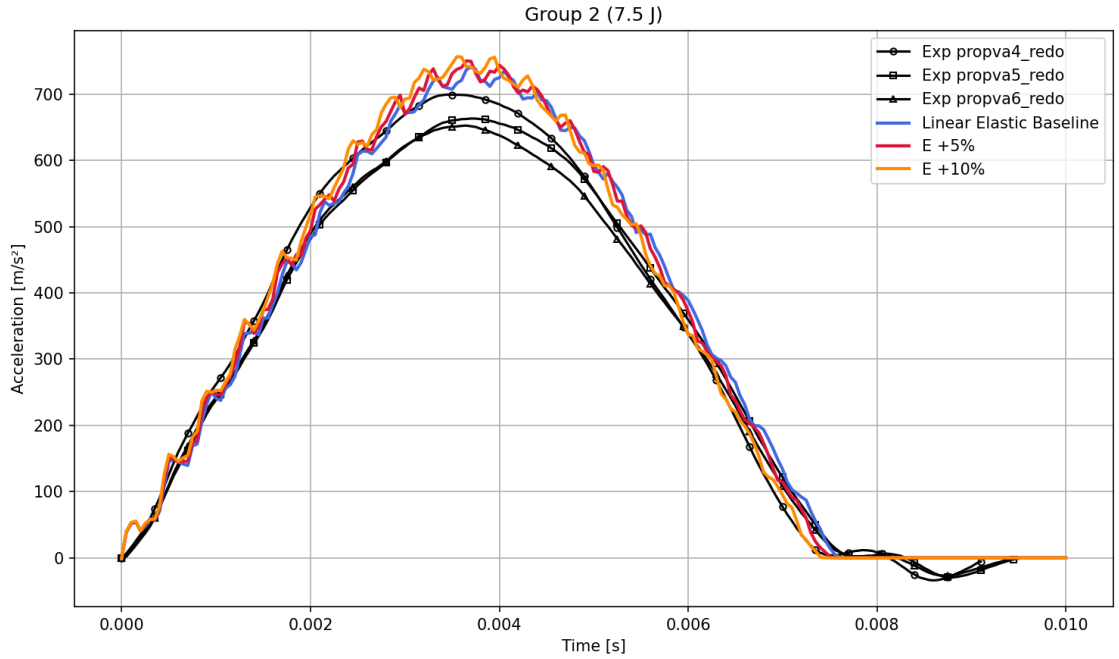


Figure 5.19: Acceleration–time response for the 7.5 J impact case: experimental data compared with the baseline elastic simulation and stiffness-increased variants.

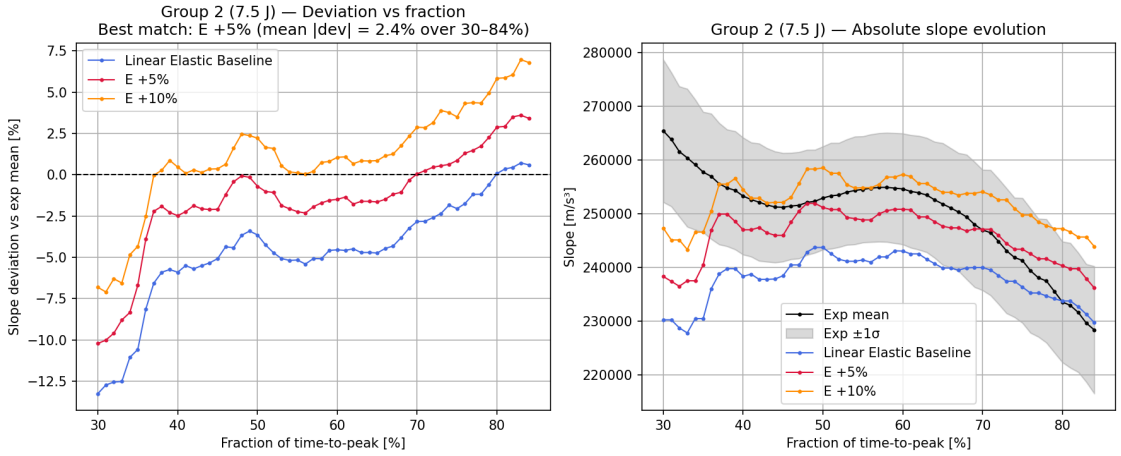


Figure 5.20: Initial slope comparison for the 7.5 J impact case: deviation of elastic simulations with respect to the experimental mean.

14.2 J impact energy

Figure 5.21 presents the acceleration-time response for the approximately 14.2 J impact case. While variations in E_1 and E_2 still affect the initial portion of the response, deviations between numerical and experimental curves become more pronounced beyond the initial peak.

The slope analysis shown in Figure 5.22 indicates that stiffness increases lead to progressively larger overestimations of the experimental response. This reflects the growing influence of damage initiation and energy dissipation mechanisms in the experimental tests, which are not captured by the purely elastic formulation. Consequently, elastic stiffness tuning alone is no longer sufficient to reproduce the experimental behaviour at this energy level.

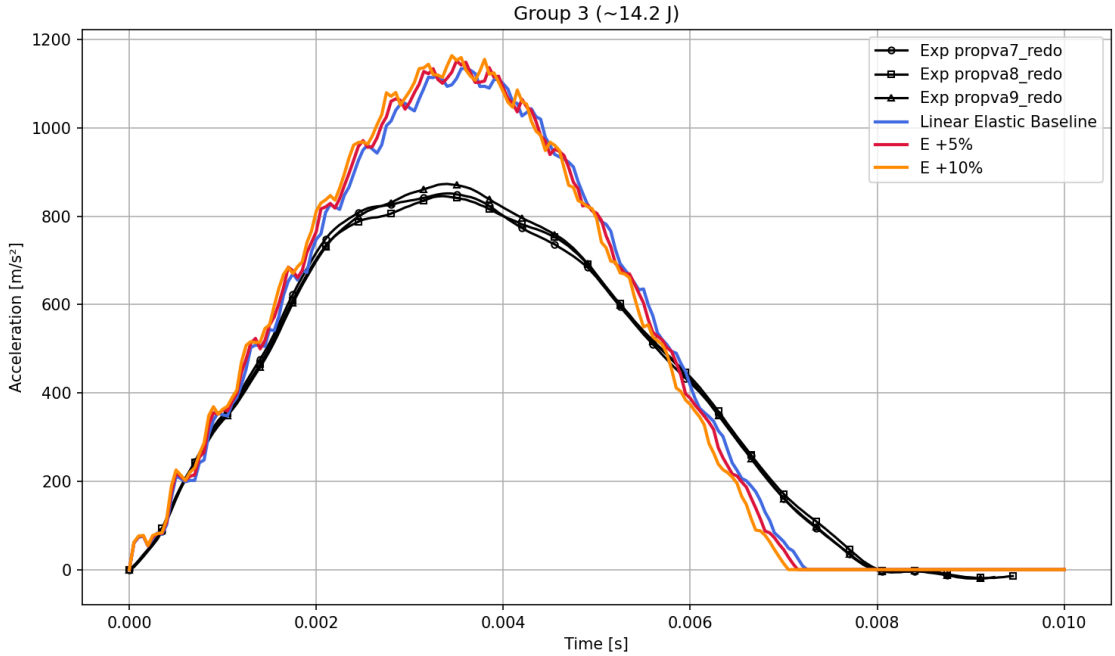


Figure 5.21: Acceleration-time response for the 14.2 J impact case: experimental data compared with elastic simulations with varied in-plane stiffness.

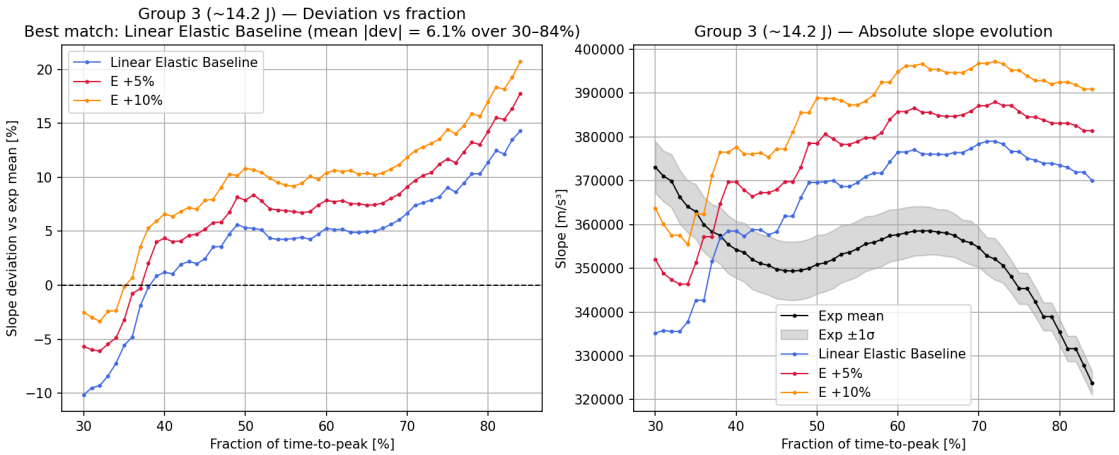


Figure 5.22: Initial slope comparison for the 14.2 J impact case: increasing influence of stiffness overestimation as damage effects become significant.

27 J impact energy

At the highest impact energy of 27 J, shown in Figure 5.23, the limitations of the linear elastic formulation become evident. The experimental response is dominated by damage initiation, stiffness degradation, and irreversible energy absorption, leading to a fundamentally different acceleration–time profile.

The slope comparison in Figure 5.24 demonstrates that variations in E_1 and E_2 have only a marginal effect on the overall correlation. Adjusting the elastic stiffness is therefore ineffective in reproducing the experimental response at this energy level, highlighting the necessity of explicit damage modelling.

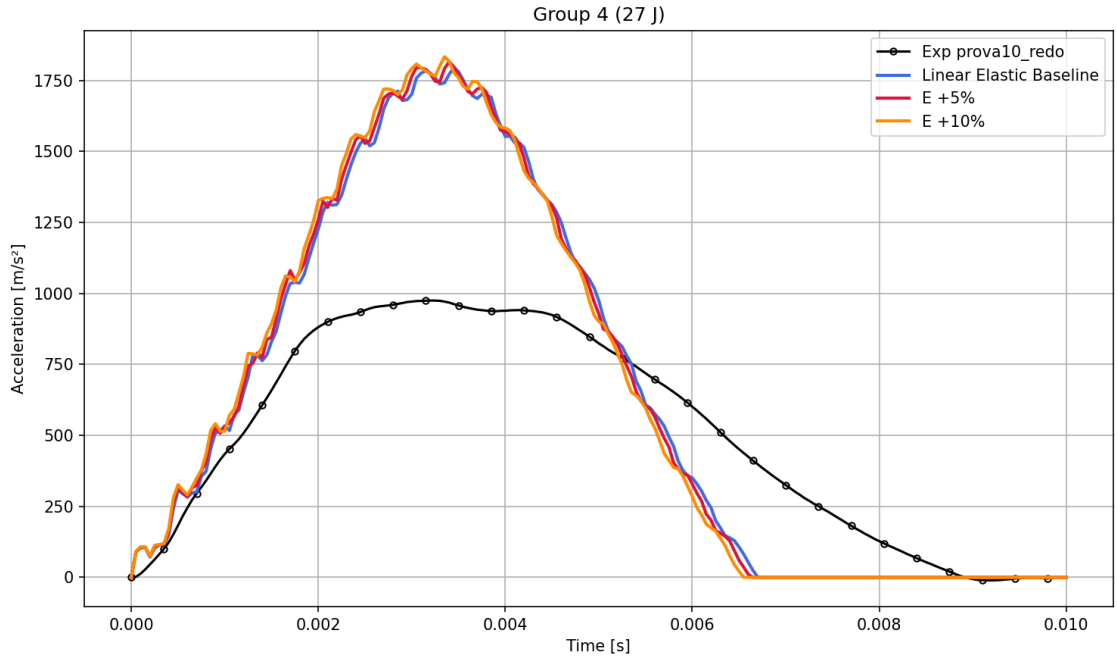


Figure 5.23: Acceleration-time response for the 27 J impact case: comparison between experimental results and elastic simulations highlighting the inadequacy of stiffness tuning in a damage-dominated regime.

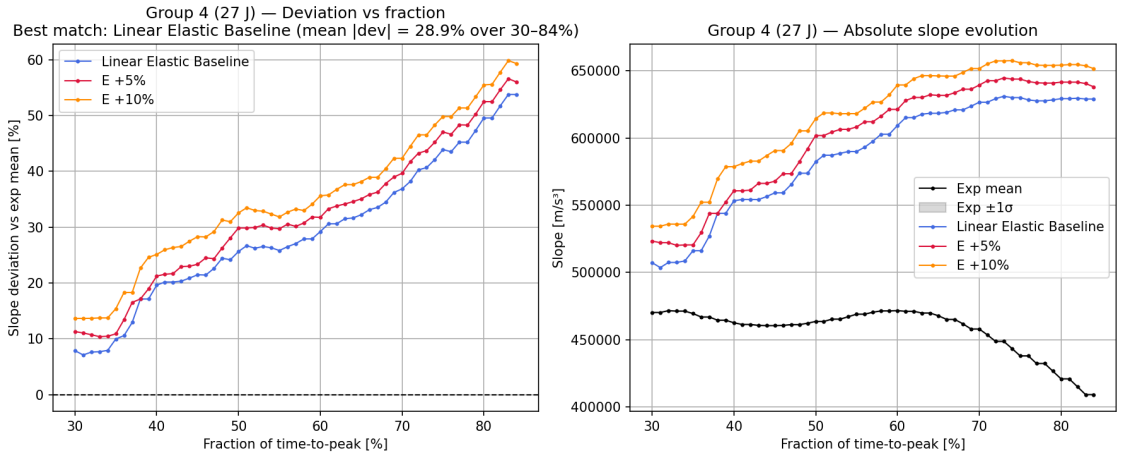


Figure 5.24: Initial slope comparison for the 27 J impact case: limited sensitivity to elastic stiffness variations due to dominant damage mechanisms.

Overall, the deterministic sensitivity analysis shows that moderate increases of the in-plane Young's moduli E_1 and E_2 effectively correct the slight compliance observed in the baseline elastic model for low-energy impacts. Although the 4.5 J case is most accurately represented by the $E + 10\%$ configuration, the $E + 5\%$ modification provides a consistently improved agreement for both the 4.5 J and 7.5 J impact cases without introducing excessive stiffness at higher energy levels. For this reason, the $+5\%$ increase in E_1 and E_2 was selected as the reference elastic configuration for the subsequent damage-inclusive numerical analyses.

5.2.4 Selection of the Reference Elastic Configuration

Based on the results of the deterministic sensitivity analysis presented in Section 5.2.3, a reference elastic configuration was selected for use in the subsequent damage-inclusive numerical models. The objective of this selection was not to achieve an exact match for a single impact condition, but rather to identify a physically consistent elastic baseline that provides a robust representation of the laminate stiffness in the elastic-dominated regime.

The baseline elastic model exhibited a slightly compliant response for the lowest impact energies, particularly in terms of the initial acceleration-time slope. Increasing the in-plane Young's moduli E_1 and E_2 by 5% and 10% progressively reduced this compliance. While the 4.5 J impact case was most accurately reproduced by the $E + 10\%$ configuration, the 7.5 J case showed the best agreement for the $E + 5\%$ increase. At higher impact energies, the experimental response was increasingly governed by damage initiation and energy dissipation mechanisms, which cannot be captured by a purely elastic formulation.

Considering all energy levels jointly, the $E + 5\%$ configuration provides the most balanced representation of the experimental response across the elastic-dominated impact regime. This choice

avoids excessive stiffening for intermediate energy cases while still correcting the baseline compliance observed at low energies. As such, it represents a conservative and physically reasonable elastic reference state, suitable for use as a baseline prior to the introduction of damage mechanisms.

It is important to emphasise that the purpose of the elastic reference model is not to reproduce the complete impact response, but rather to establish a consistent stiffness baseline. At higher impact energies, the experimental response is dominated by matrix cracking, delamination, stiffness degradation, and irreversible energy absorption, which inherently lie beyond the scope of a linear elastic material description. Consequently, deviations observed at higher energies are not addressed through further elastic stiffness tuning.

Accordingly, the elastic configuration with a simultaneous $+5\%$ increase in the in-plane Young's moduli E_1 and E_2 was adopted as the reference elastic definition for the subsequent numerical analyses incorporating intralaminar and interlaminar damage models. This ensures that the transition to damage-inclusive simulations is grounded on an elastic response that is experimentally consistent, mechanically well-posed, and free from excessive parameter tuning.

Table 5.7 summarises the elastic material properties adopted for the reference elastic configuration following the deterministic sensitivity analysis. Only the in-plane Young's moduli E_1 and E_2 were updated with respect to the baseline elastic definition, while all remaining elastic constants were kept unchanged. This update reflects the selected $+5\%$ stiffness increase, which provided the most consistent improvement in the elastic-dominated impact regime.

Table 5.7: Baseline and updated elastic material properties adopted for the reference elastic configuration following the deterministic sensitivity analysis.

Property	**Baseline value**	**Reference elastic value**
Young's modulus (warp)	$E_1 = 22.49$ GPa	$E_1 = 23.61$ GPa
Young's modulus (weft)	$E_2 = 19.29$ GPa	$E_2 = 20.25$ GPa
Poisson's ratio	$\nu_{12} = 0.113$	unchanged
In-plane shear modulus	$G_{12} = 3.60$ GPa	unchanged
Transverse shear modulus	$G_{13} = G_{23} = 3.00$ GPa	unchanged
Through-thickness modulus	$E_3 = 6.10$ GPa	unchanged
Density	$\rho = 1.95$ g/cm^3	unchanged

This updated elastic definition serves exclusively as a stiffness reference and does not alter the modelling assumptions regarding damage initiation and evolution, which are introduced and discussed in the following section.

5.3 Intralaminar Damage Model Implementation and Assessment

This section presents the implementation and assessment of the intralaminar damage model adopted in the present work. Building on the theoretical formulation introduced in Chapter 3, the focus here is placed on the practical definition of material properties, numerical implementation details, and the evaluation of model performance under impact loading. The intralaminar response is modelled using a progressive damage framework based on two-dimensional Hashin failure criteria, coupled with an energy-based damage evolution law. Model assessment is carried out through a systematic comparison with experimental results, using crack length, back-face damage morphology, and acceleration-time response as the primary correlation metrics. The following subsections describe the material parameter definition, sensitivity analyses, and the selection of a reference intralaminar configuration.

5.3.1 Intralaminar Material Properties and Model Definition

The intralaminar damage model based on Hashin failure criteria is implemented within the common CAE framework described in Section 5.1. No modifications were introduced to the geometry definition, discretisation strategy, boundary conditions, contact formulation, or numerical integration scheme. Consequently, the present subsection focuses exclusively on the definition of the material properties required to activate progressive intralaminar damage at the ply level.

The elastic properties of the laminate are identical to those adopted for the baseline elastic model presented in Section 5.2. In addition to these elastic constants, the Hashin formulation requires the definition of intralaminar strength parameters associated with fibre-dominated and matrix-dominated failure modes under tensile and compressive loading. These strength properties were defined based on the experimental testing campaign performed by [13] and are summarised in Table 5.8.

Table 5.8: Intralaminar strength properties adopted for the Hashin damage model, defined from experimental testing [13].

Property	**Symbol**	**Value [MPa]**
Longitudinal tensile strength	X_T	300.34
Longitudinal compressive strength	X_C	339.36
Transverse tensile strength	Y_T	223.65
Transverse compressive strength	Y_C	306.03
Longitudinal shear strength	S_L	54.62
Transverse shear strength	S_T	50.60

In contrast, no direct experimental measurements were available for the intralaminar fracture energies of the investigated woven Glass/PPS laminate. This limitation is not uncommon for

woven thermoplastic composites, for which intralaminar fracture energy data are rarely reported in the literature. A survey of available studies indicates that most published values concern thermoset systems, such as glass/epoxy or glass/polyester laminates, or focus on unidirectional architectures, thereby limiting their direct applicability to the present material system.

As an initial approximation, baseline intralaminar fracture energy values were adopted from a woven glass/polypropylene composite system reported in the literature. This choice was motivated by the thermoplastic nature of the polypropylene matrix, the use of glass fibre reinforcement, and the comparable scale of intralaminar damage mechanisms at the ply level. In the referenced study, the fracture energies were experimentally determined through Compact Tension and Compact Compression fracture mechanics tests following ASTM E399 and ISO 13586 procedures. These measurements correspond to macroscopic fracture energy release rates obtained from pre-cracked specimens and therefore represent the total energy dissipated during crack propagation in the tested laminate configuration. The adopted baseline values, summarised in Table 5.9, correspond to tensile and compressive intralaminar fracture energies of approximately 216 N/mm and 136 N/mm, respectively. These values were assumed to be identical in both in-plane material directions and were introduced exclusively as a reference starting point to define the sensitivity range of the intralaminar fracture energy parameters, rather than as representative properties of the investigated Glass/PPS laminate [103].

Table 5.9: Baseline intralaminar fracture energy values adopted for sensitivity analysis of the Hashin damage model [103].

Damage mode	**Symbol**	**Baseline value [N/mm]**
Tensile intralaminar fracture energy	G_f^t	216
Compressive intralaminar fracture energy	G_f^c	136

It is important to note that the baseline intralaminar fracture energy values adopted from the reference glass/polypropylene system are significantly higher than values typically reported for woven glass/epoxy and glass/polyester laminates. In hindsight, the comparison with these thermoset systems suggests that fracture energy levels on the order of several hundred N/mm may be excessive when used within a continuum damage framework for intralaminar modelling under impact loading. Although the absolute values reported for thermoplastic composites may be influenced by the specific laminate architecture, fibre configuration, and the experimental energy calculation procedure, their direct use in a ply-level damage model can lead to unrealistically high energy dissipation and overly diffuse damage patterns in numerical simulations.

The identification of more moderate intralaminar fracture energy levels for woven glass/epoxy and glass/polyester laminates provided an important reference scale for interpreting the numerical response of the present model. Rather than directly adopting rounded values reported for these

dissimilar matrix systems, the sensitivity analysis was used to progressively reduce the initial thermoplastic baseline and identify a fracture energy range that yields physically realistic damage localisation and global dynamic response for the investigated Glass/PPS laminate.

It should be emphasised that the resulting fracture energy values should be interpreted as effective parameters within the adopted continuum damage modelling framework. Since the numerical model does not explicitly represent all energy dissipation mechanisms occurring during impact—such as matrix plasticity, fibre pull-out, frictional sliding, and delamination—the calibrated intralaminar fracture energy effectively captures the combined influence of these mechanisms at the ply level. This approach ensures that the selected values remain anchored to a thermoplastic reference while avoiding the excessive dissipation associated with the original baseline estimates.

5.3.2 Intralaminar Impact Response (Hashin Baseline)

The impact response obtained using the intralaminar damage model is analysed following the same general methodology adopted for the baseline elastic formulation presented in Section 5.2.2. The acceleration-time history measured at the impactor is retained as the primary response quantity, as it provides a direct and robust basis for comparison between numerical predictions and experimental data, while avoiding uncertainties associated with signal integration.

In contrast to the elastic analysis, no slope-based correlation metric is introduced at this stage. Although the initial portion of the acceleration-time response remains governed by the elastic stiffness of the laminate, the present Hashin baseline configuration is characterised by very high intralaminar fracture energy values, which strongly delay damage evolution. As a result, the effective stiffness degradation remains negligible throughout the impact event, and the numerical response remains close to that of the purely elastic model. Consequently, quantitative slope-based metrics are not suitable for distinguishing between different intralaminar fracture energy assumptions, and alternative damage-related correlation measures are introduced in the subsequent analysis.

Baseline Hashin response at 4.5 J

The baseline intralaminar damage model was first evaluated for the lowest impact energy level ($E = 4.5$ J, Group 1), corresponding to a regime in which the experimental response is predominantly elastic. Figure 5.25 compares the experimental acceleration-time histories with the numerical response obtained using the Hashin baseline configuration.

The numerical prediction closely follows the experimental curves in terms of global shape, peak acceleration, and contact duration. No visible intralaminar damage is observed in the simulation, despite the activation of the Hashin damage initiation criteria at localised regions. This behaviour is attributed to the high intralaminar fracture energy values adopted in the baseline configuration, which strongly inhibit damage evolution and result in minimal stiffness degradation.

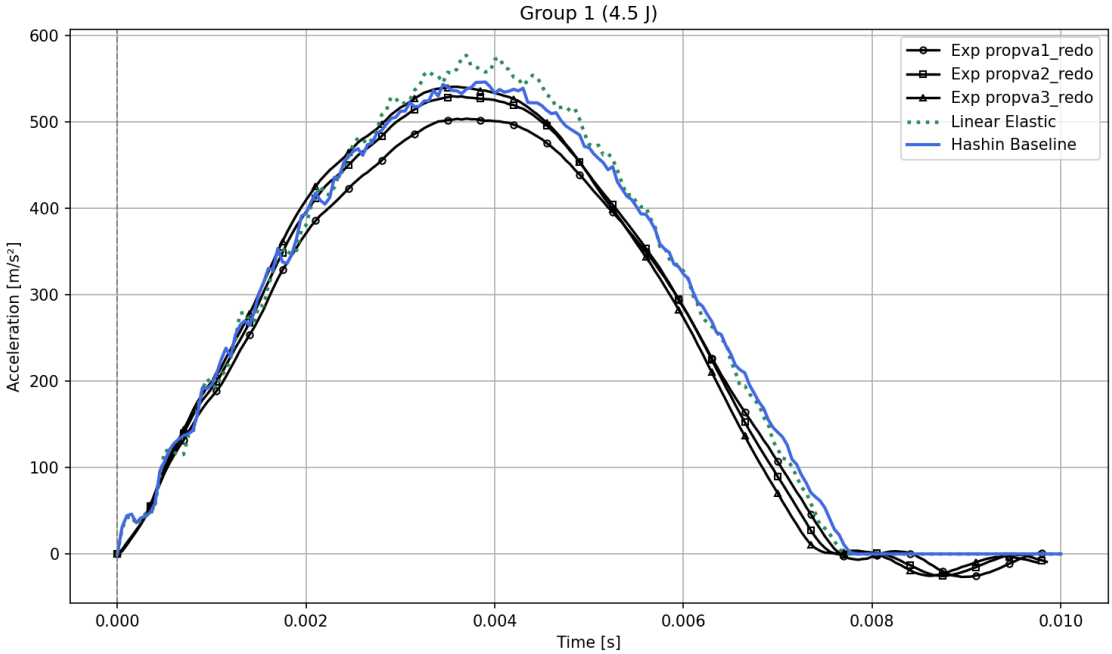


Figure 5.25: Acceleration-time response for Group 1 (4.5 J): experimental results, linear elastic model, and Hashin baseline configuration.

Baseline Hashin response at 7.5 J

The baseline Hashin model was subsequently evaluated for an intermediate impact energy of $E = 7.5$ J (Group 2). The corresponding acceleration–time responses are shown in Figure 5.26. As in the lower energy case, the numerical response exhibits good agreement with the experimental curves in terms of peak timing and overall response shape.

Despite the increased impact energy, the predicted response remains largely elastic in nature, with no significant intralaminar damage localisation observed. This indicates that, under the assumed baseline fracture energy values, intralaminar damage evolution does not contribute significantly to the global impact response at this energy level.

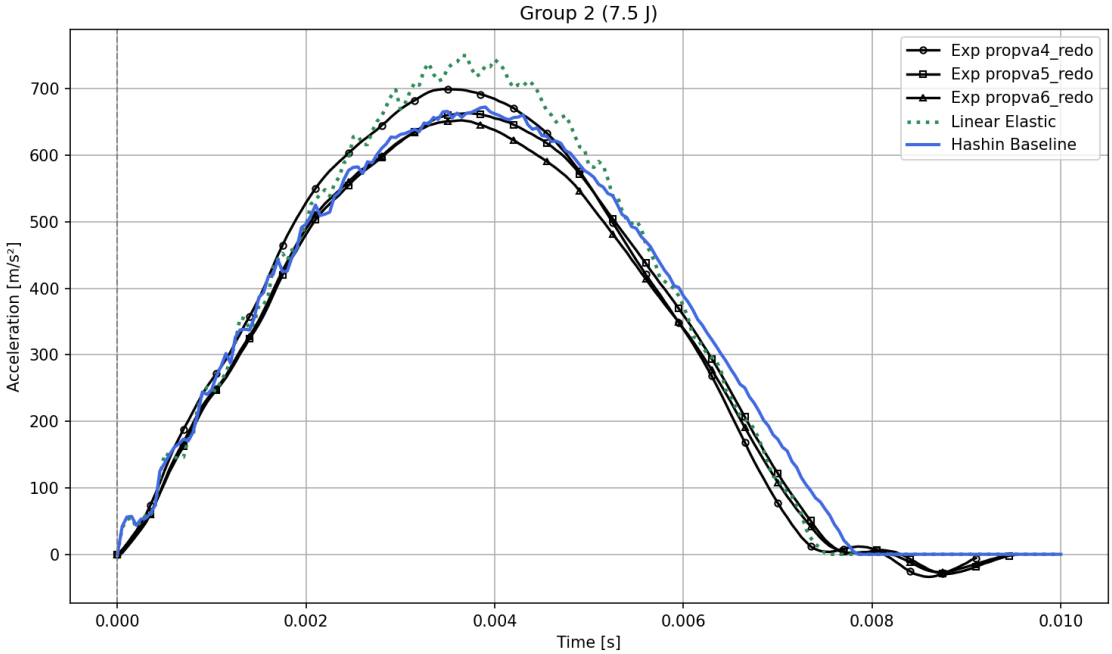


Figure 5.26: Acceleration-time response for Group 2 (7.5 J): experimental results, linear elastic model, and Hashin baseline configuration.

Baseline Hashin response at approximately 14.2 J

For the higher intermediate impact energy of approximately $E \approx 14.2$ J (Group 3), experimental observations indicate the onset of intralaminar damage mechanisms. The acceleration-time comparison for this energy level is presented in Figure 5.27.

While the Hashin baseline configuration predicts a reduction in peak acceleration relative to the purely elastic model, the numerical response still overpredicts the experimental peak and exhibits a stiffer unloading behaviour. The absence of pronounced damage-induced compliance confirms that the high baseline fracture energy values suppress intralaminar damage evolution, leading to a response that remains closer to elastic behaviour than observed experimentally.

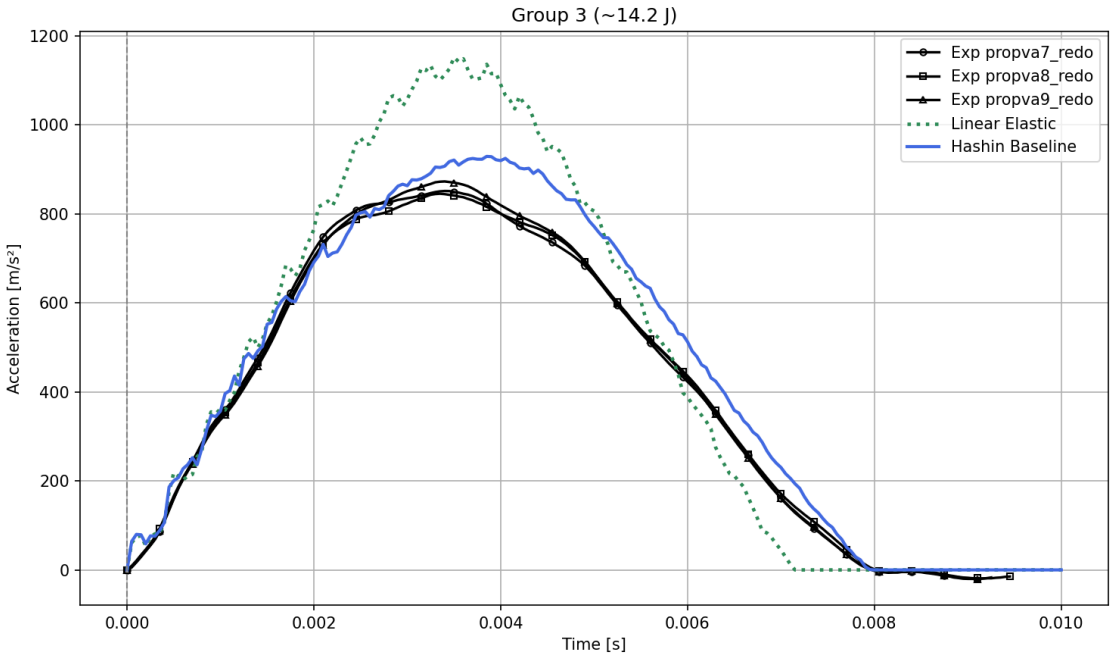


Figure 5.27: Acceleration-time response for Group 3 ($\approx$ 14.2 J): experimental results, linear elastic model, and Hashin baseline configuration.

Baseline Hashin response at 27 J

The highest impact energy considered in the present study ($E = 27$ J, Group 4) corresponds to a regime where extensive damage is experimentally observed. The comparison between experimental and numerical acceleration–time responses is shown in Figure 5.28.

At this energy level, the limitations of the baseline intralaminar fracture energy assumptions become evident. Although the Hashin formulation introduces a reduction in peak acceleration relative to the elastic model, the predicted response remains significantly stiffer than the experimental one, and the contact duration is underestimated. No localised intralaminar damage patterns comparable to the experimentally observed back-face cracking are obtained, indicating that the baseline fracture energy values lead to an over-dissipative intralaminar response within the continuum damage framework.

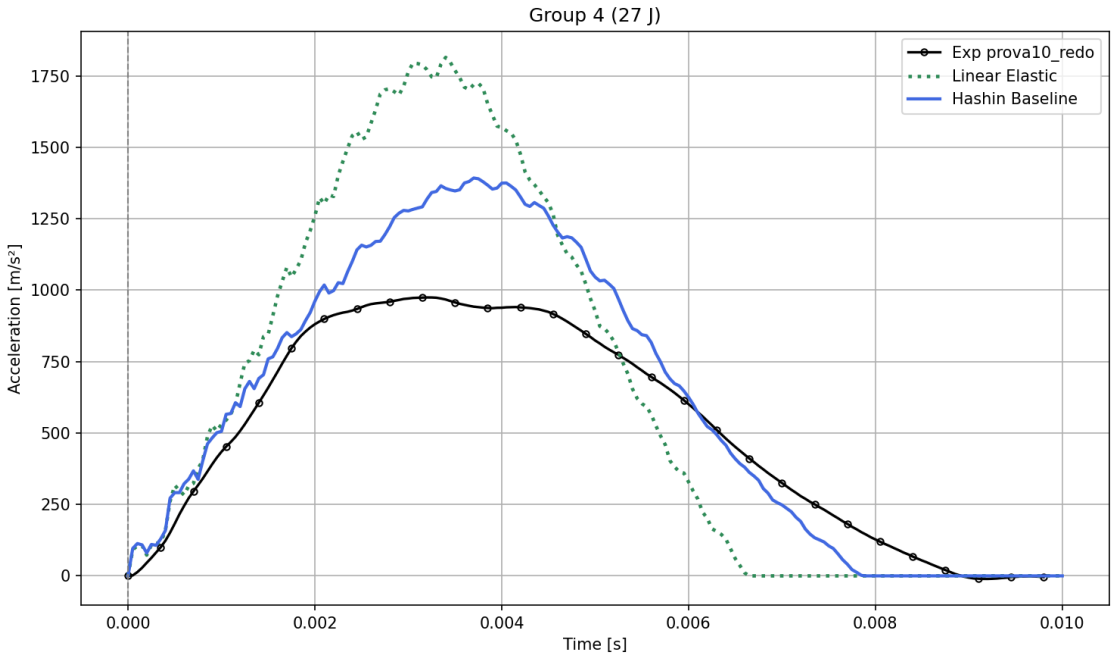


Figure 5.28: Acceleration-time response for Group 4 (27 J): experimental results, linear elastic model, and Hashin baseline configuration.

At this stage, the assessment of intralaminar damage initiation and progression is deferred, as the present baseline simulations are intended solely to characterise the global impact response prior to the introduction of damage-sensitive correlation metrics.

Overall, the results obtained with the Hashin baseline configuration demonstrate that the adopted intralaminar fracture energy values strongly suppress damage evolution, resulting in a numerical response that remains close to elastic behaviour even at high impact energies. While this configuration provides a useful upper bound for intralaminar energy dissipation, it does not reproduce the experimentally observed damage localisation. These observations motivate the subsequent sensitivity analysis of the intralaminar fracture energy parameters, in which crack length on the back face of the laminate is introduced as the primary quantitative correlation metric.

5.3.2.1 Fracture energy and strength sensitivity strategy

The baseline intralaminar fracture energy values introduced in Section 5.3.1 were adopted exclusively as reference magnitudes to define the sensitivity range of the intralaminar damage parameters. As shown in the following subsections, these baseline values lead to strongly suppressed damage evolution and do not reproduce the experimentally observed back-face cracking.

A systematic reduction of the intralaminar fracture energy revealed that physically realistic crack lengths are obtained for values on the order of approximately 10% of the adopted baseline

fracture energy. This fracture energy range was therefore identified as the primary candidate region for intralaminar damage correlation.

In addition, a limited sensitivity analysis of the intralaminar strength parameters was conducted. It was observed that a modest increase in strength leads to a more localised damage response, resulting in an increased back-face crack length for a given fracture energy. This behaviour is attributed to delayed damage initiation, which reduces diffuse early damage and promotes stress localisation along preferred damage paths.

Accordingly, a $+10\%$ adjustment of the intralaminar strength parameters was adopted for the configurations presented in the acceleration–time and damage contour comparisons. This adjustment was not used as a primary calibration parameter, but rather to ensure physically realistic damage localisation, while the fracture energy remains the dominant parameter governing crack length.

5.3.3 Correlation with Experimental Response and Back-Face Cracking

Damage metrics and interpretation

The correlation between numerical predictions and experimental observations in the present intralaminar damage analysis is based on a combination of qualitative and quantitative damage metrics. Particular care is taken in interpreting the numerical damage outputs provided by the Hashin formulation, as these variables represent continuum damage indicators rather than discrete cracks.

During the numerical modelling stage of this work, the experimental information available for correlation was limited to two primary observables: the acceleration–time response measured during the impact tests and the visible crack formation on the back face of the specimens after impact. The latter corresponds to the crack that develops in the last ply of the laminate and can be directly observed by visual inspection of the specimen surface. At that stage of the research, no additional non-destructive testing (NDT) techniques had yet been performed to characterise the internal damage state of the laminate.

It should therefore be noted that the crack length measured experimentally corresponds only to the visible crack on the back surface of the laminate. This measurement provides no direct information regarding the internal damage state, such as the depth of the crack or the extent of intralaminar or interlaminar damage within the laminate thickness. Consequently, the crack length used in this section should be interpreted strictly as a surface damage indicator representing the final manifestation of the damage process in the last ply.

Element deletion in the numerical model was observed only along the v-axis on the back face of the laminate. This direction corresponds to the most pronounced crack orientation identified in the experimental specimens and is therefore adopted as the reference direction for crack-length evaluation. No element deletion was observed along the orthogonal in-plane direction, indicating a strongly directional damage localisation consistent with the experimentally observed crack pattern.

In the damage contour plots, dark or black regions correspond to locations where one or more intralaminar damage variables reached a value of 0.997 in the last ply, which represents the maximum allowed damage level for numerical convergence purposes. This value does not imply complete material failure or full element removal; rather, it indicates near-complete stiffness degradation within the continuum damage framework. As noted above, such high damage levels were restricted to the v-axis, further supporting the observed directionality of the damage process.

The back-face crack length reported in the sensitivity tables is defined based on the extent of element deletion along the v-axis in the numerical model. This numerical crack length is compared with the experimentally measured back-face crack length obtained by visual inspection of the impacted specimens. Although this metric represents only the surface manifestation of the damage process, it provides a consistent observable quantity for establishing a first-order correlation between the numerical predictions and the experimental results available during the modelling stage of this work.

Subsequently, ultrasonic inspection using C-scan configuration was performed as part of the non-destructive testing campaign. This technique is primarily suited for identifying interlaminar damage, particularly delaminations, and therefore provides limited information regarding intralaminar damage mechanisms such as fibre rupture or matrix cracking within individual plies. For this reason, the C-scan results are mainly used to characterise the delamination patterns generated by the impact event and are discussed separately in Chapter 6.

A more complete characterisation of the intralaminar damage mechanisms represented by the Hashin formulation would require high-resolution imaging techniques capable of resolving the internal damage architecture through the laminate thickness. In this context, X-ray micro-computed tomography (micro-CT) would provide the most suitable experimental technique, as it allows the three-dimensional observation of internal damage mechanisms such as fibre rupture, matrix cracking, and delamination propagation. Such analysis was not available during the numerical modelling stage of the present work and therefore falls outside the scope of the current correlation study.

To ensure that the computed damage patterns reflect the physically relevant intralaminar damage mechanisms, damage outputs were evaluated only within the region defined by $AS = 1$, corresponding to the impact zone. This restriction avoids the inclusion of numerical artefacts outside

the contact-affected area and enables a focused assessment of damage initiation and localisation driven by the impact event. The Python script used for the automated post-processing and visualization of the intralaminar damage maps is provided in Appendix B.

Impact response at 4.5 J

The lowest impact energy level considered in the present study ($E = 4.5$ J, Group 1) corresponds to a loading regime in which the experimental response is predominantly elastic and no back-face cracking is observed. This impact case is therefore used as a control condition to verify the consistency of the intralaminar damage formulation under low-energy impact loading, rather than as a basis for intralaminar fracture energy sensitivity or parameter selection.

Crack-length assessment

For all intralaminar fracture energy configurations investigated at this energy level, the numerically predicted back-face crack length is zero, in agreement with experimental observations. No element deletion is observed along the v-axis, indicating that intralaminar damage remains confined to the impact zone and does not evolve into a crack-like pattern. As a result, crack length does not provide sensitivity to intralaminar fracture energy at 4.5 J and is not used as a discriminating metric for this impact case.

Acceleration-time response

Figure 5.29 compares the experimental acceleration-time histories with the numerical responses obtained using the linear elastic model, the Hashin baseline configuration, and a representative reduced fracture energy configuration. All numerical predictions exhibit good agreement with the experimental response in terms of global shape, peak acceleration, and contact duration.

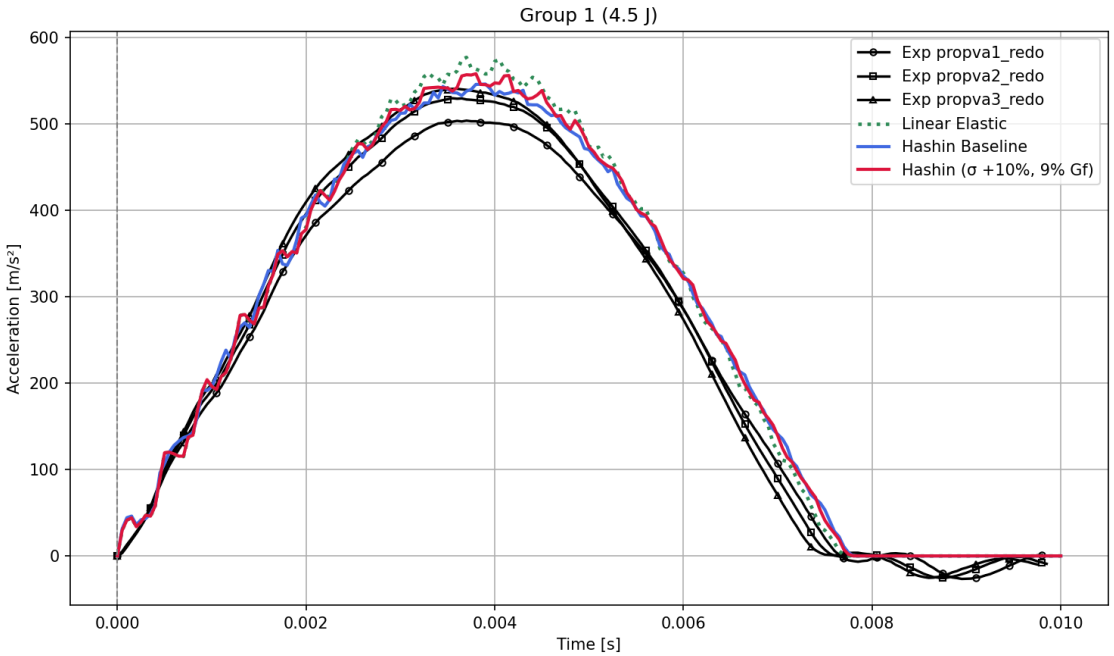
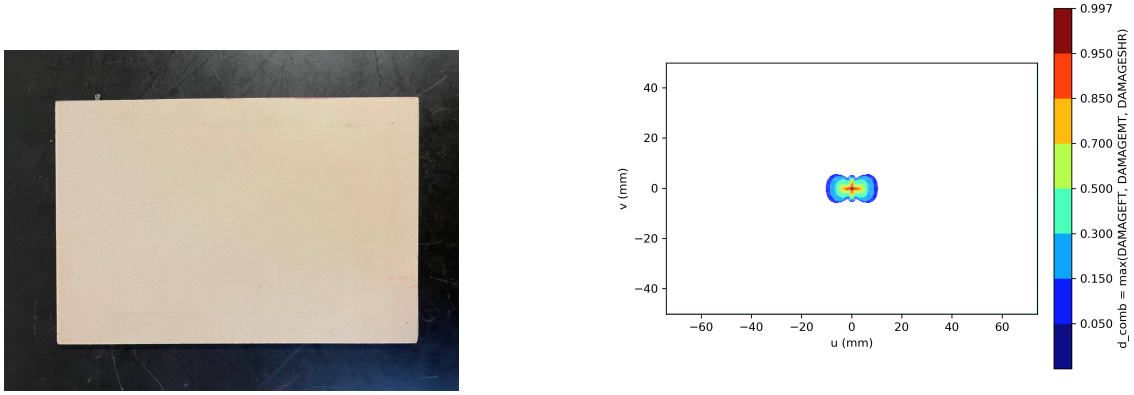


Figure 5.29: Acceleration–time response for Group 1 ($E = 4.5$ J): comparison between experimental results, linear elastic model, Hashin baseline configuration, and a representative reduced intralaminar fracture energy configuration.

At this energy level, differences between the baseline and reduced intralaminar fracture energy configurations are negligible in the acceleration response, confirming that the global dynamic behaviour is largely governed by the elastic properties of the laminate and is insensitive to intralaminar fracture energy variations under low-energy impact conditions.

Intralaminar damage distribution

A representative intralaminar damage contour for the 4.5 J case is shown in Figure 5.30. The predicted damage remains localised beneath the impactor and confined within the region defined by $AS = 1$, with no evidence of damage propagation or localisation along the v-axis. This behaviour is consistent across all fracture energy values considered and agrees with the absence of experimentally observed back-face cracking.



(a) Experimental.

(b) FEM Damage Map.

Figure 5.30: Numerical and experimental observations at $E = 4.5$ J: (a) experimental specimen back face, showing no visible cracking; (b) numerically predicted intralaminar damage contour on the back face of the laminate obtained from the Hashin damage model.

Overall, the 4.5 J impact case confirms that the intralaminar damage model does not introduce spurious damage under predominantly elastic loading conditions. While this case does not provide sensitivity to intralaminar fracture energy parameters, it establishes a physically consistent lower-bound response and supports the validity of the modelling framework prior to analysing higher-energy impact conditions.

Impact response at 7.5 J

The intermediate impact energy level $E = 7.5$ J (Group 2) represents a transition regime in which the experimental specimens exhibit the onset of back-face cracking, although the overall impact response remains predominantly elastic. This case is therefore useful to assess early damage localisation trends; however, it provides limited sensitivity when crack length is defined strictly based on element deletion.

Crack-length correlation

Table 5.10 summarises the numerically predicted back-face crack length for the different combinations of intralaminar fracture energy and strength variations investigated at $E = 7.5$ J, together with the experimentally measured back-face crack length ($L_c^{\mathrm{Exp}} = 3.23$ mm), obtained from visual inspection of the impacted specimen surface. Crack length is extracted from element deletion along the v-axis, as defined in Section 5.3.3. For several configurations, the numerical crack length remains equal to zero, indicating that no element deletion occurs under these impact conditions despite the presence of localised intralaminar damage.

Table 5.10: Back-face crack length correlation at $E = 7.5$ J for different intralaminar strength and fracture energy variations, expressed as percentages of the baseline values.

σ [%]	G_f [%]	L_c^{CAE} [mm]	L_c^{Exp} [mm]	Dev. [%]
-	5.0	11.09	3.23	243.3
-	8.0	1.16	3.23	64.1
-	9.0	0.00	3.23	-
-	10.0	0.00	3.23	-
+10	8.0	1.85	3.23	45.7
+10	9.0	0.46	3.23	85.7
+10	9.5	0.00	3.23	-
+10	10.0	0.00	3.23	-
+15	10.0	0.00	3.23	-

Here, σ and G_f are reported as relative variations with respect to the baseline intralaminar strength and fracture energy values defined in Section 5.3.1.

The results show that reducing the intralaminar fracture energy to approximately 8% of the baseline value yields the closest agreement with the experimentally observed crack length at this energy level. A further reduction of the fracture energy (e.g., 5% of the baseline) leads to a pronounced overprediction of crack extent, indicating excessive damage propagation. Although the 7.5 J case provides useful insight into the onset of crack sensitivity, it is not adopted as the primary basis for final fracture energy selection. Instead, the higher impact energy cases, for which crack length becomes a more robust and discriminating metric, are used to guide the final parameter identification.

Acceleration—time response

Figure 5.31 compares the experimental acceleration-time histories with the numerical responses obtained using the linear elastic model, the Hashin baseline configuration, and a representative reduced fracture energy configuration. At 7.5 J the acceleration-time response remains largely governed by elastic stiffness, and differences between intralaminar configurations are relatively small compared to the discrepancy observed for higher energy impact cases.

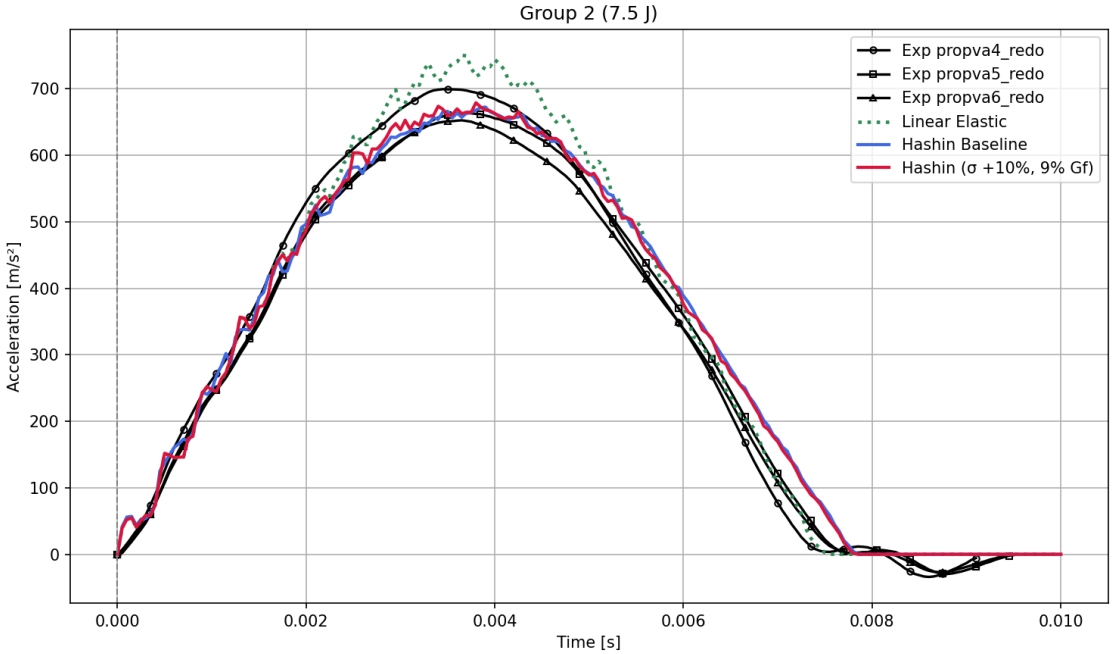
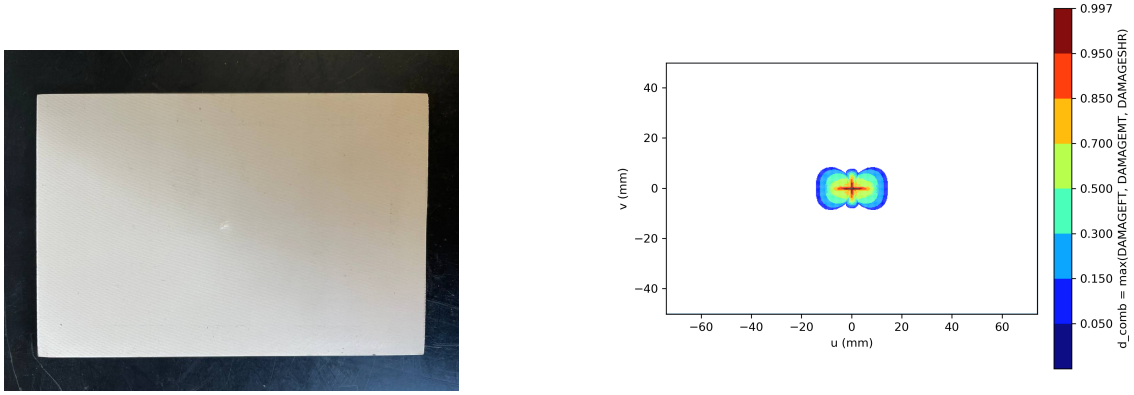


Figure 5.31: Acceleration–time response for Group 1 ($E = 7.5$ J): comparison between experimental results, linear elastic model, Hashin baseline configuration, and a representative reduced intralaminar fracture energy configuration.

Both intralaminar configurations reduce the peak acceleration relative to the purely elastic model, leading to an improved agreement with the experimental response. The baseline and reduced fracture energy cases, however, remain close in terms of global response shape and contact duration, confirming that acceleration–time is not sufficiently sensitive to discriminate between fracture energy values at this energy level.

Back-face damage comparison

A qualitative comparison between the experimental back-face surface condition and the numerically predicted intralaminar damage distribution is shown in Figure 5.32. The experimental specimen exhibits only limited visible back-face cracking at this energy level. Consistently, the numerical damage contour remains confined to a small region beneath the impactor, with no pronounced crack-like propagation.



(a) Experimental.

(b) FEM Damage Map.

Figure 5.32: Numerical and experimental observations at $E = 7.5$ J:(a) experimental specimen back face, showing limited visible cracking;(b) numerically predicted intralaminar damage contour on the back face of the laminate obtainedfrom the Hashin damage model.

Although no element deletion is observed for several configurations at this energy level, the combined damage indicator reaches the numerical upper limit ($d = 0.997$) in a localised region beneath the impactor. The corresponding damage localisation "arms" extracted along the in-plane directions exhibit a larger extension along the u-axis (3.69 mm) than along the v-axis (0.46 mm). This orientation differs from the experimentally observed crack direction, which is predominantly aligned with the v-axis.

This discrepancy indicates that, at $E = 7.5$ J, the numerical damage response is still governed by early-stage intralaminar damage initiation and local stress redistribution, rather than by fully developed crack propagation mechanisms. The absence of element deletion and the limited spatial extent of the damage confirm that damage localisation remains incipient and that the preferred experimental crack orientation is not yet fully captured at this energy level.

Overall, the 7.5 J impact case confirms the onset of intralaminar damage sensitivity to fracture energy variations but also highlights the limitations of this energy level for robust crack-orientation and crack-length correlation. As reflected in Table 5.10, realistic crack lengths are obtained only within a narrow range of reduced fracture energy values, while several configurations still result in zero or highly underpredicted crack length due to the absence of element deletion.

Consequently, the 7.5 J case is interpreted as a transitional regime, useful for identifying the lower bound of fracture energy values associated with the onset of cracking, but not adopted as the primary basis for final parameter selection. Final identification of the reference intralaminar fracture energy is therefore guided by the higher impact energy cases, for which crack length and crack orientation become more robust and physically representative metrics.

Impact response at approximately 14.2 J

The impact energy level $E \approx 14.2$ J (Group 3) corresponds to a regime in which consistently observed experimentally. In contrast to the lower-energy cases, this impact condition produces sufficiently developed intralaminar damage to enable a robust and discriminating crack-length-based correlation. For this reason, the 14.2 J case constitutes the primary reference for the sensitivity analysis and selection of the intralaminar fracture energy parameters.

Crack-length correlation

Table 5.11 reports the numerically predicted back-face crack length for the different intralaminar strength and fracture energy configurations investigated, together with the experimentally measured crack length ($L_c^{\text{Exp}} = 10.33$ mm). Crack length is extracted from element deletion along the v-axis, as defined in Section 5.3.3. For all configurations considered at this energy level, element deletion is observed, enabling a direct quantitative comparison between numerical predictions and experimental measurements.

Table 5.11: Back-face crack length correlation at $E \approx 14.2$ J for different intralaminar strength and fracture energy variations, expressed as percentages of the baseline values.

σ [%]	G_f [%]	L_c^{CAE} [mm]	L_c^{Exp} [mm]	Dev. [%]
-	5.0	22.57	10.33	118.5
-	8.0	13.38	10.33	29.5
-	9.0	10.16	10.33	1.6
-	10.0	7.16	10.33	30.7
+10	8.0	13.84	10.33	34.0
+10	9.0	10.62	10.33	2.8
+10	9.5	8.54	10.33	17.3
+10	10.0	7.62	10.33	26.2
+15	10.0	8.31	10.33	19.6

The results demonstrate a clear and systematic sensitivity of the predicted crack length to the intralaminar fracture energy. As shown in Table 5.11, the closest agreement with the experimental crack length is obtained for $G_f = 9\%$ of the baseline value, with deviations below 3%. Reducing the fracture energy further (to 8% and 5%) causes a strong overprediction of crack length, indicating excessive damage propagation. Conversely, using a higher fracture energy ($G_f = 10\%$) leads to an underprediction of crack extent, i.e., a response that is too resistant to crack growth at this energy level.

Variations in intralaminar strength have a secondary influence compared to fracture energy. At the optimal fracture energy level ($G_f = 9\%$), increasing strength by $+10\%$ produces a crack length

very similar to the baseline strength case (2.8% vs. 1.6% deviation), confirming that the fracture energy primarily governs crack propagation. At $G_f = 10\%$, increasing strength reduces the crack underprediction, but the correlation remains worse than the $G_f = 9\%$ cases. Overall, the 14.2 J results identify $G_f \approx 9\%$ of the baseline as the most consistent fracture energy level for crack-length correlation, with strength adjustments acting only as afine-tuning parameter.

Acceleration—time response

Figure 5.33 compares the experimental acceleration-time histories with the numerical responses obtained using the linear elastic model, the Hashin baseline configuration, and the selected intralaminar configuration. As expected, the purely elastic model significantly overpredicts the peak acceleration and underestimates the contact duration, reflecting the absence of damage-induced compliance.

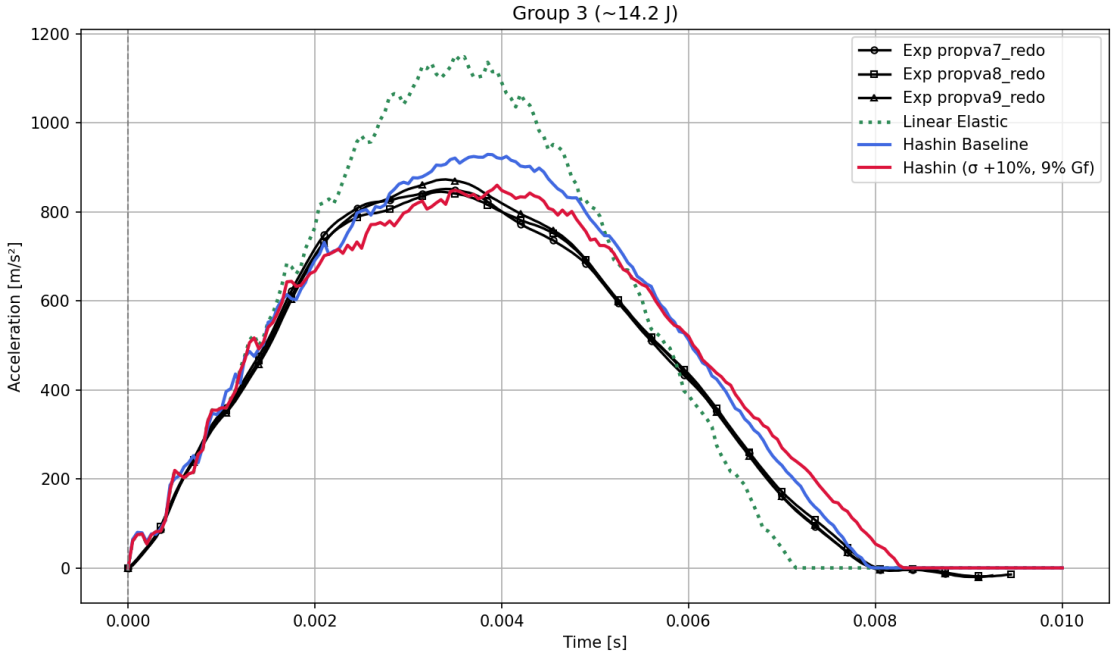
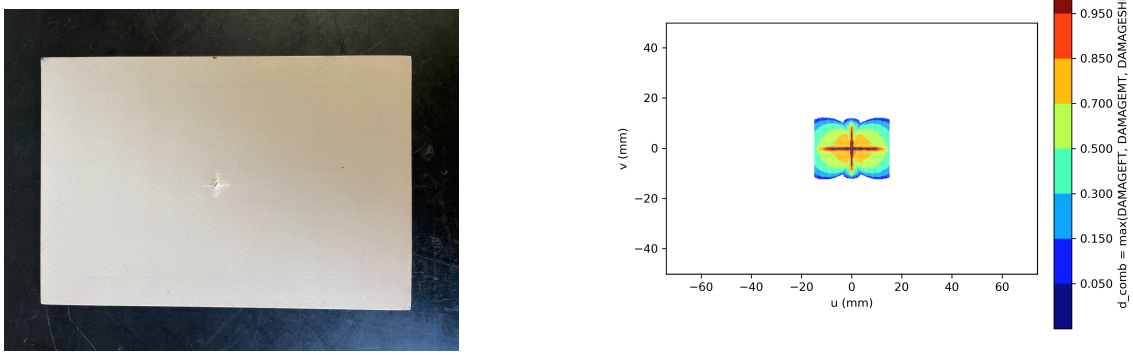


Figure 5.33: Acceleration–time response for Group 1 ($E = 14.2$ J): comparison between experimental results, linear elastic model, Hashin baseline configuration, and a representative reduced intralaminar fracture energy configuration.

The introduction of intralaminar damage leads to a substantial reduction in peak acceleration and a smoother unloading phase, resulting in markedly improved agreement with the experimental response. Differences between intralaminar configurations are less pronounced in the acceleration-time response than in the crack-length metric, confirming that acceleration is primarily sensitive to global stiffness degradation rather than to detailed variations in fracture energy.

Back-face damage and localisation

A qualitative comparison between the experimental back-face damage and the numerically predicted intralaminar damage distribution is presented in Figure 5.34. The experimental specimen exhibits a pronounced cross-shaped crack pattern centred beneath the impact point.



(a) Experimental. (b) FEM Damage Map.

Figure 5.34: Numerical and experimental observations at $E = 14.2$ J:(a) experimental specimen back face, showing a limited cross-shaped crack;(b) numerically predicted intralaminar damage contour on the back face of the laminate obtained from the Hashin damage model.

The numerical damage contour reproduces the characteristic cross-shaped damage morphology observed experimentally, with damage extensions developing along both in-plane directions. For the selected configuration, the extracted damage localisation lengths are 15.92 mm along the u-direction and 10.85 mm along the v-direction. While the numerical damage exhibits a larger extension along the u-axis, the predicted v-direction damage length is of the same order as the experimentally measured crack length and captures the dominant back-face cracking scale. This result indicates that, at this impact energy, the model reproduces the global damage morphology and extent, although some differences in directional dominance remain.

Overall, the 14.2 J impact case provides a robust and physically meaningful basis for intralaminar fracture energy calibration. Despite minor discrepancies in the relative dominance of the in-plane damage directions, the combined agreement in crack length magnitude, damage morphology, and global acceleration response supports the selection of a fracture energy level on the order of 9–10% of the baseline value for subsequent analyses. These results confirm that the calibrated intralaminar parameters capture the governing damage mechanisms at this energy level with sufficient fidelity for predictive modelling.

Impact response at $E = 27$ J

The impact energy level $E = 27$ J (Group 4) corresponds to a severe damage regime in which extensive intralaminar cracking is consistently observed experimentally. In contrast to the lower-

and intermediate-energy cases, this impact condition produces a fully developed back-face crack pattern with significant crack extension along both material directions. Consequently, the 27 J case provides the most stringent condition for assessing the robustness and physical consistency of the intralaminar damage model, and plays a key role in confirming the suitability of the selected fracture energy range.

Crack-length correlation

Table 5.12 reports the numerically predicted back-face crack length for the different intralaminar strength and fracture energy configurations investigated, together with the experimentally measured crack length ($L_c^{\text{Exp}} = 19$ mm). Crack length is extracted from element deletion along the v-axis, as defined in Section 5.3.3. For all configurations considered at this energy level, element deletion is observed, enabling a direct and quantitative comparison between numerical and experimental crack lengths.

Table 5.12: Back-face crack length correlation at $E = 27$ J for different intralaminar strength and fracture energy variations, expressed as percentages of the baseline values.

σ [%]	G_f [%]	L_c^{CAE} [mm]	L_c^{Exp} [mm]	Dev. [%]
-	8.0	18.24	19.00	4.0
-	9.0	17.53	19.00	7.7
-	10.0	13.85	19.00	27.1
+10	8.0	20.08	19.00	5.7
+10	9.0	19.38	19.00	2.0
+10	9.5	17.99	19.00	5.3
+10	10.0	15.69	19.00	17.4
+15	10.0	16.61	19.00	12.6

The results confirm a clear sensitivity of the predicted crack length to the intralaminar fracture energy also at this higher impact energy. As shown in Table 5.12, fracture energy values in the range of approximately 8-9.5% of the baseline provide the closest agreement with the experimentally measured crack length, with deviations below 6%. In contrast, the baseline fracture energy ($G_f = 10\%$ of the baseline value) consistently underpredicts crack extension, indicating an overly resistant response, while further reductions in fracture energy increase crack length and can lead to overprediction.

Variations in intralaminar strength exhibit a secondary but non-negligible influence on the predicted crack length. For fracture energy levels close to the optimal range ($G_f \approx 9\%$), a moderate increase in strength ($+10\%$) improves crack-length correlation, yielding the smallest deviation from the experimental value. However, strength variations do not alter the overall trend governed

by fracture energy. These results confirm that intralaminar fracture energy primarily controls damage propagation at $E = 27$ J, while intralaminar strength mainly modulates damage initiation and localisation.

Acceleration–time response

Figure 5.35 compares the experimental acceleration–time response with the numerical predictions obtained using the linear elastic model, the Hashin baseline configuration, and a representative reduced fracture energy configuration. As observed in the lower-energy cases, the linear elastic model significantly overpredicts the peak acceleration and underestimatesthe contact duration due to the absence of damage-induced compliance.

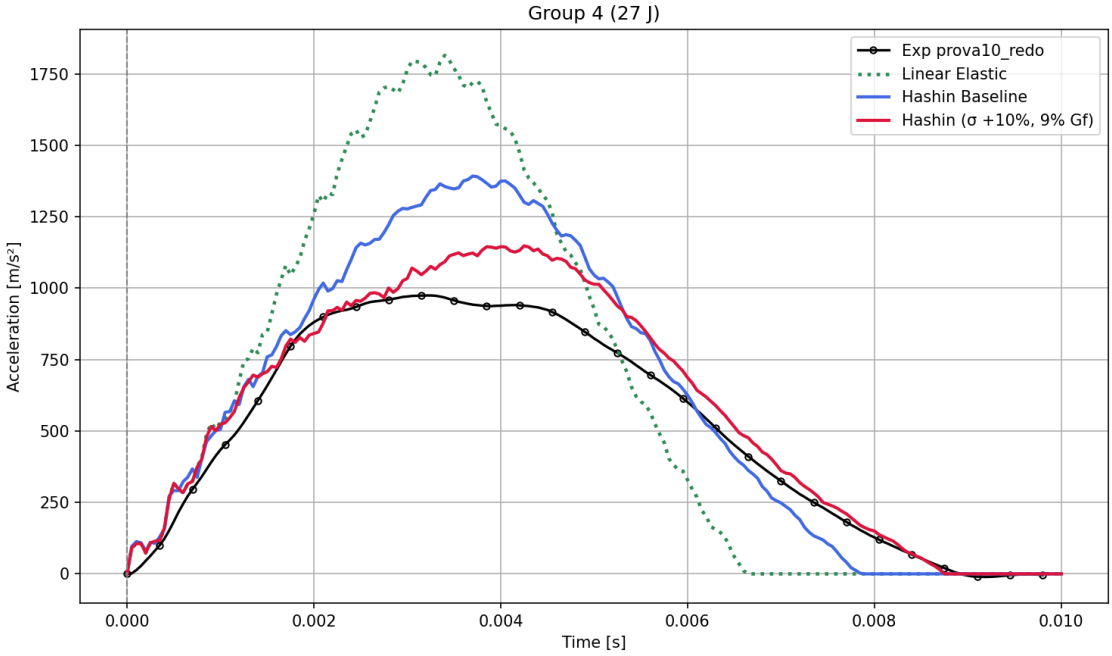


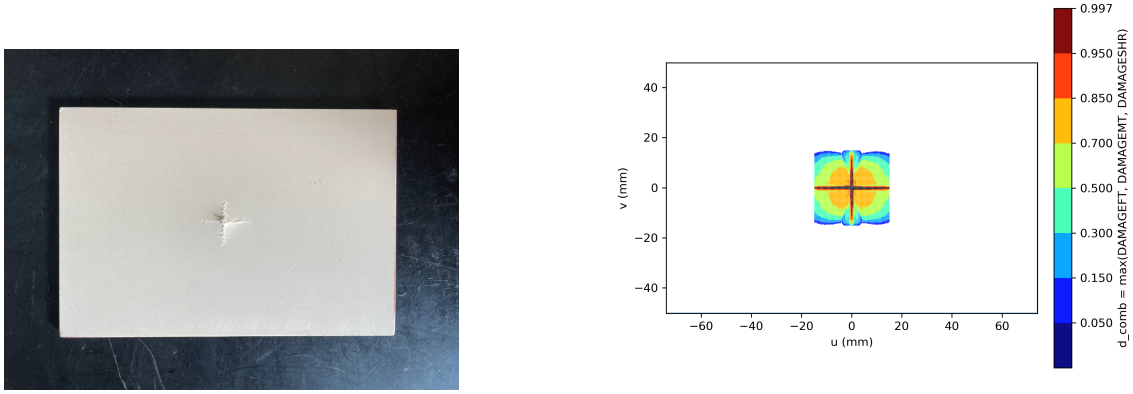
Figure 5.35: Acceleration–time response for Group 4 ($E = 27$ J): comparison between experimental results, linear elastic model, Hashin baseline configuration, and a representative reduced intralaminar fracture energy configuration.

The inclusion of intralaminar damage results in a substantial reduction of peak acceleration and a more gradual unloading phase, leading to a significantly improved agreement with the experimental response. Differences between intralaminar configurations remain less pronounced in the acceleration–time response than in the crack-length metric, reinforcing that acceleration is primarily sensitive to global stiffness degradation rather than to detailed variations in fracture energy.

Back-face damage and localisation

A qualitative comparison between the experimental back-face damage and the numerically predicted intralaminar damage distribution is shown in Figure 5.36. The experimental specimen

exhibits a well-defined cross-shaped crack pattern, with pronounced crack extension along both the u- and v-directions.



(a) Experimental.

(b) FEM damage map.

Figure 5.36: Numerical and experimental observations at $E = 27$ J: (a) experimental specimen back face, showing an extensive cross-shaped crack; (b) numerically predicted intralaminar damage contour on the back face of the laminate obtained from the Hashin damage model.

The numerical damage map reproduces the experimentally observed cross-shaped damage morphology, with a dominant crack extension along the v-direction and a longer secondary extension along the u-direction. For the selected configuration, the extracted damage localisation lengths are 29.77 mm along the u-direction and 19.85 mm along the v-direction, consistent with the anisotropic crack propagation observed experimentally. This qualitative and quantitative agreement provides strong evidence that the intralaminar damage model captures the governing damage localisation mechanisms at high impact energy.

Overall, the 27 J impact case confirms the validity of the fracture energy range identified from the 14.2 J analysis. The combined agreement in crack length, damage morphology, and global acceleration response supports the selection of intralaminar fracture energy values on the order of 8-9% of the baseline literature-derived value, with modest strength adjustments, as a robust and physically consistent reference configuration for subsequent analyses.

The results obtained across the different parameter variations reveal a pronounced sensitivity of the predicted crack length to the intralaminar fracture energy. This behaviour is inherent to progressive damage models for composite laminates. In Hashin-based formulations, the strength parameters govern damage initiation, whereas damage evolution is controlled by energy-based softening laws defined through fracture energy parameters. Consequently, the amount of energy dissipated during the damage process is directly related to the value of G_f, which strongly influences stiffness degradation and the spatial extent of intralaminar damage [15].

5.3.4 Selection of the Reference Intralaminar Configuration

The intralaminar damage analyses presented in the previous subsections enable the definition of a reference configuration for the Hashin-based damage model. In direct analogy with the procedure adopted for the elastic formulation, the selection is based on a deterministic sensitivity analysis, with emphasis on back-face crack-length correlation, crack morphology, and global acceleration–time response.

At low impact energy ($E = 4.5$ J), no element deletion and no measurable back-face crack are observed numerically for any of the investigated configurations. Consequently, this case is not discriminative with respect to the intralaminar fracture energy and is used only to verify that the damage formulation does not introduce spurious damage under predominantly elastic loading conditions. At $E = 7.5$ J, limited cracking is observed experimentally, while several numerical configurations still result in zero predicted crack length due to the absence of element deletion. This intermediate case provides qualitative insight into the onset of damage localisation but does not constitute a sufficiently robust basis for parameter selection.

The impact case at $E \approx 14.2$ J represents the first energy level at which consistently developed intralaminar cracking is observed experimentally and numerically for all configurations. At this energy level, crack length exhibits a clear and systematic sensitivity to the intralaminar fracture energy, enabling a robust quantitative correlation. Fracture energy values close to 9% of the initial baseline values provide the closest agreement with the experimentally observed back-face crack length, while both higher values ($G_f = 10\%$) and further reductions lead respectively to underprediction and excessive overprediction of crack extent. Variations in intralaminar strength act as a secondary tuning parameter, primarily influencing damage localisation, without altering the dominant role of fracture energy in governing crack propagation.

The highest impact energy investigated, $E = 27$ J, serves as a confirmation case for the selected fracture energy level. At this energy level, extensive cracking and a well-defined cross-shaped damage pattern are observed experimentally. The configurations identified as optimal at $E \approx 14.2$ J, characterised by a fracture energy of approximately 9% of the initial baseline value combined with a $+10\%$ increase in intralaminar strength, continue to provide good agreement in terms of crack length, damage morphology, and global acceleration–time response. This confirms the robustness of the selected intralaminar parameters under more severe impact conditions.

It is important to note that the calibrated intralaminar fracture energy values obtained in the present sensitivity study are significantly lower than the initial baseline values adopted from the literature. The initial reference values were taken from fracture tests performed on woven glass/polypropylene laminates and were used only as a first estimate for the numerical calibration. In such fracture tests, the measured fracture energy corresponds to the total energy dissipated during crack propagation in a specific laminate configuration and loading scenario, which may differ

significantly from the local energy dissipation represented within a ply-level continuum damage model under impact loading conditions.

Within the adopted Hashin-based formulation, the intralaminar fracture energy primarily controls the rate of stiffness degradation and crack propagation in the continuum damage framework. When additional damage mechanisms such as interlaminar delamination, matrix plasticity, or fibre pull-out are not explicitly represented in the model, the calibrated fracture energy effectively absorbs part of the global energy dissipation associated with these mechanisms. Consequently, the resulting values should be interpreted as effective intralaminar fracture energy parameters within the context of the adopted modelling framework rather than as intrinsic material fracture properties measured under standard fracture testing configurations.

To assess the physical plausibility of the calibrated parameters, the resulting values were compared with intralaminar fracture energy values adopted in numerical impact modelling studies for similar composite laminates reported in the literature. Table 5.13 summarises representative values used for woven thermoplastic and thermoset composite systems. Although direct comparison between different material systems and modelling frameworks must be interpreted cautiously, the results indicate that the calibrated values obtained in the present study fall within the same order of magnitude as those typically employed in continuum damage simulations of woven composite laminates.

Table 5.13: Representative intralaminar fracture energy values adopted in numerical impact modelling studies of woven composite laminates.

Material system	**Architecture**	G_f **(N/mm)**
Carbon/PEEK[a]	Woven laminate	$25-40$
Glass/Epoxy[b]	Woven laminate	~ 30
Glass/Elium[c]	Woven laminate	$20-35$
Glass/PPS (present work)	5HS woven laminate	19.44

[a] Intralaminar fracture energy values reported for woven CF/PEEK laminates used in impact simulations based on cohesive-zone modelling approaches [104].

[b] Representative fracture energy values used in numerical modelling of impact damage in woven glass/epoxy laminates [105].

[c] Intralaminar damage modelling parameters adopted for woven glass/Elium thermoplastic laminates subjected to impact loading [106].

It should be emphasised that the calibrated fracture energy values identified in the present intralaminar sensitivity analysis correspond to effective parameters within the adopted modelling framework. The numerical model considered in this stage includes only intralaminar damage mechanisms and does not explicitly represent other potential energy dissipation processes that may occur during impact, such as interlaminar delamination or localised matrix deformation beneath the impactor. As a result, the calibrated fracture energy should be interpreted primarily as

the parameter value required to reproduce the experimentally observed crack development within the simplified modelling assumptions. The intrinsic intralaminar fracture energy of the material system may therefore differ from the calibrated value when additional damage mechanisms are introduced in more advanced modelling formulations.

The comparison suggests that the initial glass/polypropylene-based estimate of $G_f^t = 216\,\mathrm{N/mm}$ constituted an overly conservative starting value for the investigated Glass/PPS laminate and modelling framework. The calibrated value of $G_f^t = 19.44\,\mathrm{N/mm}$ obtained through the deterministic sensitivity analysis therefore provides a more realistic representation of the effective intralaminar fracture behaviour governing crack propagation under the investigated impact conditions.

Based on the combined evidence from all impact energy levels, a reference intralaminar configuration for the Hashin damage model is defined. The resulting baseline and updated intralaminar properties adopted for the reference configuration are summarised in Table 5.14. This configuration is adopted in all subsequent analyses involving intralaminar damage.

Table 5.14: Baseline and updated intralaminar material properties adopted for the reference Hashin damage configuration following the deterministic sensitivity analysis.

Property	**Baseline value**	**Reference intralaminar value**
Longitudinal tensile strength	$X_T = 300.34$ MPa	$X_T = 330.37$ MPa
Longitudinal compressive strength	$X_C = 339.36$ MPa	$X_C = 373.30$ MPa
Transverse tensile strength	$Y_T = 223.65$ MPa	$Y_T = 246.02$ MPa
Transverse compressive strength	$Y_C = 306.03$ MPa	$Y_C = 336.63$ MPa
Longitudinal shear strength	$S_L = 54.62$ MPa	$S_L = 60.08$ MPa
Transverse shear strength	$S_T = 50.60$ MPa	$S_T = 55.66$ MPa
Tensile intralaminar fracture energy	$G_f^t = 216$ N/mm	$G_f^t = 19.44$ N/mm
Compressive intralaminar fracture energy	$G_f^c = 136$ N/mm	$G_f^c = 12.24$ N/mm

The reference intralaminar configuration reported in Table 5.14 is adopted in all subsequent simulations involving intralaminar damage. Baseline strength values are defined from experimental testing, while the initial intralaminar fracture energy values were taken from the literature only as reference starting values for the sensitivity study. The updated reference values correspond to a $+10\%$ increase in intralaminar strength and a reduction of the initial fracture energy to 9%, as identified through the deterministic sensitivity analysis. These calibrated fracture energy values are therefore interpreted in the present work as effective parameters of the adopted continuum damage formulation that provide the most consistent correlation with the experimentally observed crack development and global impact response.

5.4 Interlaminar Damage Modelling and Delamination Effects

Interlaminar damage and delamination play a central role in the impact response of laminated composite structures, particularly for thermoplastic systems where significant energy redistribution occurs through interface separation and bending-driven mechanisms. In this section, interlaminar damage is modelled using cohesive interface formulations embedded between adjacent plies, with the objective of assessing their ability to reproduce both the global dynamic response and the experimentally observed damage characteristics of the Glass/PPS laminate under impact loading.

The development of the interlaminar model follows a progressive strategy. An initial baseline cohesive formulation is first introduced to establish a reference response and identify numerical and physical limitations. This is followed by a systematic parametric study aimed at clarifying the influence of key cohesive parameters on delamination extent and global stiffness. Finally, an estimated cohesive property set is constructed based on data reported for comparable composite systems, enabling a physically informed assessment of interlaminar damage mechanisms in the absence of direct interface characterisation. Throughout this section, the evaluation of the numerical response is grounded primarily on the acceleration-time histories and the experimentally measured back-face crack length, which are treated as the most reliable quantitative metrics for model assessment.

The subsections that follow describe the modifications introduced in the finite element mesh to accommodate cohesive interfaces, the definition of the cohesive material models, and the resulting interlaminar damage predictions as a function of impact energy. The analysis ultimately highlights both the capabilities and inherent limitations of a purely interlaminar damage framework, providing the basis for the enhanced modelling strategies introduced in the subsequent sections.

5.4.1 Mesh Modification and Interface Discretization

To enable the explicit modelling of interlaminar damage through cohesive elements, the finite element mesh adopted for the elastic and intralaminar simulations was modified to include the baseline elastic and intralaminar configurations, the laminate was discretized using a ply-by-ply solid formulation without interfacial elements, which precludes the representation of delamination phenomena. Therefore, an additional discretization strategy was required to introduce cohesive behaviour at the ply interfaces.

A thin partition with a nominal thickness of 0.005 mm was introduced at each ply interface to host cohesive elements. This partition does not aim to represent a physical resin-rich interlayer, but rather serves as a numerical device to accommodate the cohesive discretization while preserving mesh conformity between neighbouring plies. The selected thickness issufficiently small to avoid

any significant modification of the global laminate geometry, while ensuring numerical stability and compatibility with the surrounding continuum shell elements.

The cohesive interfaces were inserted between all adjacent plies within the laminate stacking sequence, allowing interlaminar separation to develop at any interface subjected to sufficient normal or shear loading. Node-to-node compatibility was maintained between the cohesive elements a consistent transfer of tractions and displacements across the interfaces. The structured meshing strategy adopted for the plies was preserved, with the introduction of the interfacial partitions resulting in a local refinement exclusively in the through-thickness direction.

Figure 5.37 illustrates the modified ply discretization strategy, highlighting the introduction of the interfacial partitions used to accommodate the cohesive elements. This mesh modification enables the explicit representation of delamination initiation and propagation within the laminate, while maintaining consistency with the discretization strategy employed in the elastic and intralaminar damage models. The resulting mesh constitutes the basis for the subsequent definition of the cohesive element formulation and interlaminar material properties.

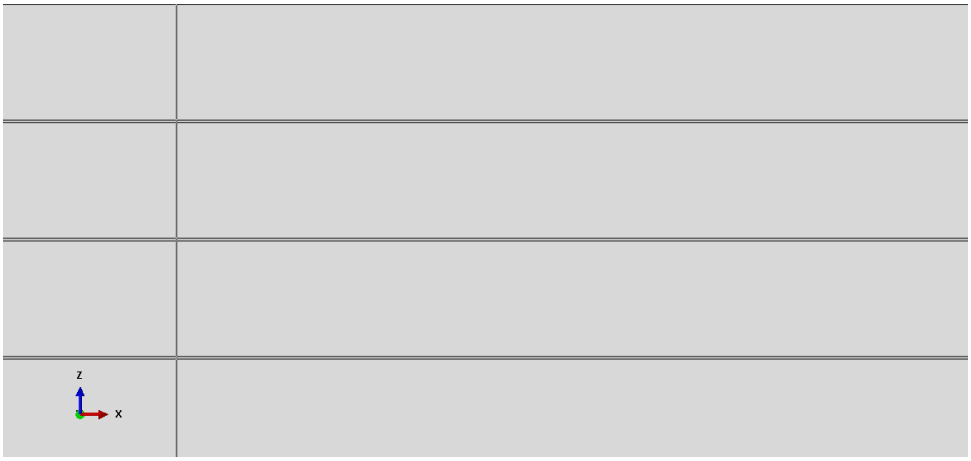


Figure 5.37: Modified ply discretization adopted for the interlaminar damage model, showing the introduction of thin interfacial partitions between adjacent plies to accommodate cohesive elements.

5.4.2 Cohesive Element Formulation and Element Type

Interlaminar damage was modelled using three-dimensional cohesive elements inserted at the ply interfaces defined by the discretization strategy described in Section 5.4.1. The objective of this formulation is to explicitly represent ply-to-ply separation while preserving a clear distinction between intralaminar damage mechanisms, governed by the continuum damage formulation adopted for the composite plies, and interlaminar damage, governed by the cohesive law.

The composite plies were discretized using linear hexahedral continuum-shell elements of type SC8R, which enable an efficient through-thickness representation of the laminate while retaining a three-dimensional stress state. Interlaminar interfaces were discretized using linear eight-node three-dimensional cohesive elements of type COH3D8, which are fully compatible with the surrounding continuum-shell formulation. This combination ensures a consistent transfer of normal and shear tractions across the ply interfaces prior to damage initiation, followed by progressive interfacial stiffness degradation associated with delamination.

The cohesive elements were defined using a traction–separation response with default Abaqus settings. In this configuration, the constitutive (original) thickness of the cohesive elements is assumed equal to unity ($T_0 = 1.0$), such that the nominal strains employed internally by Abaqus correspond directly to the relative separation components. Consequently, the 0.005 mm interfacial partition introduced to accommodate the cohesive elements serves exclusively as a geometric discretization device and does not prescribe the constitutive thickness of the cohesive law.

The composite plies were discretized using continuum-shell elements of type SC8R, which employ reduced integration to improve computational efficiency while maintaining an accurate representation of the laminate kinematics. The cohesive elements of type COH3D8 adopt the default integration scheme implemented in Abaqus for three-dimensional cohesive elements. This formulation is fully compatible with the explicit time integration framework and the contact interactions employed in the impact simulations.

Table 5.15 summarizes the element discretization adopted for the interlaminar damage model, highlighting the relative contribution of continuum-shell and cohesive elements to the overall mesh. The introduction of cohesive elements resulted in an increase in the total element count relative to the intralaminar damage model, while preserving the in-plane discretization of the laminate.

Table 5.15: Element discretization of the composite panel for the interlaminar damage model.

Element type	**Number of elements**
Continuum-shell elements (SC8R)	705 280
Cohesive elements (COH3D8)	661 200
Total elements	**1 366 480**

5.4.3 Baseline Interlaminar Properties and Material Assumptions

The definition of interlaminar cohesive properties constitutes a key modelling aspect of the numerical framework, as these parameters directly control the initiation and propagation of delamination within the laminate. Owing to the limited availability of experimentally characterized

cohesive parameters for Glass/PPS systems under impact loading, a baseline set of interlaminar properties was established based on a combination of literature data and physically motivated assumptions. This baseline configuration serves as a reference model for the parametric studies and sensitivity analyses presented in the subsequent sections.

Elastic interfacial stiffness.

The elastic response of the cohesive interfaces was defined using a traction-separation elastic law. The normal interfacial stiffness was estimated from the Young's modulus of the PPS matrix, assuming that the elastic behaviour prior to damage initiation is primarily governed by the polymer-rich interlaminar region. The shear stiffness components were derived assuming isotropic behaviour of the PPS matrix in the transverse directions, using the corresponding shear modulus. This approach provides a physically reasonable starting point while avoiding excessive artificial compliance of the laminate prior to damage initiation.

Damage initiation properties.

Damage initiation at the ply interfaces was governed by a quadratic nominal stress criterion. The interlaminar shear strength was taken from End-Notched Flexure (ENF) experimental data reported for Glass/PPS laminates [107]. In the absence of direct experimental measurements for the normal interfacial strength of the studied laminate, the mode I strength was estimated based on double cantilever beam (DCB) studies on PPS-based thermoplastic composite laminates [108]. Equal shear strengths were assumed in the two transverse directions due to the lack of directional interfacial strength data.

Damage evolution properties.

The evolution of interlaminar damage was modelled using an energy-based softening law defined in terms of critical fracture energies for mode I and mode II separation. The mode II fracture energy was selected based on ENF testing of Glass/PPS laminates reported in the literature [107]. The mode I fracture energy was obtained from DCB studies on PPS-based thermoplastic laminates [108]. Mixed-mode damage evolution was described using the Benzeggagh–Kenane (BK) interaction criterion, with an exponent representative of thermoplastic composite systems.

The baseline cohesive material parameters implemented in Abaqus are summarized in Table 5.16. These values were not intended to represent a fully validated material characterization, but rather to provide a consistent and physically motivated reference configuration for assessing the sensitivity of the numerical response to interlaminar property variations.

Table 5.16: Baseline cohesive material properties adopted for the interlaminar damage model.

Property	**Symbol**	**Value**
Normal stiffness[a]	K_n	4000 N/mm^3
Shear stiffness (1–2)[a]	K_s	1493 N/mm^3
Shear stiffness (1–3)[a]	K_t	1493 N/mm^3
Normal strength (Mode I)[b]	σ_n^0	9.53 MPa
Shear strength (Mode II)[c]	$\tau_s^0 = \tau_t^0$	5.5 MPa
Mode I fracture energy[b]	G_{Ic}	1.0 N/mm
Mode II fracture energy[c]	$G_{IIc} = G_{IIIc}$	3.0 N/mm
Mixed-mode exponent (BK law)[d]	η	1.3
Density[e]	ρ	1.95×10^{-9} tonne/mm^3

[a] Estimated from the elastic properties of the PPS matrix material [98].

[b] Estimated from double cantilever beam (DCB) testing of PPS-based thermoplastic laminates [108].

[c] Taken from end-notched flexure (ENF) testing of Glass/PPS laminates [107].

[d] Typical value reported for thermoplastic composite laminates.

[e] Adopted for numerical consistency in explicit dynamic simulations.

5.4.4 Interlaminar Damage Quantification and Post-Processing Strategy

The assessment of interlaminar damage in the present work required the definition of quantitative damage metrics capable of capturing both the extent and the spatial characteristics of delamination. At the stage when the interlaminar modelling study was conducted, no ultrasonic C-scan or X-ray computed tomography data were yet available for the tested specimens. Consequently, the evaluation of the interlaminar numerical response relied on a combination of global dynamic indicators and numerically derived delamination metrics extracted directly from the finite element results.

Interlaminar damage was quantified using the scalar damage variable `SDEG` associated with the cohesive elements. This variable represents the degradation state of the cohesive traction capacity, ranging from zero for an undamaged interface to unity for a fully failed interface. To distinguish between damage onset and fully developed delamination, two threshold values were considered throughout the post-processing:

- `SDEG` ≥ 0.9, corresponding to the onset of significant interfacial damage and the development of a delamination process zone;
- `SDEG` ≥ 0.99, corresponding to fully degraded cohesive elements and physically open delamination cracks.

A dedicated Python-based post-processing routine was developed to extract delamination metrics from the Abaqus output database. The routine identifies cohesive elements exceeding the selected `SDEG` thresholds and computes a series of damage descriptors, including the total number of damaged cohesive elements, the projected delamination footprint on the laminate back face, and the distribution of delaminated interfaces through the laminate thickness. The use of two damage thresholds enables a clear distinction between diffuse interfacial damage and fully developed delamination, providing additional insight into the sensitivity of the model response to the interlaminar material parameters. The Python script used for the extraction and visualization of the delamination maps from the cohesive element damage variable `SDEG` is provided in Appendix C.

To facilitate comparison with experimental observations, particular emphasis was placed on damage metrics associated with the back face of the laminate. The delamination footprint was evaluated by projecting the damaged cohesive elements onto the laminate surface, yielding a two-dimensional representation comparable to experimentally observed back-face cracking patterns. In addition, the number of plies involved in interlaminar damage was determined by counting the distinct ply interfaces containing cohesive elements exceeding the defined damage thresholds.

The interlaminar damage metrics extracted from the cohesive elements were analysed in conjunction with the global acceleration–time response and the experimentally measured back-face crack length. This combined assessment allowed the identification of physically plausible parameter sets that simultaneously reproduce the global dynamic response and yield realistic interlaminar damage patterns. The adopted post-processing strategy therefore provides a consistent framework for evaluating the influence of cohesive stiffness, strength, fracture energy, and mixed-mode interaction parameters during the parametric study, particularly in the absence of direct experimental delamination measurements at the time this modelling stage was performed.

5.4.5 Baseline Interlaminar Model Response: Energy-Based Assessment

The numerical response of the baseline interlaminar cohesive model was evaluated for each impact energy level considered in the experimental campaign. The results are presented and discussed on a case-by-case basis as a function of the impact energy, allowing the evolution of the global dynamic response and the predicted interlaminar damage patterns to be analysed under increasing loading severity. For each energy level, the assessment focuses on the acceleration-time response and the associated interlaminar damage features, with particular emphasis on the back-face cracking and delamination characteristics.

The discussion presented in the following sub-sections refers exclusively to the baseline cohesive property set defined in Section 5.4.3. Quantitative damage metrics extracted from the cohesive elements, together with the experimentally measured back-face crack length, are used to support the interpretation of the numerical results at each impact energy.

Baseline Cohesive response at 4.5 J

Figure 5.38 presents the acceleration-time response predicted by the baseline interlaminar cohesive model at an impact energy of $E = 4.5$ J, together with the corresponding experimental measurements. The numerical response exhibits a significantly more compliant behaviour than observed experimentally, as indicated by a reduced initial slope and a delayed development of the acceleration peak. While the experimental curves reach peak accelerations of approximately 500 $\mathrm{m/s^2}$ at $t \approx 3.5$–4.0 ms, the baseline cohesive model predicts a lower peak value and a broader acceleration pulse, suggesting excessive interfacial deformation prior to effective load transfer.

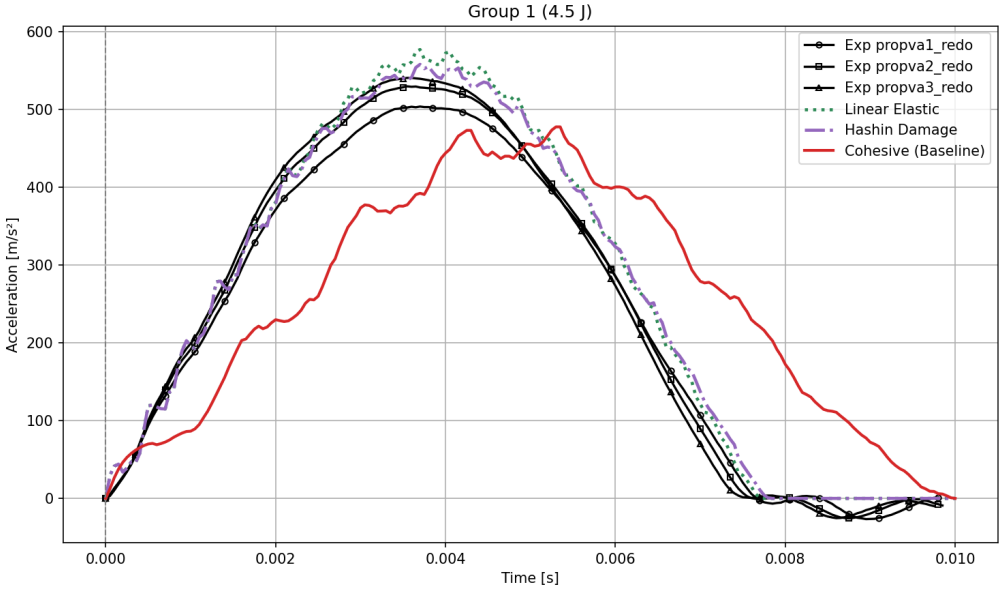


Figure 5.38: Acceleration-time response at $E = 4.5$ J comparing experimental data with the baseline interlaminar cohesive model.

At this energy level, no measurable back-face crack length is predicted by the numerical model, with a computed crack length of 0 mm. This result is consistent with the experimental observations, which also indicate the absence of a clearly defined macroscopic crack on the back face of the laminate at $E = 4.5$ J.

The corresponding interlaminar damage distribution is shown in Figure 5.39, obtained from the cohesive damage variable `SDEG`. Damage is observed exclusively in the range $0.9 \leq \texttt{SDEG} < 0.99$, indicating the formation of a delamination process zone without the development of fully failed cohesive elements. The damage footprint is localized around the impact point and exhibits an approximately axisymmetric distribution, consistent with subcritical interlaminar damage under low-energy impact loading.

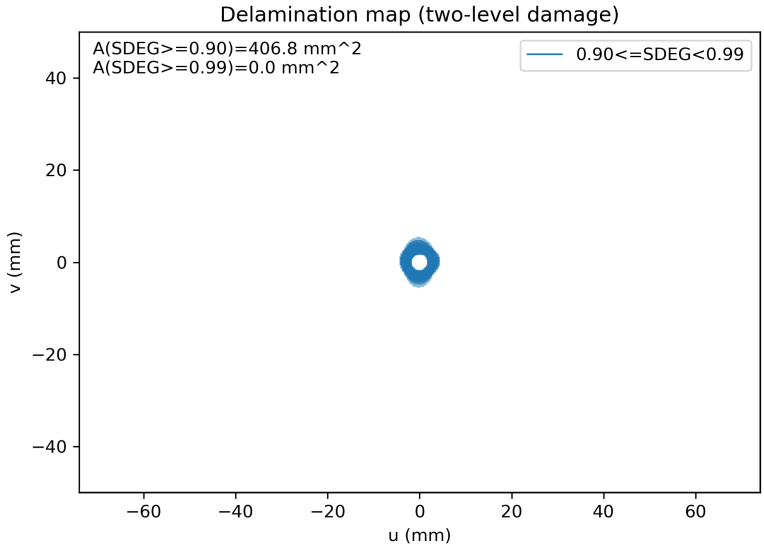


Figure 5.39: Back-face interlaminar damage map at $E = 4.5$ J for the baseline cohesive model, showing regions with $0.9 \leq \texttt{SDEG} < 0.99$.

Quantitative interlaminar damage metrics extracted using the post-processing strategy described in Section 5.4.4 are summarized in Table 5.17. The projected delaminated area associated with $\texttt{SDEG} \geq 0.9$ is approximately 406.8 mm^2, corresponding to 0.18% of the total laminate area. No elements exceed the $\texttt{SDEG} \geq 0.99$ threshold, confirming the absence of a fully developed interlaminar crack. Despite the limited spatial extent of the damage, the results indicate that multiple ply interfaces are affected, with an average of approximately seven interfaces exhibiting interlaminar damage above the $\texttt{SDEG} \geq 0.9$ threshold.

Table 5.17: Interlaminar damage metrics for the baseline cohesive model at $E = 4.5$ J.

Metric	**Value**
Back-face crack length (CAE)	0 mm
Back-face crack length (Exp)	0 mm
Delaminated area ($\texttt{SDEG} \geq 0.9$)	406.8 mm^2
Delaminated area ($\texttt{SDEG} \geq 0.99$)	0.0 mm^2
Process zone area ($0.9 \leq \texttt{SDEG} < 0.99$)	406.8 mm^2
Footprint area ($SDEG \geq 0.90$)	60.0 mm^2
Footprint area ($SDEG \geq 0.99$)	0.0 mm^2
Plies with interlaminar damage ($\texttt{SDEG} \geq 0.9$)	6.78
Plies with fully failed interfaces ($\texttt{SDEG} \geq 0.99$)	0

Overall, the baseline interlaminar cohesive model at $E = 4.5$ J predicts diffuse interlaminar damage in the form of a localized process zone, while failing to reproduce the experimental acceleration peak and global stiffness. Although no macroscopic crack is predicted at this energy level, the presence of interlaminar damage across multiple ply interfaces, combined with the overly compliant global response, indicates that the baseline interlaminar properties do not provide sufficient interfacial stiffness to accurately capture the dynamic response of the laminate under low-energy impact conditions.

Baseline Cohesive response at 7.5 J

Figure 5.40 presents the acceleration–time response predicted by the baseline interlaminar cohesive model at an impact energy of $E = 7.5$ J, together with the corresponding experimental measurements. As observed at lower impact energy, the numerical response remains significantly more compliant than the experimental response. The baseline cohesive model underpredicts the initial stiffness and peak acceleration, while exhibiting a broader acceleration pulse and a delayed unloading phase. Experimentally, peak accelerations of approximately 650–700 $\mathrm{m/s^2}$ are reached at around $t \approx 3.5$–4.0 ms, whereas the numerical prediction remains markedly lower and more spread in time, indicating excessive interfacial deformation during the impact event.

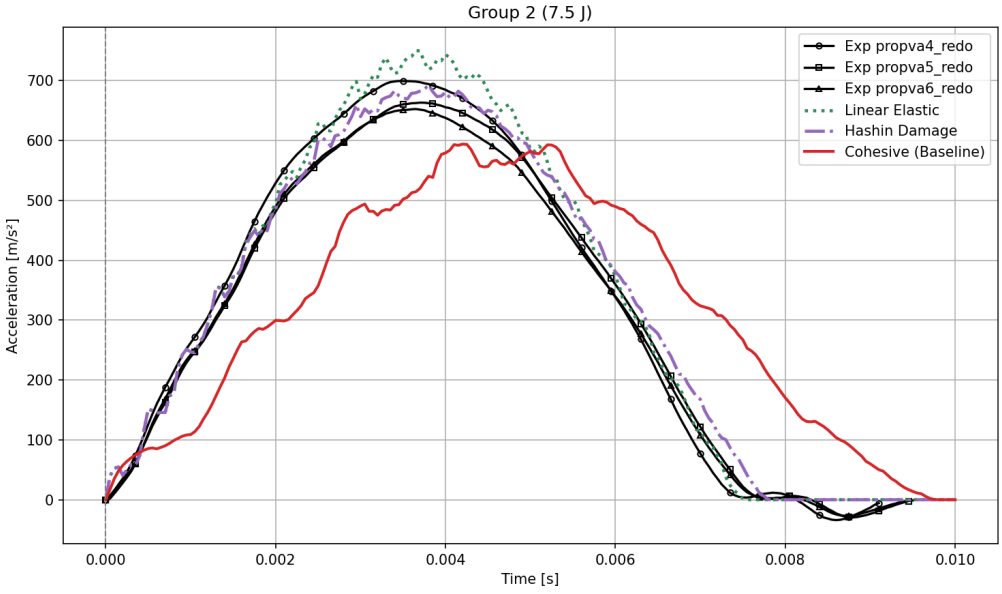


Figure 5.40: Acceleration-time response at $E = 7.5$ J comparing experimental data with the baseline interlaminar cohesive model.

At this energy level, the experimental specimens exhibit a measurable back-face crack length of approximately 3.23 mm. In contrast, the baseline cohesive model predicts no back-face crack development, yielding a numerical crack length of 0 mm. This discrepancy highlights the inability of the baseline interlaminar properties to reproduce the onset of macroscopic damage observed experimentally at intermediate impact energies.

The corresponding interlaminar damage distribution obtained from the cohesive damage variable SDEG is shown in Figure 5.41. As in the 4.5 J case, damage is confined to the range $0.9 \leq \text{SDEG} < 0.99$, indicating the formation of an extended delamination process zone without the development of fully degraded cohesive elements. The damage footprint remains localized around the impact point but exhibits a noticeably larger spatial extent than at lower energy, reflecting the increased severity of interlaminar damage with increasing impact energy.

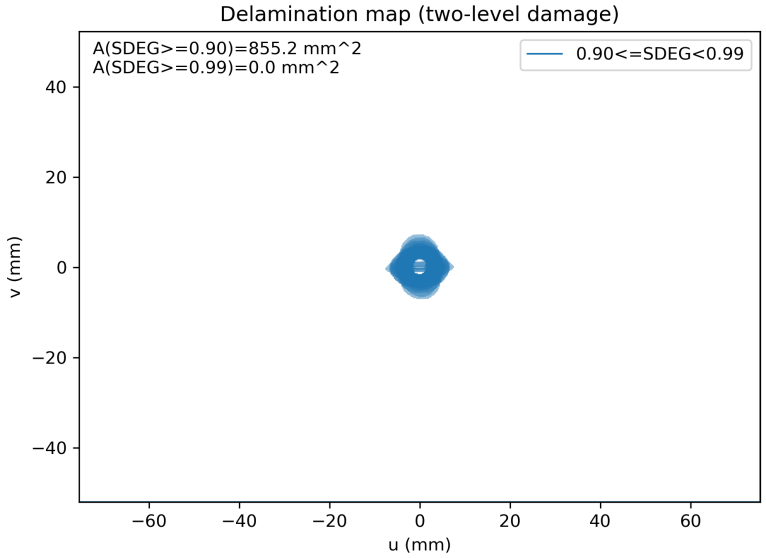


Figure 5.41: Back-face interlaminar damage map at $E = 7.5$ J for the baseline cohesive model, showing regions with $0.9 \leq \text{SDEG} < 0.99$.

Quantitative interlaminar damage metrics extracted from the post-processing routine are summarized in Table 5.18. The projected delaminated area associated with $\text{SDEG} \geq 0.9$ reaches approximately 855.2 mm^2, corresponding to 0.39% of the interlaminate area and 0.91% of the equivalent back-face footprint area. No cohesive elements exceed the $\text{SDEG} \geq 0.99$ threshold, confirming the absence of a fully developed interlaminar crack in the numerical model. Despite the absence of fully failed interfaces, interlaminar damage is predicted across multiple ply interfaces, with an average of approximately six interfaces exhibiting damage above the $\text{SDEG} \geq 0.9$ threshold.

Table 5.18: Interlaminar damage metrics for the baseline cohesive model at $E = 7.5$ J.

Metric	**Value**
Back-face crack length (CAE)	0 mm
Back-face crack length (Exp)	3.23 mm
Delaminated area ($\texttt{SDEG} \geq 0.9$)	855.2 mm^2
Delaminated area ($\texttt{SDEG} \geq 0.99$)	0.0 mm^2
Process zone area ($0.9 \leq \texttt{SDEG} < 0.99$)	855.2 mm^2
Footprint area ($SDEG \geq 0.90$)	134.0 mm^2
Footprint area ($SDEG \geq 0.99$)	0.0 mm^2
Plies with interlaminar damage ($\texttt{SDEG} \geq 0.9$)	6.38
Plies with fully failed interfaces ($\texttt{SDEG} \geq 0.99$)	0

Overall, at $E = 7.5$ J the baseline interlaminar cohesive model predicts a progressive increase in diffuse interlaminar damage relative to the lower-energy case, while still failing to reproduce the experimentally observed back-face crack formation and global stiffness. The absence of fully degraded cohesive elements and the persistent underestimation of the acceleration peak indicate that the baseline interlaminar properties remain overly compliant and do not provide sufficient constraint to capture the transition from subcritical interlaminar damage to macroscopic cracking observed experimentally at this impact energy.

Baseline Cohesive response at 14.2 J

Figure 5.42 presents the acceleration-time response predicted by the baseline interlaminar cohesive model at an impact energy of approximately $E \approx 14.2$ J, together with the corresponding experimental measurements. At this energy level, the numerical response continues to exhibit a more compliant behaviour than the experimental response, particularly during the initial loading phase and unloading stage. While the experimental curves reach peak accelerations of approximately 850–880 m/s^2 at $t \approx 3.5$–4.0 ms, the baseline cohesive model predicts a reduced peak value and a broader acceleration pulse, indicating excessive energy dissipation and delayed load transfer associated with interfacial deformation.

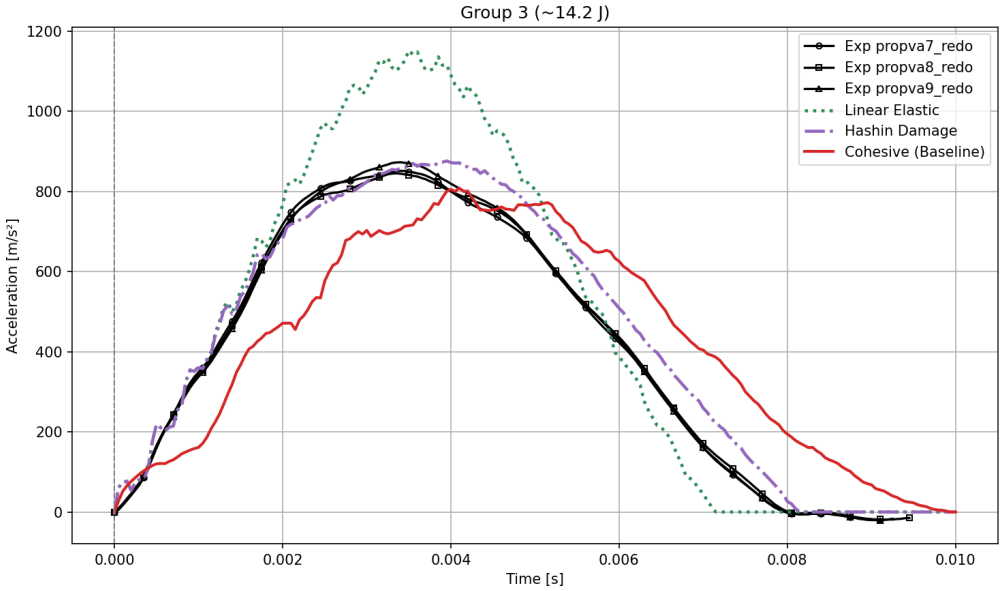


Figure 5.42: Acceleration-time response at $E \approx 14.2$ J comparing experimental data with the baseline interlaminar cohesive model.

In contrast with the lower-energy cases, a measurable back-face crack length is predicted by the baseline cohesive model at this impact energy. The numerical crack length is approximately 11.08 mm, in close agreement with the experimentally measured value of 10.33 mm, corresponding to a deviation of 7.3%. This result indicates that the baseline interlaminar model is capable of capturing the onset and extent of macroscopic interlaminar cracking at higher impact energy levels.

The corresponding interlaminar damage distribution obtained from the cohesive damage variable `SDEG` is shown in Figure 5.43. In this case, damage is observed both in the range $0.9 \leq$ `SDEG` < 0.99 and for `SDEG` ≥ 0.99, indicating the development of a fully degraded interlaminar crack surrounded by a more extended delamination process zone. The damage footprint is elongated along the longitudinal direction of the specimen and remains centred around the impact location, reflecting the increased severity of interlaminar damage under higher impact energy.

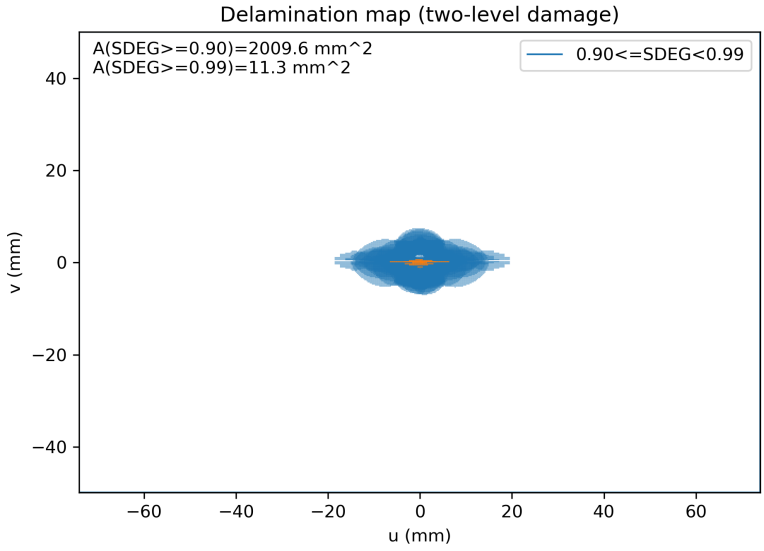


Figure 5.43: Back-face interlaminar damage map at $E \approx 14.2$ J for the baseline cohesive model, showing regions with $0.9 \leq \texttt{SDEG} < 0.99$ and $\texttt{SDEG} \geq 0.99$.

Quantitative interlaminar damage metrics extracted from the post-processing routine are summarized in Table 5.19. The projected delaminated area associated with $\texttt{SDEG} \geq 0.9$ reaches approximately 2009.6 mm^2, corresponding to 0.91% of the interlaminate area and 2.15% of the equivalent back-face footprint area. Fully failed cohesive elements ($\texttt{SDEG} \geq 0.99$) cover an area of approximately 11.3 mm^2, confirming the presence of a physically open interlaminar crack. The damage extends through several ply interfaces, with an average of approximately six interfaces exhibiting interlaminar damage above the $\texttt{SDEG} \geq 0.9$ threshold and nearly two interfaces containing fully failed cohesive elements.

Table 5.19: Interlaminar damage metrics for the baseline cohesive model at $E \approx 14.2$ J.

Metric	**Value**
Back-face crack length (CAE)	11.08 mm
Back-face crack length (Exp)	10.33 mm
Delaminated area ($\texttt{SDEG} \geq 0.9$)	2009.6 mm^2
Delaminated area ($\texttt{SDEG} \geq 0.99$)	11.3 mm^2
Process zone area ($0.9 \leq \texttt{SDEG} < 0.99$)	1998.3 mm^2
Footprint area ($SDEG \geq 0.90$)	318.0 mm^2
Footprint area ($SDEG \geq 0.99$)	6.0 mm^2
Plies with interlaminar damage ($\texttt{SDEG} \geq 0.9$)	6.32
Plies with fully failed interfaces ($\texttt{SDEG} \geq 0.99$)	1.89

Overall, at $E \approx 14.2$ J the baseline interlaminar cohesive model predicts the transition from diffuse interlaminar damage to the formation of a fully developed back-face crack, with a crack length in close agreement with the experimental measurements. Nevertheless, despite the improved correlation in terms of crack length, the global acceleration response remains overly compliant, indicating that the baseline interlaminar stiffness and energy parameters do not simultaneously capture both the dynamic response and the interlaminar damage evolution at this impact energy.

Baseline Cohesive response at 27 J

Figure 5.44 presents the acceleration-time response predicted by the baseline interlaminar cohesive model at an impact energy of $E = 27$ J, together with the corresponding experimental measurements. At this energy level, the baseline cohesive model continues to exhibit an overly compliant response during the loading stage, with a reduced initial slope and a broader acceleration pulse compared to the experimental curve. While the experiment reaches a peak acceleration close to 1000 $\mathrm{m/s^2}$ at approximately $t \approx 3.0$–3.5 ms, the numerical response shows a lower and more sustained peak, followed by a delayed unloading phase, indicating excessive deformation and energy dissipation associated with the interfacial response.

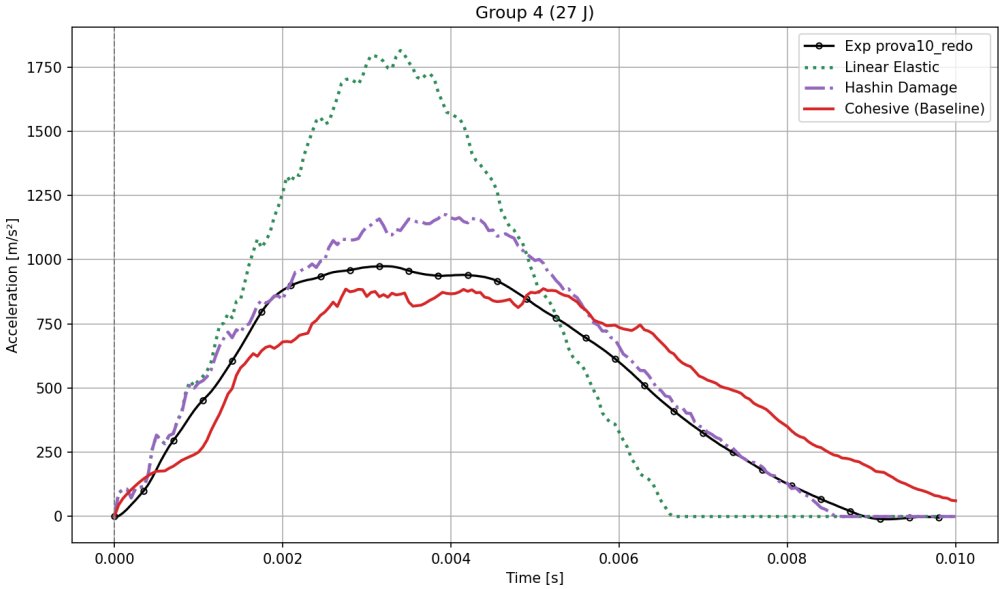


Figure 5.44: Acceleration-time response at $E = 27$ J comparing experimental data with the baseline interlaminar cohesive model.

At $E = 27$ J, the experimental specimen exhibits a measurable back-face crack length of approximately 19 mm. The baseline cohesive model predicts a crack length of 26.08 mm, corresponding to a deviation of 37.3%. Although the model captures the presence of a pronounced macroscopic crack at this energy level, the predicted crack extent is overestimated relative to the experiment.

The corresponding interlaminar damage distribution obtained from the cohesive damage variable SDEG is shown in Figure 5.45. In this case, extensive delamination is predicted, with both a large process zone ($0.9 \leq \texttt{SDEG} < 0.99$) and a substantial region of fully failed cohesive elements ($\texttt{SDEG} \geq 0.99$). The back-face damage pattern is characterized by a pronounced cross-shaped feature around the impact point, indicating a strong directional propagation of interlaminar damage consistent with the dominant bending and shear-driven separation at higher impact energy. The presence of a sizeable fully degraded region suggests the formation of an open delamination crack accompanied by extensive surrounding interfacial degradation.

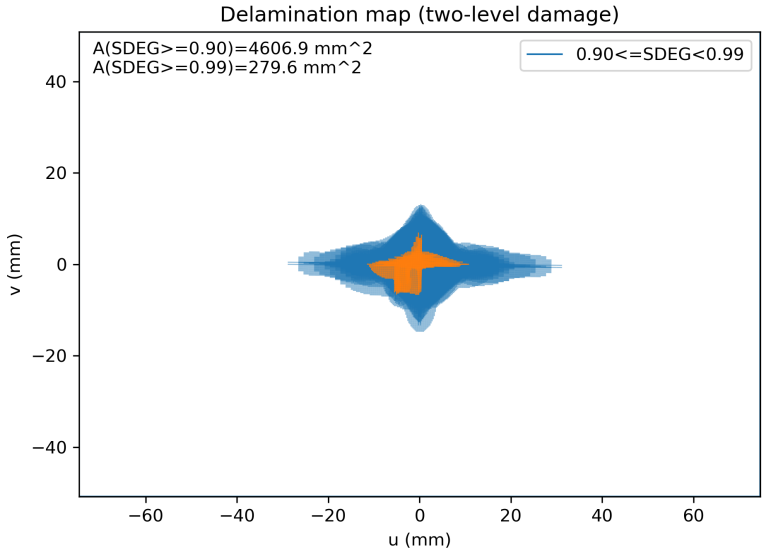


Figure 5.45: Back-face interlaminar damage map at $E = 27$ J for the baseline cohesive model, showing regions with $0.9 \leq \texttt{SDEG} < 0.99$ and $\texttt{SDEG} \geq 0.99$.

Quantitative interlaminar damage metrics extracted from the post-processing routine are summarized in Table 5.20. The projected delaminated area associated with $\texttt{SDEG} \geq 0.9$ reaches approximately 4606.9 mm^2, corresponding to 2.08% of the interlaminate area and 4.11% of the equivalent back-face footprint area. Fully failed cohesive elements ($\texttt{SDEG} \geq 0.99$) cover an area of approximately 279.6 mm^2, confirming the presence of an extended physically open interlaminar crack. The damage involves a larger number of interfaces than observed at lower energies, with an average of approximately 7.6 interfaces affected above $\texttt{SDEG} \geq 0.9$ and approximately 2.8 interfaces containing fully failed cohesive elements.

Table 5.20: Interlaminar damage metrics for the baseline cohesive model at $E = 27$ J.

Metric	**Value**
Back-face crack length (CAE)	26.08 mm
Back-face crack length (Exp)	19.00 mm
Delaminated area ($\texttt{SDEG} \geq 0.9$)	4606.9 mm^2
Delaminated area ($\texttt{SDEG} \geq 0.99$)	279.6 mm^2
Process zone area ($0.9 \leq \texttt{SDEG} < 0.99$)	4327.2 mm^2
Footprint area ($SDEG \geq 0.90$)	608.0 mm^2
Footprint area ($SDEG \geq 0.99$)	100.0 mm^2
Plies with interlaminar damage ($\texttt{SDEG} \geq 0.9$)	7.58
Plies with fully failed interfaces ($\texttt{SDEG} \geq 0.99$)	2.80

Overall, at $E = 27$ J the baseline interlaminar cohesive model predicts extensive delamination and a fully developed back-face crack, capturing the qualitative transition to severe interlaminar damage at high impact energy. However, the numerical crack length is overestimated and the global acceleration response remains overly compliant, indicating that the baseline interlaminar stiffness and damage parameters do not simultaneously reproduce both the dynamic response and the observed macroscopic damage extent at this energy level.

Summary of baseline interlaminar response

The energy-based assessment of the baseline interlaminar cohesive model highlights clear and systematic limitations of the adopted property set across the full range of impact energies. At low and intermediate energies ($E = 4.5$ and 7.5 J), the model predicts diffuse interlaminar damage in the form of extended process zones, while failing to reproduce the experimentally observed global stiffness and the onset of macroscopic back-face cracking. As the impact energy increases ($E \approx$ 14.2 and 27 J), the model captures the transition towards fully developed interlaminar cracking and provides a reasonable estimate of the crack length at intermediate energy. However, at the highest energy level, the crack extent is overestimated and the numerical response remains excessively compliant.

Overall, these results indicate that, while the baseline cohesive parameters are able to qualitatively reproduce the progression from subcritical interlaminar damage to severe delamination, they do not simultaneously capture the global dynamic response and the spatial characteristics of interlaminar damage. This behaviour suggests that the elastic stiffness, damage initiation strengths, and fracture energy parameters require systematic calibration to achieve a consistent and predictive representation of the interlaminar response under impact loading.

5.4.6 Global overview of interlaminar cohesive response regimes

Before analysing the influence of individual cohesive parameters in a systematic manner, it is instructive to examine the global response of the interlaminar model across the full set of simulations performed in this study. The objective of this subsection is not to validate the cohesive formulation against experimental measurements, but rather to provide a qualitative overview of the distinct damage regimes generated by different combinations of cohesive properties.

To this end, a global cohesive-regime map is constructed by combining two complementary damage indicators: the normalized deviation of the numerically predicted back-face crack length relative to the experimental value, and the delamination footprint expressed as a percentage of the projected 2D panel area. The latter is evaluated using the post-processing procedure described in Section 5.4.4, based on the scalar damage variable SDEG and the thresholds SDEG ≥ 0.90 and SDEG ≥ 0.99.

Figure 5.46 summarizes the response of all cohesive simulations performed at impact energies of 4.5 J, 7.5 J, 14.2 J and 27 J. Each marker represents one simulation, with colours indicating the impact energy and marker shapes identifying the simulation family (baseline, strength-scaled, stiffness-modified, mixed, suppressed, or runaway configurations). The vertical dashed line denotes zero crack-length deviation, corresponding to perfect agreement with the experimental reference.

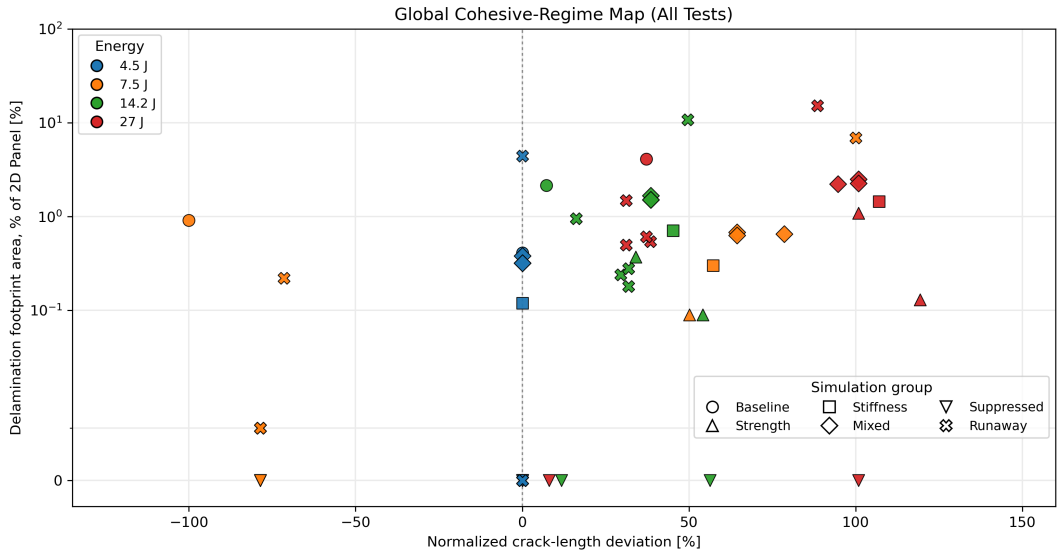


Figure 5.46: Global cohesive-regime map summarizing all interlaminar simulations. The delamination footprint (percentage of 2D panel area) is plotted against the normalized crack-length deviation relative to experimental measurements. Marker colour denotes impact energy, while marker shape identifies the simulation family. Distinct response regimes, including baseline, suppressed and runaway delamination behaviours, can be clearly distinguished.

Several physically distinct response regimes can be identified. Baseline and moderately modified cohesive configurations generally exhibit localized delamination footprints that increase monotonically with impact energy, consistent with contrast, simulations employing very high cohesive stiffness values (e.g. $K = 10^5$) produce disproportionately large delamination footprints at all energy levels, including low-energy impacts, indicating a nonphysical "runaway" delamination regime. At the opposite extreme, simulations with strongly increased cohesive strength (e.g. $2000\times$ nominal strength), particularly when combined with calibrated stiffness values, suppress delamination almost entirely while preserving a realistic global dynamic response.

Importantly, this regime map demonstrates that the same cohesive formulation can transition between fundamentally different damage behaviours depending on parameter selection. This observation motivates the detailed parametric sensitivity analysis presented in the following subsections, where the individual influence of cohesive strength, elastic stiffness, fracture energy and mode-mixity parameters on the delamination footprint is examined quantitatively. The regime map thus serves as a qualitative framework that contextualizes the subsequent tornado-style sensitivity plots, rather than as a validation tool.

5.4.7 Parametric Sensitivity Analysis: Overview and Modelling Strategy

Following the assessment of the baseline interlaminar cohesive response, a parametric sensitivity analysis was conducted to investigate how individual cohesive parameters influence the predicted delamination footprint. The objective of this analysis is not to calibrate a unique set of cohesive properties, but rather to identify the relative importance of each parameter and to reveal distinct response regimes of the interlaminar model under impact loading.

The parametric study was organised according to the chronological modelling strategy adopted during the development of the numerical model. Starting from the initial reference cohesive configuration, individual parameters were varied in a controlled manner in order to isolate their influence on the delamination footprint. The parameters investigated include the interlaminar strength, the elastic cohesive stiffness of the cohesive interfaces, the fracture energy, and the mixed-mode interaction parameter of the Benzeggagh–Kenane (BK) law. In addition to single-parameter variations, selected combined modifications were examined to capture potential interaction effects and non-linear responses.

A key aspect of the adopted strategy is the distinction between two fundamentally different sources of structural compliance: damage-driven compliance associated with cohesive degradation, and undamaged interface compliance arising from the elastic stiffness of the cohesive formulation. To separate these effects, dedicated high-strength configurations were introduced to suppress interlaminar damage initiation, thereby allowing the influence of cohesive stiffness on the global response to be examined independently of damage evolution.

The sensitivity analysis was performed independently for each impact energy level ($E = 4.5$, 7.5, 14.2, and 27 J). For each energy case, the effect of parameter variations was evaluated relative to the reference cohesive configuration using delamination footprint metrics extracted from the cohesive damage variable `SDEG`. The results are therefore interpreted in terms of relative changes and qualitative trends, rather than absolute predictions of physical delamination extent.

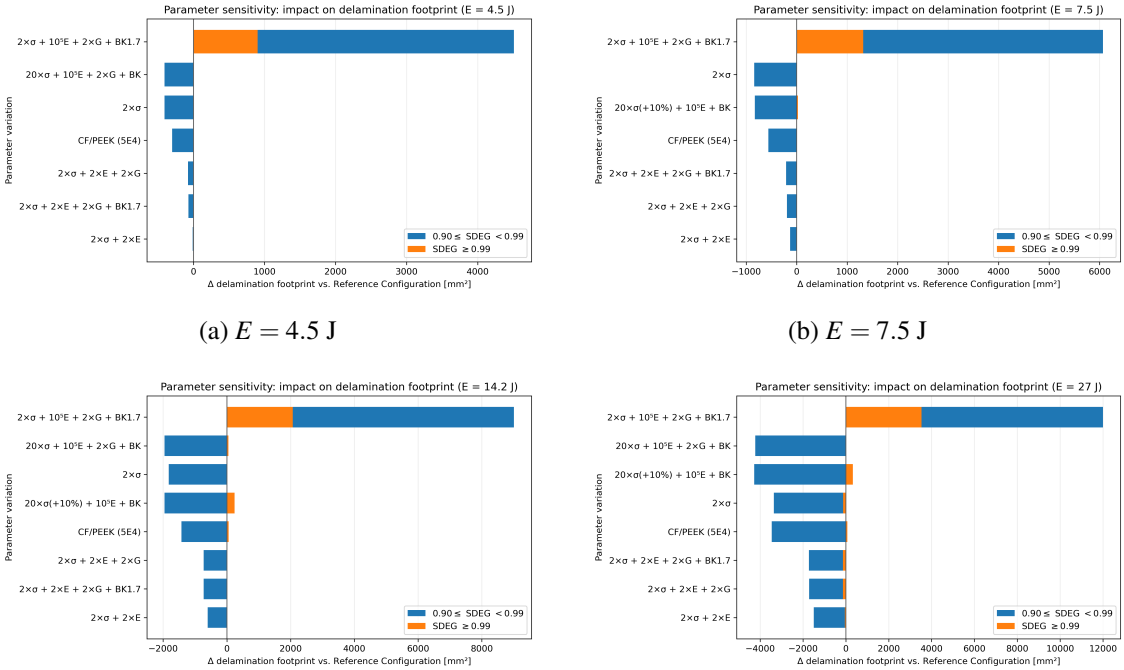
The outcomes of this parametric investigation are presented in the following subsection using tornado-style sensitivity diagrams, which provide a direct ranking of the influence of each cohesive parameter on the delamination footprint. This analysis forms the basis for the subsequent identification of non-physical response regimes and motivates the dedicated calibration of cohesive stiffness parameters discussed later in this section.

5.4.8 Sensitivity of Delamination Footprint to Cohesive Parameters

The influence of the interlaminar cohesive parameters on the predicted delamination footprint is assessed through a parametric sensitivity analysis. Given the absence of experimental delamination footprint measurements for direct validation, the objective of this analysis is not to identify an optimal parameter set, but rather to understand how variations in individual cohesive properties affect the extent and character of interlaminar damage. The results are therefore interpreted in a relative sense, focusing on trends, parameter ranking, and the identification of distinct response regimes.

To this end, tornado-style sensitivity diagrams are employed to quantify the variation in delamination footprint induced by changes in cohesive parameters with respect to the reference configuration. The diagrams report the change in footprint area associated with elements exceeding the damage threshold $\texttt{SDEG} \geq 0.90$, thereby capturing both damage initiation and early propagation effects. Separate diagrams are constructed for each impact energy level ($E = 4.5$, 7.5, 14.2, and 27 J), allowing the consistency of parameter influence across energy regimes to be evaluated.

Figure 5.47 presents the tornado-style sensitivity diagrams for all investigated impact energies. In each diagram, the horizontal bars represent the variation in delamination footprint relative to the reference cohesive configuration, while the vertical ordering of the bars provides a direct ranking of the influence of each parameter. Positive and negative variations indicate an increase or decrease in delamination footprint, respectively. These diagrams form the basis for the parameter-by-parameter discussion presented in the following subsections.



(a) $E = 4.5$ J

(b) $E = 7.5$ J

(c) $E = 14.2$ J

(d) $E = 27$ J

Figure 5.47: Tornado-style sensitivity diagrams illustrating the influence of interlaminar cohesive parameter variations on the delamination footprint for different impact energy levels. The bars represent the change in footprint area relative to the reference cohesive configuration for elements exceeding $\texttt{SDEG} \geq 0.90$.

Effect of Interlaminar Strength Scaling

The interlaminar cohesive strength primarily governs the onset of damage initiation and therefore acts as a threshold parameter controlling whether delamination is triggered under a given loading condition. This behaviour is consistently reflected across the tornado-style sensitivity diagrams, where strength scaling produces the most pronounced reductions in delamination footprint at low impact energies.

At $E = 4.5$ and 7.5 J, moderate increases in cohesive strength (e.g., $2\times$ the reference values) frequently suppress delamination entirely above the adopted damage threshold, resulting in negligible footprint areas. This indicates that, in the low-energy regime, the response is predominantly initiation-controlled, with the applied stresses remaining close to the nominal interlaminar strength limits. Under these conditions, strength scaling is highly effective in limiting damage development.

As the impact energy increases, the effectiveness of strength scaling diminishes. At $E = 14.2$ and 27 J, increased cohesive strength continues to reduce the delamination footprint relative to the reference configuration, but does not fully prevent damage initiation or propagation. This suggests that, once interlaminar damage is activated under severe loading, additional mechanisms—most

notably the elastic stiffness of the cohesive interfaces—play a more dominant role in governing damage extent.

It is further observed that large strength scaling factors (e.g., $20\times$ the reference values) do not constitute a clean "damage-off" condition in terms of global structural response. Although delamination footprint may be significantly reduced or locally suppressed, residual compliance and non-linear behaviour persist in the acceleration–time response. This highlights that cohesive strength alone controls damage activation but does not eliminate the influence of cohesive elements on the undamaged structural stiffness.

Overall, the sensitivity analysis demonstrates that interlaminar strength is a critical parameter for controlling damage initiation, particularly at low impact energies. However, it is insufficient as a standalone control variable for achieving a physically consistent interlaminar response across all energy levels, thereby motivating the separate investigation of cohesive stiffness effects presented in the following subsection.

Effect of Cohesive Elastic Stiffness

The elastic penalty stiffness of the cohesive interfaces is identified as the most influential parameter governing the predicted delamination footprint and the qualitative response of the interlaminar model. Unlike cohesive strength, which primarily controls damage initiation, cohesive stiffness affects both the undamaged compliance of the laminate and the stress distribution driving damage evolution. As a result, its influence extends across all impact energy levels and parameter combinations considered.

Moderate increases in cohesive stiffness are observed to counteract the footprint reduction induced by strength scaling. In several cases, configurations combining elevated strength with increased stiffness restore delamination footprints comparable to, or exceeding, those of the reference configuration. This behaviour highlights the strong coupling between elastic stiffness and damage initiation criteria, whereby stiffer interfaces promote higher interlaminar stress transfer and facilitate damage activation.

When excessively high stiffness values are adopted, a qualitatively distinct response regime emerges. This regime is characterised by a dramatic increase in delamination footprint, including the appearance of fully separated regions ($\texttt{SDEG} \geq 0.99$) even at low impact energies. Such behaviour is consistently observed across all energy levels and does not follow the expected scaling of damage extent with impact energy. This phenomenon, referred to here as a *runaway delamination regime*, is indicative of non-physical stress concentrations and artificial damage amplification driven by unrealistically stiff cohesive interfaces.

Importantly, the stiffness sensitivity observed in the tornado diagrams demonstrates that cohesive stiffness governs not only damage evolution but also the global structural compliance of the model. Even in configurations where interlaminar damage is largely suppressed, variations in cohesive stiffness significantly alter the acceleration–time response, revealing the contribution of undamaged cohesive interfaces to the overall deformation behaviour.

These findings establish cohesive elastic stiffness as a critical parameter requiring dedicated calibration prior to any interpretation of interlaminar damage predictions. The dominance of stiffness effects and the identification of non-physical response regimes directly motivate the damage-suppression and stiffness-isolation strategies examined in the subsequent subsections.

Effect of Fracture Energy

The influence of the cohesive fracture energy on the predicted delamination footprint is found to be secondary when compared to the effects of interlaminar strength and cohesive elastic stiffness. Across all impact energy levels, variations in fracture energy do not induce qualitative changes in damage morphology or response regime, but instead modulate the extent of damage propagation once initiation has occurred.

At low impact energies ($E = 4.5$ and 7.5 J), changes in fracture energy produce only minor variations in delamination footprint. In this regime, damage initiation and early process-zone development dominate the response, limiting the opportunity for fracture energy to significantly influence damage growth. Consequently, increases in fracture energy do not prevent damage initiation but may slightly reduce the size of the partially degraded region.

At intermediate and high impact energies ($E = 14.2$ and 27 J), fracture energy variations exert a more noticeable effect on delamination propagation. Increased fracture energy generally leads to moderate reductions in delamination footprint, particularly within the full separated region.$\texttt{SDEG} \geq 0.99$). Nevertheless, these reductions remain modest and do not alter the overall ranking of parameter influence observed in the sensitivity diagrams.

Overall, the parametric study indicates that cohesive fracture energy primarily governs the extent of delamination growth rather than its initiation or the global structural response. Its role is therefore subordinate to that of cohesive stiffness and interlaminar strength, and fracture energy variations alone are insufficient to resolve non-physical response regimes or stiffness-related discrepancies observed in the cohesive model.

Effect of Mixed-Mode Interaction Parameter

The influence of the mixed-mode interaction parameter of the BK law was assessed through a limited number of simulations, in which the BK exponent was varied from its reference value to

a higher value representative of increased mode II dominance. Within the scope of the performed analyses, this variation resulted in only negligible changes in the predicted delamination footprint.

No clear or systematic effect of the BK exponent on either the extent of the delaminated area or the balance between partially degraded and fully separated regions was observed. The overall damage patterns and global response remained essentially unchanged, suggesting that the interlaminar behaviour under the investigated impact conditions is not strongly sensitive to the assumed mixed-mode interaction formulation.

Given the limited sensitivity observed and the restricted number of simulations performed for this parameter, the BK exponent is not considered a dominant driver of the interlaminar response in the present study. Consequently, it is fixed in subsequent analyses in order to reduce model complexity and focus attention on parameters with a more pronounced influence.

Energy Dependence and Regime Consistency

Although the absolute magnitude of the delamination footprint increases with impact energy for most cohesive configurations, the relative influence of the investigated parameters remains consistent across all energy levels. The sensitivity analysis reveals a stable hierarchy of parameter effects, indicating that the observed trends are intrinsic to the cohesive formulation rather than artefacts of a specific loading case.

At low impact energies ($E = 4.5$ and 7.5 J), the interlaminar response is predominantly initiation-controlled. In this regime, cohesive strength strongly governs whether delamination is activated, while fracture energy and mixed-mode interaction effects play a secondary role. As impact energy increases ($E = 14.2$ and 27 J), damage propagation becomes increasingly relevant, leading to larger delamination footprints and the appearance of fully separated regions. Despite this shift in damage severity, the qualitative ranking of cohesive parameters remains unchanged.

Across all energy levels, cohesive elastic stiffness consistently emerges as the dominant parameter controlling both delamination extent and global structural response. Excessively stiff interfaces give rise to non-physical behaviour characterised by disproportionate delamination footprints that do not scale with impact energy, whereas insufficient stiffness introduces artificial compliance even in configurations where damage is largely suppressed. This behaviour contrasts with the more localized and energy-consistent response observed for moderate stiffness values.

The consistency of these trends across energy levels supports the interpretation of the interlaminar response in terms of distinct parameter-driven regimes rather than isolated case-by-case outcomes. In particular, the sensitivity analysis highlights the need to explicitly separate damage-driven effects from stiffness-related compliance before attempting any further interpretation or

refinement of interlaminar damage predictions. This observation directly motivates the damage suppression and stiffness isolation strategies examined in the following subsection.

Summary of Interlaminar Parameter Effects

To consolidate the observations drawn from the individual parameter studies presented in this subsection, Table 5.21 provides a compact overview of the qualitative influence of each interlaminar cohesive parameter on the predicted delamination response. Rather than reporting numerical values, the table summarizes the dominant physical trends observed across all impact energies, with particular emphasis on the delamination footprint and crack-length metrics. This summary is intended to support model interpretation and decision-making, while avoiding redundant repetition of results already discussed.

Table 5.21: Summary of interlaminar cohesive parameter effects on delamination behaviour.

Cohesive stiffness (K)	
Primary effect	Controls elastic response and energy storage
Sensitivity level	Moderate-High
Effect on footprint	Strong increase; runaway regimes observed
Effect on crack metric	Indirect influence
Interlaminar strength (σ)	
Primary effect	Delays damage initiation
Sensitivity level	Low-Moderate
Effect on footprint	Minor reduction
Effect on crack metric	Weak influence
Fracture energy (G_{Ic}, G_{IIc})	
Primary effect	Governs damage propagation
Sensitivity level	High
Effect on footprint	Strong increase
Effect on crack metric	Limited correlation
BK exponent (η)	
Primary effect	Mode-mix weighting
Sensitivity level	Low
Effect on footprint	Negligible
Effect on crack metric	Negligible

5.4.9 Damage Suppression and Isolation of Cohesive Stiffness Effects

The parametric sensitivity analysis presented in the previous subsection demonstrated that cohesive elastic stiffness exerts a dominant influence on both the predicted delamination footprint and the global structural response. However, the coupled nature of damage evolution and elastic compliance in cohesive formulations complicates the interpretation of these effects, as changes in stiffness simultaneously alter stress transfer and damage activation.

To decouple these mechanisms, a series of dedicated simulations was performed in which interlaminar damage initiation was intentionally suppressed through the use of artificially high cohesive strength values. By elevating the nominal interlaminar strengths well beyond physically realistic levels, cohesive degradation is effectively prevented, allowing the response of the model to be examined in the absence of interlaminar damage. This approach is adopted here as a numerical diagnostic tool rather than a physically representative material model.

The primary objective of this damage-suppressed configuration is to isolate the contribution of undamaged cohesive interfaces to the overall structural compliance. In a laminate containing multiple interlaminar interfaces, even thin cohesive layers introduce additional compliance that accumulates through the thickness. As a result, the global response of the model may be altered significantly even when no cohesive damage develops. The damage-suppressed simulations therefore provide a direct means of assessing whether discrepancies in acceleration-time response and crack length predictions originate from damage evolution or from the elastic formulation of the cohesive interfaces themselves.

The results of these simulations reveal that suppressing damage alone is insufficient to recover a physically consistent response. Even in the absence of interlaminar degradation, the model may exhibit excessive compliance or altered dynamic behaviour if the cohesive elastic stiffness is not appropriately selected. This observation confirms that the cohesive interfaces influence the structural response independently of damage and that stiffness-related effects must be addressed explicitly.

It is emphasised that the damage-suppression strategy employed in this subsection does not aim to identify realistic interlaminar strength values, nor to represent a physically valid laminate configuration. Instead, it serves as an essential intermediate step to disentangle stiffness-driven effects from damage-driven phenomena. The insights gained from this analysis form the basis for the subsequent calibration of cohesive stiffness parameters and the identification of physically admissible response regimes, which are discussed in the following subsection.

5.4.10 Cohesive Stiffness Calibration and Regime Selection

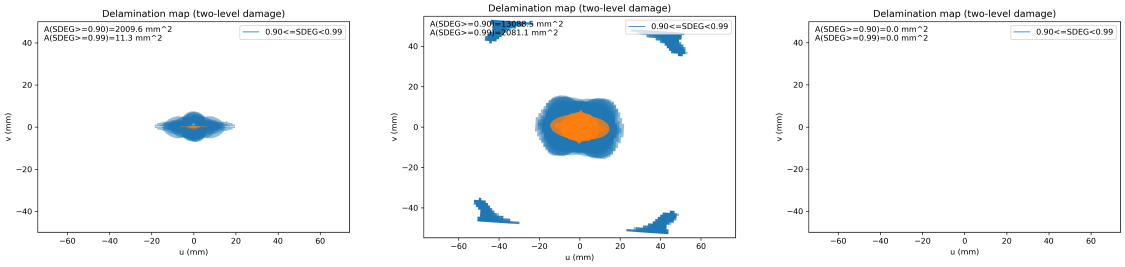
The damage suppression strategy introduced in the previous subsection enables a clear separation between damage-driven effects and stiffness-related compliance introduced by the cohesive interfaces. Building upon this separation, the present subsection aims to identify physically admissible response regimes and to guide the selection of cohesive stiffness parameters suitable for subsequent analyses.

The focus is placed on the elastic contribution of the cohesive interfaces to the global structural response, as the parametric sensitivity analysis demonstrated that inappropriate stiffness values can lead either to excessive compliance or to non-physical amplification of interlaminar damage. Rather than seeking a unique calibration of interlaminar properties, the objective here is to exclude non-physical regimes and retain cohesive configurations that preserve the structural response characteristics of the laminate while enabling meaningful interpretation of interlaminar damage.

Representative Damage Patterns and Regime Identification

While the quantitative sensitivity metrics presented in the previous subsection allow a systematic comparison of interlaminar parameter variations, a visual inspection of representative delamination patterns is essential to confirm that the identified regimes correspond to physically distinct damage mechanisms. For this purpose, a reduced set of delamination maps is presented for the $E = 14.2$ J impact case, which is representative of the intermediate energy regime and exhibits clear differentiation between parameter configurations.

Figure 5.48 compares three characteristic interlaminar modelling regimes: (i) the reference cohesive configuration, (ii) a high-stiffness configuration leading to a runaway delamination response, and (iii) a damage-suppressed configuration used for stiffness calibration. All maps are extracted at the same analysis time, employ identical damage thresholds ($0.90 \leq \text{SDEG} < 0.99$ and $\text{SDEG} \geq 0.99$), and share the same spatial scale and colour conventions.



(a) Reference cohesive model. (b) Runaway delamination. (c) Damage-suppressed regime.

Figure 5.48: Representative delamination patterns at $E = 14.2$ J illustrating distinct interlaminar response regimes. Identical damage thresholds and visual scaling are used in all cases.

The reference cohesive model exhibits a compact and symmetric delamination footprint, with a limited fully degraded core surrounded by a broader process zone. This pattern is consistent with the quantitative footprint metrics and represents a physically plausible delamination morphology for the considered impact energy.

In contrast, the high interfacial stiffness configuration (10^5 scaling) produces an extensive delamination footprint with pronounced propagation away from the impact zone, accompanied by a substantial region of fully degraded cohesive elements. This behaviour confirms the existence of a distinct runaway delamination regime, previously identified through the sensitivity analysis, and highlights the non-physical nature of such parameter combinations despite their apparent numerical stability.

Finally, the damage-suppressed configuration ($2000\times$ strength scaling combined with calibrated cohesive stiffness) results in the complete suppression of delamination above the selected damage thresholds. The absence of interlaminar damage in this case, together with the improved dynamic stiffness response discussed in subsequent sections, supports its use as a reference configuration for isolating the influence of cohesive elastic compliance independently of damage evolution.

Overall, the comparison confirms that the parametric variations explored in this study lead to clearly distinguishable interlaminar response regimes. These regimes provide a consistent physical interpretation of the quantitative trends observed in the sensitivity analysis and motivate the subsequent stiffness calibration and model refinement steps.

Strength Substitution Using Carbon/PEEK Reference Values

In addition to the damage-suppressed configurations, an alternative approach was explored in which interlaminar strength values representative of higher-performance composite systems were adopted. Specifically, cohesive strength values typical of Carbon/PEEK laminates were introduced in an attempt to suppress damage initiation while retaining physically realistic material parameters.

Despite the increased strength levels, the resulting simulations continued to exhibit discrepancies in crack length prediction and global response when compared to the experimentally observed behaviour. In particular, the back-face crack length remained overestimated, and the acceleration-time response retained excessive compliance relative to the calibrated intralaminar model.

These results indicate that the observed compliance and crack-length discrepancies are not governed solely by interlaminar strength levels, even when values characteristic of tougher composite systems are employed. This finding further reinforces the conclusion that cohesive elastic stiffness, rather than strength, is the dominant parameter controlling the undamaged structural response of the laminate. Consequently, increasing strength alone is insufficient to resolve stiffness-related artefacts introduced by the cohesive formulation.

Summary of Cohesive Regimes and Modelling Implications

The parametric investigation of the interlaminar cohesive model revealed that different combinations of cohesive stiffness, strength, and fracture energy lead to markedly distinct response regimes. Among the extracted metrics, the back-face crack length and the acceleration-time response were identified as the most meaningful indicators for quantitative comparison with experimental results, as both are directly measurable and reflect the global structural response under impact loading. Conversely, the delamination footprint was employed as a complementary, qualitative indicator to support interpretation of interlaminar damage trends, given the absence of experimental reference data for this quantity.

Table 5.22 synthesizes the main cohesive response regimes identified during the parametric study. Rather than ranking configurations by numerical accuracy, the table summarizes their characteristic behaviour with respect to crack-length prediction, global stiffness response inferred from the acceleration-time curves, and qualitative delamination extent. This summary condenses the outcomes of previous subsections and highlights the limitations encountered when tuning individual cohesive parameters within the reference Glass/PPS formulation.

Table 5.22: Summary of cohesive response regimes identified in the parametric study. Back-face crack length and acceleration-time response are treated as the primary comparison metrics, while delamination footprint is used as a qualitative indicator.

Reference cohesive configuration	
Crack-length prediction	Overestimated at intermediate and high energies
Acceleration-time response	Excessively compliant compared to experiment
Delamination footprint	Localised and energy-consistent
High stiffness configuration ($K \sim 10^5$)	
Crack-length prediction	Improved agreement for selected energy cases
Acceleration-time response	Comparable to elastic and intralaminar models
Delamination footprint	Non-physical spreading (runaway regime)
Damage-suppressed configurations ($2000\times$ **strength)**	
Crack-length prediction	Still overestimated
Acceleration–time response	Remains overly compliant
Delamination footprint	Fully suppressed above damage thresholds
Strength substitution (Carbon/PEEK reference values)	
Crack-length prediction	No systematic improvement
Acceleration–time response	Inconsistent across energy levels
Delamination footprint	Reduced but not eliminated

The comparison of cohesive response regimes indicates that improvements in either crack-length prediction or global stiffness response can be achieved locally through targeted parameter adjustments, but not simultaneously within the current cohesive formulation. In particular, configurations that recover a realistic acceleration-time response tend to promote non-physical delamination growth, whereas damage suppression strategies fail to resolve discrepancies in crack length and global compliance. These observations suggest that the limitations identified in the present parametric study are not solely associated with the selection of cohesive properties, but are instead linked to the underlying representation of energy absorption mechanisms within the model. The implications of these findings are further examined following the introduction of an estimation-based cohesive formulation in the subsequent section.

5.4.11 Estimation of Interlaminar Cohesive Properties for Glass/PPS

5.4.11.1 Motivation for an Estimation-Based Approach

Unlike elastic and intralaminar damage properties, interlaminar cohesive parameters for glass-fibre reinforced PPS laminates are not directly available in the open literature, particularly for woven thermoplastic systems subjected to impact loading. As demonstrated in the preceding parametric study, variations of cohesive stiffness, strength and fracture energy strongly affect the predicted impact response, yet no unique parameter set could be identified solely through calibration against a single numerical metric.

For this reason, an estimation-based strategy was adopted. Rather than fitting cohesive parameters to the present experimental data, reference values were inferred from published studies on mechanically comparable material systems. These literature anchors provide physically justified bounds and trends, allowing the construction of a Glass/PPS cohesive law that is consistent with known thermoplastic GFRP behaviour while remaining compatible with the Abaqus cohesive element formulation used in this work.

5.4.11.2 Literature Anchors for Cohesive Property Estimation

Three classes of material systems were identified as relevant anchors for the estimation of Glass/PPS interlaminar cohesive properties: thermoplastic GFRPs laminates, conventional glass/epoxy systems, and high-performance thermoplastic carbon-fibre composites.

Glass/Elium (Thermoplastic GFRP — Primary Anchor)

Among the available references, glass fibre reinforced Elium resin laminates represent the closest match to the present material system. The study reports a traction-separation law implemented using cohesive elements in Abaqus/Explicit, with stiffness expressed in MPa/mm and fracture energies defined separately for Mode I and Mode II using the Benzeggagh–Kenane criterion [106].

Reported values include a cohesive stiffness of approximately 1.75×10^5 MPa/mm, normal and shear strengths in the range of 70–80 MPa, Mode I fracture energy of $G_{IC} \approx 1.9$ N/mm, and Mode II fracture energy of $G_{IIC} \approx 3.8$ N/mm with a BK exponent close to $\eta = 1.45$ [106].

This system constitutes the primary anchor for the Glass/PPS estimation, as it combines a thermoplastic matrix with glass reinforcement and employs a cohesive formulation directly compatible with the present modelling framework.

Glass/Epoxy (Secondary Anchor — Strength and Toughness Bounds)

Glass/epoxy systems were considered as a lower-bound reference for cohesive strength and Mode I fracture energy. Reported normal and shear strengths are typically in the range of 45–50 MPa, with Mode I fracture energies close to 1.0 N/mm and Mode II values around 3.7 N/mm. Although stiffness values are reported in terms of surface-based cohesive contact with different unit conventions, these data provide useful guidance for relative trends rather than absolute stiffness magnitudes [105].

Carbon/PEEK (Upper-Bound Thermoplastic Reference)

Carbon/PEEK cohesive data were used exclusively as an upper-bound thermoplastic reference. These studies report very high penalty stiffness values (up to 10^6 MPa/mm) together with strengths comparable to or slightly above those of thermoplastic GFRP systems. However, due to the use of unidirectional carbon reinforcement and a high-performance thermoplastic matrix, these values were not directly transferred, but instead used to validate the plausibility of the stiffness range adopted [104].

5.4.11.3 Estimated Glass/PPS Cohesive Property Set

Based on the trends identified from the literature anchors, a cohesive parameter set for Glass/PPS was defined that reflects the thermoplastic nature of the matrix while remaining conservative relative to the more ductile Elium system. The parameters were implementedusing Abaqus default cohesive elements, where the normal and shear penalty stiffnesses are defined directly in MPa/mm.

Table 5.23: Estimated interlaminar cohesive properties for Glass/PPS based on literature anchors.

Parameter	**Estimated value**
Elastic Cohesive stiffness (normal) K_n	1.8×10^5 MPa/mm
Elastic Cohesive stiffness (shear) $K_s = K_t$	1.8×10^5 MPa/mm
Normal strength σ_{n0}	60 MPa
Shear strengths $\tau_{s0} = \tau_{t0}$	70 MPa
Mode I fracture energy G_{IC}	1.65 N/mm
Mode II fracture energy $G_{IIC} = G_{IIIC}$	3.80 N/mm
BK exponent η	1.45

5.4.11.4 Rationale for Parameter Selection

The cohesive stiffness was selected slightly above the value reported for Glass/Elium, ensuring sufficient numerical stability and stiffness representation without approaching the excessively high values typical of carbon/PEEK systems. This choice reflects the thermoplastic nature of PPS while

accounting for the lower ductility typically observed in glass-fibre reinforced PPS compared to Elium-based laminates.

The nominal strengths were chosen as intermediate values between glass/epoxy and glass/Elium systems. This reflects the improved interfacial strength expected from thermoplastic matrices while remaining conservative with respect to damage initiation.

Fracture energies were selected to preserve the characteristic increase in Mode I toughness associated with thermoplastic matrices, while maintaining Mode II values close to those reported consistently across glass-fibre composite systems. The Benzeggagh-Kenane exponent was fixed at $\eta = 1.45$, consistent with both thermoplastic GFRP and classical epoxy-based cohesive models.

The estimated cohesive property set defined in this section is evaluated in the following subsection through a direct comparison with experimental impact results. Using the same set of numerical metrics adopted for the baseline cohesive model, the influence of the estimated parameters on acceleration-time response, back-face crack length and delamination patterns is assessed on an energy-by-energy basis.

5.4.12 Estimated Interlaminar Cohesive Model Response: Energy-Based Assessment

The numerical response of the estimated interlaminar cohesive model was evaluated for each impact energy level considered in the experimental campaign. In this formulation, the cohesive properties were not directly calibrated against experimental impact data, but instead estimated based on published results for comparable thermoplastic composite systems, including Glass/Elium, Glass/Epoxy, and Carbon/PEEK laminates. This approach was adopted in order to define a physically informed and internally consistent set of cohesive parameters representative of Glass/PPS interfaces.

The results are presented and discussed on a case-by-case basis as a function of the impact energy, enabling the evolution of the global dynamic response and the predicted interlaminar damage mechanisms to be analysed under increasing loading severity. For each energy level, the assessment focuses on the acceleration-time response and the associated interlaminar damage patterns, with particular emphasis on the back-face crack length, which is treated as the most reliable quantitative experimental metric for model evaluation. Predicted delamination footprints and damage distributions are reported to support the qualitative interpretation of interlaminar failure modes, while recognising the absence of direct experimental measurements for delaminated area.

The discussion presented in the following sub-sections refers exclusively to the estimated cohesive property set defined in Section 5.4.11.2. Quantitative damage metrics extracted from the

cohesive elements, together with the experimentally measured back-face crack length and the measured acceleration-time histories, are used to assess the capabilities and limitations of the estimated interlaminar cohesive formulation at each impact energy.

Estimated Cohesive Response at 4.5 J

Figure 5.49 presents the acceleration–time response predicted by the estimated interlaminar cohesive model at an impact energy of $E = 4.5$ J, together with the corresponding experimental measurements. At this low energy level, the estimated cohesive model exhibits a markedly improved dynamic response compared to the baseline cohesive configuration. The initial loading slope and peak acceleration are closely aligned with the experimental curves, indicating that the selected cohesive stiffness provides an appropriate representation of the global structural response without introducing excessive compliance.

The numerical response reaches a peak acceleration of approximately 500–520 $\mathrm{m/s^2}$ at $t \approx 3.5$–4.0 ms, in good agreement with the experimental measurements. The unloading phase is also well captured, with a similar pulse duration and decay rate, confirming that the estimated cohesive parameters do not artificially prolong the contact event or over-dissipate energy at this impact energy.

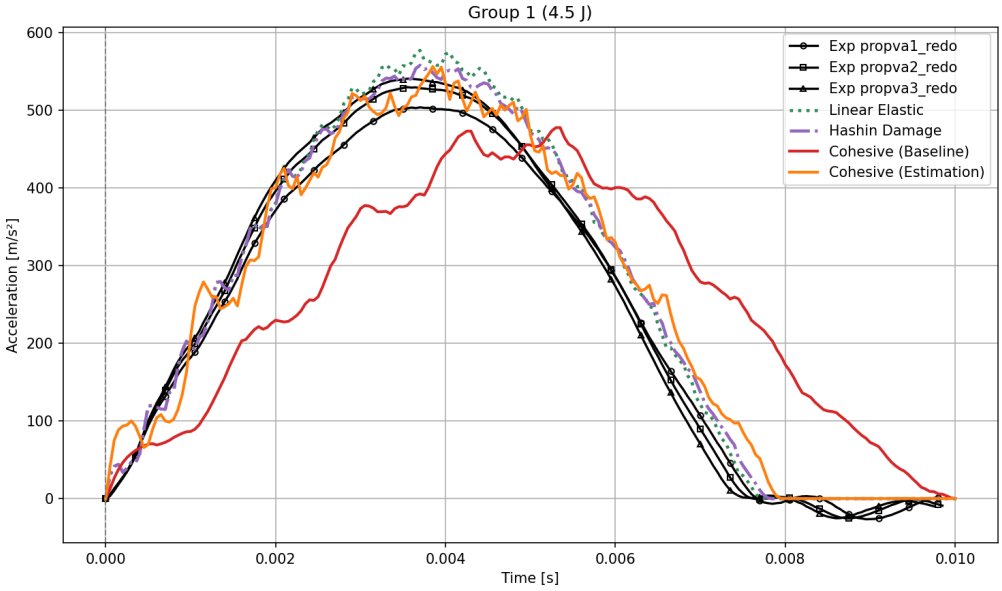


Figure 5.49: Acceleration—time response at $E = 4.5$ J comparing experimental data with the estimated interlaminar cohesive model.

Experimentally, no measurable back-face crack is observed at $E = 4.5$ J. Consistently, the estimated cohesive model predicts no macroscopic crack formation, with a computed back-face crack length of 0 mm. This result represents a significant improvement over the baseline cohesive

configuration and indicates that the estimated fracture initiation and propagation parameters are sufficiently restrictive to prevent non-physical crack growth at low impact energy.

The corresponding interlaminar damage distribution obtained from the cohesive damage variable SDEG is shown in Figure 5.50. Damage is confined to a small and highly localised region surrounding the impact point, with no fully failed cohesive elements (SDEG ≥ 0.99) detected. The predicted damage is limited to a minor process zone ($0.9 \leq$ SDEG < 0.99), reflecting the onset of interlaminar degradation without the formation of an open delamination crack.

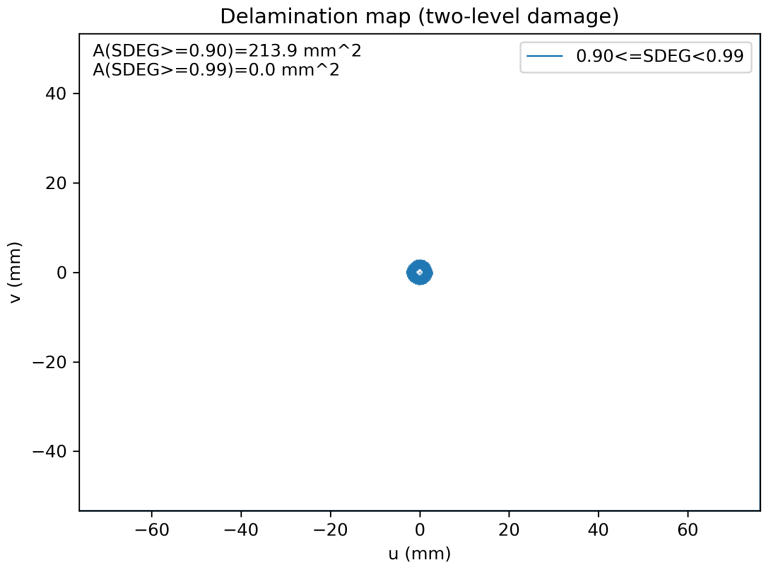


Figure 5.50: Back-face interlaminar damage map at $E = 4.5$ J for the estimated cohesive model, showing regions with $0.9 \leq$ SDEG < 0.99.

Quantitative interlaminar damage metrics extracted from the post-processing routine are summarised in Table 5.24. The projected delaminated area associated with SDEG ≥ 0.9 is approximately 213.9 mm^2, corresponding to only 0.10% of the interlaminate area. No fully failed cohesive elements are observed, and the predicted damage is distributed over a limited number of interfaces, consistent with the absence of a macroscopic crack.

Table 5.24: Interlaminar damage metrics for the estimated cohesive model at $E = 4.5$ J.

Metric	**Value**
Back-face crack length (CAE)	0.0 mm
Back-face crack length (Exp)	0.0 mm
Delaminated area ($\texttt{SDEG} \geq 0.9$)	213.9 mm^2
Delaminated area ($\texttt{SDEG} \geq 0.99$)	0.0 mm^2
Footprint area ($SDEG \geq 0.90$)	24.0 mm^2
Footprint area ($SDEG \geq 0.99$)	0.0 mm^2
Plies with interlaminar damage ($\texttt{SDEG} \geq 0.9$)	8.91
Plies with fully failed interfaces ($\texttt{SDEG} \geq 0.99$)	0.0

Overall, at $E = 4.5$ J the estimated interlaminar cohesive model provides an accurate representation of both the global dynamic response and the observed damage state. The acceleration–time history is well reproduced, and the absence of a back-face crack is correctly predicted. These results confirm that the estimated cohesive stiffness, strength, and fracture energy parameters are appropriate for low-energy impact conditions, and do not introduce spurious interlaminar damage or excessive compliance at the lower end of the energy range.

Estimated Cohesive Response at 7.5 J

Figure 5.51 shows the acceleration-time response predicted by the estimated interlaminar cohesive model at an impact energy of $E = 7.5$ J, together with the corresponding experimental measurements. At this intermediate energy level, the estimated cohesive model continues to provide a good representation of the global dynamic response. The initial loading slope and peak acceleration are in close agreement with the experimental curves, indicating that the selected cohesive stiffness does not introduce excessive compliance into the system.

The numerical response reaches a peak acceleration of approximately 650-700 m/s^2 at $t \approx 3.5$-4.0 ms, closely matching the experimental range. The duration and decay of the acceleration pulse are also well captured, with no pronounced tail or delayed unloading phase, confirming that the estimated cohesive parameters preserve the correct impact event timing and stiffness characteristics.

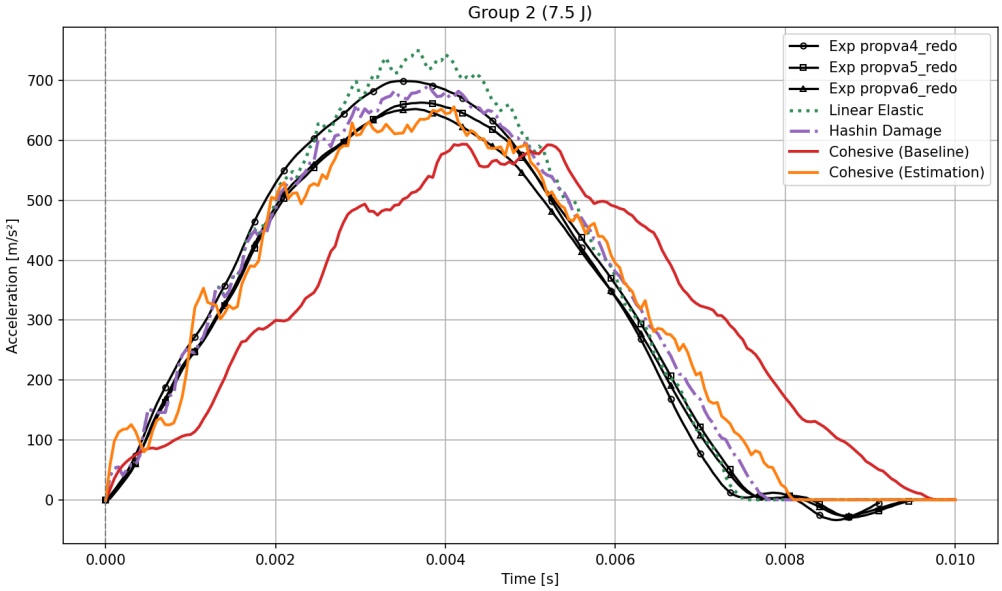


Figure 5.51: Acceleration—time response at $E = 7.5$ J comparing experimental data with the estimated interlaminar cohesive model.

Experimentally, a measurable back-face crack length of approximately 3.23 mm is observed at $E = 7.5$ J. The estimated cohesive model predicts a back-face crack length of 2.54 mm, corresponding to a deviation of 21.4%. While the crack length is slightly underestimated, the model correctly captures the onset of macroscopic cracking at this energy level, in contrast to the baseline cohesive configuration which failed to reproduce this transition.

The interlaminar damage distribution derived from the cohesive damage variable `SDEG` is presented in Figure 5.52. Damage remains relatively localised around the impact region, with a limited process zone ($0.9 \leq$ `SDEG` < 0.99) and a small but non-zero region of fully failed cohesive elements (`SDEG` ≥ 0.99). The back-face damage pattern exhibits a compact, cross-shaped morphology, indicating controlled interlaminar crack initiationwithout extensivee delamination propagation.

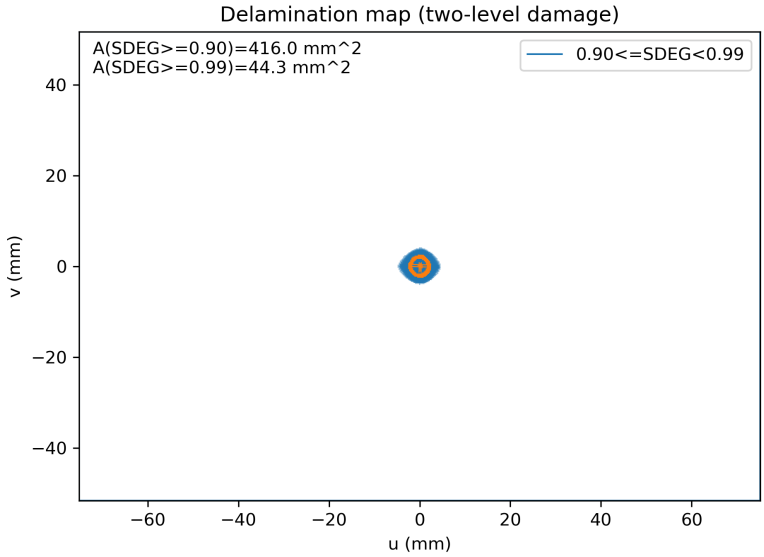


Figure 5.52: Back-face interlaminar damage map at $E = 7.5$ J for the estimated cohesive model, showing regions with $0.9 \leq \texttt{SDEG} < 0.99$ and $\texttt{SDEG} \geq 0.99$.

Quantitative interlaminar damage metrics are summarised in Table 5.25. The projected delaminated area associated with $\texttt{SDEG} \geq 0.9$ is approximately 416.0 mm^2, corresponding to 0.19% of the interlaminate area. Fully failed cohesive elements occupy a limited area off approximately 44.3 mm^2, confirming that damage remains largely confined to a narrow region surrounding the back-face crack. On average, approximately 7.7 interfaces exhibit interlaminar degradation above $\texttt{SDEG} \geq 0.9$, while fewer than three interfaces contain fully failed cohesive elements.

Table 5.25: Interlaminar damage metrics for the estimated cohesive model at $E = 7.5$ J.

Metric	**Value**
Back-face crack length (CAE)	2.54 mm
Back-face crack length (Exp)	3.23 mm
Delaminated area ($\texttt{SDEG} \geq 0.9$)	416.0 mm^2
Delaminated area ($\texttt{SDEG} \geq 0.99$)	44.3 mm^2
Process zone area ($0.9 \leq \texttt{SDEG} < 0.99$)	371.7 mm^2
Footprint area ($SDEG \geq 0.90$)	54.0 mm^2
Footprint area ($SDEG \geq 0.99$)	16.0 mm^2
Plies with interlaminar damage ($\texttt{SDEG} \geq 0.9$)	7.70
Plies with fully failed interfaces ($\texttt{SDEG} \geq 0.99$)	2.77

In summary, at $E = 7.5$ J the estimated interlaminar cohesive model successfully captures the transition from purely localised damage to the onset of macroscopic interlaminar cracking. The

global acceleration response is accurately reproduced, and the predicted crack length is in reasonable agreement with the experimental observation. While some discrepancies remain in the exact crack extent, the overall damage morphology and energy-dependent behaviour are physically consistent, supporting the suitability of the estimated cohesive parameters at intermediate impact energies.

Estimated Cohesive Response at 14.2 J

Figure 5.53 presents the acceleration-time response predicted by the estimated interlaminar cohesive model at an impact energy of $E = 14.2$ J, together with the corresponding experimental measurements. At this energy level, the estimated cohesive parameters continue to reproduce the global dynamic response with good fidelity. The initial loading slope and peak acceleration are closely aligned with the experimental curves, indicating that the cohesive stiffness calibration successfully preserves the laminate bending stiffness and impact event duration.

The numerical response reaches a peak acceleration of approximately 800-850 $\mathrm{m/s^2}$ at $t \approx 3.0$-3.5 ms, in agreement with the experimental range. The unloading phase shows a comparable decay rate, with no excessive tail or artificial compliance introduced by the cohesive interfaces. Overall, the acceleration-time response confirms that the estimated cohesive formulation remains dynamically consistent at this intermediate-to-high energy level.

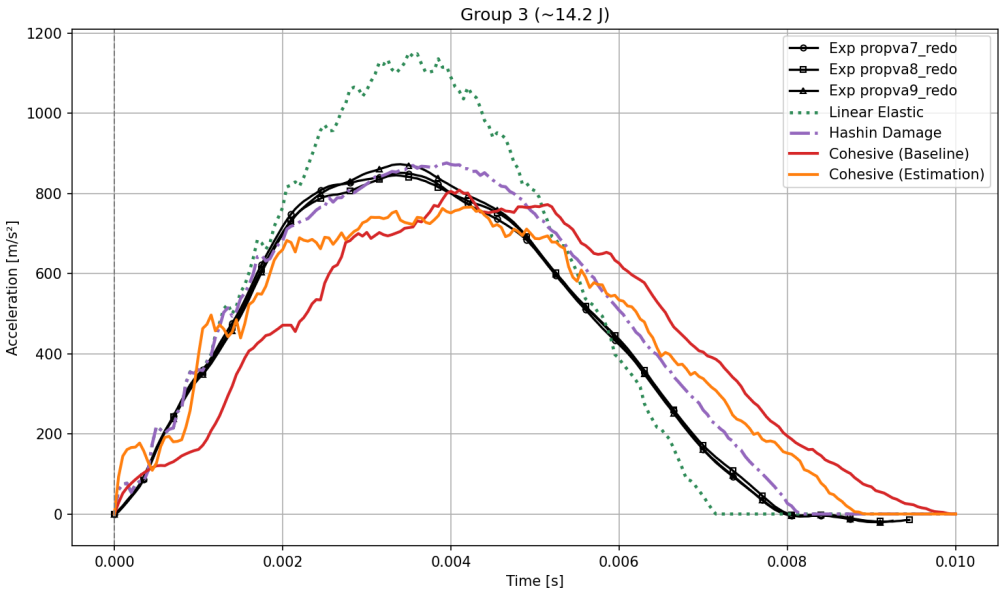


Figure 5.53: Acceleration–time response at $E = 14.2$ J comparing experimental data with the estimated interlaminar cohesive model.

At $E = 14.2$ J, the experimental specimen exhibits a back-face crack length of approximately 10.33 mm. The estimated cohesive model predicts a crack length of 14.31 mm, corresponding to a deviation of 38.5%. Although the model successfully captures the formation of a pronounced

macroscopic crack, the crack extent is significantly overestimated. This discrepancy indicates that, while the interlaminar damage mechanisms are activated at the correct energy level, the cohesive formulation alone is unable to accurately constrain crack growth once substantial inelastic deformation and distributed damage processes develop within the laminate.

The interlaminar damage distribution obtained from the cohesive damage variable SDEG is shown in Figure 5.54. Compared to lower energy cases, a larger delamination footprint is predicted, with an extended process zone ($0.9 \leq \text{SDEG} < 0.99$) and a clearly identifiable region of fully failed cohesive elements ($\text{SDEG} \geq 0.99$). The back-face damage pattern exhibits a well-defined cross-shaped morphology, indicating directional propagation of interlaminar damage driven by bending and shear-dominated separation mechanisms.

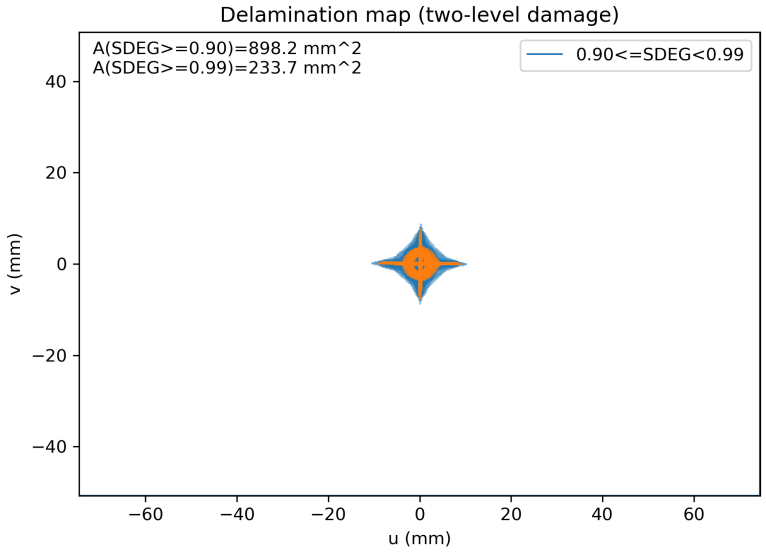


Figure 5.54: Back-face interlaminar damage map at $E = 14.2$ J for the estimated cohesive model, showing regions with $0.9 \leq \text{SDEG} < 0.99$ and $\text{SDEG} \geq 0.99$.

Quantitative interlaminar damage metrics extracted from the post-processing routine are summarised in Table 5.26. The delaminated area associated with $\text{SDEG} \geq 0.9$ reaches approximately 898.2 mm^2, corresponding to 0.40% of the interlaminate area and 0.80% of the equivalent back-face footprint. Fully failed cohesive elements occupy an area of approximately 233.7 mm^2, confirming the formation of an open interlaminar crack accompanied by a surrounding process zone. On average, approximately 7.6 interfaces exhibit interlaminar degradation above $\text{SDEG} \geq 0.9$, while nearly four interfaces contain fully failed cohesive elements.

Table 5.26: Interlaminar damage metrics for the estimated cohesive model at $E = 14.2$ J.

Metric	**Value**
Back-face crack length (CAE)	14.31 mm
Back-face crack length (Exp)	10.33 mm
Delaminated area ($\texttt{SDEG} \geq 0.9$)	898.2 mm^2
Delaminated area ($\texttt{SDEG} \geq 0.99$)	233.7 mm^2
Process zone area ($0.9 \leq \texttt{SDEG} < 0.99$)	664.5 mm^2
Footprint area ($SDEG \geq 0.90$)]	118.0 mm^2
Footprint area ($SDEG \geq 0.99$)	61.0 mm^2
Plies with interlaminar damage ($\texttt{SDEG} \geq 0.9$)	7.61
Plies with fully failed interfaces ($\texttt{SDEG} \geq 0.99$)	3.83

Overall, at $E = 14.2$ J the estimated interlaminar cohesive model reproduces the global dynamic response in good agreement with the experimental measurements. The predicted delamination morphology also appears qualitatively consistent with patterns commonly reported in the literature for impacted composite laminates. At the stage when this modelling study was conducted, no ultrasonic inspection data were yet available to quantify the actual delamination footprint. Consequently, the assessment of the cohesive model was primarily based on the global dynamic response and the predicted crack morphology. However, the persistent overestimation of back-face crack length highlights a limitation of relying primarily on interlaminar damage to constrain crack propagation. These observations suggest that additional deformation and damage mechanisms—such as localized matrix deformation and indentation beneath the impactor, together with their interaction with intralaminar damage processes—may contribute to the impact response at this energy level and should be represented to improve the quantitative agreement with the experimentally observed crack growth.

Estimated Cohesive Response at 27 J

Figure 5.55 presents the acceleration–time response predicted by the estimated interlaminar cohesive model at an impact energy of $E = 27$ J, together with the corresponding experimental measurement. At this highest energy level, the estimated cohesive formulation continues to reproduce the initial stiffness and peak acceleration with good agreement. The numerical response captures both the rising slope and the peak magnitude of the experimental curve, indicating that the calibrated cohesive stiffness does not introduce artificial compliance even under severe loading conditions.

The predicted acceleration history reaches a peak of approximately 950-1000 $\mathrm{m/s^2}$ at $t \approx 3.0$-3.5 ms, consistent with the experimental response. The unloading phase is also reasonably captured, with no excessively prolonged tail. These results confirm that, from a purely dynamic standpoint, the estimated cohesive model remains stable and mechanically consistent at 27 J.

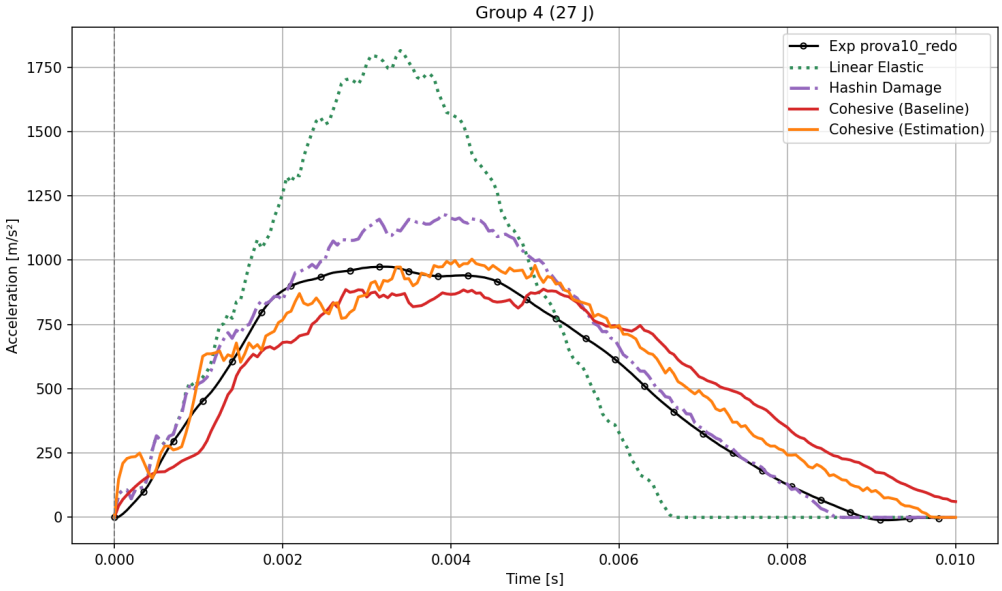


Figure 5.55: Acceleration—time response at $E = 27$ J comparing experimental data with the estimated interlaminar cohesive model.

At $E = 27$ J, the experimental specimen exhibits a pronounced back-face crack with a measured length of approximately 19 mm. In contrast, the estimated cohesive model predicts a back-face crack length of approximately 37 mm, corresponding to a deviation of 94.7%. This substantial overestimation demonstrates that, although interlaminar damage is correctly triggered and evolves extensively, the cohesive formulation alone is unable to constrain crack propagation once large-scale inelastic deformation and distributed damage processes occur within the laminate.

The corresponding interlaminar damage map obtained from the cohesive damage variable `SDEG` is shown in Figure 5.56. A wide delamination footprint is predicted, with both an extended process zone ($0.9 \leq$ `SDEG` < 0.99) and a significant region of fully failed cohesive elements (`SDEG` ≥ 0.99). The damage pattern retains the characteristic cross-shaped morphology observed at lower energies, but with increased spatial extent and a higher proportion of fully degraded interfaces, indicating the formation of an open interlaminar crack surrounded by a large degraded zone.

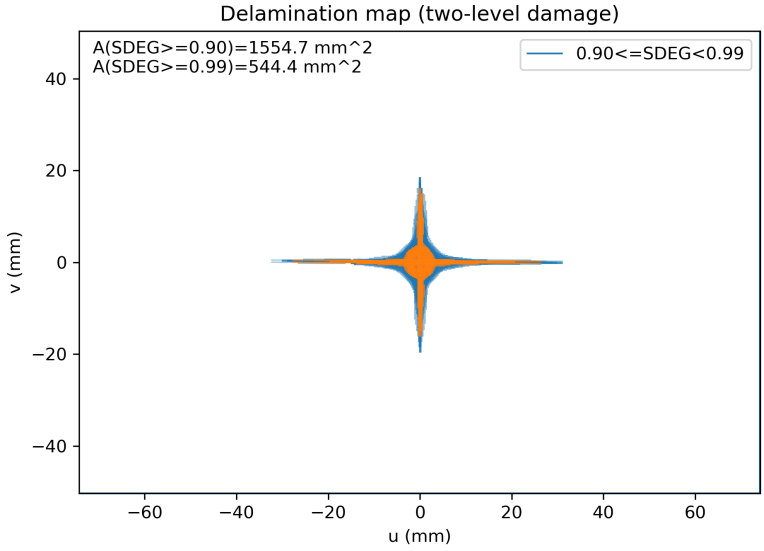


Figure 5.56: Back-face interlaminar damage map at $E = 27$ J for the estimated cohesive model, showing regions with $0.9 \leq \texttt{SDEG} < 0.99$ and $\texttt{SDEG} \geq 0.99$.

Quantitative interlaminar damage metrics extracted from the post-processing routine are summarised in Table 5.27. The delaminated area associated with $\texttt{SDEG} \geq 0.9$ reaches approximately 1554.7 mm^2, corresponding to 0.70% of the interlaminate area and 1.46% of the equivalent back-face footprint. Fully failed cohesive elements cover an area of approximately 544.4 mm^2, confirming the presence of an extensive open delamination crack. On average, more than seven interfaces exhibit interlaminar degradation above $\texttt{SDEG} \geq 0.9$, with over five interfaces containing fully failed cohesive elements.

Table 5.27: Interlaminar damage metrics for the estimated cohesive model at $E = 27$ J.

Metric	**Value**
Back-face crack length (CAE)	37.0 mm
Back-face crack length (Exp)	19.0 mm
Delaminated area ($\texttt{SDEG} \geq 0.9$)	1554.7 mm^2
Delaminated area ($\texttt{SDEG} \geq 0.99$)	544.4 mm^2
Process zone area ($0.9 \leq \texttt{SDEG} < 0.99$)	1010.3 mm^2
Footprint area ($SDEG \geq 0.90$)	216.0 mm^2
Footprint area ($SDEG \geq 0.99$)	105.0 mm^2
Plies with interlaminar damage ($\texttt{SDEG} \geq 0.9$)	7.20
Plies with fully failed interfaces ($\texttt{SDEG} \geq 0.99$)	5.19

Overall, at $E = 27$ J the estimated interlaminar cohesive model preserves a physically realistic global dynamic response and predicts extensive interlaminar damage. However, the severe overestimation of back-face crack length demonstrates that interlaminar damage alone cannot fully account for the experimentally observed impact response at higher energy levels. The simulations suggest that additional deformation and damage mechanisms—such as localized matrix deformation beneath the impactor and its interaction with intralaminar damage processes—may contribute to the structural response and influence crack propagation despite the presence of significant delamination.

5.4.13 Summary and Transition to Bi-phasic Modelling

The interlaminar modelling strategy investigated in this section was developed through three successive stages: an initial baseline cohesive formulation, a systematic parametric sensitivity study, and a final estimation of cohesive properties based on published data for comparable thermoplastic composite systems. This structured approach allowed the influence and limitations of interlaminar damage modelling to be assessed independently of numerical artefacts and stiffness calibration effects.

The baseline cohesive formulation was shown to capture qualitative delamination patterns but introduced excessive global compliance, leading to poor agreement with the experimental acceleration-time response at all impact energies. The subsequent parametric study demonstrated that cohesive stiffness and nominal strength are the most influential parameters governing the predicted delamination footprint, while the mixed-mode interaction parameter exhibited only a marginal effect within the investigated range. Importantly, extreme stiffness values were found to induce non-physical damage regimes, whereas strength scaling primarily acted as a damage-suppression mechanism without resolving the underlying compliance issue.

A physically informed estimation of Glass/PPS cohesive properties was then constructed using reference data from Glass/Elium, Glass/Epoxy, and Carbon/PEEK systems reported in the literature. This estimated cohesive set successfully preserved the laminate stiffness and produced acceleration-time responses in close agreement with experimental measurements across all impact energies. The dynamic response consistency confirms that, when properly calibrated, cohesive interfaces do not artificially alter the global structural behaviour of the laminate.

Despite these improvements, the estimated cohesive formulation systematically overpredicted the back-face crack length at intermediate and high impact energies. While the model correctly captured the initiation and qualitative morphology of interlaminar damage, the crack length—identified as the most reliable quantitative experimental metric—was increasingly overestimated as impact energy increased. This discrepancy persisted even when the acceleration response and delamination footprint remained physically reasonable.

The main capabilities and limitations of the interlaminar cohesive approach are summarised in Table 5.28. The table highlights that, although interlaminar damage modelling is essential for reproducing delamination morphology and energy scaling trends, it is insufficient to constrain macroscopic crack growth at higher impact energies. This observation indicates that additional energy dissipation mechanisms are active in the experiments but not explicitly represented in the cohesive framework.

Table 5.28: Summary of interlaminar cohesive modelling capabilities and limitations.

Aspect	**Baseline**	**Estimated Cohesive**	**Outcome**
Acceleration-time response	Poor	Good	Calibrated stiffness
Back-face crack length	Overpredicted	Strongly overpredicted	Not controlled
Delamination morphology	Qualitative	Qualitative	Captured
Energy scaling consistency	Partial	Good	Preserved
Dominant dissipation mode	Interlaminar	Interlaminar	Incomplete
Overall suitability	–	–	**Requires enhancement**

The combined evidence from the baseline, parametric, and estimated cohesive analyses demonstrates that interlaminar damage alone cannot account for the experimentally observed impact response of the Glass/PPS laminate. In particular, the inability of the cohesive model to limit back-face crack growth at higher energies strongly suggests that additional energy dissipation mechanisms, including indentation-driven matrix plasticity beneath the impactor and its interaction with intralaminar damage processes, play an important role in the experimental energy absorption and crack suppression. These mechanisms are not explicitly represented within a purely interlaminar damage framework.

Consequently, while cohesive modelling provides essential insight into delamination initiation and propagation, it cannot, by itself, constitute a complete digital twin of the experimental response. This limitation motivates the introduction of a bi-phasic modelling strategy, in which a matrix-dominated phase capable of representing plasticity and indentation effects is explicitly coupled with the intralaminar and interlaminar damage models. The formulation and assessment of this bi-phasic approach are presented in the following section.

5.5 Bi-phasic Modelling Strategy for Matrix Impact Response

5.5.1 Motivation for the Bi-phasic Modelling Strategy

The numerical framework developed in the previous sections progressively incorporated increasing levels of physical realism. The purely elastic model provided a baseline representation of the global dynamic stiffness of the laminate, while the intralaminar damage implementation

enabled the simulation of fibre- and matrix-dominated cracking mechanisms. The introduction of cohesive interfaces further allowed the prediction of interlaminar delamination and its influence on the global impact response.

Although this combined intralaminar-interlaminar formulation captured some key damage mechanisms observed experimentally, a persistent limitation remained in reproducing the local plastic indentation observed on the impacted (front) face of the specimens.

Experimental inspection of the impacted plates revealed the presence of a permanent compressive dent in the vicinity of the contact zone at all levels of impact energy tested. This indentation is characteristic of thermoplastic-matrix composites, where the polymer matrix exhibits inelastic deformation prior to extensive fibre fracture or delamination propagation. In contrast, the previously implemented numerical models treated the laminate behaviour as predominantly elastic–brittle, with damage initiation governed by failure criteria but without an explicit representation of matrix plastic flow. As a result:

- The contact interaction remained excessively stiff;
- Local compressive crushing beneath the impactor was underpredicted;
- Permanent indentation was not reproduced;
- Energy dissipation associated with matrix yielding was not explicitly accounted for.

These discrepancies suggest that, despite the inclusion of intralaminar damage and delamination, the matrix-dominated compressive response in the impact zone is insufficiently represented within a homogenised laminate formulation.

To address this limitation, a bi-phasic modelling strategy is introduced. The objective is not to replace the previous intralaminar and interlaminar formulations, but rather to enhance the local constitutive representation in the impact region by allowing controlled inelastic deformation of a matrix-dominated phase. This approach enables the investigation of how matrix plasticity influences:

- Contact stiffness and contact duration;
- Peak acceleration levels;
- Local energy dissipation;
- Surface indentation development;
- Interaction with fibre failure and delamination mechanisms.

In order to systematically evaluate the contribution of each physical mechanism, the bi-phasic implementation is developed incrementally through a sequence of modelling stages, beginning with an elastic phase-separated baseline and progressively incorporating intralaminar damage and matrix plasticity effects. This staged approach ensures that the influence of each modelling refinement can be clearly isolated and assessed against experimental observations.

5.5.2 Incremental Model Development

In order to evaluate the influence of matrix-dominated inelastic deformation on the impact response in a controlled and systematic manner, the bi-phasic framework is developed through a sequence of incremental modelling stages. This approach ensures that the contribution of each physical mechanism can be clearly isolated and assessed against both the previously validated numerical models and the experimental observations.

Rather than introducing all constitutive refinements simultaneously, the model complexity is progressively increased. At each stage, a single additional mechanism is activated while the remaining modelling assumptions are kept consistent. This methodology allows direct comparison between configurations and prevents ambiguity in attributing variations in the global response to specific constitutive modifications.

The incremental development is structured as follows:

- **Phase I — Elastic Biphasic Baseline:** The laminate is represented using a phase-separated description, while retaining purely elastic behaviour in both phases. The objective of this stage is to verify that the elastic bi-phasic representation preserves the global stiffness and dynamic response of the previously validated monolithic elastic model.
- **Phase II — Intralaminar Damage Activation:** Intralaminar failure mechanisms are introduced within the fibre-dominated phase, consistent with the Hashin-based formulation previously implemented. The matrix-dominated phase remains elastic at this stage. This configuration allows assessment of fibre and matrix cracking within a phase-separated elastic framework.
- **Phase III - Intralaminar Damage with Matrix Plasticity:** Inelastic deformation is introduced in the matrix-dominated solid elements located in the impact region. This stage enables the representation of compressive yielding and permanent indentation beneath the impactor, while retaining intralaminar damage mechanisms in the fibre-dominated phase.

By structuring the development in this manner, the influence of matrix plasticity on contact stiffness, energy dissipation, peak acceleration levels and indentation formation can be rigorously quantified.

In addition to the constitutive evolution described above, two geometrical implementations of the bi-phasic discretisation are investigated in parallel. The first configuration introduces matrix-dominated solid elements across the full laminate panel, while the second confines the solid discretisation to the refined impact zone beneath the impactor. This dual approach enables evaluation of the trade-off between modelling fidelity and computational efficiency, while maintaining identical constitutive assumptions across all development phases.

5.5.3 Geometrical Implementation and Computational Strategy

The bi-phasic formulation was implemented through a superimposed discretisation strategy in which matrix-dominated solid elements are introduced using the same nodal connectivity as the existing continuum-shell mesh. This approach preserves the established mesh topology and impact refinement strategy while enabling independent assignment of constitutive behaviour to the fibre-dominated and matrix-dominated phases.

Superimposed Discretisation Procedure

The superposition procedure consists of duplicating the element connectivity definitions of the reference SC8R continuum-shell elements and generating a corresponding C3D8R solid element block within the Abaqus input file. To ensure uniqueness of element identifiers, the duplicated elements are assigned new IDs exceeding the maximum existing element number.

Because both element types share identical nodal connectivity, they experience the same kinematic field during deformation. This superposition enables the total stress within the laminate to be expressed as the sum of the stress contributions from the fibre-dominated shell phase and the matrix-dominated solid phase, consistent with the bi-phasic theoretical framework presented in Section 5.5.

The procedure was automated through dedicated Python scripts operating directly on the Abaqus input file. This automation ensured consistency across multiple simulations and eliminated the need for remeshing or manual redefinition of element sets.

Full-Panel and Localised Impact-Zone Configurations

The introduction of solid elements increases the number of degrees of freedom and may reduce the stable time increment in explicit integration due to the smaller characteristic element length associated with three-dimensional solid discretisation. To evaluate the trade-off between modelling fidelity and computational efficiency, two geometrical configurations were investigated:

- **Full-panel bi-phasic configuration:** Matrix-dominated solid elements are superimposed across the entire laminate panel. This configuration provides a uniform phase-separated representation of the composite structure.

- **Localised impact-zone configuration:** Solid elements are introduced only within the refined region beneath the impactor, while the remainder of the panel retains the original continuum-shell discretisation. This strategy aims to capture local matrix-dominated inelastic deformation responsible for indentation while limiting the computational cost associated with a full three-dimensional discretisation.

In both configurations, identical material properties, boundary conditions and loading definitions are maintained. This ensures that differences in numerical response arise solely from the spatial extent of the matrix- dominated solid phase and not from changes in constitutive assumptions.

To facilitate the interpretation of the bi-phasic modelling approach adopted in this work, Figure 5.57 illustrates the two discretisation strategies investigated for the superposed solid phase. In the bi-phasic formulation, the laminate is primarily represented by a continuum shell phase capturing the global fibre-dominated in-plane response, while an additional solid phase is superposed to represent matrix-dominated through-thickness deformation mechanisms associated with impact loading.

(a) Full-panel (FP) bi-phasic discretisation. (b) Impact-zone (IZ) bi-phasic discretisation.

Figure 5.57: Three-dimensional visualisation of the two discretisation strategies considered for the superposed solid phase in the bi-phasic modelling framework. In the full-panel (FP) configuration, solid elements are distributed across the entire laminate surface. In the impact-zone (IZ) configuration, the solid phase is restricted to the refined central region beneath the impactor, reducing the computational cost while preserving the ability to capture local matrix-dominated deformation mechanisms associated with impact loading.

The comparative assessment of these two geometrical implementations provides insight into the minimum discretisation complexity required to reproduce the experimentally observed indentation behaviour under impact loading.

5.5.4 Elastic Material Partitioning and Model Definition

The first stage of the bi-phasic modelling strategy consists of establishing a mechanically consistent elastic baseline configuration. The objective of this configuration is to verify that the introduction of superimposed matrix-dominated solid elements does not modify the global stiffness

and dynamic response of the previously validated monolithic elastic laminate model presented in Section 5.2.

Stiffness Superposition Framework

As described in Section 5.5, the bi-phasic formulation represents the composite laminate as the superposition of two mechanically coupled phases: a fibre-dominated continuum-shell phase and a matrix-dominated solid phase. Since both element types share identical nodal connectivity, they undergo the same strain field during deformation. The total stress in the laminate is therefore expressed as

$$\boldsymbol{\sigma}^{\text{lam}} = \boldsymbol{\sigma}^{\text{shell}} + \boldsymbol{\sigma}^{\text{solid}}, \tag{5.19}$$

which implies the stiffness consistency condition

$$\mathbf{C}^{\text{lam}} = \mathbf{C}^{\text{shell}} + \mathbf{C}^{\text{solid}}. \tag{5.20}$$

To prevent artificial stiffness amplification, the laminate stiffness was partitioned using a rule-of-mixtures (RoM) approach based on a fibre volume fraction $V_f = 0.5$ and matrix volume fraction $V_m = 0.5$.

The PPS matrix properties were defined as

$$E_m = 4000 \text{ MPa}, \qquad \nu_m = 0.36, \tag{5.21}$$

with corresponding shear modulus

$$G_m = \frac{E_m}{2(1+\nu_m)} = 1470.6 \text{ MPa}. \tag{5.22}$$

The effective matrix stiffness contribution is therefore scaled by V_m, leading to the isotropic solid definition

$$E^{\text{solid}} = 2000 \text{ MPa}, \qquad \nu^{\text{solid}} = 0.36. \tag{5.23}$$

The fibre-dominated shell phase is obtained by subtracting the scaled matrix contribution from the calibrated laminate stiffness components,

$$E_i^{\text{shell}} = E_i^{\text{lam}} - V_m E_m, \qquad G_{ij}^{\text{shell}} = G_{ij}^{\text{lam}} - V_m G_m. \tag{5.24}$$

The resulting baseline elastic properties for both phases are summarised in Table 5.29.

Table 5.29: Baseline elastic properties for the bi-phasic elastic configuration obtained through stiffness partitioning with $V_f = V_m = 0.5$.

Property	**Continuum Shell Phase [MPa]**	**Solid Phase [MPa]**
E_1	21614.5	–
E_2	18254.5	–
E_3	4100.0	–
G_{12}	2864.7	–
G_{23}	2264.7	–
G_{13}	2264.7	–
E (isotropic)	–	2000.0
ν_{12}	0.113	–
ν_{23}	0.32	–
ν_{13}	0.32	–
ν (isotropic)	–	0.36

5.5.5 Baseline Bi-Phasic Elastic Impact Response

The elastic bi-phasic baseline configuration was evaluated for all impact energy levels considered in the experimental campaign. The results are presented and discussed energy-by-energy, following the same methodology adopted for the monolithic elastic model in Section 5.2. The comparison focuses on the acceleration–time response and the initial slope evolution relative to the experimental mean response.

Baseline bi-phasic elastic response at 4.5 J

The elastic bi-phasic configuration was first evaluated for the lowest impact energy level ($E = 4.5$ J), corresponding to Group 1 of the experimental campaign. At this energy level, the structural response remains predominantly elastic, making it particularly suitable for verifying that the stiffness partitioning strategy preserves the global laminate behaviour prior to the introduction of damage or inelastic matrix mechanisms.

Figure 5.58 compares the experimental acceleration-time histories with the responses predicted by the monolithic linear elastic model and the two bi-phasic elastic implementations (full-panel and localised impact-zone).

Both bi-phasic configurations reproduce the global shape of the response, including the initial acceleration rise, the peak acceleration region and the unloading phase associated with impactor rebound. The numerical signals remain nearly coincident with the monolithic elastic reference, indicating that the superposition of shell and solid phases does not modify the global bending

stiffness of the laminate. The predicted peak acceleration is slightly higher than the experimental mean, consistent with the behaviour previously observed for the monolithic elastic model.

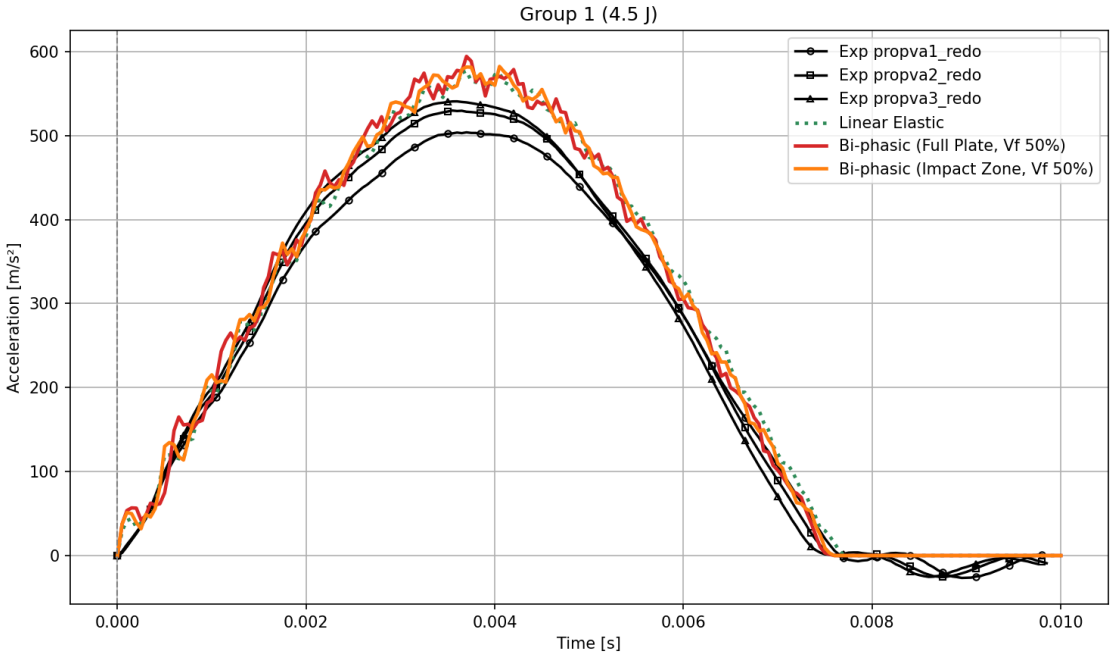


Figure 5.58: Acceleration-time response for Group 1 (4.5 J): experimental repetitions compared with the monolithic linear elastic model and the elastic bi-phasic configurations (full-panel and localised impact-zone).

Quantitative correlation was assessed through the slope-based comparison in the 30-84% time-to-peak interval. The deviation of each numerical configuration relative to the experimental mean response is shown in Figure 5.59.

The full-panel bi-phasic configuration yields the lowest mean absolute deviation (approximately 1.6%), while the impact-zone configuration produces a similarly small deviation. Both responses remain very close to the monolithic elastic prediction throughout the analysed interval, confirming that the stiffness partitioning procedure does not introduce artificial stiffness amplification or numerical inconsistencies.

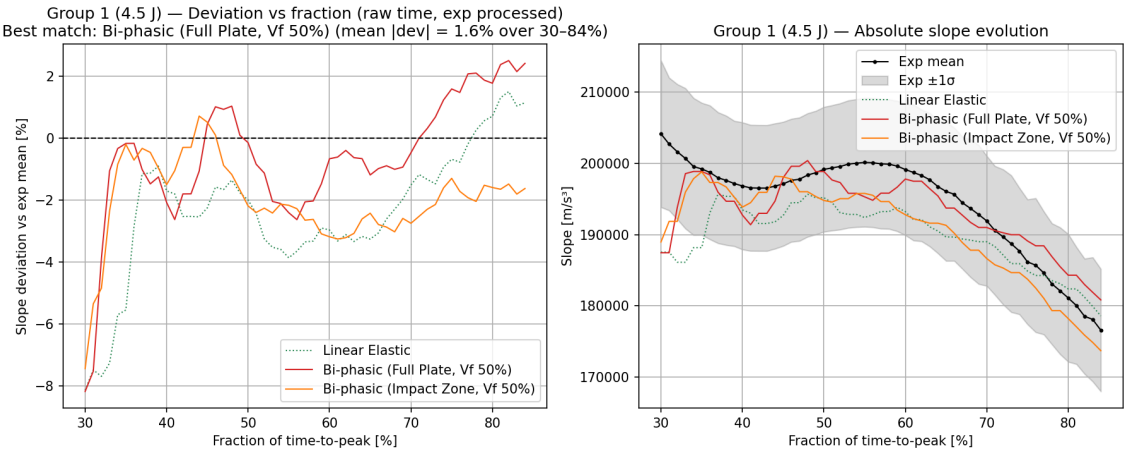


Figure 5.59: Slope deviation relative to the experimental mean respons for Group 1 (4.5 J): comparison between the monolithic linear elastic model and the elastic bi-phasic configurations.

Overall, the results obtained at 4.5 J demonstrate that the elastic bi-phasic formulation preserves the effective laminate stiffness and global impact response of the monolithic elastic model. The negligible difference between the full-panel and localised impact-zone implementations further indicates that, at this energy level, the spatial extent of the matrix-dominated solid phase does not influence the global elastic response.

Baseline bi-phasic elastic response at 7.5 J

The elastic bi-phasic configuration was next evaluated for the impact energy level of ($E = 7.5$ J), corresponding to Group 2 of the experimental campaign. At this energy level, the laminate still exhibits a predominantly elastic structural response, allowing further assessment of the mechanical consistency of the stiffness partitioning strategy under increased loading severity.

Figure 5.60 compares the experimental acceleration-time histories with the predictions obtained from the monolithic linear elastic model and the two bi-phasic elastic configurations.

The overall numerical response remains nearly coincident for all elastic formulations. The initial acceleration rise, peak acceleration region and unloading behaviour are consistently reproduced. The bi-phasic configurations closely follow the monolithic elastic prediction, confirming that the superimposed discretisation does not modify the effective bending stiffness of the laminate. Minor differences observed near the peak acceleration remain within the experimental scatter band.

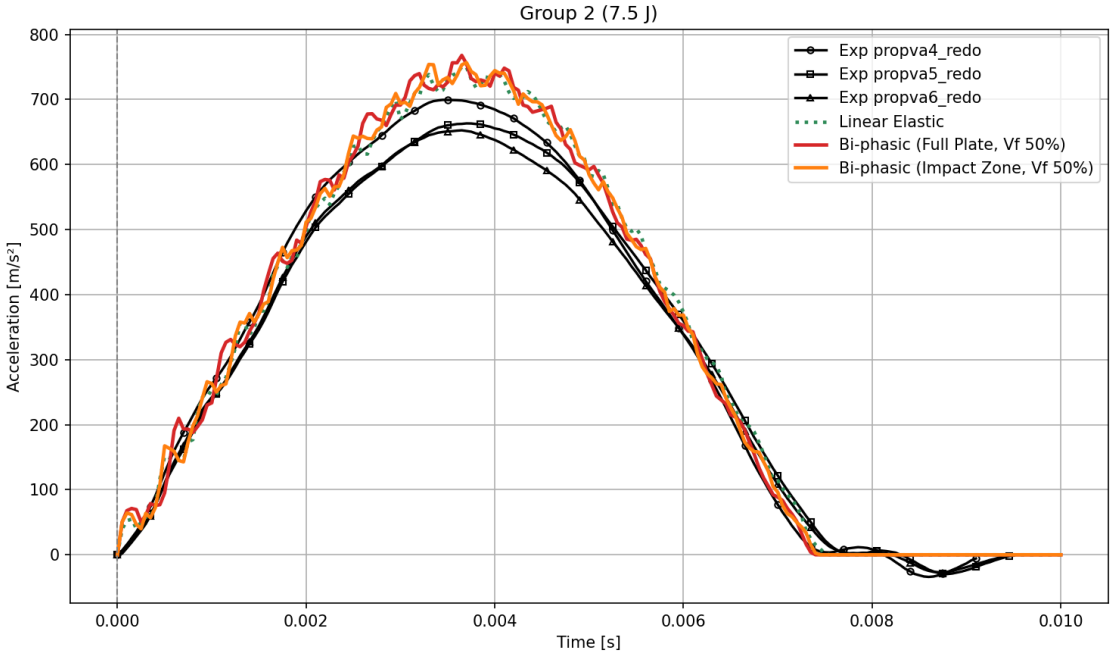


Figure 5.60: Acceleration-time response for Group 2 (7.5 J): experimenta repetitions compared with the monolithic linear elastic model and the elastic bi-phasic configurations (full-panel and localised impact-zone).

The slope-based deviation analysis in the 30-84% time-to-peak interval is shown in Figure 5.61. The impact-zone bi-phasic configuration yields the lowest mean absolute deviation, approximately 2.2%, while the full-panel configuration exhibits a similarly small deviation. The linear elastic monolithic model remains very close to both bi-phasic responses.

These results indicate that, even with increased impact energy, the stiffness superposition strategy preserves the global elastic response without introducing artificial stiffness effects.

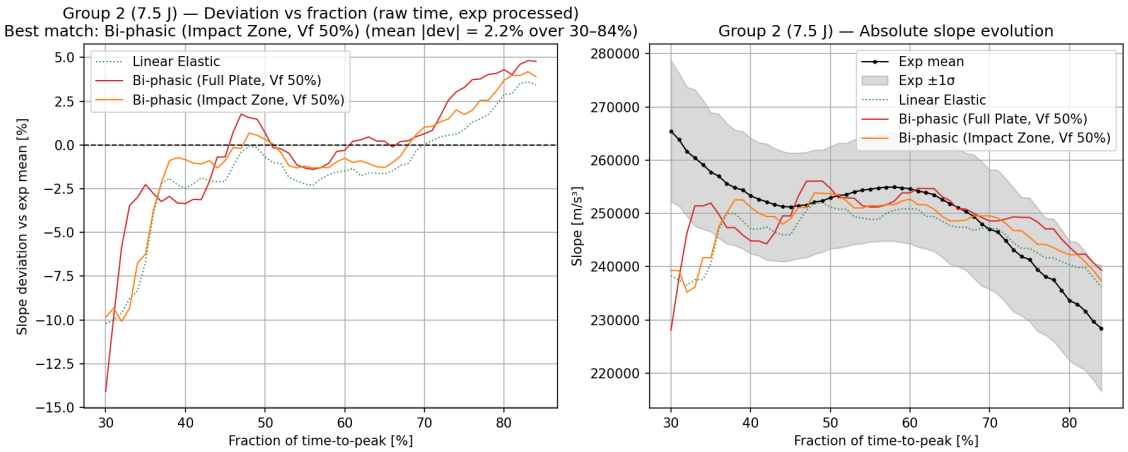


Figure 5.61: Slope deviation relative to the experimental mean respons for Group 2 (7.5 J): comparison between the monolithic linear elastic model and the elastic bi-phasic configurations.

Overall, the results at 7.5 J confirm that the elastic bi-phasic formulation remains mechanically consistent under this impact energy. The negligible differences between the full-panel and localised impact-zone implementations indicate that, in the absence of damage or inelastic deformation, the spatial extent of the matrix-dominated solid phase does not significantly influence the global elastic impact response.

Baseline bi-phasic elastic response at 14.2 J

The elastic bi-phasic configuration was subsequently evaluated for the impact energy level of ($E \approx 14.2$ J), corresponding to Group 3 of the experimental campaign. At this energy, the structural response begins to deviate more significantly from purely elastic behaviour experimentally, although the numerical formulation remains strictly elastic.

Figure 5.62 presents the acceleration–time comparison between the experimental repetitions, the monolithic linear elastic model and the two elastic bi-phasic configurations.

All elastic numerical formulations predict higher peak accelerations than observed experimentally, consistent with the absence of damage and energy dissipation mechanisms. The bi-phasic configurations remain almost coincident with the monolithic elastic response throughout the entire impact event, including the initial acceleration rise and the peak region. This confirms that the stiffness partitioning does not modify the global elastic response even at increased loading severity.

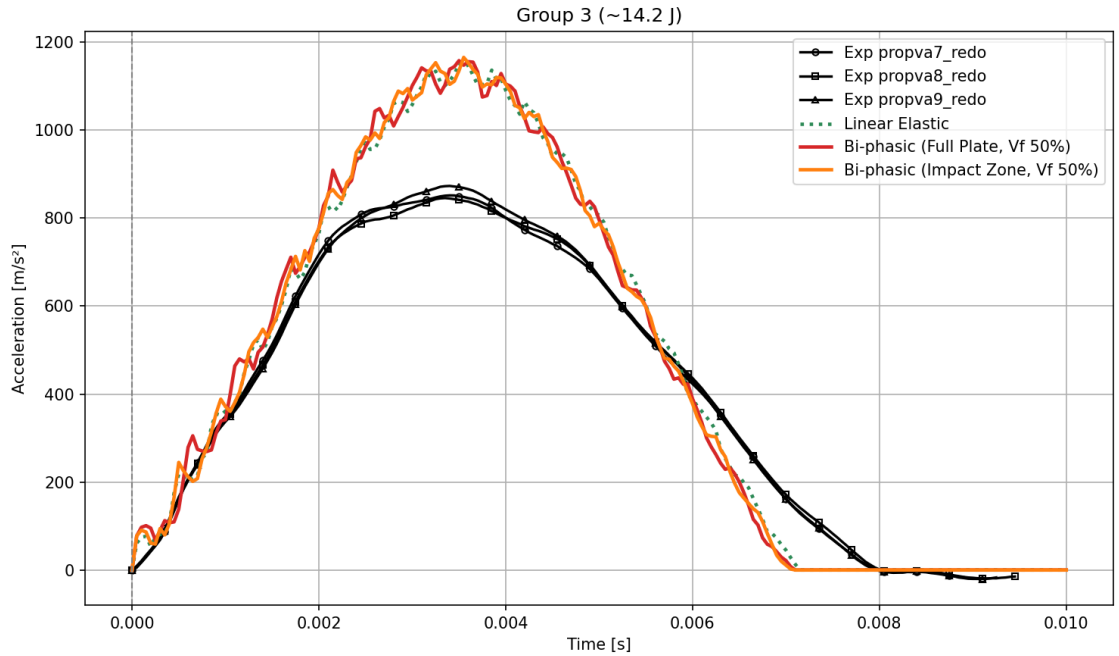


Figure 5.62: Acceleration-time response for Group 3 (∼14.2 J): experimental repetitions compared with the monolithic linear elastic model and the elastic bi-phasic configurations (full-panel and localised impact-zone).

The slope-based deviation analysis in the 30–84% time-to-peak interval is shown in Figure 5.63. The monolithic linear elastic model provides the lowest mean absolute deviation (approximately 7.9%), while both bi-phasic configurations exhibit slightly larger deviations. Nevertheless, the difference between the monolithic and bi-phasic predictions remains small, and the general trend of the slope evolution is consistently reproduced.

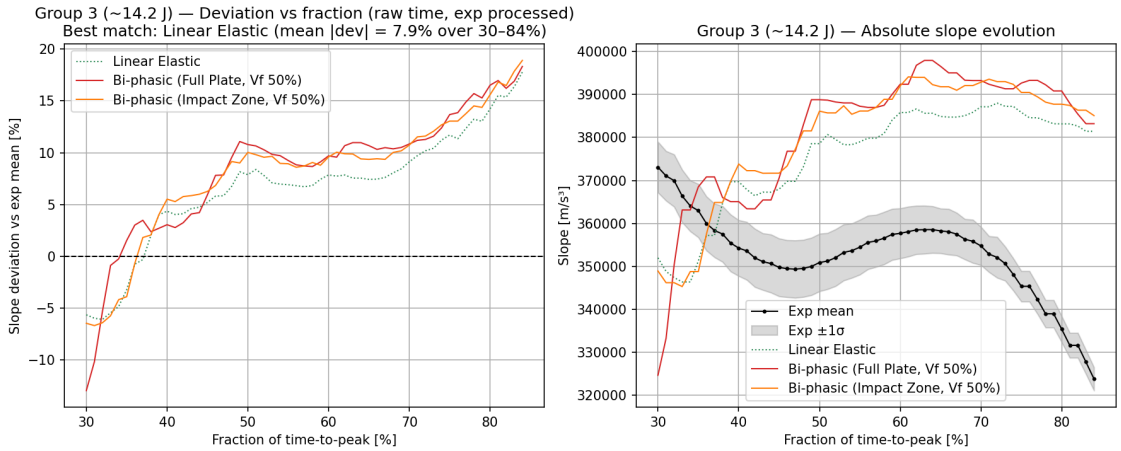


Figure 5.63: Slope deviation relative to the experimental mean response for Group 3 (∼14.2 J): comparison between the monolithic linear elastic model and the elastic bi-phasic configurations.

Overall, the results at $\sim$14.2 J indicate that the elastic bi-phasic formulation preserves the global dynamic stiffness of the laminate and does not introduce artificial stiffness amplification. The increasing discrepancy between numerical and experimental responses is attributed to the absence of damage and inelastic mechanisms in the purely elastic formulation, rather than to the phase-separated representation itself.

Baseline bi-phasic elastic response at 27 J

The highest impact energy level ($E = 27$ J), corresponding to Group 4 of the experimental campaign, was finally analysed using the elastic bi-phasic configuration. At this energy level, significant damage and inelastic mechanisms are observed experimentally, while the numerical formulation remains strictly elastic.

Figure 5.64 presents the comparison between the experimental acceleration histories, the monolithic linear elastic model and the two bi-phasic elastic configurations.

All elastic numerical formulations substantially overestimate the experimental peak acceleration and exhibit a steeper initial slope. The predicted response remains highly symmetric and fully reversible, reflecting the absence of energy dissipation mechanisms. Both the full-panel and localised impact-zone bi-phasic configurations remain nearly coincident with the monolithic elastic response throughout the entire impact event, confirming that the stiffness partitioning procedure does not introduce artificial stiffness amplification even under more severe loading conditions.

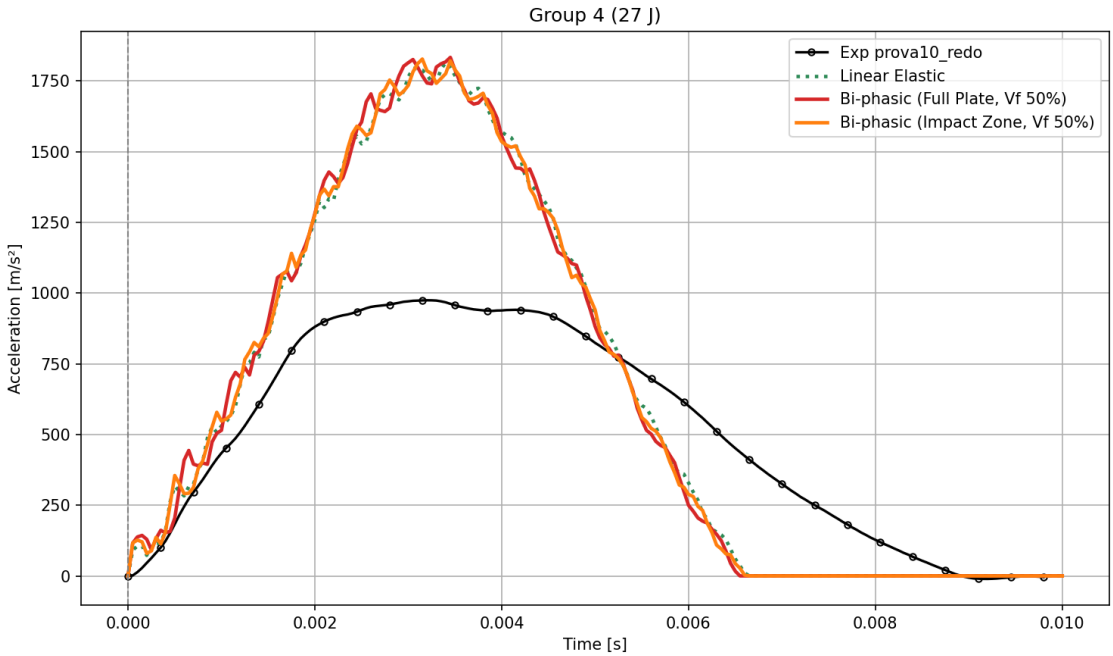


Figure 5.64: Acceleration-time response for Group 4 (27 J): experimental repetitions compared with the monolithic linear elastic model and the elastic bi-phasic configurations (full-panel and localised impact-zone).

The slope-based deviation analysis in the 30–84% time-to-peak interval is shown in Figure 5.65. The monolithic elastic model exhibits a mean absolute deviation of approximately 31.8%, and both bi-phasic configurations follow the same trend with nearly identical deviations.

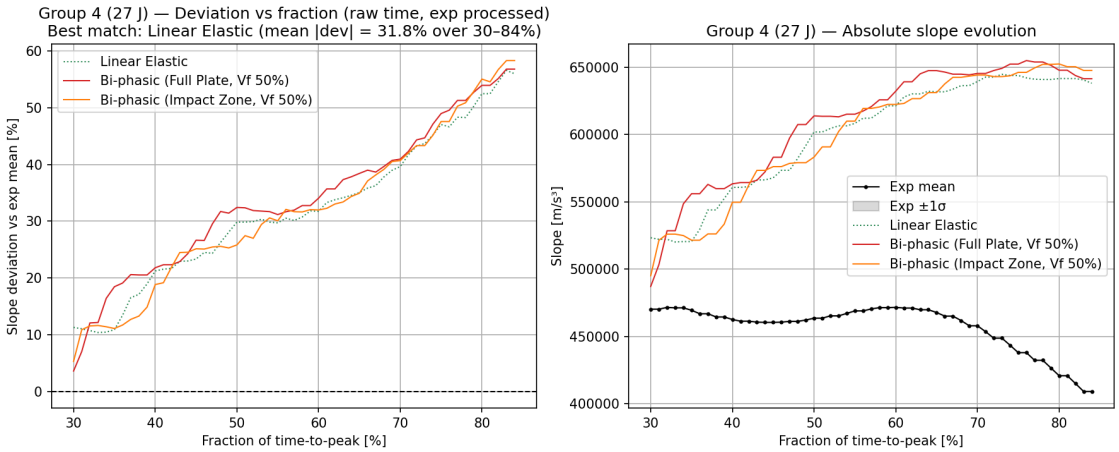


Figure 5.65: Slope deviation relative to the experimental mean response for Group 4 (27 J): comparison between the monolithic linear elastic model and the elastic bi-phasic configurations.

At this energy level, the discrepancy between numerical and experimental responses is clearly associated with the absence of damage and inelastic deformation in the purely elastic formulation.

Importantly, the bi-phasic elastic configurations remain mechanically consistent with the monolithic elastic model, indicating that the phase-separated representation does not alter the global laminate stiffness. The observed deviation therefore reinforces the need to introduce intralaminar damage and matrix-dominated inelastic mechanisms insubsequent modelling stages.

Across all impact energy levels, the elastic bi-phasic configurations produce responses that are nearly indistinguishable from the monolithic linear elastic laminate model. The acceleration histories, peak values, contact duration and slope evolution remain essentially unchanged, demonstrating that the stiffness partitioning strategy successfully preserves the global dynamic behaviour of the laminate.

The primary objective of this stage was to verify the mechanical consistency of the phase-separated representation and to ensure that no artificial stiffness amplification was introduced through the superposition of continuum shell and solid elements. Given the negligible differences observed between the bi-phasic and monolithic elastic responses, no additional sensitivity analysis was performed at this stage. The elastic bi-phasic configuration is therefore adopted as a mechanically consistent baseline for the subsequent introduction of intralaminar damage and matrix-dominated inelastic mechanisms.

5.5.6 Intralaminar Impact Response (Bi-phasic Hashin Baseline)

The intralaminar impact response of the bi-phasic formulation is assessed following the same methodology previously adopted for the calibrated continuum-shell Hashin model. In the present configuration, the Hashin damage criteria are activated exclusively in the continuum shell phase, while the superimposed solid phase remains purely elastic. No matrix plasticity is introduced at this stage.

The intralaminar fracture energy values used in the present simulations correspond exactly to the calibrated parameters obtained from the deterministic sensitivity study of the full continuum-shell model (Section 5.3.2). The objective of this first assessment is therefore to isolate the effect of introducing the superimposed matrix-dominated solid phase while retaining the same intralaminar parameter set.

As in the previous sections, the acceleration–time history measured at the impactor is adopted as the primary response metric, providing a robust basis for comparison between numerical predictions and experimental results. In addition, the surface crack length observed on the back face of the laminate is used as a secondary damage metric. As discussed previously in Section 5.3.3, this measurement corresponds only to the visible surface manifestation of the crack and does not provide information regarding the internal crack depth or the complete three-dimensional damage state of the laminate.

Bi-phasic Hashin response at 4.5 J

The lowest impact energy level ($E = 4.5$ J, Group 1) corresponds experimentally to a predominantly elastic regime. The acceleration–time comparison between experimental results and the bi-phasic Hashin baseline configuration is presented in Figure 5.66.

The numerical prediction closely follows the experimental curves in terms of global shape, peak acceleration, and contact duration. The response is very similar to that obtained with the calibrated continuum-shell Hashin model, indicating that the introduction of the elastic solid phase does not significantly alter the global stiffness at this low energy level. Although local damage initiation may occur according to the Hashin criteria, the overall structural response remains largely elastic.

No intralaminar crack formation is predicted on the back face of the laminate, in agreement with the limited experimental damage observed in this regime.

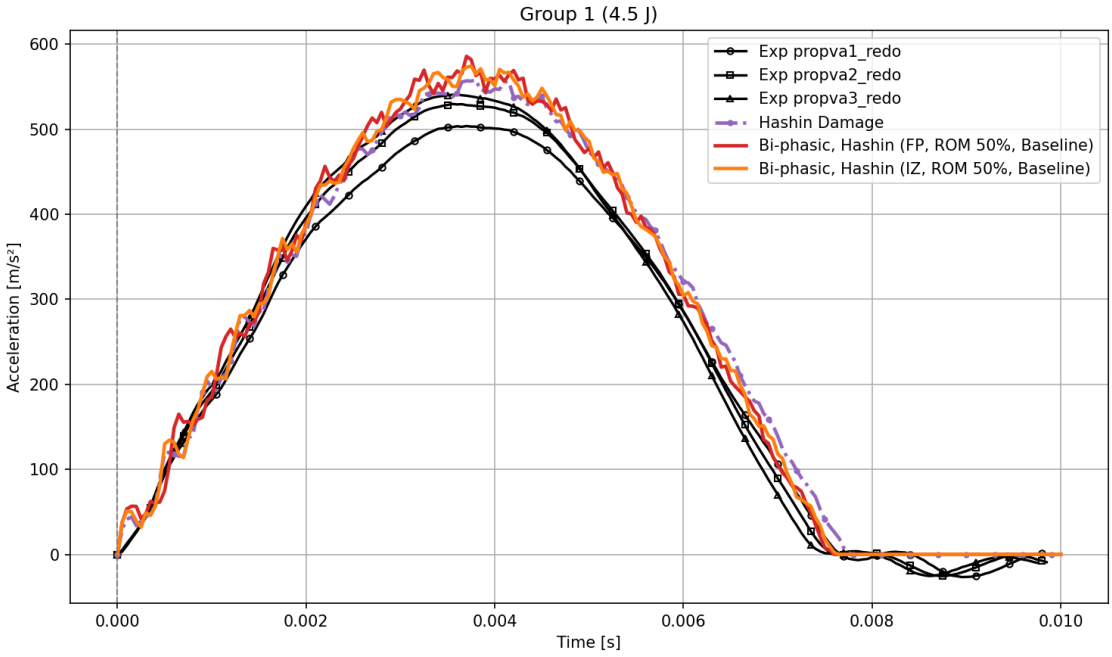


Figure 5.66: Acceleration-time response for Group 1 (4.5 J): experimental results compared with the bi-phasic Hashin baseline configuration.

Bi-phasic Hashin response at 7.5 J

The intermediate impact energy level ($E = 7.5$ J, Group 2) was subsequently analysed. The corresponding acceleration–time histories are shown in Figure 5.67.

The predicted response remains close to that of the calibrated continuum-shell Hashin model, with only moderate differences in peak acceleration and contact duration. However, a slight increase in peak acceleration can already be observed, indicating a tendency toward a stiffer global response compared to the single-phase configuration.

Despite the activation of intralaminar damage in the shell phase, the superimposed elastic solid phase continues to sustain compressive loads without stiffness degradation. As a result, the overall structural response remains predominantly elastic in character.

No visible intralaminar crack formation is predicted on the back face of the laminate at this energy level.

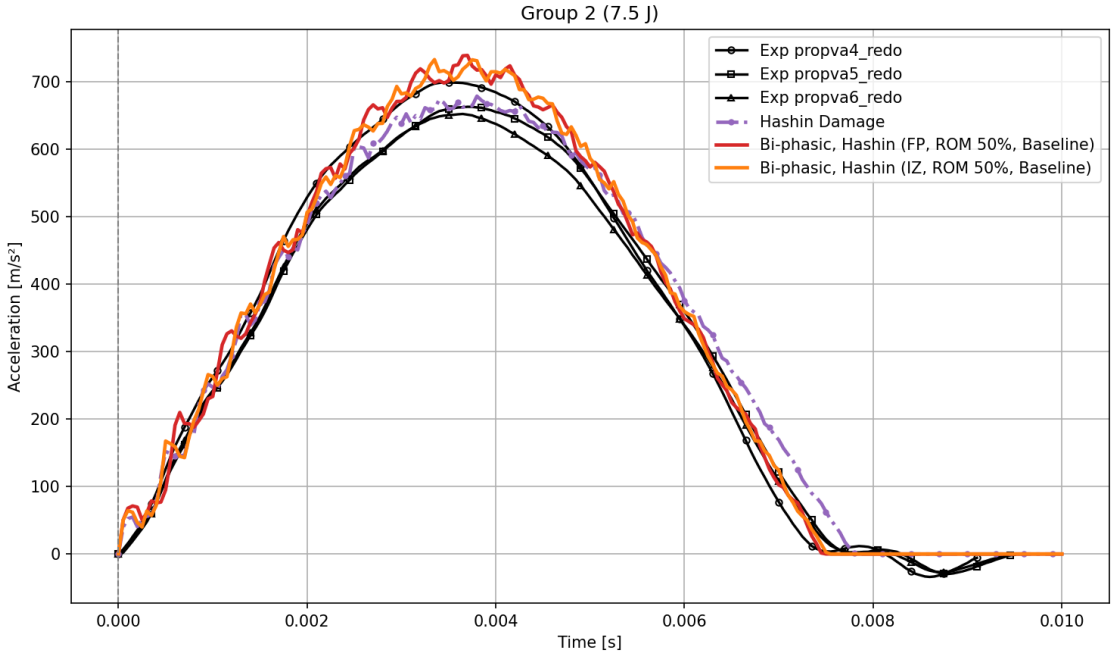


Figure 5.67: Acceleration-time response for Group 2 (7.5 J): experimental results compared with the bi-phasic Hashin baseline configuration.

Bi-phasic Hashin response at approximately 14.2 J

For the higher intermediate impact energy ($E \approx 14.2$ J, Group 3), experimental observations indicate the onset of significant intralaminar damage and the development of back-face cracking. The acceleration–time comparison is shown in Figure 5.68.

At this energy level, a clear deviation from both the experimental response and the calibrated continuum-shell Hashin model becomes evident. The bi-phasic configuration predicts a noticeably higher peak acceleration and a shorter contact duration, resulting in a stiffer global response. The

unloading branch also exhibits a steeper decay, indicating limited energy dissipation during the impact event.

Although intralaminar damage is activated within the shell phase, the superimposed solid phase remains purely elastic and continues to carry compressive stresses without degradation. This additional elastic constraint limits the effective stiffness reduction induced by the Hashin damage evolution, leading to an overestimation of the structural rigidity.

No localised intralaminar crack formation is predicted on the back face of the laminate, despite the experimental evidence of cracking at this energy level.

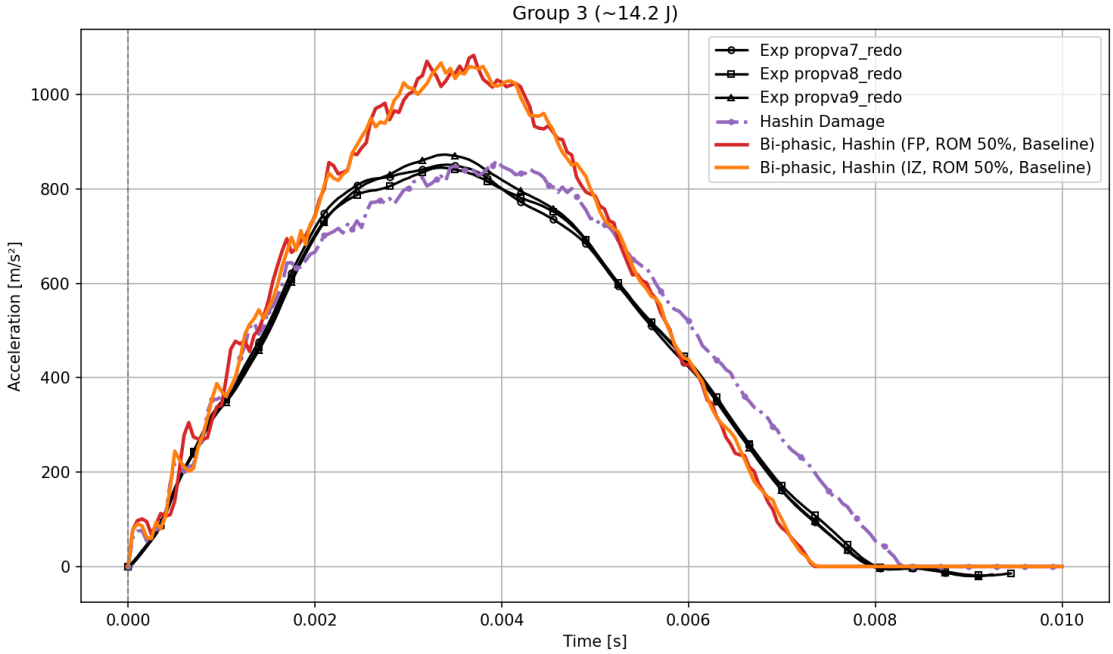


Figure 5.68: Acceleration-time response for Group 3 ($\approx$ 14.2 J): experimental results compared with the bi-phasic Hashin baseline configuration.

Bi-phasic Hashin response at 27 J

The highest impact energy considered ($E = 27$ J, Group 4) corresponds experimentally to a regime characterised by extensive damage and pronounced back-face cracking. The numerical comparison is shown in Figure 5.69.

The discrepancy between the bi-phasic baseline configuration and the experimental response becomes more pronounced at this energy level. The predicted peak acceleration is significantly higher than the experimental mean, and the contact duration is clearly underestimated, indicating an overly stiff structural behaviour.

While intralaminar damage is triggered in the shell phase, the elastic solid phase continues to resist indentation without any inelastic mechanism, thereby maintaining a high compressive stiffness. This behaviour prevents the progressive softening observed experimentally and in the calibrated continuum-shell model.

No pronounced back-face crack development is obtained in the simulation, highlighting the inability of the present configuration to reproduce the experimentally observed damage localisation.

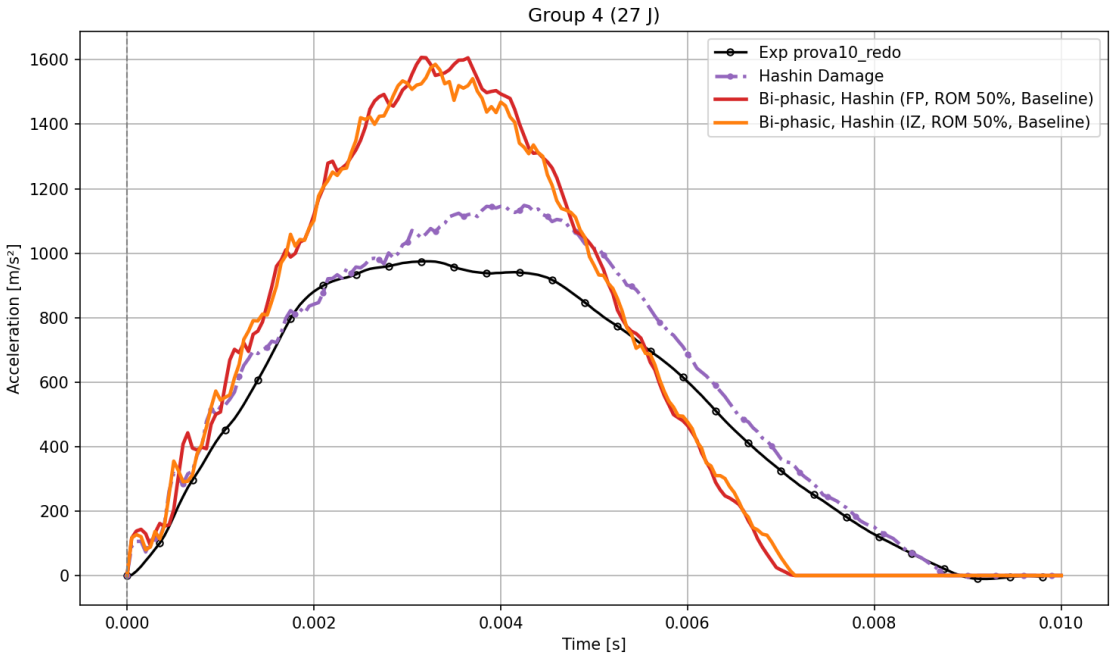


Figure 5.69: Acceleration–time response for Group 4 (27 J): experimental results compared with the bi-phasic Hashin baseline configuration.

Overall, the bi-phasic Hashin formulation, when adopting the fracture energy parameters calibrated for the continuum-shell model, produces a systematically stiffer impact response, particularly as the impact energy increases and damage mechanisms become dominant.

While the low-energy regime remains largely unaffected, the deviation from the experimental response becomes increasingly significant at intermediate and high energies. The introduction of the superimposed elastic solid phase modifies the compressive load transfer within the laminate and limits the effective stiffness degradation associated with intralaminar damage in the shell phase.

Most importantly, no back-face crack formation is predicted for any impact energy level. This indicates that the fracture energy parameters calibrated for the single-phase continuum-shell model are not directly transferable to the bi-phasic framework under the present modelling assumptions.

These observations motivate a dedicated sensitivity study of the intralaminar fracture energy parameters within the bi-phasic formulation, which is addressed in the following subsection.

5.5.7 Deterministic Sensitivity Analysis of Intralaminar Fracture Energy in the Bi-phasic Framework

A dedicated deterministic sensitivity study was performed to evaluate the influence of intralaminar fracture energy on crack propagation within the bi-phasic Hashin formulation.

In this analysis, only the intralaminar fracture energy parameters were varied, while strength values were kept unchanged. Fracture energy values are expressed as percentages of the original baseline reference defined in the continuum-shell study (Section 5.3.1), for which the previously calibrated value corresponded to 9% of the baseline.

Crack length, extracted from element deletion along the v-direction, is adopted as the primary quantitative metric. For each impact energy level, both the Full Plate (FP) and Impact Zone (IZ) configurations are considered to ensure that the reduced modelling strategy does not affect crack-length prediction. As previously discussed in Section 5.3.3, this metric corresponds to the visible crack extension on the back face of the laminate and therefore represents only the surface projection of the damage, without providing direct information about the internal crack depth or the full three-dimensional damage state of the laminate.

The following subsections present the results for each impact energy level.

Impact response at 4.5 J

The lowest impact energy level $E = 4.5$ J (Group 1) corresponds to a predominantly elastic regime in which the experimental specimens do not exhibit visible back-face cracking. This case therefore represents the lower bound of the fracture energy sensitivity study and provides a useful reference for verifying that the numerical model does not predict spurious crack formation under low-energy impact conditions.

Crack-length correlation

Table 5.30 summarises the numerically predicted back-face crack length for the different intralaminar fracture energy variations investigated at $E = 4.5$ J, for both the Impact Zone (IZ) and Full Plate (FP) configurations. The fracture energy values are reported as percentages relative to the original baseline intralaminar fracture energy defined in the continuum-shell study (Section 5.3.1). The calibrated value identified in the previous intralaminar sensitivity analysis corresponds to 9% of this original baseline reference.

The experimentally measured crack length at this energy level is $L_c^{\mathrm{Exp}} = 0$ mm. Crack length is extracted from element deletion along the v-axis, as defined in Section 5.3.3.

Table 5.30: Back-face crack length correlation at $E = 4.5$ J for different intralaminar fracture energy variations (relative to the original baseline value).

G_f [%]	$L_c^{CAE,IZ}$ [mm]	$L_c^{CAE,FP}$ [mm]	L_c^{Exp} [mm]	Dev.
5.0	0.00	0.00	0.00	0
2.5	0.00	0.00	0.00	0
2.0	0.00	0.00	0.00	0
1.8	0.00	0.00	0.00	0
1.7	0.00	0.00	0.00	0
1.6	6.00	6.46	0.00	overprediction
1.5	6.46	6.46	0.00	overprediction
1.0	11.31	11.08	0.00	overprediction

For fracture energy values down to approximately 1.7% of the original baseline value, no element deletion occurs in either the IZ or FP configuration, in full agreement with the experimental observation of no back-face crack. When the fracture energy is reduced further (between 1.6% and 1.0% of the baseline), pronounced crack propagation is triggered, leading to a clear and non-physical overprediction of crack length.

The IZ and FP models exhibit nearly identical crack-length predictions for all fracture energy values considered, indicating that the global plate extent does not significantly influence crack initiation at this low energy level.

Acceleration-time response

Figure 5.70 compares the experimental acceleration–time histories with the numerical responses obtained using representative bi-phasic Hashin configurations.

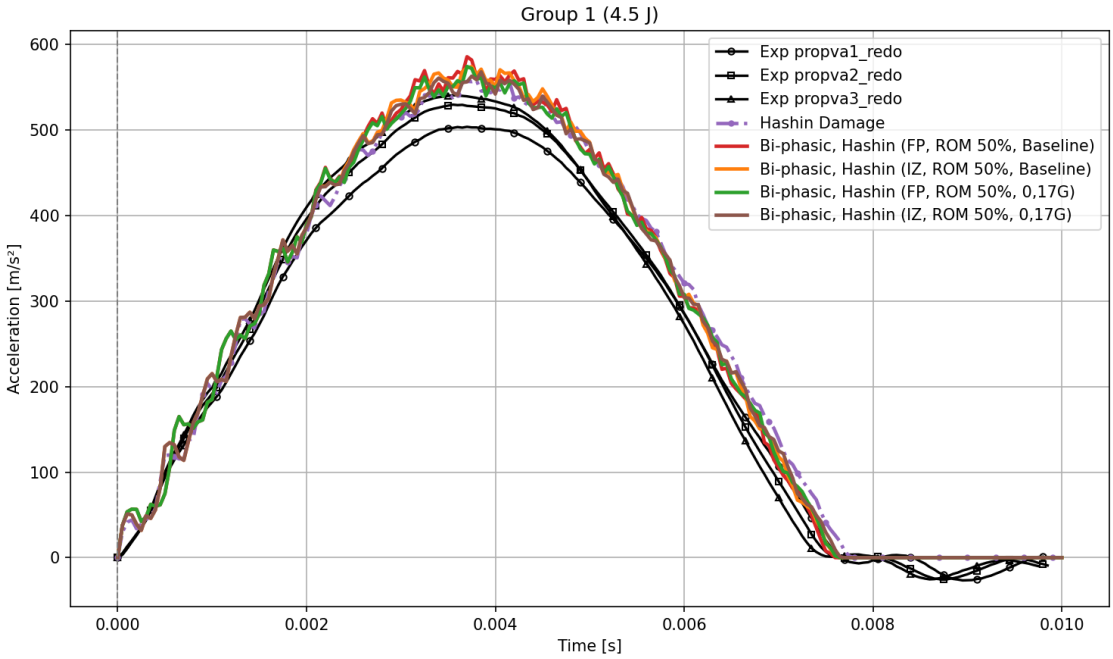
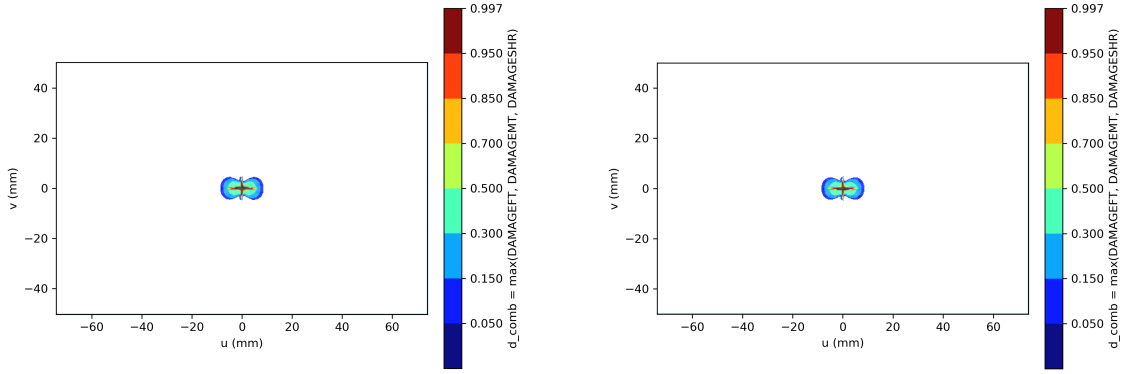


Figure 5.70: Acceleration–time response for Group 1 ($E = 4.5$ J): comparison between experimental results and representative bi-phasic Hashin configurations.

At 4.5 J the acceleration–time response remains governed by elastic stiffness. Differences between fracture energy configurations are minimal, and all numerical curves remain close in terms of peak acceleration and contact duration. This confirms that the global impact response at this energy level is not sensitive to intralaminar fracture energy variations.

Back-face damage comparison

A qualitative comparison between the Impact Zone (IZ) and Full Plate (FP) damage distributions for a representative reduced fracture energy case is shown in Figure 5.71. Both configurations exhibit a localised cross-shaped damage pattern beneath the impactor, with very similar spatial extent and orientation.



(a) Full Plate (FP). (b) Impact Zone (IZ).

Figure 5.71: Comparison of back-face intralaminar damage distribution at $E = 4.5$ J between the Full Plate (FP) and Impact Zone (IZ) configurations for a representative reduced fracture energy case.

The extracted damage "arms" show very similar extensions in both models, with u-direction lengths of approximately 4.39 mm and v-direction lengths between 5.31 and 5.54 mm. The close agreement between IZ and FP results confirms that the reduced modelling domain does not introduce significant bias in the predicted damage morphology at this energy level.

Overall, the 4.5 J case confirms that fracture energy values above approximately 1.7% of the original baseline reference correctly suppress crack formation, consistent with the experimental observation of no back-face cracking. Conversely, excessively reduced fracture energy values lead to unrealistic crack propagation. This impact level therefore serves primarily as a lower-bound validation scenario, eliminating overly small fracture energy values while offering limited discriminatory power for final parameter selection.

Impact response at 7.5 J

The intermediate impact energy level $E = 7.5$ J (Group 2) represents a transitional regime in which the experimental specimens exhibit the onset of back-face cracking, with an average crack length of $L_c^{\text{Exp}} = 3.23$ mm. This case therefore provides a first meaningful metric for fracture energy identification, as crack propagation becomes experimentally measurable while the overall impact response remains largely stiffness-dominated.

Crack-length correlation

Table 5.31 summarises the numerically predicted back-face crack length for the different intralaminar fracture energy variations investigated at $E = 7.5$ J, for both the Impact Zone (IZ) and Full Plate (FP) configurations. The fracture energy values are reported as percentages relative

to the original baseline intralaminar fracture energy defined in the continuum-shell study (Section 5.3.1). The previously calibrated continuum-shell value corresponds to 9% of this baseline reference.

Crack length is extracted from element deletion along the v-axis, as defined in Section 5.3.3.

Table 5.31: Back-face crack length correlation at $E = 7.5$ J for different intralaminar fracture energy variations (relative to the original baseline value).

G_f [%]	$L_c^{CAE,IZ}$ [mm]	$L_c^{CAE,FP}$ [mm]	L_c^{Exp} [mm]	Dev.
5.0	0.00	0.00	3.23	100.0%
2.5	0.00	0.00	3.23	100.0%
2.0	0.00	0.00	3.23	100.0%
1.8	0.69	1.15	3.23	78.6% / 64.4%
1.7	2.31	2.31	3.23	28.5%
1.6	11.08	11.31	3.23	243.0% / 250.2%
1.5	11.54	11.77	3.23	257.3% / 264.4%
1.0	17.08	17.31	3.23	428.8% / 435.9%

For fracture energy values equal to or above 2.0% of the original baseline, no element deletion occurs in either model, resulting in a complete underprediction of crack length. As the fracture energy is reduced to 1.8%, crack initiation becomes detectable but remains significantly underpredicted relative to the experimental value.

The closest agreement with the experimental crack length is obtained at 1.7% of the original baseline value, for which both IZ and FP configurations predict $L_c^{\mathrm{CAE}} = 2.31$ mm, corresponding to a deviation of approximately 28.5%.

A further reduction of the fracture energy to 1.6% and below results in excessive crack propagation, with crack lengths more than three times the experimental value. This behaviour indicates a sharp transition in damage sensitivity within a narrow fracture energy interval.

Importantly, the IZ and FP configurations exhibit very similar crack length predictions for all fracture energy values considered, confirming that the reduced modelling domain does not significantly affect crack length estimation at this energy level.

Acceleration–time response

Figure 5.72 compares the experimental acceleration–time histories with representative biphasic Hashin configurations.

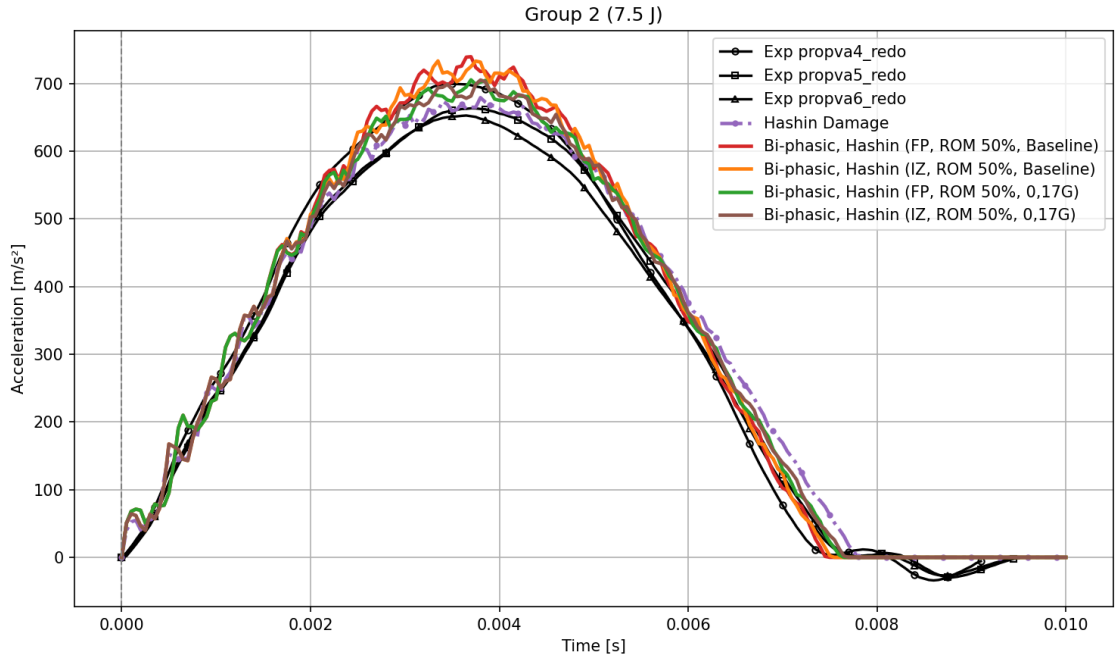


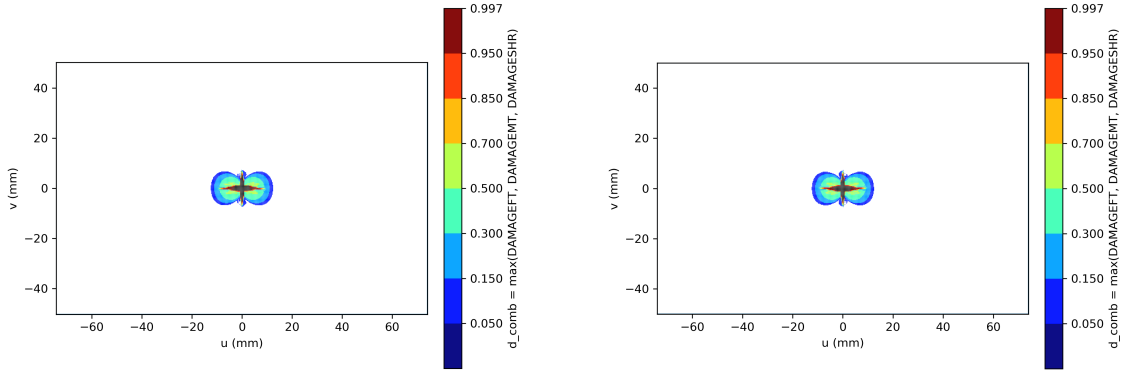
Figure 5.72: Acceleration–time response for Group 2 ($E = 7.5$ J): comparison between experimental results and representative bi-phasic Hashin configurations.

At 7.5 J the acceleration–time response remains primarily governed by global stiffness. Differences between fracture energy configurations are comparatively small relative to the variation observed in crack length. Although reduced fracture energy values slightly decrease peak acceleration due to earlier damage activation, the overall response shape and contact duration remain similar across configurations.

This confirms that acceleration–time is not sufficiently sensitive to discriminate fracture energy values at this energy level and that crack length provides a more robust metric for parameter identification.

Back-face damage comparison

Figure 5.73 presents a qualitative comparison of the back-face intralaminar damage distribution between the Full Plate (FP) and Impact Zone (IZ) configurations for a representative fracture energy value near the optimal range.



(a) Full Plate (FP).

(b) Impact Zone (IZ).

Figure 5.73: Comparison of back-face intralaminar damage distribution at $E = 7.5$ J between the Full Plate (FP) and Impact Zone (IZ) configurations.

The damage contours exhibit a cross-shaped pattern beneath the impactor, with comparable spatial extent and orientation in both configurations. The extracted damage "arms" show very similar lengths, with u-direction extensions of approximately 9.92–10.39 mm and v-direction extensions of approximately 10.39 mm.

The close agreement between IZ and FP results further confirms that the reduced model does not introduce significant distortion in the predicted damage morphology.

Overall, the 7.5 J case demonstrates a strong sensitivity of crack length to intralaminar fracture energy variations within a narrow range between approximately 1.8% and 1.6% of the original baseline value. The best compromise between underprediction and excessive crack growth is obtained near 1.7% of the baseline reference.

This impact level therefore provides the first robust calibration point for fracture energy identification, as crack length becomes a discriminating and physically meaningful metric.

Impact response at ≈ 14.2 J

The higher intermediate impact energy level $E \approx 14.2$ J (Group 3) corresponds experimentally to a regime characterised by clearly developed back-face cracking, with an average measured crack length of $L_c^{\text{Exp}} = 10.33$ mm. At this energy level, intralaminar fracture energy becomes a strongly discriminating parameter, as crack propagation is well established and no longer limited to incipient damage initiation.

Crack-length correlation

Table 5.32 summarises the predicted crack lengths for the different intralaminar fracture energy values considered at $E \approx 14.2$ J, for both the Impact Zone (IZ) and Full Plate (FP) configurations. The fracture energy values are expressed as percentages of the original baseline intralaminar value defined in the continuum-shell study (Section 5.3.1). The calibrated continuum-shell value corresponds to 9% of this reference baseline.

Crack length is extracted from element deletion along the v-axis, as defined previously.

Table 5.32: Back-face crack length correlation at $E \approx 14.2$ J for different intralaminar fracture energy variations (relative to the original baseline value).

G_f [%]	$L_c^{CAE,IZ}$ [mm]	$L_c^{CAE,FP}$ [mm]	L_c^{Exp} [mm]	Dev.
5.0	0.00	0.00	10.33	100.0%
2.5	0.00	0.00	10.33	100.0%
2.0	5.54	5.76	10.33	46.4% / 44.2%
1.8	7.61	7.85	10.33	26.3% / 24.0%
1.7	9.23	9.69	10.33	10.6% / 6.2%
1.6	19.38	19.38	10.33	87.6%
1.5	20.31	20.77	10.33	96.6% / 101.1%
1.0	25.38	25.38	10.33	145.7%

For fracture energy values equal to or above 2.5% of the original baseline, no crack propagation is predicted, resulting in complete underprediction of the experimental crack length. As the fracture energy is reduced to 2.0% and 1.8%, crack growth becomes progressively more realistic but remains significantly underestimated.

The closest agreement with the experimental value is obtained at 1.7% of the original baseline, for which the IZ configuration predicts $L_c^{\mathrm{CAE}} = 9.23$ mm (deviation 10.6%) and the FP configuration predicts 9.69 mm (deviation 6.2%). This represents the best overall compromise between underprediction and excessive crack propagation.

A further reduction of fracture energy below 1.7% leads to a sharp increase in crack length, with values nearly doubling the experimental measurement. This confirms a strong nonlinear sensitivity of crack propagation to intralaminar fracture energy at this energy level.

As observed for 7.5 J, the IZ and FP configurations produce very similar crack-length predictions across all fracture energy values, indicating that the reduced modelling domain does not significantly influence crack-length estimation.

Acceleration–time response

Figure 5.74 compares the experimental acceleration–time histories with representative bi-phasic Hashin configurations.

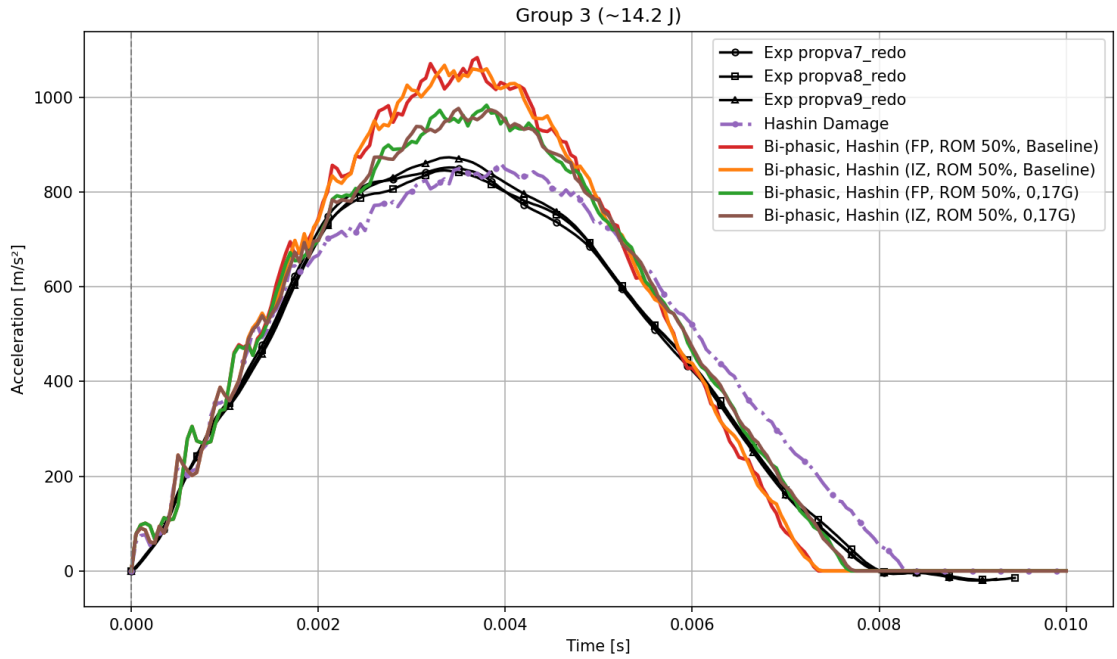


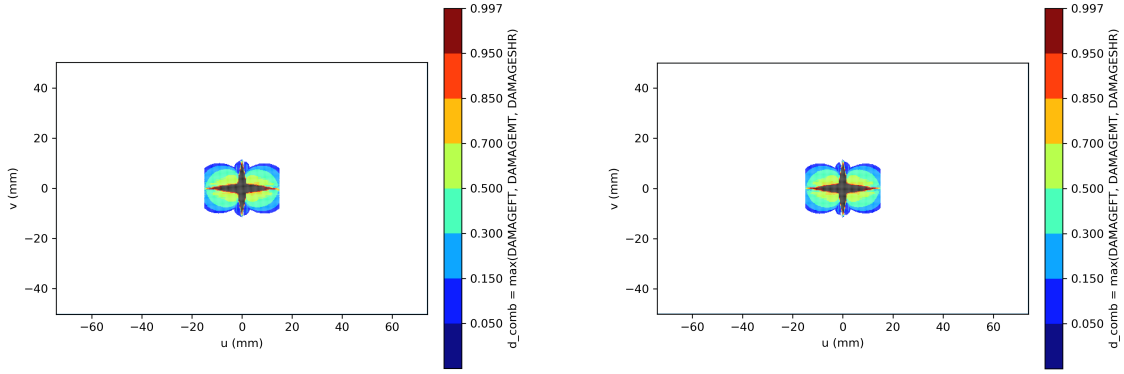
Figure 5.74: Acceleration–time response for Group 3 ($E \approx 14.2$ J): comparison between experimental results and representative bi-phasic Hashin configurations.

At 14.2 J the acceleration–time response exhibits greater sensitivity to fracture energy than at lower impact energies. Nevertheless, even for fracture energy values that provide accurate crack length prediction, the numerical peak acceleration remains higher than the experimental mean response, and the unloading phase retains a relatively stiff character.

This indicates that although intralaminar fracture energy governs crack extent, additional dissipative mechanisms may be required to fully capture the experimentally observed global compliance.

Back-face damage comparison

Figure 5.75 presents the predicted back-face damage distribution for the IZ and FP configurations.



(a) Full Plate (FP). (b) Impact Zone (IZ).

Figure 5.75: Comparison of back-face intralaminar damage distribution a $E \approx 14.2$ J between the Full Plate (FP) and Impact Zone (IZ) configurations.

The damage contours exhibit a well-developed cross-shaped morphology, with significantly larger spatial extension than observed at lower impact energies. The extracted damage arms show u-direction extensions of approximately 21.0–21.5 mm and v-direction extensions of approximately 18.7–19.2 mm, again with very close agreement between IZ and FP configurations.

The orientation and spatial development of the crack are consistent between both modelling approaches, confirming that the reduced Impact Zone formulation preserves the essential crack morphology.

Overall, the 14.2 J impact case provides a robust and highly discriminating calibration point for intralaminar fracture energy identification. The optimal fracture energy is consistently located near 1.7% of the original baseline reference value, for which crack length prediction falls within approximately 6–11% of the experimental measurement.

This impact level therefore serves as the primary basis for final fracture energy selection within the bi-phasic framework.

Impact response at 27 J

The highest impact energy level considered, $E = 27$ J (Group 4), corresponds experimentally to a regime characterised by extensive intralaminar cracking and pronounced back-face crack propagation. The average measured crack length is $L_c^{\text{Exp}} = 19$ mm, making this case a highly discriminating calibration point for intralaminar fracture energy identification.

Crack-length correlation

Table 5.33 summarises the predicted crack lengths for the different intralaminar fracture energy values investigated at $E = 27$ J, for both the Impact Zone (IZ) and Full Plate (FP) configurations. Fracture energy values are expressed as percentages of the original baseline intralaminar value defined in the continuum-shell study (Section 5.3.1). The calibrated continuum-shell value corresponds to 9% of this reference baseline.

Crack length is extracted from element deletion along the v-axis.

Table 5.33: Back-face crack length correlation at $E = 27$ J for different intralaminar fracture energy variations (relative to the original baseline value).

G_f [%]	$L_c^{CAE,IZ}$ [mm]	$L_c^{CAE,FP}$ [mm]	L_c^{Exp} [mm]	Dev.
5.0	0.00	0.00	19.00	100.0%
2.5	6.46	7.38	19.00	66.0% / 61.2%
2.0	11.08	12.00	19.00	41.7% / 36.8%
1.8	13.85	14.31	19.00	27.1% / 24.7%
1.7	16.15	16.62	19.00	15.0% / 12.5%
1.6	26.31	26.76	19.00	38.5% / 40.8%
1.5	27.23	27.00	19.00	43.3% / 42.1%
1.0	30.00	32.33	19.00	57.9% / 70.2%

For fracture energy values above 2.5% of the original baseline, crack propagation is severely underpredicted or completely absent. As the fracture energy is reduced to 2.0% and 1.8%, crack growth increases progressively but still remains below the experimentally observed extent.

The best overall agreement is obtained at 1.7% of the original baseline value. At this level, the IZ configuration predicts $L_c^{\mathrm{CAE}} = 16.15$ mm (deviation 15.0%), while the FP configuration predicts 16.62 mm (deviation 12.5%). This represents the closest match across the entire parameter range.

A further reduction to 1.6% and below results in excessive crack propagation, with predicted crack lengths significantly exceeding the experimental measurement. The sensitivity trend observed at 14.2 J is therefore confirmed: crack length exhibits a strong nonlinear dependence on intralaminar fracture energy within a narrow interval.

The IZ and FP configurations again show very close agreement across all fracture energy values, confirming that the reduced Impact Zone model preserves crack-length accuracy even under severe damage conditions.

Acceleration-time response

Figure 5.76 compares the experimental acceleration–time history with representative bi-phasic Hashin configurations.

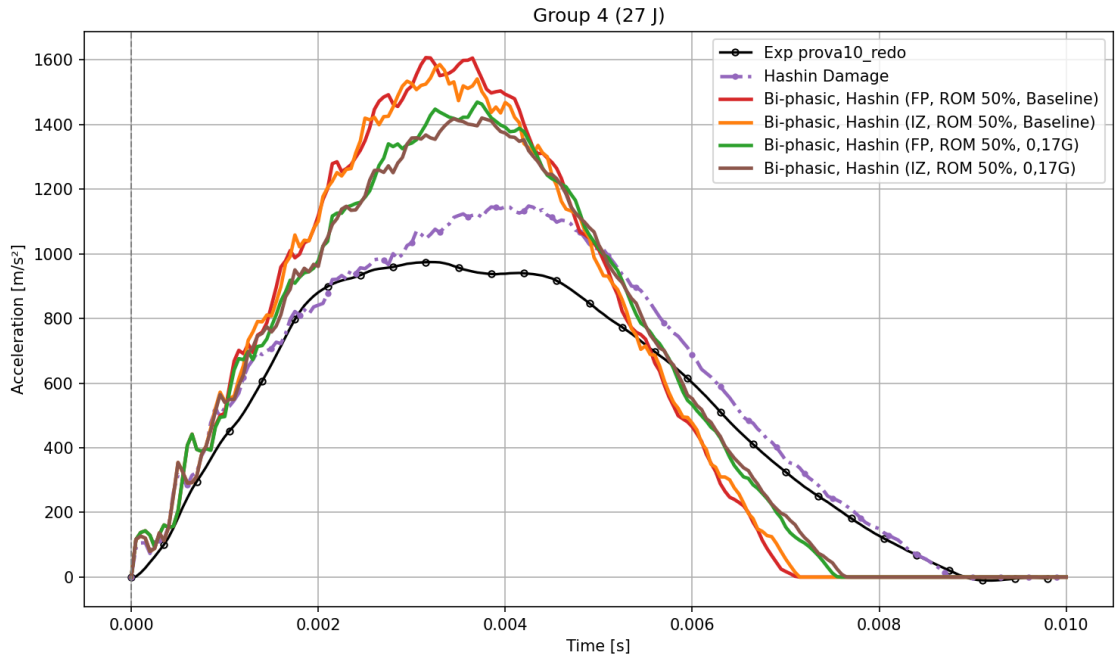


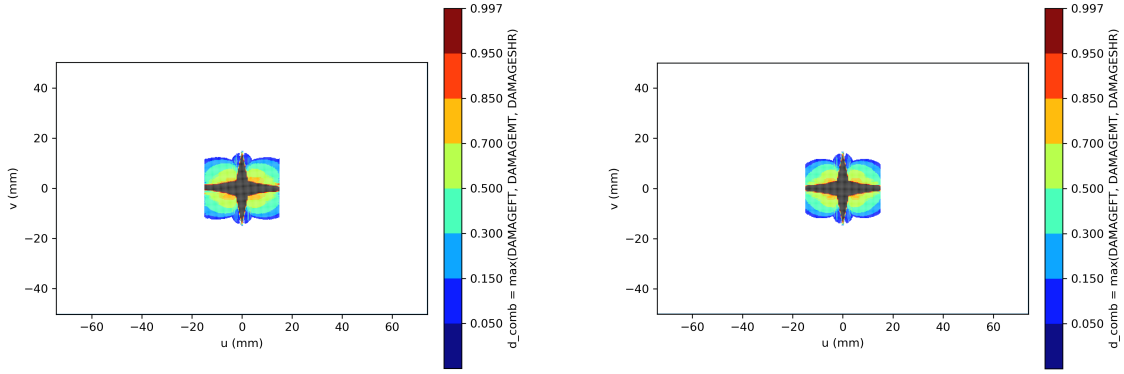
Figure 5.76: Acceleration–time response for Group 4 ($E = 27$ J): comparison between experimental results and representative bi-phasic Hashin configurations.

At 27 J the acceleration–time response remains noticeably stiffer than the experimental curve for all fracture energy values considered. Although reduced fracture energy values slightly decrease peak acceleration due to earlier and more extensive damage activation, the predicted peak remains significantly higher than the experimental measurement, and the unloading phase retains a comparatively stiff character.

This confirms that intralaminar fracture energy alone is insufficient to fully capture the global compliance observed experimentally at high impact energy. Additional dissipative mechanisms are likely required to reproduce the experimentally observed softening behaviour.

Back-face damage comparison

Figure 5.77 presents the predicted back-face damage distributions for the FP and IZ configurations at a representative fracture energy level near the optimal range.



(a) Full Plate (FP).

(b) Impact Zone (IZ).

Figure 5.77: Comparison of back-face intralaminar damage distribution at $E = 27$ J between the Full Plate (FP) and Impact Zone (IZ) configurations.

The damage contours exhibit a fully developed cross-shaped morphology with significant spatial extension. The extracted damage arms show u-direction extensions of approximately 29.77 mm and v-direction extensions of approximately 25.6–26.1 mm, with negligible differences between IZ and FP configurations.

The agreement in crack morphology further validates the use of the reduced Impact Zone model for computational efficiency without compromising damage prediction accuracy.

Overall, the 27 J case confirms the strong and consistent fracture energy sensitivity identified at 14.2 J. The optimal fracture energy remains located near 1.7% of the original baseline value, providing the best compromise between underprediction and overprediction of crack length.

Across all three energy levels (7.5, 14.2, and 27 J), $G_f \approx 1.7\%$ of the original baseline emerges as the most consistent and physically representative value for the bi-phasic Hashin formulation.

Nevertheless, the persistent overestimation of peak acceleration and the relatively stiff unloading response indicate that intralaminar fracture energy reduction alone cannot fully reproduce the experimentally observed global compliance. This observation motivates the introduction of additional inelastic mechanisms within the matrix-dominated phase, which is addressed in the subsequent subsection.

5.5.7.1 Overall Assessment of the Bi-phasic Hashin Fracture Energy Study

The parametric investigation performed across the four impact energy levels (4.5, 7.5, 14.2, and 27 J) provides a comprehensive assessment of the role of intralaminar fracture energy within the bi-phasic Hashin framework.

A consistent and robust trend emerges across all energy levels: the optimal crack-length correlation is obtained for fracture energy values close to 1.7% of the original baseline intralaminar value defined in the continuum-shell study. This consistency confirms that the bi-phasic formulation exhibits stable and coherent fracture-energy sensitivity behaviour, and that crack length is governed by a narrow and well-defined parameter window.

However, the calibrated fracture energy value required to reproduce experimental crack lengths is significantly lower than values typically reported in the literature for Glass/epoxy and Glass/thermoplastic systems under impact loading. Such a drastic reduction suggests that the identified value does not represent a physically realistic material parameter, but rather compensates for modelling assumptions introduced in the current bi-phasic configuration.

The introduction of the superimposed solid phase, while intended to represent matrix-dominated behaviour, increases the global stiffness of the laminate, particularly in the out-of-plane direction. In the present formulation, the solid phase remains purely elastic and therefore stores and releases energy without contributing to permanent dissipation. This additional constraint enhances transverse stiffness and stress confinement, leading to:

- Higher peak accelerations,
- Shorter contact durations,
- A relatively stiff unloading response,
- Delayed or suppressed crack propagation for moderate fracture energy values.

Consequently, intralaminar fracture energy must be artificially reduced to unrealistically low levels in order to compensate for the lack of dissipative mechanisms within the solid phase. In this sense, fracture energy becomes a numerical tuning parameter rather than a purely material property.

Importantly, the comparison between the Full Plate (FP) and Impact Zone (IZ) configurations demonstrates excellent agreement in terms of crack length prediction and damage morphology across all energy levels. This confirms that the reduced IZ modelling strategy preserves the essential fracture mechanisms while offering improved computational efficiency.

Although crack-length correlation improves significantly with reduced fracture energy, the acceleration -time response remains consistently stiffer than the experimental curves, particularly at higher impact energies. This indicates that intralaminar damage evolution alone is insufficient to reproduce the experimentally observed global compliance.

Overall, the results of this study demonstrate that the insertion of a purely elastic solid phase accentuates the stiffness response of the panel and shifts the dissipation demand toward intralaminar fracture mechanisms. The unrealistically low fracture energy values required for crack calibration provide clear evidence that additional energy-dissipation mechanisms are necessary within the matrix-dominated phase.

This finding directly motivates the next modelling stage, in which plasticity is introduced in the solid phase in order to provide a physically meaningful dissipation mechanism capable of reproducing both crack propagation and global impact compliance.

5.6 Plasticity Modelling of the Matrix Phase

Although the bi-phasic modelling strategy introduced in the previous section improved the representation of the through-thickness behaviour of the laminate, the numerical response remained excessively stiff in the impact region. In particular, the simulations were unable to reproduce the indentation behaviour observed experimentally, suggesting that an important energy dissipation mechanism was still missing from the model.

Under localised impact loading, thermoplastic matrices such as PPS may undergo significant plastic deformation, especially in regions subjected to high compressive stresses beneath the impactor. This plastic flow contributes to the dissipation of impact energy and plays an important role in the formation of the indentation profile. Since the solid elements introduced in the bi-phasic approach represent the matrix-dominated behaviour of the laminate, incorporating plasticity in this phase provides a physically consistent way to account for matrix deformation during impact.

To address this limitation, plasticity models available in Abaqus were introduced in the solid elements representing the matrix phase within the impact zone. Several constitutive formulations were investigated, including classical von Mises plasticity, pressure-dependent Drucker-Prager plasticity, and the crushable foam plasticity model. The objective of this phase is to evaluate the ability of these formulations to reproduce the experimentally observed indentation behaviour and to improve the representation of energy dissipation mechanisms in the numerical model.

In order to limit the computational cost associated with the large number of parametric simulations required for the plasticity investigation, only the impact energy levels of 14.2 J and 27 J were considered in this phase. These two cases correspond to representative intermediate and high-energy impact conditions within the experimental campaign and were therefore considered sufficient to assess the influence of the different plasticity formulations on the predicted structural response.

5.6.1 Impact Zone Solid Element Strategy

In the bi-phasic modelling approach adopted in this work, solid elements were introduced to represent the matrix-dominated behaviour of the laminate in the vicinity of the impact region. Two alternative discretisation strategies were initially considered: a full solid representation of the laminate and a localised solid element patch restricted to the impact zone.

As discussed in the previous section, the comparison between these two approaches showed that the predicted global response of the structure was practically identical. In particular, the acceleration-time histories and the overall deformation patterns presented negligible differences between the full-solid and localised solid discretisations. This indicates that the highly localised nature of the impact stresses allows the use of solid elements only in the vicinity of the contact region without affecting the global dynamic response of the panel.

Since the full-solid representation significantly increases the number of degrees of freedom and the associated computational cost, the localised impact-zone strategy was adopted for the remainder of the plasticity investigations. This approach allows the matrix behaviour in the contact region to be captured using solid elements while maintaining a computationally efficient hybrid continuum shell-solid discretisation of the laminate.

5.6.2 Mechanical Behaviour of the PPS Matrix

The solid elements introduced in the bi-phasic modelling strategy represent the matrix- dominated response of the composite laminate in the vicinity of the impact region. Consequently, the constitutive behaviour assigned to these elements should reflect the mechanical response of the polymer matrix phase. PPS is a thermoplastic polymer known to exhibit nonlinear deformation and plastic flow when subjected to high localised stresses, particularly under compressive loading conditions such as those occurring beneath the impactor during impact events.

Table 5.34 summarises the mechanical properties adopted for the PPS matrix in the present work. These values were obtained from literature sources and represent typical properties for unfilled PPS materials. It is worth noting that thermoplastic polymers generally exhibit different strengths in tension and compression, with compressive strengths typically exceeding tensile strengths. This tension–compression asymmetry is particularly relevant for impact problems, where the material beneath the impactor is predominantly subjected to compressive stresses.

Table 5.34: Mechanical properties adopted for the PPS matrix phase.

Property	**Symbol**	**Value**
Elastic modulus[a]	E	2000 MPa
Poisson's ratio[b]	ν	0.36
Tensile strength[b]	σ_t	102 MPa
Compressive strength[b]	σ_c	148 MPa

[a] Young's modulus derived from the PPS material data reported by MCAM [98]. The effective value used in the solid elements was estimated using a Rule of Mixtures (RoM) approach to account for the matrix contribution within the composite laminate, assuming an approximate fibre volume fraction of $V_f \approx 50\%$.

[b] Mechanical properties of PPS obtained from the material data reported by MCAM [98].

5.6.3 Perfect Plasticity Implementation

As a first step in introducing plastic behaviour in the solid elements representing the matrix phase, an elastic–perfectly plastic constitutive model was adopted. This simplified formulation was selected in order to introduce plastic energy dissipation in the impact zone while avoiding the need for detailed calibration of a full stress-strain curve for the PPS matrix material.

In this approach, the material is assumed to behave linearly elastic up to the yield stress, after which plastic deformation occurs at constant stress without strain hardening. The yield stress was taken as the compressive strength of the PPS matrix, $\sigma_c = 148$ MPa, since the material directly beneath the impactor is predominantly subjected to compressive stresses during the impact event.

This simplified plasticity representation provides a baseline description of matrix yielding in the impact region and allows the influence of plastic deformation on the structural response to be investigated before introducing more advanced pressure-dependent plasticity formulations.

Figure 5.78 presents a schematic stress–strain representation of the perfectly plastic material behaviour adopted for the matrix.

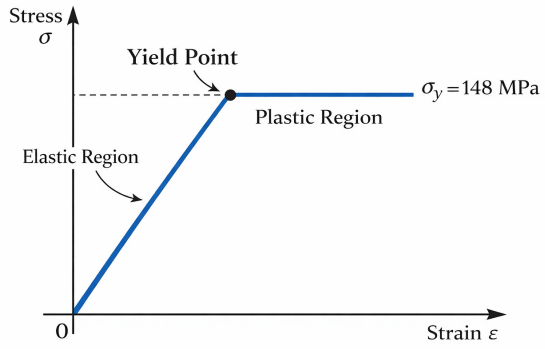


Figure 5.78: Idealized elastic-perfectly plastic stress–strain response used to represent the matrix material behaviour, with yielding occurring at $\sigma_y = 148$ MPa.

5.6.4 Von Mises Perfect Plasticity Formulation

To represent permanent deformation of the polymer matrix in the impact zone, a von Mises plasticity formulation with perfectly plastic behaviour was adopted. The von Mises criterion is a pressure-independent yield model in which yielding is governed by the distortional component of the stress state. Consequently, the hydrostatic pressure does not influence the onset of plastic deformation. This formulation is widely used to represent isotropic yielding behaviour and provides a simple yet robust framework for modelling permanent deformation in numerical simulations [20].

The yield condition is expressed as

$$F = f(\sigma - \alpha) - \sigma_0 = 0, \tag{5.25}$$

where $f(\sigma - \alpha)$ denotes the equivalent von Mises stress, σ_0 is the uniaxial yield stress, and α represents the backstress tensor used in kinematic hardening formulations. In the present work, perfect plasticity was assumed and therefore the yield stress remains constant once yielding occurs [20].

The equivalent von Mises stress q is obtained from the deviatoric stress tensor $\mathbf{S}$ according to

$$q = \sqrt{\frac{3}{2}\mathbf{S} : \mathbf{S}}, \tag{5.26}$$

where the deviatoric stress tensor is defined as

$$\mathbf{S} = \sigma - p\mathbf{I}, \tag{5.27}$$

with σ denoting the Cauchy stress tensor, p the hydrostatic pressure stress, and $\mathbf{I}$ the identity tensor. Since the yield function depends exclusively on the deviatoric stress components, hydrostatic pressure does not affect the yielding condition [20].

Plastic deformation is governed by an associated flow rule, meaning that the direction of plastic strain evolution is normal to the yield surface in stress space. As a result, the plastic deformation predicted by the von Mises model is volume invariant, implying that plastic strains do not produce volumetric changes in the material [20].

Under the assumption of perfect plasticity, the yield surface does not evolve with increasing plastic strain. Consequently, once the yield stress is reached the material continues to deform plastically at constant stress.

5.6.5 Plastic Material Parameters and Numerical Implementation

The plastic response of the matrix material was implemented in Abaqus using the classical von Mises plasticity formulation with perfectly plastic behaviour. In this framework, the onset of plastic deformation is defined through the uniaxial yield stress, while no hardening evolution is considered after yielding.

The yield stress of the PPS matrix was defined based on the compressive strength of the material, which was taken as $\sigma_y = 148$ MPa. This value was selected because the stress state in the impact contact region is predominantly compressive, particularly for the energy levels considered in the present study where perforation does not occur.

In Abaqus, the perfectly plastic behaviour was defined through the `*Plastic` material option by specifying the yield stress and the corresponding equivalent plastic strain. For perfect plasticity, the yield stress remains constant with increasing plastic strain, resulting in the following material definition

```
*Plastic
148., 0.
148., 1.
```

where the first pair corresponds to the onset of yielding and the second entry enforces constant stress during further plastic deformation.

The equivalent plastic strain (PEEQ) output variable was requested in order to monitor the spatial distribution and magnitude of plastic deformation within the impact region [20].

5.6.6 Numerical Response of the von Mises Plasticity Model

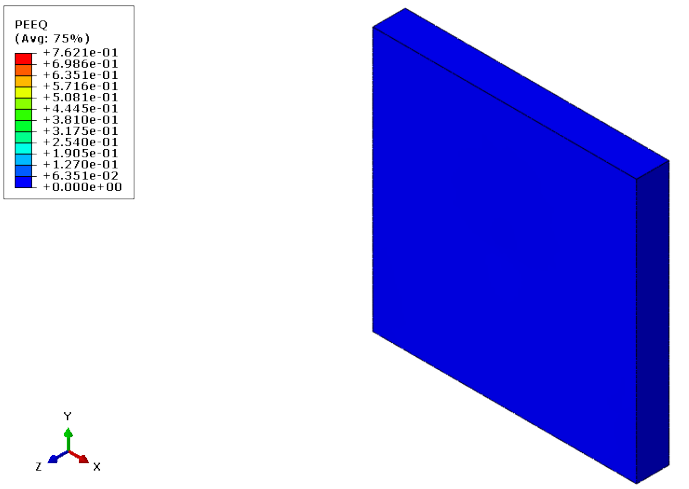
The von Mises plasticity formulation was first introduced to investigate the effect of matrix yielding on the predicted impact response of the laminate. The compressive strength of the PPS matrix, $\sigma_y = 148$ MPa, was adopted as the yield stress, assuming that the stress state in the impact region would be primarily governed by compressive stresses during the contact event.

Figure 5.79 presents the predicted equivalent plastic strain (PEEQ) distribution for the 27 J impact case. Three views are shown in order to illustrate the spatial distribution of plastic deformation across the laminate thickness: the front surface (impact side), a through-thickness section, and the back surface.

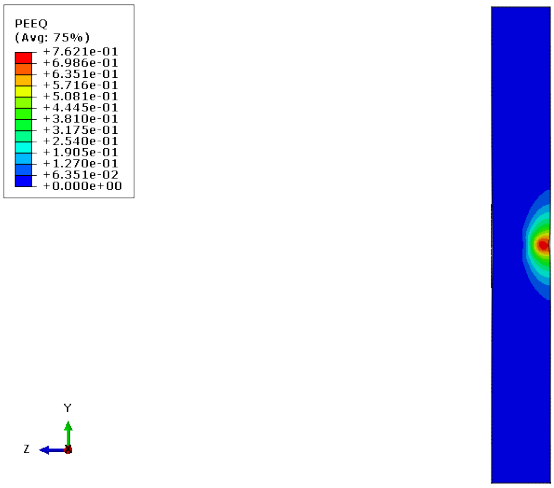
The results indicate that plastic deformation does not develop in the impact contact region beneath the impactor. Instead, plastic strains appear predominantly on the tensile back face of the laminate, where bending-induced tensile stresses reach their maximum values during the impact event. The through-thickness view further confirms that the plastic zone originates near the tensile surface and gradually decreases towards the interior of the laminate.

This behaviour can be attributed to the pressure-independent nature of the von Mises yield criterion. Since yielding depends only on the deviatoric component of the stress tensor, the hydrostatic pressure generated beneath the impactor does not contribute to the yielding condition. Consequently, the compressive stress state present in the impact contact region does not trigger plastic deformation in the matrix material when using the von Mises formulation.

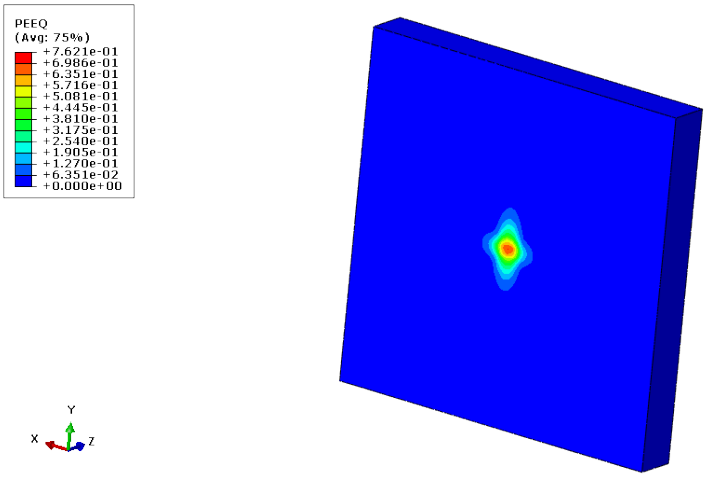
As a result, the introduction of von Mises plasticity does not produce any indentation behaviour on the impacted surface. The predicted plastic deformation is restricted to the tensile back face of the laminate, indicating that a pressure-dependent plasticity formulation is required to properly represent the mechanical response of the PPS matrix under impact loading.



(a) Front surface (impact side)



(b) Through-thickness section



(c) Back surface

Figure 5.79: Equivalent plastic strain (PEEQ) distribution predicted using the von Mises perfectly plastic model for the 27 J impact case. Plastic deformation develops primarily on the tensile back face of the laminate, while negligible plastic strain is observed on the impact surface. The through-thickness view confirms that plasticity is concentrated near the tensile side of the laminate.

5.6.7 Drucker-Prager Plasticity Formulation

To account for the pressure sensitivity of polymer materials and overcome the limitations observed with the von Mises formulation, a Drucker-Prager plasticity model was investigated. Unlike the von Mises criterion, which is pressure independent, the Drucker–Prager model introduces a dependence on the hydrostatic stress component, allowing the material strength to increase with confining pressure. This behaviour is characteristic of frictional materials such as soils, rocks, and several polymeric materials [20].

The extended Drucker-Prager models available in Abaqus define yielding through a yield surface expressed in the meridional $(p\text{–}t)$ stress plane, where p represents the equivalent pressure stress and t corresponds to the deviatoric stress measure. In the linear formulation adopted in this work, the yield criterion is expressed as

$$F = t - p\tan\beta - d = 0 \tag{5.28}$$

where β is the friction angle defining the slope of the yield surface in the p–t plane and d is the material cohesion. The pressure stress p is defined as

$$p = -\frac{1}{3}\mathrm{tr}(\sigma) \tag{5.29}$$

where σ is the Cauchy stress tensor. The variable t represents the deviatoric stress component and therefore accounts for the shear contribution of the stress state [20].

The parameters β and d can be estimated directly from the tensile and compressive strengths of the material. For the PPS matrix considered in this work, the tensile and compressive strengths were taken as $\sigma_t = 102$ MPa and $\sigma_c = 148$ MPa, respectively. The friction angle is obtained from

$$\tan\beta = \frac{\sigma_c + \sigma_t}{3(\sigma_c - \sigma_t)} \tag{5.30}$$

which yields

$$\tan\beta = \frac{148 + 102}{3(148 - 102)} = 0.552 \tag{5.31}$$

resulting in a friction angle of

$$\beta \approx 28.9^\circ. \tag{5.32}$$

The cohesion parameter d can be obtained from the compressive yield stress using

$$d = \left(1 - \frac{\tan\beta}{3}\right)\sigma_c \tag{5.33}$$

which leads to

$$d \approx 120.8 \text{ MPa}. \tag{5.34}$$

In the deviatoric stress plane, the Drucker-Prager yield surface is controlled by the parameter K, defined as the ratio between the yield stress in triaxial tension and triaxial compression. For the yield surface to remain convex, this parameter must satisfy $0.778 \leq K \leq 1.0$. In the present study, $K = 1.0$ was assumed, resulting in a circular yield surface in the deviatoric plane [20].

Plastic flow is governed by a flow potential of the form

$$G = t - p \tan \psi \tag{5.35}$$

where ψ is the dilation angle controlling the volumetric component of plastic deformation. For polymeric materials, it is common to assume nonassociated flow where $\psi < \beta$. In the present work a dilation angle of $\psi = 0^\circ$ was adopted, corresponding to incompressible plastic flow [20].

The material parameters required for the linear Drucker–Prager formulation were derived from the tensile and compressive strengths of the PPS matrix material. Table 5.35 summarizes the parameters adopted in the numerical simulations.

Table 5.35: Drucker-Prager plasticity parameters derived from PPS strength values.

Parameter	Value
Tensile strength σ_t	102 MPa
Compressive strength σ_c	148 MPa
Friction angle β	28.9°
Cohesion d	120.8 MPa
Deviatoric plane parameter K	1.0
Dilation angle ψ	0°

5.6.8 Numerical Response of the Drucker-Prager Plasticity Model

The Drucker-Prager plasticity model was introduced to account for the pressure sensitivity of the polymer matrix material and to overcome the limitations observed with the von Mises formulation. In contrast to the pressure-independent von Mises criterion, the Drucker-Prager model allows the yielding condition to depend on the hydrostatic stress component, which is expected to be significant in the compressive contact region beneath the impactor.

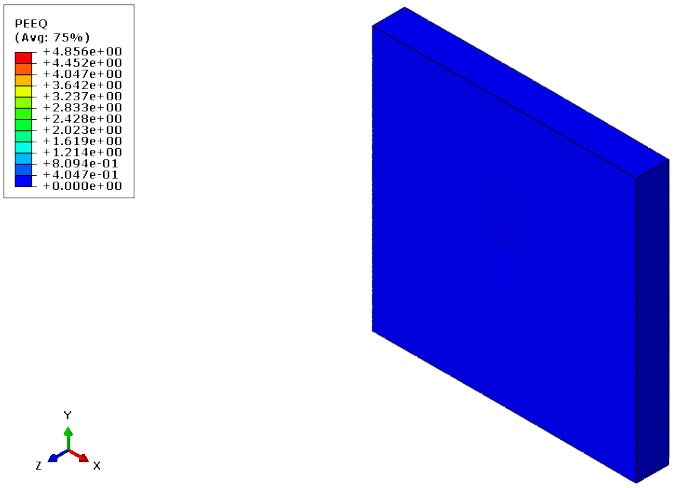
Figure 5.80 presents the predicted equivalent plastic strain (PEEQ) distribution obtained using the baseline parameter set defined in Table 5.35. Three views are shown to illustrate the spatial

distribution of plastic deformation: the front surface (impact side), a through-thickness section, and the back surface of the laminate.

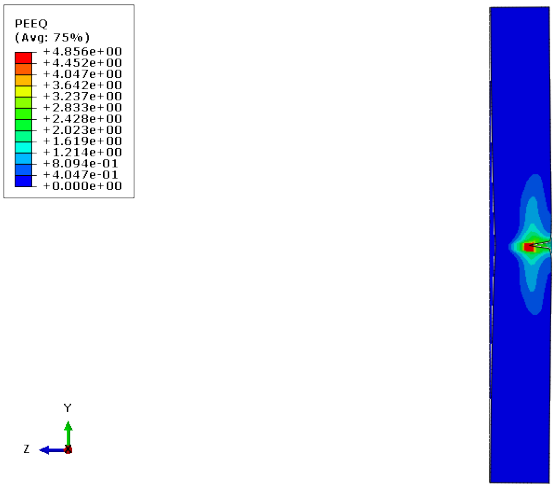
The results indicate that plastic deformation remains primarily localized on the tensile back face of the laminate, where bending-induced tensile stresses reach their maximum values during the impact event. In contrast, no plastic deformation is observed on the impact surface beneath the impactor. The through-thickness section further confirms that the plastic zone originates near the tensile side of the laminate and gradually decreases toward the interior.

Additional simulations were performed to evaluate the sensitivity of the response to variations in the Drucker-Prager parameters, including different friction and dilation angles. These variations produced minor changes in the shape of the plastic strain distribution on the back surface, but the overall structural response remained essentially unchanged and no plastic zone developed beneath the impactor.

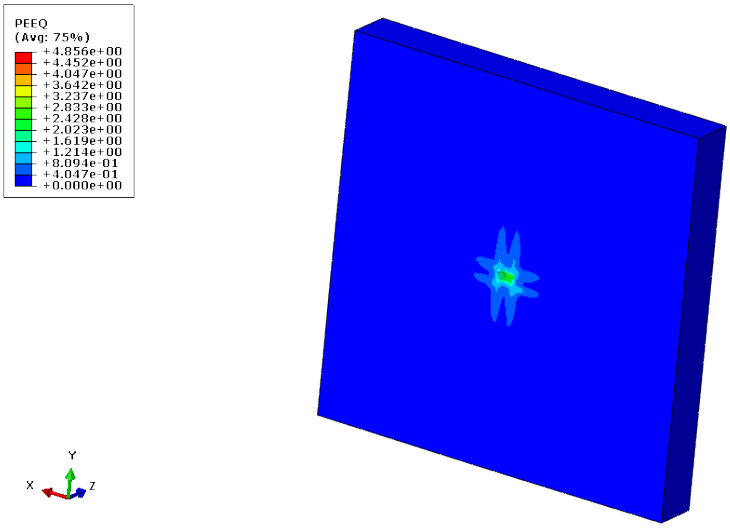
The results therefore indicate that, although the Drucker-Prager model introduces pressure-dependent yielding, it remains unable to reproduce the indentation behaviour expected in the impact region. The predicted plastic deformation continues to be dominated by bending-induced tensile stresses on the back face of the laminate.



(a) Front surface (impact side)



(b) Through-thickness section



(c) Back surface

Figure 5.80: Equivalent plastic strain (PEEQ) distribution predicted using the Drucker–Prager plasticity model for the 27 J impact case. Plastic deformation develops primarily on the tensile back face of the laminate, while no plastic zone forms beneath the impactor. The through-thickness view confirms that the plastic strain is concentrated near the tensile side of the laminate.

5.6.9 Verification of Contact and Element Formulations

Before introducing alternative plasticity formulations, additional verification analyses were performed in order to ensure that the absence of indentation behaviour was not caused by numerical modelling artefacts. Two aspects of the finite element model were investigated: the element formulation used for the solid elements in the impact zone and the contact formulation between the impactor and the laminate.

First, the influence of the element integration scheme was evaluated by comparing reduced integration elements (C3D8R) with fully integrated elements (C3D8). The numerical results showed negligible differences in the predicted structural response and plastic strain distribution, indicating that the element formulation did not influence the development of plastic deformation in the impact region.

Second, the contact formulation was investigated by comparing the general contact algorithm with an explicit contact pair definition between the impactor and the laminate surface. In addition, the penalty stiffness scaling used in the contact definition was varied in order to evaluate its influence on the predicted response. These modifications resulted in significantly increased computational time when using contact pairs, but produced essentially identical impact responses and plastic strain distributions.

The verification analyses therefore confirmed that the absence of indentation behaviour observed with the von Mises and Drucker–Prager plasticity models was not caused by numerical artefacts related to the contact modelling or the element formulation. The results suggest that the underlying limitation originates from the plasticity formulation itself rather than from the numerical implementation of the finite element model.

Table 5.36: Verification tests performed to evaluate the influence of numerical modelling parameters.

Test	Modification	Observation
Element formulation	C3D8R → C3D8	No change in plastic strain distribution
Contact algorithm	General contact → Contact pair	Identical structural response
Penalty stiffness	Increased scaling factor	No influence on indentation behaviour

5.6.10 Crushable Foam Plasticity Model

Since both the von Mises and Drucker–Prager plasticity formulations were unable to reproduce the indentation behaviour observed experimentally, a crushable foam plasticity model was investigated. This constitutive model is designed for materials that undergo permanent deformation through volumetric compaction, such as polymeric foams, metallic foams, or cellular materials. Unlike classical plasticity models, which assume plastic flow occurs at constant volume,

the crushable foam formulation allows for significant volumetric plastic strain associated with material densification under compressive loading [20].

The model defines a yield surface that is circular in the deviatoric stress plane and elliptical in the meridional $(p–q)$ stress plane, where p is the hydrostatic pressure stress and q is the von Mises equivalent stress. This representation enables the model to capture both shear-dominated yielding and pressure-driven compaction mechanisms. Two hardening formulations are available in Abaqus: isotropic hardening and volumetric hardening. In the present work, the volumetric hardening formulation was adopted, where the evolution of the yield surface is governed by the volumetric plastic strain associated with progressive compaction of the material [20].

In the volumetric hardening formulation, the yield surface can be expressed as

$$F = \sqrt{q^2 + \alpha^2 (p - p_0)^2} - B = 0, \tag{5.36}$$

where q is the equivalent deviatoric stress, p is the hydrostatic pressure stress, p_0 represents the hydrostatic tensile strength, and α and B control the shape and size of the yield surface. As plastic deformation progresses, the yield surface expands due to the accumulation of volumetric plastic strain, representing the densification behaviour typical of compressible materials [20].

The shape of the yield surface is governed by strength ratios defined from the compressive and hydrostatic yield stresses. The compression yield stress ratio k is defined as the ratio between the initial yield stress in uniaxial compression σ_c^0 and the hydrostatic compressive yield stress p_c^0,

$$k = \frac{\sigma_c^0}{p_c^0}. \tag{5.37}$$

Since hydrostatic test data were not available, it was assumed that the hydrostatic compressive strength is approximately equal to the uniaxial compressive yield stress of the PPS matrix. This assumption results in a value of $k = 1.0$, which corresponds to a circular yield surface in the deviatoric plane [20].

A second parameter controlling the yield surface is the hydrostatic yield stress ratio k_t, defined as

$$k_t = \frac{p_t}{p_c^0}, \tag{5.38}$$

where p_t is the hydrostatic tensile yield stress. Using the measured tensile and compressive strengths of the PPS matrix ($\sigma_t = 102$ MPa and $\sigma_c = 148$ MPa), this ratio can be approximated as

$$k_t \approx \frac{102}{148} \approx 0.689. \tag{5.39}$$

Plastic hardening in the crushable foam model is defined through a tabular relationship between the uniaxial compressive yield stress and the corresponding plastic strain. The first data pair corresponds to the onset of yielding and is defined as $(148\ \text{MPa}, 0)$, while subsequent values describe the progressive compaction and densification behaviour of the material under increasing compressive strain [20].

By allowing plastic deformation to occur through volumetric compaction rather than purely distortional deformation, the crushable foam formulation provides a more realistic representation of the compressive behaviour of the polymer matrix in the impact contact region beneath the impactor.

The material parameters adopted for the crushable foam plasticity model are summarized in Table 5.37.

Table 5.37: Crushable foam plasticity parameters adopted for the matrix material.

Parameter	Value
Compressive yield stress σ_c^0	148 MPa
Tensile strength σ_t	102 MPa
Compression yield stress ratio k	1.0
Hydrostatic yield stress ratio k_t	0.689

5.6.11 Crushable Foam Plasticity Model Response

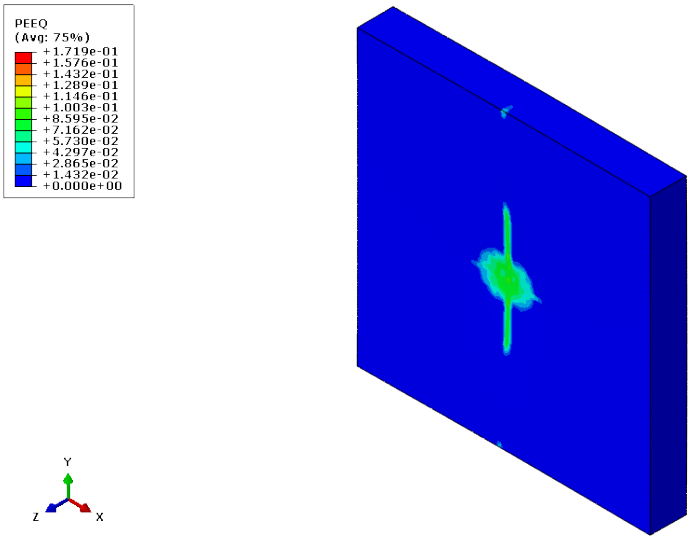
The crushable foam plasticity formulation was implemented in order to capture the pressure-dependent yielding behaviour of the PPS matrix under the highly compressive stress state generated during impact. Unlike the von Mises model, which assumes pressure-independent yielding, and the Drucker–Prager formulation, which introduces a linear pressure dependence, the crushable foam model employs an elliptical yield surface in the p–q stress space that allows the material to undergo volumetric compaction during plastic deformation.

Within the present bi-phasic modelling approach, plasticity is restricted to the solid elements representing the matrix phase. This enables the model to capture the permanent deformation associated with matrix yielding in the impact region while maintaining the intralaminar damage description provided by the Hashin failure criterion in the continuum-shell elements representing the fibre-dominated laminate.

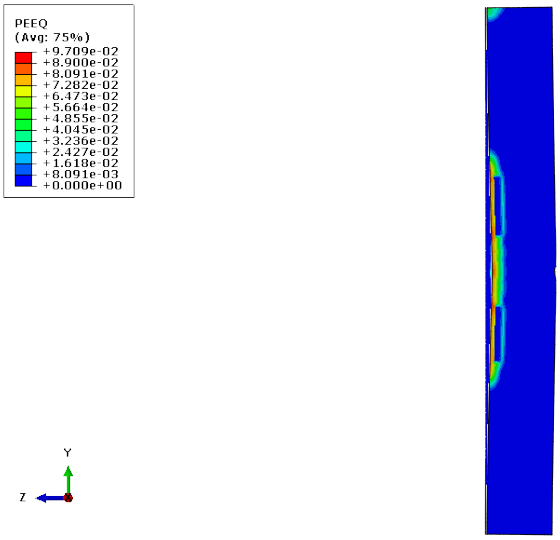
Figure 5.81 illustrates the equivalent plastic strain (PEEQ) distribution obtained for the 27 J impact case. The results show that plastic deformation is highly localised beneath the impactor,

forming a concentrated plastic zone in the matrix-dominated region directly below the contact area. This behaviour is consistent with the development of high hydrostatic compressive stresses during the early stages of the impact event.

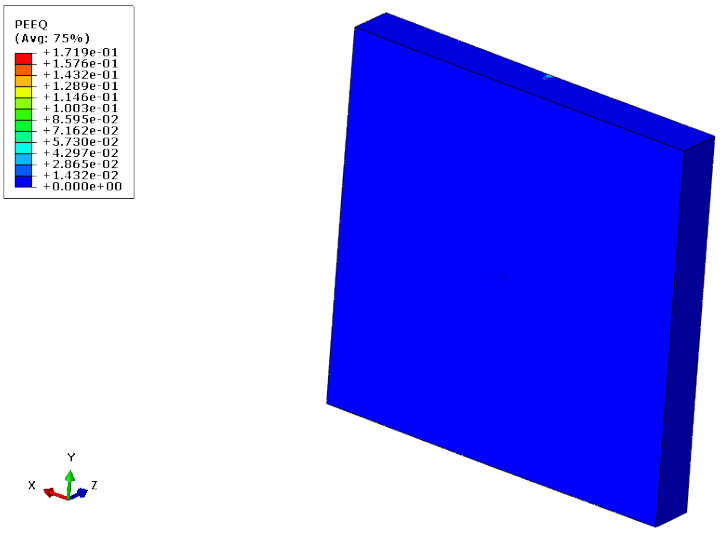
The through-thickness view further highlights the formation of a vertical plastic band extending across the thickness of the specimen, indicating the propagation of compressive stresses through the laminate during indentation. This behaviour is characteristic of pressure-sensitive plasticity models and reflects the volumetric compaction mechanism captured by the crushable foam formulation. As a result, the model is able to reproduce the localised indentation behaviour observed experimentally while maintaining a physically consistent representation of matrix yielding in the impact zone.



(a) Front view



(b) Through-thickness view



(c) Back view

Figure 5.81: Equivalent plastic strain (PEEQ) distribution obtained with the crushable foam plasticity model for the 27 J impact case. Plastic deformation localises beneath the impactor and forms a through-thickness plastic zone associated with the indentation mechanism.

5.6.12 Deterministic Sensitivity Analysis of the Crushable Foam Plasticity Model

Following the implementation of the crushable foam plasticity formulation, a sensitivity analysis was performed in order to evaluate the influence of the matrix yield strength on the predicted impact response of the laminate. The objective of this study is to assess how variations in the compressive yielding threshold of the matrix phase affect the development of indentation, the evolution of back-face cracking, and the global dynamic response of the structure.

Within the present bi-phasic modelling framework, plasticity is confined to the solid elements representing the matrix phase in the impact region. As a result, the yield strength parameter directly controls the onset of permanent deformation beneath the impactor and therefore influences the extent of the indentation and the stress distribution transmitted through the laminate thickness.

In order to isolate the influence of matrix plasticity, the intralaminar damage properties were kept constant throughout this analysis. The Hashin fracture energies adopted correspond to the calibrated values previously obtained from the sensitivity study performed using the continuum-shell laminate model.

The sensitivity study was conducted for two representative impact energy levels, namely 14.2 J and 27 J. These cases correspond to distinct damage regimes observed in the experimental campaign and allow the influence of plasticity to be analysed under different loading severities. For each energy level, the compressive yield strength of the matrix phase was varied and the resulting simulations were evaluated in terms of indentation development, back-face crack length, and the acceleration-time response of the impactor.

Impact Response at 14.2 J

The influence of the matrix yield strength on the impact response was first investigated for the 14.2 J impact case. In this sensitivity study, the compressive yield strength defined in the crushable foam plasticity model was progressively reduced (baseline, -10%, -20%, and -30%) while keeping all remaining modelling parameters unchanged. In particular, the intralaminar Hashin fracture energies were fixed to the calibrated values previously obtained from the sensitivity study performed using the continuum-shell model, so that the observed changes can be attributed primarily to the matrix plasticity response.

Table 5.38 summarises the predicted back-face crack length and indentation depth. For all yield-strength levels, the model does not predict back-face crack propagation ($L_c^{\mathrm{CAE}} = 0$), despite the measured experimental crack length of 10.3 mm. In contrast, the predicted indentation is clearly sensitive to the yield strength: decreasing the yielding threshold increases the permanent deformation beneath the impactor, leading to a progressive increase of the indentation depth from 3.92 mm

(baseline) up to 5.54 mm (-30% σ_y), therefore reducing the deviation to the experimental indentation (7.4 mm) from approximately 47% to 25%. This behaviour indicates that, at 14.2 J, matrix plasticity primarily governs the local indentation response, whereas the current configuration is still unable to trigger the tensile-dominated back-face cracking observed experimentally.

Table 5.38: Sensitivity of the crushable foam plasticity model to the matrix yield strength for the 14.2 J impact case.

σ_y variation	L_c^{CAE} [mm]	L_c^{Exp} [mm]	$\mathrm{Indentation}_{\mathrm{CAE}}$ [mm]	$\mathrm{Indentation}_{\mathrm{Exp}}$ [mm]
Baseline	0.00	10.3	3.92	7.40
-10%	0.00	10.3	4.38	7.40
-20%	0.00	10.3	4.84	7.40
-30%	0.00	10.3	5.54	7.40

The global dynamic response is shown in Figure 5.82, where the acceleration–time histories for the different yield-strength levels are compared with the experimental records. The plasticity-enabled bi-phasic responses exhibit a higher peak acceleration than the experiments, and the differences between the yield-strength variants are relatively small at the global scale. Nevertheless, the decreasing yield strength leads to a slightly more compliant response and a marginally earlier unloading, consistent with the increased permanent deformation predicted in Table 5.38. Overall, the 14.2 J results suggest that yield-strength variations are effective in tuning the indentation magnitude, while the back-face crack response remains controlled by other damage and energy dissipation mechanisms.

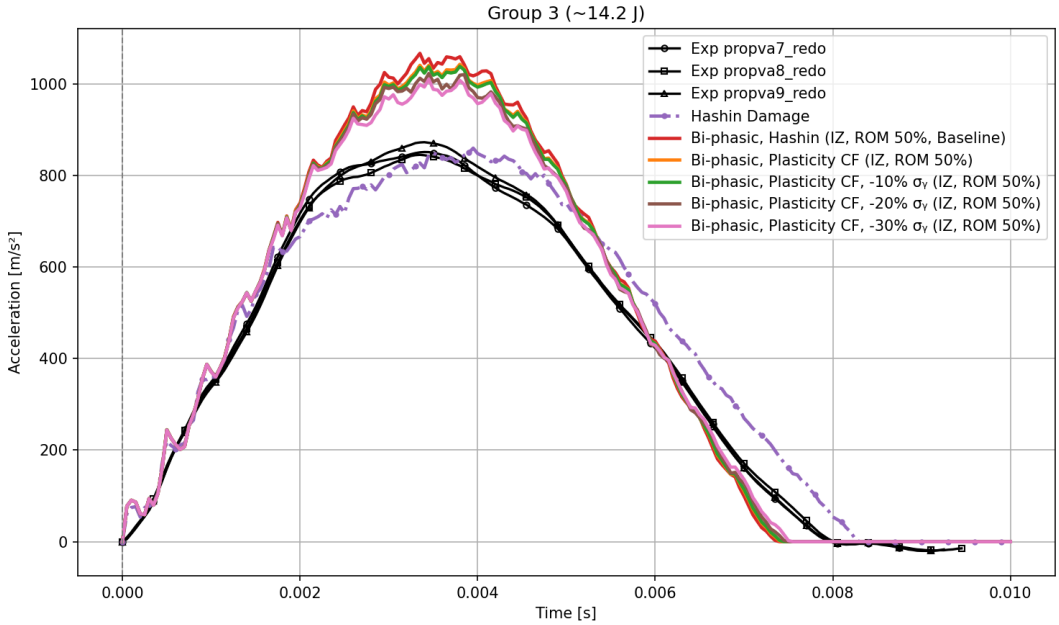


Figure 5.82: Acceleration-time response for the 14.2 J impact case (Group 3), comparing the experimental curves with the bi-phasic model predictions for the baseline crushable foam plasticity configuration and reduced yield strength levels (-10%, -20%, and -30% σ_y).

Impact Response at 27 J

The influence of the matrix yield strength on the impact response was further investigated for the 27 J impact case, which corresponds to the highest energy level considered in the experimental campaign. At this impact severity, both significant indentation and pronounced back-face cracking are observed experimentally, making this case particularly relevant for evaluating the interaction between matrix plasticity and intralaminar damage mechanisms.

As in the previous subsection, the compressive yield strength of the matrix phase defined in the crushable foam plasticity model was progressively reduced from the baseline value to -10%, -20%, and -30% in order to evaluate the sensitivity of the predicted response. All other modelling parameters remained unchanged, including the Hashin fracture energies obtained from the previous continuum-shell sensitivity study.

Table 5.39 summarises the predicted back-face crack length and indentation depth for the different yield-strength configurations. In contrast to the 14.2 J case, crack propagation is now predicted by the model. For the baseline configuration, the numerical crack length reaches 9.23 mm, which corresponds to a deviation of approximately 51% relative to the experimental value of 19 mm. Reducing the matrix yield strength progressively increases the predicted crack length, reaching 12.92 mm for the -30% σ_y case and reducing the deviation to approximately 32%.

Table 5.39: Sensitivity of the crushable foam plasticity model to the matrix yield strength for the 27 J impact case.

σ_y variation	L_c^{CAE} [mm]	L_c^{Exp} [mm]	$\mathrm{Indentation}_{\mathrm{CAE}}$ [mm]	$\mathrm{Indentation}_{\mathrm{Exp}}$ [mm]
Baseline	9.23	19	6.00	9.00
-10%	10.62	19	5.77	9.00
-20%	12.00	19	5.30	9.00
-30%	12.92	19	6.00	9.00

The indentation response shows a more moderate sensitivity to the yield strength. The baseline model predicts an indentation of 6 mm, compared with the experimental value of 9 mm. Variations in the yield strength lead to relatively small changes in the predicted indentation, indicating that, at this higher impact energy, a significant portion of the impact energy is dissipated through damage propagation rather than purely through plastic deformation.

The global dynamic response is illustrated in Figure 5.83, where the acceleration–time histories for the different yield-strength levels are compared with the experimental record. The baseline elastic bi-phasic configuration produces the highest peak acceleration, indicating an overly stiff response when plasticity is not included. Introducing matrix plasticity reduces the peak acceleration and brings the numerical curves closer to the experimental response.

The influence of yield-strength variation on the global acceleration response remains relatively limited compared with the change introduced by the inclusion of plasticity itself. However, the reduction of the yield strength slightly decreases the peak acceleration and promotes a marginally earlier unloading, which is consistent with the increased damage propagation observed in the crack length predictions.

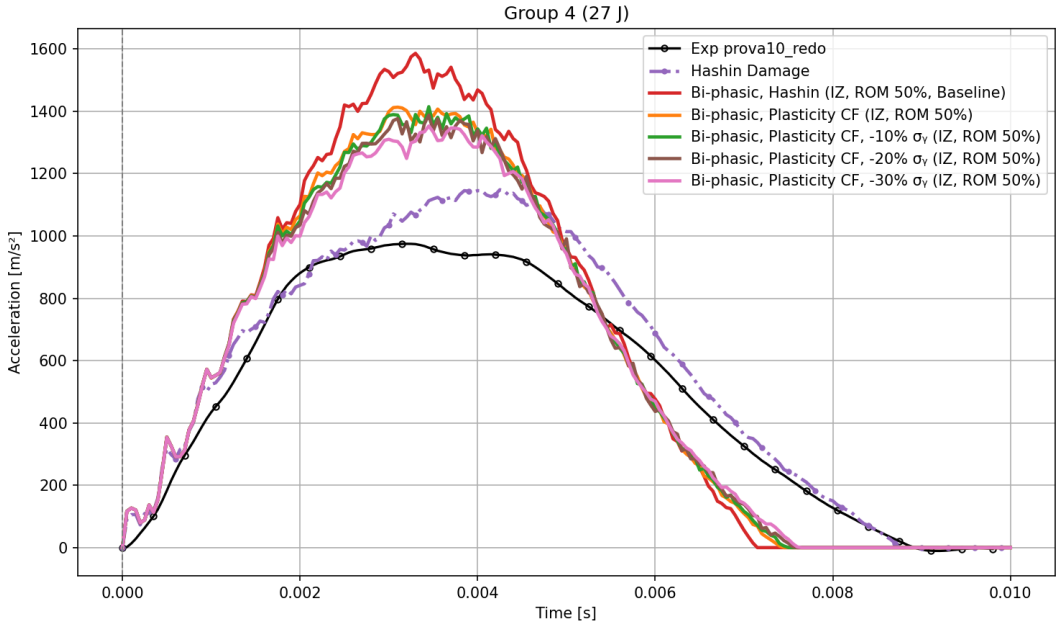


Figure 5.83: Acceleration-time response for the 27 J impact case (Group 4), comparing the experimental measurement with the predictions obtained using the bi-phasic model with crushable foam plasticity and reduced yield strength levels (-10%, -20%, and -30% σ_y).

5.6.13 Plastic Zone Evolution and Model Limitations

The evolution of the equivalent plastic strain distribution obtained from the crushable foam plasticity simulations provides further insight into the role of matrix plasticity in the impact response of the laminate. Figure 5.84 presents the predicted PEEQ distributions for the 27 J impact case for different reductions of the matrix yield strength.

For the baseline configuration, plastic deformation remains strongly localised beneath the impactor, forming a concentrated plastic zone aligned with the through-thickness direction. As the yield strength is progressively reduced (-10%, -20%, and -30% σ_y), the plastic deformation spreads further away from the contact region. In particular, the plastic zone develops a characteristic cross-shaped morphology around the indentation site, whose lateral arms progressively extend along the in-plane directions of the laminate.

This behaviour indicates that decreasing the matrix yield strength promotes the propagation of plastic deformation within the matrix-dominated region before the onset of intralaminar damage. As a result, the plastic zone becomes less confined to the immediate contact area and extends along preferential directions within the laminate. The enlargement of the cross-shaped plastic zone observed in Figure 5.84 is consistent with the increase in predicted crack length reported in the previous subsection, suggesting that the redistribution of stresses associated with matrix plasticity influences the development of tensile damage at the back surface.

Nevertheless, the interpretation of these results must consider the underlying assumptions of the constitutive model. The crushable foam plasticity model implemented in Abaqus assumes an isotropic material response and was originally developed for materials whose deformation is governed by cell-wall collapse, such as polymeric or metallic foams. In contrast, the matrix phase of fibre-reinforced thermoplastic composites exhibits a complex mechanical response influenced by the surrounding fibre architecture, leading to anisotropic or orthotropic behaviour at the mesoscale.

Consequently, the plastic strain distributions predicted by the present model should be interpreted primarily as a parametric representation of the influence of pressure-sensitive plasticity rather than a fully calibrated constitutive description of the PPS matrix. As highlighted in previous studies, reliable identification of the parameters governing pressure-dependent plasticity models such as Drucker-Prager or Crushable Foam requires dedicated experimental characterisation, including uniaxial compression tests and potentially hydrostatic loading experiments. Without such data, the yield strength used in the present work represents an approximate value derived from available material properties.

Despite these limitations, the sensitivity study demonstrates that matrix plasticity plays a key role in controlling the indentation behaviour and the distribution of stresses within the impact zone. The progressive enlargement of the plastic zone observed in Figure 5.84 highlights the importance of accurately characterising the compressive behaviour of the matrix phase in order to improve the predictive capability of bi-phasic impact models.

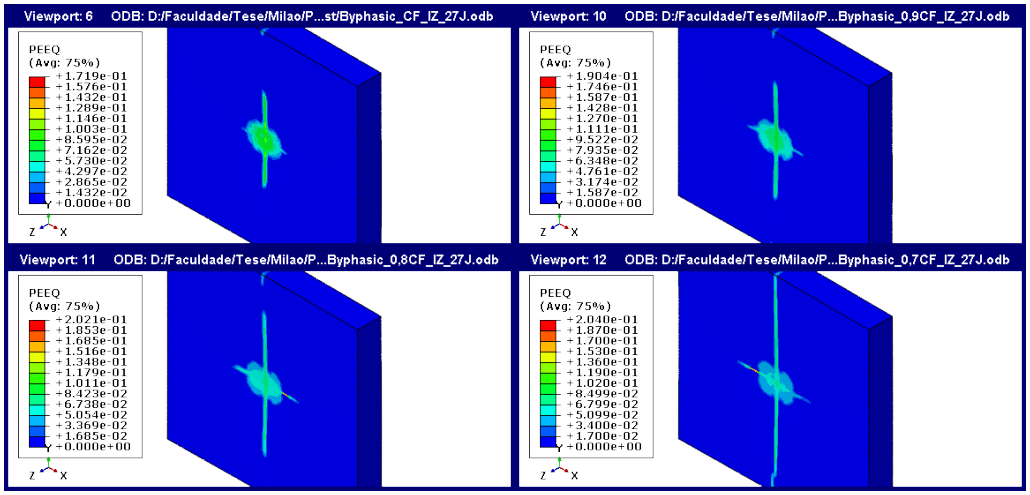


Figure 5.84: Equivalent plastic strain (PEEQ) distribution for the 27 J impact case obtained with the crushable foam plasticity model for different matrix yield strengths (baseline, -10%, -20%, and -30% σ_y). The plastic zone develops a cross-shaped morphology around the indentation region, whose lateral extension increases as the yield strength is reduced.

5.6.14 Influence of Intralaminar Fracture Energy in the Bi-Phasic Plastic Model

The intralaminar fracture energy values obtained from the sensitivity analysis of the elastic bi-phasic model were previously identified as unrealistically low for glass fibre reinforced thermoplastic composites. As discussed in the previous section, these values were therefore considered non-physical and were not adopted for the reference simulations presented in this work. Nevertheless, additional simulations were performed using these parameters in order to investigate the numerical behaviour of the model under such conditions. These analyses were conducted exclusively as a numerical investigation of the model behaviour and not as part of the calibration process.

The objective of this analysis was to evaluate the sensitivity of the impact response to unrealistically low fracture energy values and to better understand the interaction between intralaminar damage and matrix plasticity within the bi-phasic modelling framework. The resulting responses were analysed for both impact energies considered in this study (14.2 J and 27 J), allowing the influence of these non-physical parameters on crack propagation, indentation behaviour and global dynamic response to be assessed.

Impact Response at 14.2 J

Table 5.40 summarises the predicted crack length and indentation obtained with the reduced fracture energy. The numerical model predicts a crack length of 25.85 mm, which significantly exceeds the experimental value of 10.3 mm. At the same time, the predicted indentation is considerably reduced, reaching only 2.08 mm compared with the experimental value of 7.4 mm.

These results indicate that the reduction of the intralaminar fracture energy promotes premature damage propagation, leading to excessive crack growth while limiting the development of permanent indentation. In this configuration, damage propagation becomes the dominant energy dissipation mechanism during the impact event.

Table 5.40: Predicted crack length and indentation for the 14.2 J impact case using the reduced intralaminar fracture energy derived from the elastic bi-phasic model.

Energy [J]	L_c^{CAE} [mm]	L_c^{Exp} [mm]	$\mathrm{Indentation}_{\mathrm{CAE}}$ [mm]	$\mathrm{Indentation}_{\mathrm{Exp}}$ [mm]
14.2	25.85	10.3	2.08	7.40

Figure 5.85 presents the corresponding acceleration-time response. In contrast with the previous simulations performed with elastic solid elements, where the bi-phasic model exhibited a significantly stiffer response than the continuum-shell configuration, the introduction of matrix plasticity combined with the reduced fracture energy produces a markedly softer dynamic response. Interestingly, for the 14.2 J impact case, the resulting acceleration history closely resembles the behaviour

previously obtained with the continuum-shell model governed solely by the Hashin damage criterion. This observation suggests that the inclusion of plasticity partly compensates for the additional stiffness introduced by the solid elements of the bi-phasic formulation. However, this agreement should not be interpreted as a physical calibration of the model. The fracture energy values used in this configuration were previously identified as unrealistically low, and significant discrepancies remain regarding the experimental crack length data.

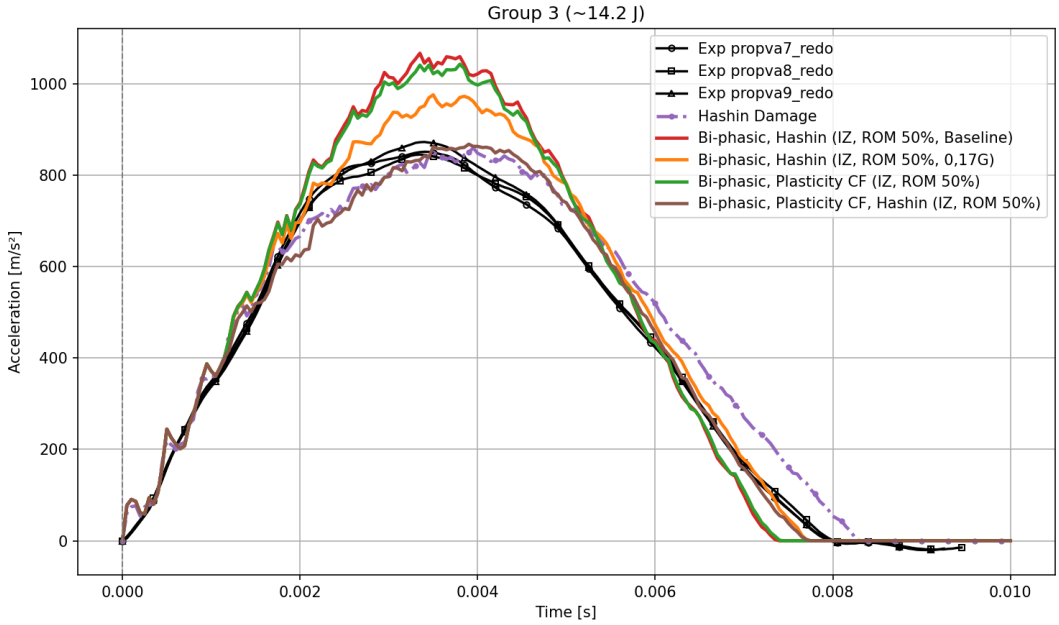


Figure 5.85: Acceleration-time response for the 14.2 J impact case using the reduced intralaminar fracture energy.

Impact Response at 27 J

A similar behaviour is observed for the 27 J impact case. As shown in Table 5.41, the predicted crack length increases to 32.33 mm, considerably exceeding the experimental value of 19 mm. At the same time, the predicted indentation remains limited to 2.08 mm, whereas the experimental indentation reaches 9 mm.

Table 5.41: Predicted crack length and indentation for the 27 J impact case using the reduced intralaminar fracture energy derived from the elastic bi-phasic model.

Energy [J]	L_c^{CAE} [mm]	L_c^{Exp} [mm]	$\mathrm{Indentation}_{\mathrm{CAE}}$ [mm]	$\mathrm{Indentation}_{\mathrm{Exp}}$ [mm]
27	32.33	19.00	2.08	9.00

A different behaviour is observed for the 27 J impact case, as shown in Figure 5.86. Although the introduction of plasticity reduces the stiffness of the bi-phasic model compared with the elastic configuration, the acceleration–time response no longer coincides with the continuum-shell

Hashin model as observed for the 14.2 J case. The dynamic response still shows similarities with the continuum-shell behaviour, but the interaction between matrix plasticity and damage propagation becomes more pronounced at the higher impact energy. Consequently, while the inclusion of plasticity improves the dynamic response compared with the purely elastic bi-phasic model, the results remain inconsistent with the experimental curves due to the excessive damage propagation associated with the unrealistically low fracture energy values.

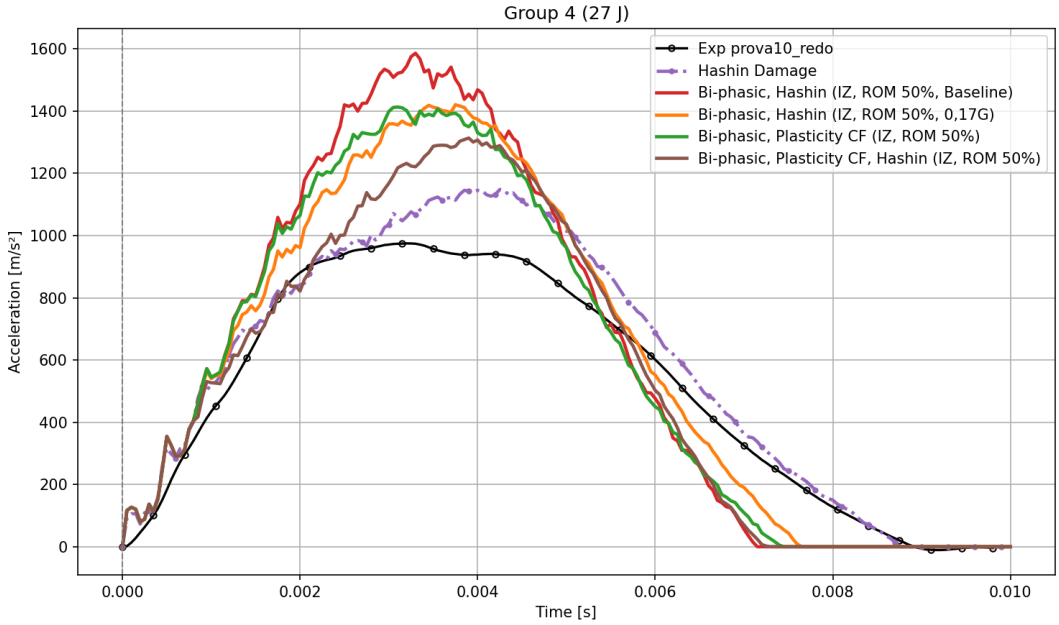


Figure 5.86: Acceleration-time response for the 27 J impact case using the reduced intralaminar fracture energy.

Plastic Deformation Patterns

The corresponding equivalent plastic strain distributions are illustrated in Figure 5.87. The results reveal that the plastic zone beneath the impactor develops a pronounced cross-shaped morphology whose arms extend along the in-plane laminate directions.

The extent of these plastic zones increases significantly when the fracture energy is reduced. Instead of remaining localised beneath the impactor, the plastic deformation propagates along the laminate directions, producing elongated plastic bands that resemble wrinkling or creasing patterns in the matrix material.

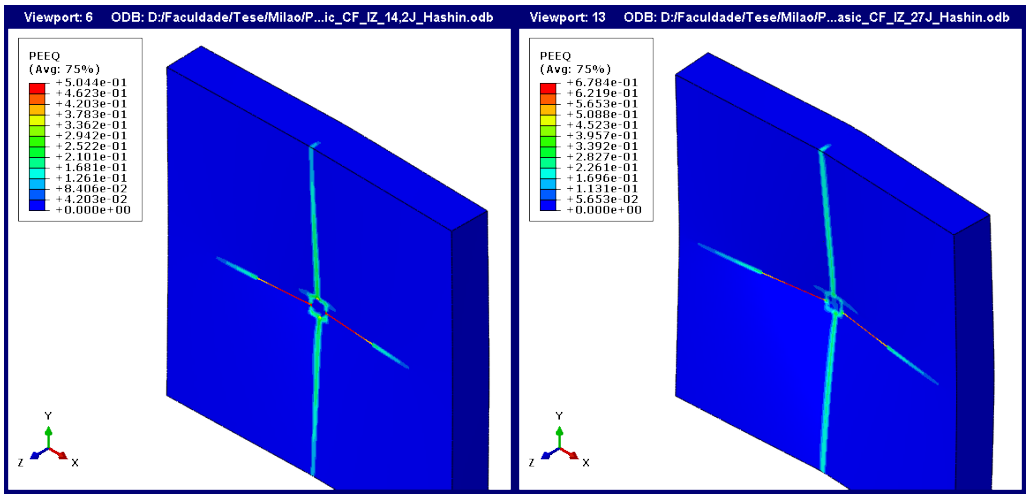


Figure 5.87: Equivalent plastic strain (PEEQ) distributions obtained using the reduced intralaminar fracture energy for the 14.2 J and 27 J impact cases.

The plastic deformation patterns predicted by the model further highlight the limitations of these parameters. However, such deformation patterns were not observed in the experimental specimens. In particular, the extended cross-shaped plastic zones predicted by the numerical model resemble wrinkling or creasing mechanisms developing within the matrix-dominated region of the laminate. This discrepancy suggests that the plastic response predicted by the crushable foam formulation may not fully represent the mechanical behaviour of the PPS matrix within the composite laminate. The crushable foam model assumes isotropic material behaviour, whereas the matrix phase in fibre-reinforced thermoplastic composites is constrained by the surrounding fibre architecture and may exhibit anisotropic or orthotropic characteristics at the mesoscale.

Furthermore, the results show that when the intralaminar fracture energy is reduced to these non-physical values, the indentation behaviour becomes nearly identical for both impact energies considered in this study. This indicates that damage propagation dominates the energy dissipation process, leading to premature stiffness degradation of the laminate and limiting the development of permanent indentation beneath the impactor.

Overall, these results confirm that the intralaminar fracture energy values derived from the elastic bi-phasic sensitivity study are not physically representative for the present composite system. While the reduction of fracture energy promotes crack propagation, it also produces excessive damage growth and suppresses the indentation behaviour observed experimentally. These simulations therefore serve primarily as a numerical investigation of the model behaviour under unrealistically low fracture energy values. Additional experimental characterisation of the compressive behaviour of the PPS matrix and thermoplastic composite system would therefore be required in order to properly calibrate pressure-dependent plasticity models and obtain physically representative parameters for the bi-phasic impact simulations.

For completeness, the computational cost associated with the different numerical models investigated in this chapter is reported in Appendix E. The appendix summarises the wall-clock simulation times obtained for each impact case and modelling strategy, providing an overview of the relative computational effort required by the elastic, intralaminar damage, cohesive, and bi-phasic formulations considered in this work.

Chapter 6

Post-Impact Non-Destructive Testing and Delamination Assessment

Following the experimental impact campaign and the development of the numerical modelling framework presented in the previous chapters, an additional experimental investigation was conducted to characterize the internal damage state of the impacted composite laminates. In fibre-reinforced composite structures, low-velocity impact events often generate complex damage mechanisms that develop beneath the laminate surface and may remain undetectable through visual inspection alone. NDT techniques therefore play an important role in post-impact assessment by enabling the identification and characterization of internal damage features such as delamination.

The objective of this chapter is to document the post-impact inspection campaign performed in the present work and to provide an experimental assessment of the internal damage generated by the impact events. Particular attention is given to the use of ultrasonic inspection for mapping the spatial distribution of delamination within the specimens. The inspection methodology, experimental setup, and processing procedures used to quantify the projected delamination footprint are presented in the following sections. The experimental observations obtained in this chapter provide an additional reference for interpreting the damage mechanisms discussed in the numerical modelling studies.

6.1 Motivation for Post-Impact Non-Destructive Evaluation

Low-velocity impact in laminated composite structures often generates complex internal damage mechanisms, including matrix cracking, fibre breakage and interlaminar delamination. A significant portion of this damage develops beneath the laminate surface and may therefore remain invisible during visual inspection. As a consequence, post-impact characterization techniques capable of probing the internal state of the laminate are required in order to properly assess the extent and spatial distribution of the induced damage.

In the present work, NDT is employed to experimentally characterize the internal damage produced by the impact events described in Chapter 4. The objective of this investigation is to provide an experimental assessment of the internal delamination footprint generated by the impacts and to document the actual damage state of the tested laminates.

It should be noted that the numerical simulations presented in Chapter 5 were developed prior to the ultrasonic inspection campaign and therefore do not rely on the NDT measurements for calibration. The ultrasonic results presented in this chapter should therefore be interpreted as an independent experimental observation of the internal damage distribution rather than a direct input used during model development.

Among the available NDT techniques for composite inspection, ultrasonic testing represents one of the most widely used methods for the detection of impact-induced delamination. Ultrasonic C-scan inspection enables the spatial mapping of internal damage through variations in the reflected signal amplitude, allowing the damaged region to be visualized and quantified over the inspected surface area. Although this technique does not directly resolve the full three-dimensional damage morphology, it provides a consistent planar projection of the delaminated region.

Advanced volumetric characterization techniques such as X-ray micro-computed tomography (μCT) can provide a more detailed three-dimensional representation of internal damage. However, such measurements were not available within the time frame of this work. The present chapter therefore focuses on ultrasonic inspection, which provides a practical approach for assessing the extent of impact-induced delamination and documenting the internal damage state of the tested specimens.

6.2 Ultrasonic Testing Campaign

Ultrasonic inspection was conducted to evaluate the internal delamination damage induced by impact loading. The primary objective of this experimental campaign was to obtain a projected planar footprint of the delaminated region suitable for quantitative comparison with the cohesive damage predictions obtained from the finite element simulations.

All specimens were inspected using an immersion ultrasonic configuration. During scanning, the laminate was fully submerged in water to ensure stable acoustic coupling and improved signal repeatability. The inspection system consisted of an Olympus OmniScan X3 acquisition unit equipped with a 5 MHz, 64-element phased-array probe (Olympus 5L64-NW1) mounted on a manual scanning fixture. The probe was positioned normal to the laminate surface and raster scanning was performed across the impacted region.

For clarity, the main ultrasonic inspection parameters adopted in the present experimental campaign are summarized in Table 6.1. These parameters define the acquisition configuration used during the immersion ultrasonic inspection and were kept consistent for all specimens analysed in this study.

Table 6.1: Main ultrasonic inspection parameters used in the present study.

Parameter	Value
Inspection mode	Immersion pulse–echo
Acquisition unit	Olympus OmniScan X3
Probe type	Phased array (Olympus 5L64-NW1)
Nominal frequency	5 MHz
Number of elements	64
Coupling medium	Water immersion
Scanning method	Manual raster scan
Damage metric	Back-face C-scan amplitude projection

Figure 6.1 presents the ultrasonic acquisition system used in this study, while Figure 6.2 shows the immersion scanning configuration adopted during inspection.

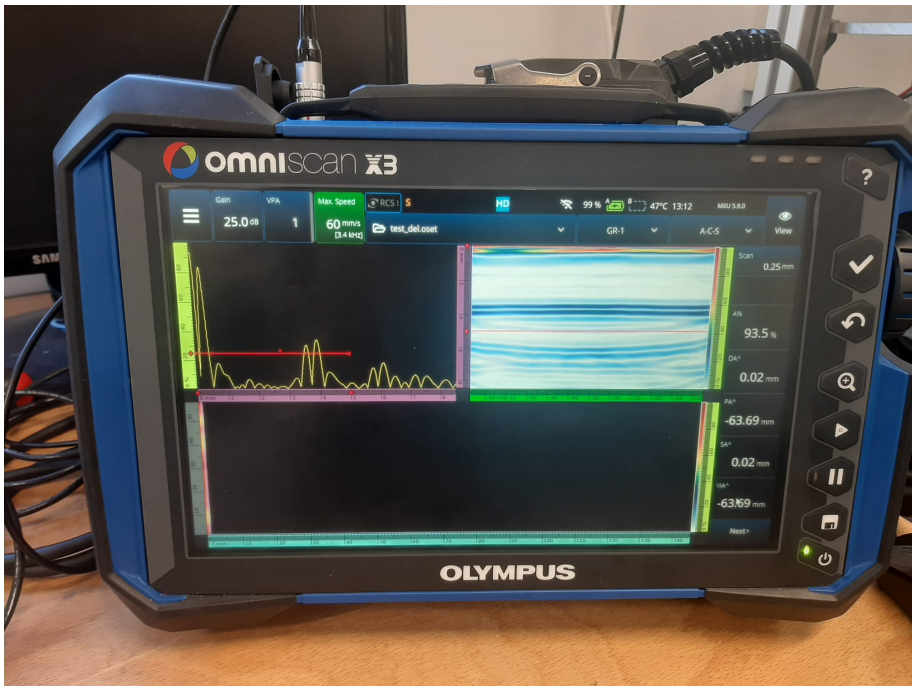


Figure 6.1: Olympus OmniScan X3 ultrasonic acquisition unit used for immersion inspection.

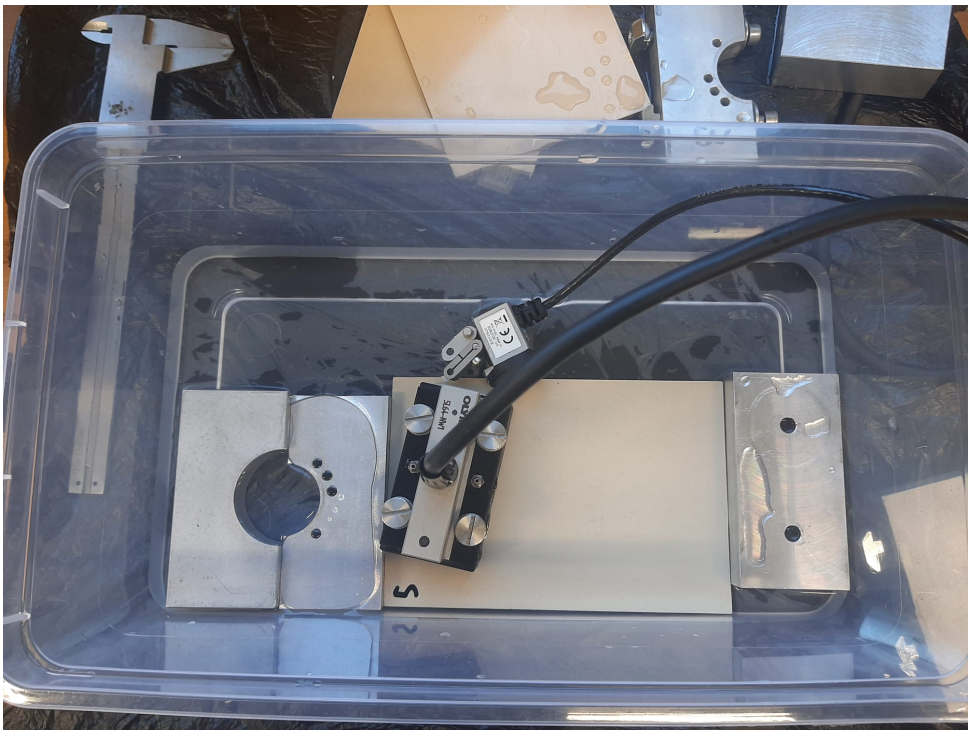


Figure 6.2: Immersion ultrasonic inspection setup with the specimen fully submerged in water. The probe was mounted on a manual scanning fixture and positioned normal to the laminate surface.

A pulse–echo configuration was employed. A back-wall gated acquisition strategy was adopted to monitor variations in the reflected signal amplitude associated with internal damage. In undamaged regions, a stable and coherent back-wall echo was observed, whereas impacted areas exhibited modifications in amplitude and waveform characteristics due to the presence of internal delamination and matrix cracking.

6.2.1 Representative A-Scan Response

The A-scan signal was used to qualitatively compare the ultrasonic response between healthy and impacted regions. In undamaged areas, the signal exhibited a well-defined front-surface reflection followed by a strong back-wall echo, indicating uninterrupted wave propagation through the laminate thickness. In contrast, impacted regions showed intermediate reflections within the laminate and a reduction or distortion of the back-wall echo, consistent with internal delamination acting as an acoustic impedance discontinuity.

Figure 6.3 illustrates representative A-scan signals obtained in both undamaged and damaged regions.

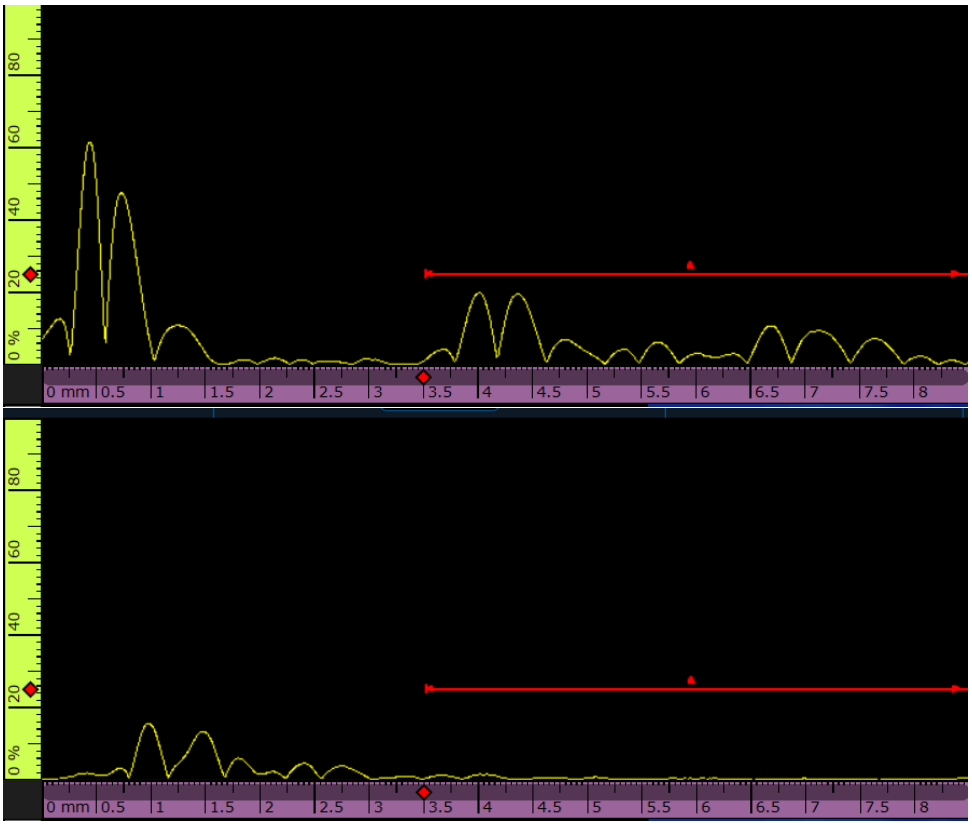


Figure 6.3: Representative A-scan signals: (top) undamaged region showing a stable back-wall echo; (bottom) impacted region exhibiting altered waveform characteristics and internal reflections due to delamination.

6.2.2 C-Scan Damage Mapping

C-scan maps were generated to visualize the spatial distribution of impact-induced damage. Two projections were analyzed: the top surface response and the back-face projection.

Figure 6.4 presents a representative surface C-scan. In this representation, the central region exhibits a colour variation associated with the indentation produced by the impact event. The red outer region corresponds to undamaged laminate, while the central colour contrast is primarily related to surface indentation effects rather than interlaminar delamination.

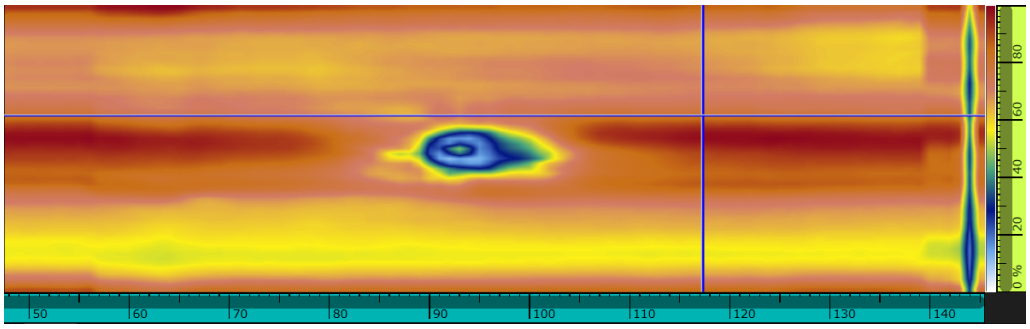


Figure 6.4: Surface C-scan representation of the impacted specimen. The central colour variation corresponds primarily to surface indentation effects, while the surrounding region represents undamaged material.

For quantitative analysis, the back-face projection was selected. This representation provides a clearer two-dimensional projection of the internal delamination footprint. In the adopted colour scale, undamaged regions appear predominantly blue, while delaminated areas are represented by high-contrast bright regions.

Figure 6.5 shows a representative back-face C-scan used for footprint quantification.

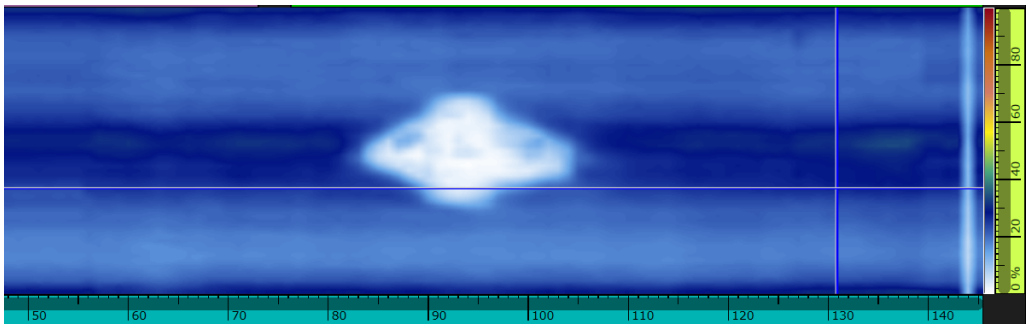


Figure 6.5: Back-face C-scan projection used for delamination footprint quantification. Bright regions correspond to internal delamination, while blue areas represent undamaged laminate.

The back-face C-scan therefore provides a two-dimensional projection of the internal delamination extent. This planar footprint is directly comparable to the cohesive damage area predicted by the numerical simulations.

6.3 C-Scan Results at Different Impact Energies

Back-face C-scan projections were obtained for all impacted specimens in order to evaluate the qualitative evolution of the delamination footprint with increasing impact energy. Three energy levels were investigated: 7.5 J, 14.2 J and 27 J. For each specimen, the scan was repeated three times to assess repeatability of the manually performed immersion inspection.

Impact Energy: 7.5 J

Figure 6.6 shows the representative back-face C-scan obtained for the specimen impacted at 7.5 J. A localized high-contrast region is observed at the impact location, corresponding to the internal delamination footprint. The affected area remains relatively compact and approximately elliptical, with limited lateral propagation.

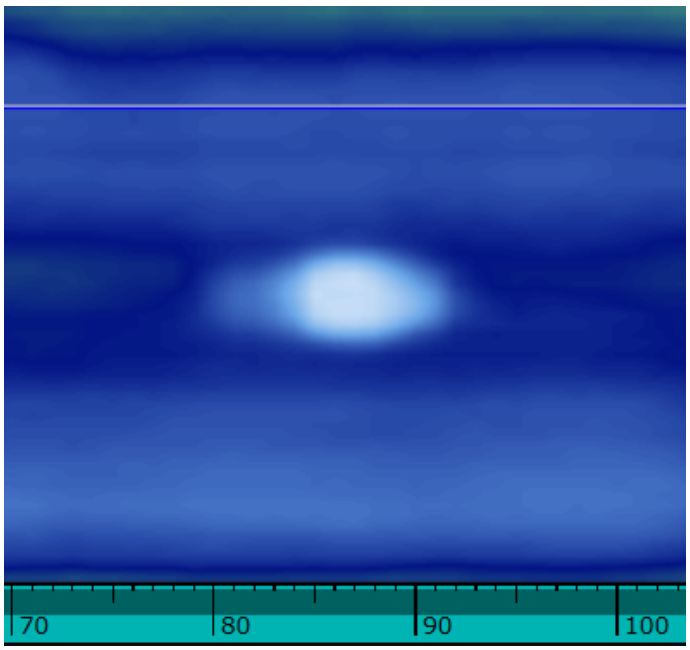


Figure 6.6: Back-face C-scan projection for the specimen impacted at 7.5 J. The bright region indicates the projected delamination footprint.

Impact Energy: 14.2 J

The C-scan corresponding to 14.2 J is presented in Figure 6.7. Compared to the 7.5 J case, the delamination footprint is visibly larger and exhibits increased lateral extension. The central region appears more diffused, indicating a broader internal damage zone.

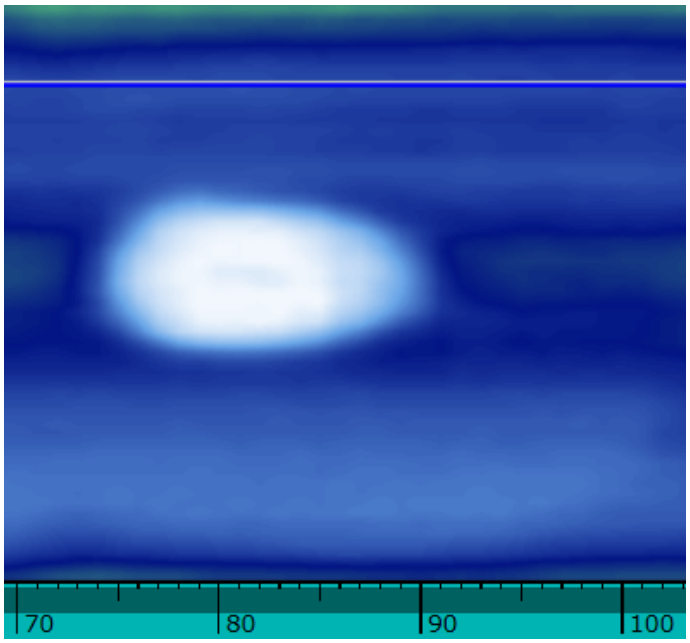


Figure 6.7: Back-face C-scan projection for the specimen impacted at 14.2 J. An increased delamination footprint is observed relative to the 7.5 J case.

Impact Energy: 27 J

The highest impact energy case (27 J) is shown in Figure 6.8. A significantly larger and more irregular delamination footprint is observed. The damage region extends further in the in-plane directions and exhibits a less uniform morphology, suggesting increased energy dissipation through interlaminar separation and matrix cracking mechanisms.

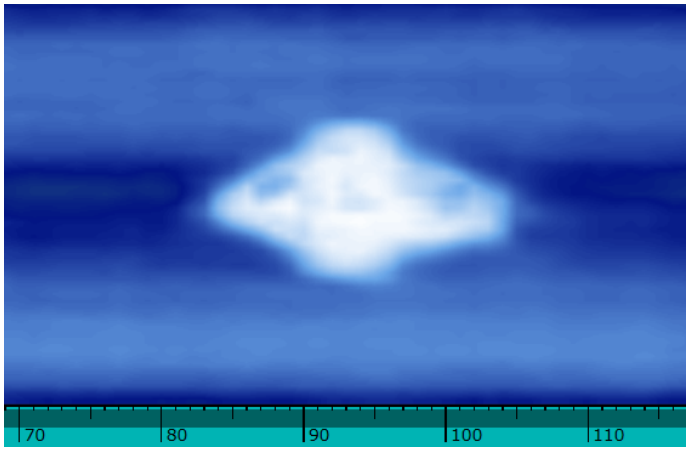


Figure 6.8: Back-face C-scan projection for the specimen impacted at 27 J. A substantially larger and more irregular delamination footprint is observed.

6.3.1 Qualitative Trend with Impact Energy

A clear increase in the projected delamination footprint is observed with increasing impact energy. The 7.5 J case exhibits a compact and localized damage region, the 14.2 J case shows moderate lateral propagation, and the 27 J case presents extensive in-plane delamination with increased morphological complexity.

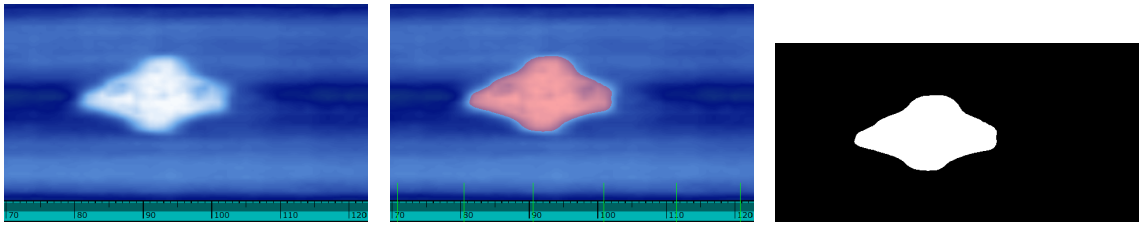
This qualitative trend is consistent with the expected increase in absorbed energy and interlaminar damage mechanisms as impact severity increases.

6.4 Quantitative Delamination Footprint and Repeatability

The projected delamination footprint was extracted from the back-face C-scan images using an image-based segmentation procedure implemented in Python, available in Appendix D. For each impact energy level, three independent immersion scans were performed in order to assess repeatability of the manually executed inspection and the robustness of the adopted damage metric.

6.4.1 Illustrative Example of the Segmentation Output (27 J)

To illustrate the output of the automated processing workflow, Figure 6.9 shows a representative segmentation result for the 27 J specimen. The original back-face C-scan is presented together with the corresponding overlay (detected footprint highlighted) and the final binary mask used for area computation.



(a) Back-face C-scan. (b) Overlay output. (c) Binary mask.

Figure 6.9: Example of the Python-based delamination footprint extraction applied to the 27 J specimen: original back-face C-scan, overlay highlighting the detected footprint, and the corresponding binary mask used for pixel counting.

6.4.2 Repeatability of Ultrasonic Measurements

Table 6.2 summarizes the extracted projected delamination area obtained from three repeated scans for each impact energy level.

Table 6.2: Extracted projected delamination area from repeated back-face C-scan measurements (three scans per specimen).

Impact Energy [J]	**Mean Area [mm^2]**	**Std. Dev. [mm^2]**	**Rel. Dev. [%]**
7.5	19.93	2.88	14.4
14.2	80.04	4.93	6.2
27	142.70	13.31	9.3

The variability between repeated scans remains within acceptable limits for all energy levels. The intermediate energy case (14.2 J) exhibits the lowest relative deviation, indicating high repeatability of both the immersion scanning procedure and the segmentation process. Slightly increased scatter is observed for the lowest and highest energy levels, which may be attributed to the smaller footprint sensitivity at 7.5 J and the increased irregularity of the delamination morphology at 27 J.

6.4.3 Energy Dependence of Delamination Area

A clear monotonic increase in projected delamination area is observed with increasing impact energy. The 7.5 J case results in a compact footprint of approximately 20 mm^2, while the 14.2 J impact produces a fourfold increase to approximately 80 mm^2. The highest energy level (27 J) leads to a substantially larger footprint of approximately 143 mm^2.

This quantitative trend is fully consistent with the qualitative observations from the C-scan maps and reflects the increased interlaminar damage propagation activated at higher impact severities. The extracted footprint therefore provides a consistent planar damage metric suitable for comparison with cohesive damage predictions from the numerical simulations.

6.4.4 Limitations of the Ultrasonic Damage Metric

It should be emphasized that the extracted delamination area represents a two-dimensional projection derived from amplitude-based C-scan contrast. The method does not resolve the through-thickness distribution of delamination or distinguish between multiple interlaminar interfaces.

The reported values therefore correspond to an apparent planar damage footprint rather than a full three-dimensional characterization of internal damage. Despite this limitation, the metric provides a consistent and repeatable comparative measure suitable for evaluating the evolution of impact-induced delamination and for subsequent correlation with numerical cohesive damage predictions.

6.5 Non-Destructive Testing Summary

The ultrasonic inspection performed in this work provided a qualitative and quantitative characterization of the internal damage generated by the impact events. The C-scan maps revealed a clear increase in delamination footprint with increasing impact energy, while the automated image-processing procedure enabled a consistent extraction of the projected damage area.

Although the ultrasonic measurements were not used during the development of the numerical model, they provide an independent experimental observation of the internal damage state of the tested laminates and support the interpretation of the impact response discussed in the previous chapters.

6.6 Interpretation of Energy Dissipation Mechanisms in the Glass/PPS Impact Response

The combined analysis of the numerical simulations and the ultrasonic inspection results provides additional insight into the distribution of energy dissipation mechanisms governing the impact response of the investigated Glass/PPS laminate.

During the development of the numerical models, the acceleration-time response was initially reproduced through calibration of the intralaminar fracture energy parameters. While this approach provided excellent agreement with the experimentally measured global dynamic response and produced accurate predictions of the back-face crack length, the corresponding simulations did not reproduce several damage mechanisms later observed experimentally, including permanent indentation beneath the impactor and significant interlaminar delamination.

The introduction of the matrix plasticity formulation within the bi-phasic modelling framework enabled the representation of localized indentation beneath the impactor and improved the physical consistency of the predicted deformation mechanisms. Although this formulation captured an important deformation mode of the PPS matrix, the ultrasonic inspection presented in in this chapter demonstrates that substantial delamination is present in the impacted specimens even at intermediate impact energies. This observation indicates that the total impact energy absorbed by the laminate cannot be attributed solely to matrix plastic deformation and fiber rupture.

These findings suggest that the intralaminar fracture energy values previously calibrated in simplified modelling configurations may partially compensate for damage mechanisms that were not yet explicitly represented in the numerical model. In particular, when the intralaminar fracture energy was tuned to reproduce the experimental acceleration response, the resulting formulation effectively absorbed energy associated with indentation and delamination processes that were later introduced through the plasticity and cohesive modelling frameworks. Consequently, the effective

intralaminar fracture energy required to reproduce the global impact response in the simplified model may be slightly underestimated when additional physical mechanisms are explicitly represented.

Nevertheless, the intralaminar damage formulation consistently provided accurate predictions of the back-face crack length and preserved the global acceleration response across the investigated impact energies. This observation suggests that fibre-dominated intralaminar fracture remains one of the primary mechanisms governing crack formation in the investigated Glass/PPS laminate under low-velocity impact loading.

Overall, the results of the present work indicate that the impact response of the Glass/PPS laminate is governed by a coupled interaction between intralaminar fracture, interlaminar delamination, and localized matrix deformation beneath the impactor. While the calibrated intralaminar fracture energy values provide an effective representation of the global impact behaviour within the adopted modelling framework, the combined numerical and experimental analysis highlights the importance of explicitly representing multiple damage mechanisms in order to obtain a physically consistent description of the impact process.

For clarity, the main energy dissipation mechanisms identified in the present study and their representation within the numerical modelling framework are summarized in Table 6.3.

Table 6.3: Qualitative summary of the main energy dissipation mechanisms identified in the Glass/PPS laminate during low-velocity impact.

Mechanism	**Experimental Evidence**	**Model Representation**	**Role in Impact Response**
Intralaminar fibre fracture	Back-face crack observed in all impact tests; crack length well correlated with simulations	Hashin intralaminar damage model implemented in continumm-shell elements	Primary mechanism governing crack formation and global damage propagation
Interlaminar delamination	Ultrasonic C-scan measurements revealing internal delamination patterns	Cohesive interface elements (COH3D8) inserted between laminate plies	Secondary energy dissipation mechanism influencing damage footprint and energy scaling
Matrix plasticity and indentation	Permanent indentation beneath the impactor observed experimentally	Bi-phasic formulation with pressure-dependent plasticity in localized solid elements	Localized deformation mechanism contributing to impact energy absorption

6.7 Additional Numerical Validation Using NDT Damage Measurements

Following the ultrasonic inspection campaign presented in the previous sections, an additional numerical validation study was performed in order to assess the capability of the cohesive model to reproduce the experimentally observed internal damage state.

The motivation for this additional study arises from the comparison between the experimentally measured delamination areas obtained from C-scan inspection and the predictions of the numerical model developed in Chapter 5. In particular, the experimental observations suggested that the previously calibrated intralaminar fracture energy could underestimate the contribution of interlaminar delamination mechanisms.

To investigate this effect, an additional cohesive configuration was analysed, referred to as the *Estimation_2* model. In this configuration, the intralaminar fracture energy was increased from

$G_f = 20$ N/mm to $G_f = 30$ N/mm, while maintaining all other modelling parameters identical to those previously introduced for the cohesive formulation.

The validation was conducted for the two highest impact energies investigated experimentally, namely $E = 14.2$ J and $E = 27$ J, as these cases exhibit clearly measurable internal delamination damage in the NDT inspection.

Estimated Cohesive Response at 14.2 J

Figure 6.10 presents the comparison between the experimental acceleration–time histories and the numerical response obtained using the updated cohesive configuration.

The predicted dynamic response shows a clear improvement relative to the previously analysed cohesive configurations. The peak acceleration and overall pulse shape are reproduced with good agreement, indicating that the increased intralaminar fracture energy leads to a more realistic balance between stiffness and damage dissipation mechanisms.

The numerical peak acceleration occurs at approximately $t \approx 3.5$–4.0 ms, closely matching the experimental measurements. Furthermore, the unloading phase and total contact duration are well captured, confirming that the updated fracture energy does not introduce excessive structural compliance.

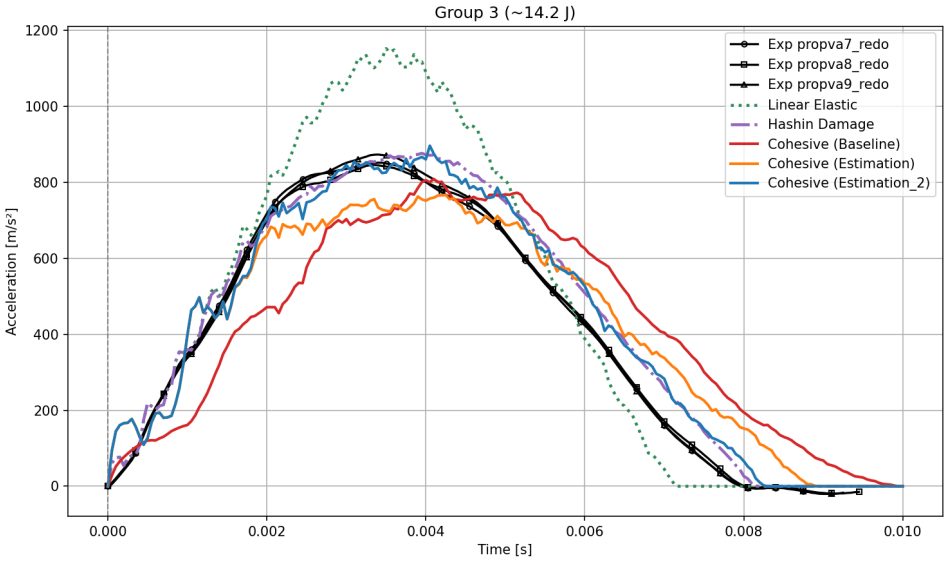


Figure 6.10: Acceleration–time response at $E = 14.2$ J comparing experimental data with the updated cohesive model (*Estimation_2*).

The predicted interlaminar damage distribution extracted from the cohesive damage variable SDEG is compared with the experimental C-scan image in Figure 6.11. The numerical model reproduces the compact delamination footprint observed experimentally, with damage concentrated around the impact region.

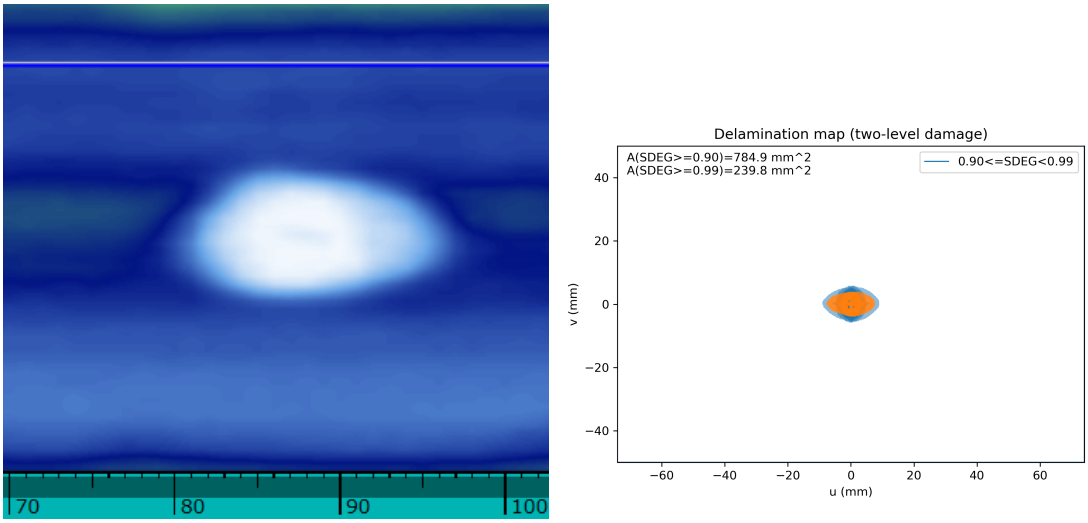


Figure 6.11: Comparison between experimental C-scan damage and numerical cohesive delamination map at $E = 14.2$ J.

Quantitative damage metrics extracted from the numerical post-processing routine are summarised in Table 6.4. The predicted delamination footprint area for SDEG ≥ 0.99 is approximately 90 mm^2, which compares well with the experimentally measured delamination area of approximately 80 mm^2 obtained from the C-scan inspection.

Table 6.4: Comparison between experimental and numerical damage metrics at $E = 14.2$ J.

Metric	**Value**
Back-face crack length (Exp)	10.3 mm
Back-face crack length (CAE)	6.0 mm
Delaminated area (NDT)	80 mm^2
Footprint area (SDEG ≥ 0.99)	90 mm^2
Footprint area (SDEG ≥ 0.90)	149 mm^2
Cumulative delamination area (SDEG ≥ 0.99)	239.8 mm^2
Cumulative delamination area (SDEG ≥ 0.90)	784.9 mm^2

Estimated Cohesive Response at 27 J

A similar validation analysis was conducted for the highest impact energy investigated experimentally. Figure 6.12 shows the comparison between experimental acceleration histories and the numerical predictions obtained with the updated cohesive configuration.

The numerical response again shows good agreement with the experimental measurements, reproducing both the peak acceleration and the global shape of the acceleration pulse. Compared with the previous cohesive configurations, the updated model provides a closer match to the experimental curve, particularly in the peak region.

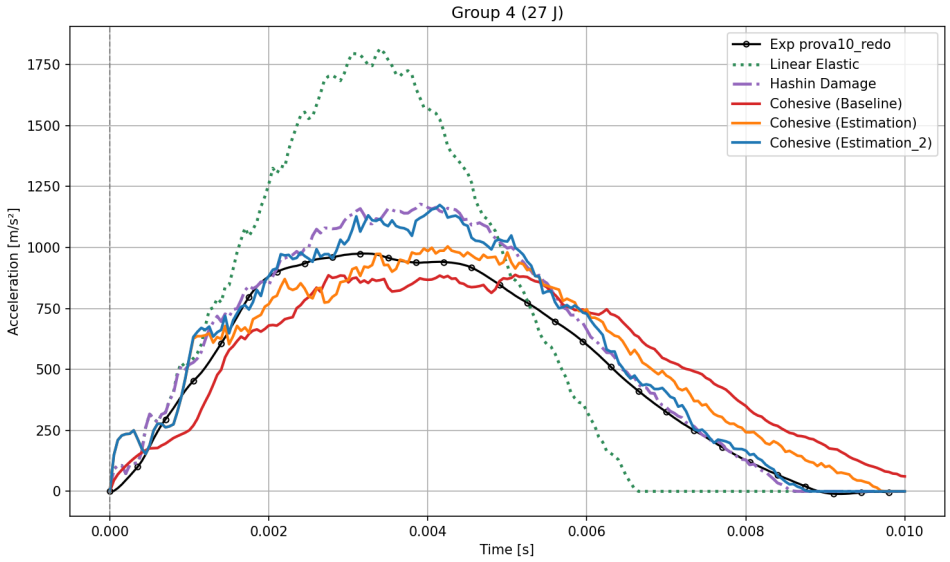


Figure 6.12: Acceleration-time response at $E = 27$ J comparing experimental data with the updated cohesive model (*Estimation_2*).

The predicted delamination morphology is compared with the C-scan inspection results in Figure 6.13. At this higher impact energy, the numerical model predicts a larger delamination footprint with a characteristic cross-shaped pattern aligned with the principal material directions. This morphology is consistent with the experimental observations obtained from ultrasonic inspection.

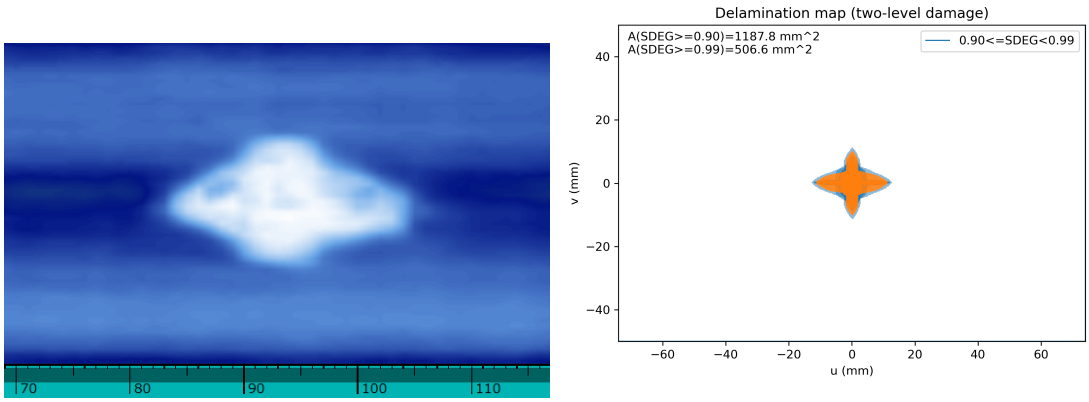


Figure 6.13: Comparison between experimental C-scan damage and numerical cohesive delamination map at $E = 27$ J.

Table 6.5 summarises the quantitative comparison between experimental measurements and

numerical predictions. The predicted delamination footprint for SDEG ≥ 0.99 reaches approximately 165 mm^2, which is reasonably close to the experimentally measured value of 143 mm^2.

Table 6.5: Comparison between experimental and numerical damage metrics at $E = 27$ J.

Metric	**Value**
Back-face crack length (Exp)	19 mm
Back-face crack length (CAE)	16 mm
Delaminated area (NDT)	143 mm^2
Footprint area (SDEG ≥ 0.99)	165 mm^2
Footprint area (SDEG ≥ 0.90)	224 mm^2
Cumulative delamination area (SDEG ≥ 0.99)	506.6 mm^2
Cumulative delamination area (SDEG ≥ 0.90)	1187.8 mm^2

Overall, the additional validation analyses indicate that increasing the intralaminar fracture energy leads to a more realistic prediction of the internal damage mechanisms observed experimentally. The updated cohesive configuration improves both the global dynamic response and the agreement with NDT measurements of delamination area.

These results further support the interpretation that the intralaminar fracture energy calibrated in the absence of explicit interlaminar modelling effectively acted as a numerical parameter compensating for missing delamination mechanisms. When cohesive interfaces are introduced, higher fracture energy values become necessary to correctly represent the interaction between intralaminar fracture and interlaminar delamination processes.

Furthermore, the adopted value of $G_f = 30$ N/mm is also closer to the fracture energy values reported in the literature for woven glass fibre composites discussed in the previous sections. This suggests that the updated parameter not only improves the numerical correlation with the experimental results but may also represent a more realistic material property for the investigated laminate compared with the lower value identified in the previous calibration study.

Chapter 7

Conclusions

This chapter summarizes the main findings and contributions of the present work. The conclusions are drawn from the combined experimental observations and numerical investigations performed throughout this study on the low-velocity impact behaviour of Glass/PPS thermoplastic composite laminates. The key results obtained from the experimental characterization, the progressive development of the numerical modelling framework, and the evaluation of different damage and plasticity formulations are discussed. The chapter also outlines the principal limitations of the current study and proposes directions for future research aimed at improving the predictive capability and applicability of numerical models for impact damage in thermoplastic composite structures.

7.1 Summary of the Work

This thesis investigated the behaviour of Glass/PPS thermoplastic composite laminates subjected to low-velocity impact (LVI), a loading scenario that is critical for the structural integrity of aerospace structures. The main objective of this work was to develop and assess a numerical modelling framework capable of reproducing the impact response and damage mechanisms of these laminates using exclusively commercial finite element tools, namely Abaqus/Explicit, supported by experimental observations of post-impact damage.

The adopted methodology combined the processing of existing experimental reference data with additional post-impact inspections performed using non-destructive testing (NDT) based on ultrasonic techniques in order to map the internal delamination damage. These experimental observations provided qualitative and quantitative information regarding the evolution of damage as a function of the impact energy.

From the numerical perspective, the modelling strategy evolved progressively from simple linear elastic representations towards increasingly complex formulations incorporating damage and

inelastic behaviour. Intralaminar damage was modelled using the Hashin failure criterion implemented with continuum shell elements, while interlaminar damage was introduced through cohesive interface elements to simulate delamination propagation.

In order to address the limitations of conventional laminate damage models in reproducing indentation behaviour, the modelling framework was further extended through the development of a bi-phasic strategy. In this approach, the matrix-dominated behaviour in the impact region was represented using solid elements with pressure-dependent plasticity, allowing the numerical model to capture localized deformation mechanisms associated with indentation beneath the impactor.

In addition to the modelling framework itself, the present work also contributes to improving the reproducibility and efficiency of the numerical-experimental comparison process. Dedicated Python scripts were developed for the automated post-processing of both numerical simulation results and experimental inspection data. These tools enable the automated extraction of damage metrics, including delamination areas obtained from finite element output databases (ODB) and ultrasonic inspection images. The automation procedures documented in the appendices enable rapid and consistent quantitative comparison between numerical predictions and experimental observations, improving both the reliability and reproducibility of the analysis workflow.

7.2 Main Experimental Findings

The experimental observations confirmed that the damage mechanisms in the Glass/PPS laminate are strongly dependent on the impact energy level. A clear correlation was observed between the impact energy and the severity of the resulting damage, particularly in terms of indentation depth and crack length on the back face of the specimens.

Low-energy impacts resulted primarily in shallow surface indentations with limited visible damage on the rear surface. As the impact energy increased, the extent of damage became progressively more pronounced. At the highest energy level considered in this study (27 J), extensive cross-shaped cracks were observed on the back face of the laminate, indicating significant internal damage and fibre fracture.

From a dynamic perspective, the experimental acceleration–time responses indicated that both the peak acceleration and the contact duration increase systematically with the impact energy. These measurements provided an important reference for the validation of the numerical simulations.

Post-impact ultrasonic inspections performed using C-scan mapping confirmed the presence of internal delamination damage within the laminate structure. The estimated delamination footprint increased significantly with the impact energy, ranging from approximately 20 mm^2 for the

7.5 J impact to approximately 143 mm^2 for the 27 J case. These results illustrate the progressive development of interlaminar damage mechanisms within the thermoplastic composite laminate.

The comparison between these experimentally measured delamination areas and the numerical predictions obtained in the cohesive simulations provided an important additional validation opportunity. As discussed in the final section of Chapter 6, an additional numerical investigation was conducted after the completion of the NDT campaign in order to directly compare the predicted delamination morphology with the experimentally observed C-scan damage patterns.

These additional simulations confirmed that the cohesive modelling framework is capable of reproducing the experimentally observed delamination footprints when appropriate intralaminar fracture energy values are adopted. In particular, the numerical delamination areas predicted for the 14.2 J and 27 J impacts showed good agreement with the experimentally measured values obtained from ultrasonic inspection.

7.3 Main Numerical Modelling Findings

The numerical modelling campaign demonstrated that the progressive introduction of increasingly complex damage and deformation mechanisms is required to reproduce the experimental impact response of the laminate.

The baseline linear elastic model was able to reproduce the initial stiffness of the panel and the general chronology of the impact event, particularly for the lower energy levels. However, this simplified representation was not capable of capturing the permanent damage observed experimentally.

Among the modelling strategies investigated, the intralaminar damage formulation based on the Hashin failure criterion implemented with continuum shell elements provided the most consistent correlation with the experimental observations. This modelling approach successfully reproduced both the acceleration–time response and the experimentally observed crack lengths while maintaining relatively low computational cost when compared with more complex modelling strategies.

The calibration of the intralaminar fracture energy parameters proved to be particularly influential in reproducing the experimentally observed crack morphology and global dynamic response. In simplified modelling configurations where delamination and plastic deformation mechanisms were not explicitly represented, the intralaminar fracture energy effectively absorbed part of the impact energy associated with multiple damage mechanisms.

The introduction of cohesive interface elements enabled the explicit simulation of interlaminar delamination within the laminate. Cohesive models using properties estimated from similar

thermoplastic composite systems were initially able to preserve the global dynamic response and capture the general orientation of damage, although they slightly underestimated the total delamination area compared to NDT measurements. However, when combined with the previously calibrated intralaminar damage formulation, the cohesive model tended to overestimate back-face crack propagation at higher impact energies. This behaviour indicates that interlaminar damage alone is not sufficient to constrain crack growth and that the overall impact response is governed by the interaction between multiple damage mechanisms within the laminate.

The bi-phasic modelling strategy incorporating localized solid elements and pressure-dependent plasticity was introduced to represent indentation beneath the impactor. The Crushable Foam formulation allowed the numerical model to reproduce the compressive deformation of the PPS matrix and the permanent indentation observed experimentally.

The combined analysis of the numerical simulations and the ultrasonic inspection results indicates that the impact response of the Glass/PPS laminate is governed by a coupled interaction between intralaminar fracture, interlaminar delamination, and localized matrix deformation beneath the impactor. While plastic deformation contributes to the overall energy absorption, the numerical results suggest that fibre-dominated intralaminar fracture remains one of the primary mechanisms governing crack formation in the investigated laminate configuration.

The results also indicate that the intralaminar fracture energy values calibrated in simplified modelling configurations may represent effective parameters that partially compensate for damage mechanisms not explicitly represented in the model. When additional mechanisms such as delamination and matrix plasticity are introduced, the effective energy balance within the laminate may change, suggesting that the intrinsic intralaminar fracture energy of the material system may be higher than the values required in simplified numerical calibrations.

This hypothesis, which emerged from the analysis of the modelling strategies discussed in Chapter 5, was further confirmed by the additional validation simulations performed after the NDT campaign. By increasing the intralaminar fracture energy to values closer to those reported in the literature for woven glass fibre composites, the numerical model achieved improved agreement with both the experimentally measured acceleration histories and the delamination areas obtained from ultrasonic inspection. These results support the interpretation that the lower fracture energy values previously identified in simplified modelling configurations primarily acted as numerical compensation parameters rather than intrinsic material properties.

Overall, the additional validation analyses performed after the NDT inspection campaign provide further confidence in the modelling framework developed in this work. The agreement between the experimentally observed delamination footprints and the numerical predictions obtained with the updated cohesive configuration confirms the earlier interpretation that the impact response

of the laminate is governed by the interaction between multiple damage mechanisms. In particular, the results reinforce the conclusion that intralaminar fracture energy values calibrated in simplified numerical models should be interpreted cautiously, as they may implicitly account for damage mechanisms that are not explicitly represented in the model.

7.4 Limitations of the Present Study

Despite the advances achieved in this work, several limitations should be acknowledged.

Firstly, several key material parameters required for the numerical simulations were not directly measured for the investigated material system. In particular, the interlaminar fracture energies and the pressure-dependent plastic behaviour of PPS were estimated from values reported in the literature for similar thermoplastic composite systems.

Secondly, the numerical models rely on simplified representations of complex damage mechanisms. The bi-phasic modelling approach introduces plastic deformation of the matrix but still represents the laminate using homogenized ply properties and simplified damage formulations.

Finally, the absence of full three-dimensional experimental characterization techniques such as X-ray micro-computed tomography (μCT) limits the ability to directly validate the predicted internal damage distribution within the laminate.

7.5 Recommendations for Future Work

Several directions for future research can be identified based on the findings of this work.

Firstly, dedicated experimental characterization of the material system should be performed in order to obtain more reliable input parameters for the numerical models. In particular, Double Cantilever Beam (DCB) and End-Notched Flexure (ENF) tests should be conducted to directly determine the interlaminar fracture energies and interface strengths of the Glass/PPS laminate. In addition, intralaminar fracture energy could be experimentally characterized through fracture mechanics tests on notched composite specimens, such as Single Edge Notched Tension (SENT) or Compact Tension (CT) tests, which allow the in-plane fracture toughness and the associated fracture energy of the laminate to be determined.

Secondly, further investigation of the pressure-dependent plastic behaviour of the PPS matrix is recommended. Uniaxial compression and hydrostatic compression tests would provide valuable data for calibrating advanced plasticity models.

Thirdly, the use of X-ray micro-computed tomography (μCT) could enable volumetric visualization of internal damage mechanisms and provide a more rigorous validation of the delamination patterns predicted by the numerical models.

Finally, the implementation of probabilistic modelling approaches, such as Monte Carlo simulations, could allow the influence of modelling parameters and their uncertainties to be quantified, improving the robustness of impact damage predictions.

7.6 Final Remarks

The main contributions of this thesis can be summarized as follows:

- Development of a progressive finite element modelling framework for the simulation of low-velocity impact on Glass/PPS thermoplastic laminates using commercial software tools.
- Experimental characterization of the impact response and damage evolution in Glass/PPS laminates, including the correlation between impact energy, indentation behaviour, delamination development, and back-face crack morphology.
- Demonstration that intralaminar damage modelling using the Hashin failure criterion implemented with continuum shell elements provides an efficient and accurate representation of crack formation in the investigated laminate configuration.
- Assessment of cohesive interface elements for modelling interlaminar delamination and evaluation of the influence of cohesive parameters on predicted damage patterns.
- Development of automated Python-based post-processing tools enabling efficient extraction and comparison of numerical and experimental damage metrics.
- Development and evaluation of a localized bi-phasic modelling strategy incorporating pressure-dependent matrix plasticity in order to reproduce indentation behaviour and improve the representation of impact energy dissipation mechanisms.
- Additional validation of the cohesive modelling framework through direct comparison between predicted delamination maps and ultrasonic C-scan damage measurements, confirming that higher intralaminar fracture energy values consistent with literature ranges for woven composites provide improved agreement with experimental observations.

Overall, this work demonstrates that reliable numerical prediction of low-velocity impact behaviour in Glass/PPS thermoplastic laminates can be achieved using commercial finite element tools when appropriate modelling strategies are adopted. The progressive modelling framework developed in this study highlights the importance of representing multiple interacting damage mechanisms in order to reproduce the complex impact response of thermoplastic composite laminates.

The combined experimental and numerical analysis presented in this thesis contributes to improving the understanding of impact damage mechanisms in thermoplastic composite laminates and provides a practical modelling methodology that can be extended to similar material systems and structural configurations.

The main contribution of this work is the development and validation of a numerical modelling framework capable of reproducing the key impact damage mechanisms of Glass/PPS thermoplastic laminates using commercial finite element tools. The results demonstrate that accurate prediction of low-velocity impact behaviour requires the explicit representation of interacting mechanisms including intralaminar fracture, interlaminar delamination, and localized matrix deformation.

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Appendix A

Mesh Sensitivity Study and Energy-Based Verification

This appendix documents the mesh sensitivity analyses performed during the development of the numerical model. Several alternative partition strategies and mesh configurations were investigated in order to assess their influence on numerical stability and global energy balance in the explicit dynamic simulations. The objective of this study was not to optimise the mesh for accuracy, but to verify that the final discretization adopted in Chapter 5 provides stable and physically consistent results without excessive hourglass or artificial dissipation effects.

Unless otherwise stated, all simulations were performed using identical material models, boundary conditions, contact definitions, and loading conditions. Differences between cases arise solely from the adopted partitioning and mesh strategies.

A.1 Cross-partition baseline mesh

The first configuration corresponds to a cross-partition strategy, in which the composite panel is divided into four regions intersecting at the panel centre. This partition represents the simplest structured subdivision of the laminate and was initially considered as a baseline configuration.

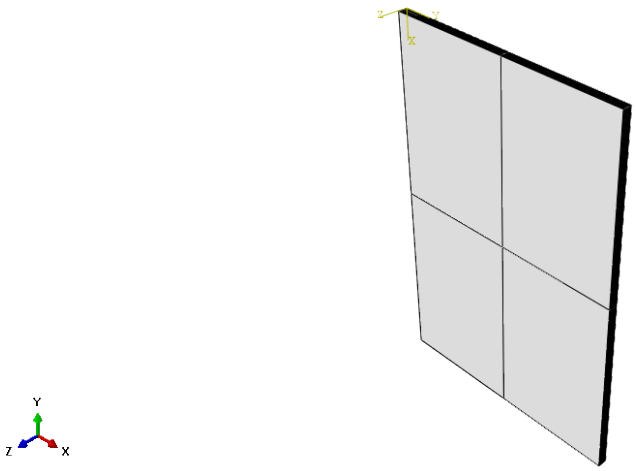

Figure A.1: Part partition of the composite panel using the cross-partition strategy.

In the cross-partition configuration, no dedicated local refinement region was defined at the impact location. As a result, the element aspect ratio varies across the entire panel. The best in-plane to through-thickness aspect ratio observed in this configuration is approximately AS ≈ 2.25, while the average value over the full laminate is AS ≈ 8.08. The worst-case aspect ratio reaches AS ≈ 15.22, occurring in regions far from the panel centre.

Table A.1: Mesh characteristics for the cross-partition configuration.

Number of nodes	113 908
Number of elements	103 129
Element types	SC8R, C3D8R, R3D4

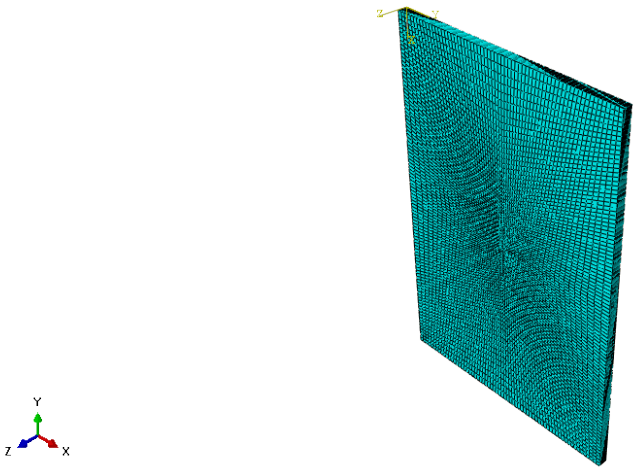

Figure A.2: Finite element mesh corresponding to the cross-partition configuration.

The corresponding energy evolution is shown in Figure A.3. While the global energy balance is preserved throughout the simulation, a non-negligible artificial strain energy contribution

is observed, indicating sensitivity to hourglass modes associated with the global aspect-ratio distribution. Owing to the biased seeding towards the panel centre, the mesh exhibits comparatively favourable element shapes in the immediate vicinity of the impact location, resulting in a slightly reduced hourglass contribution relative to some square-partition configurations.

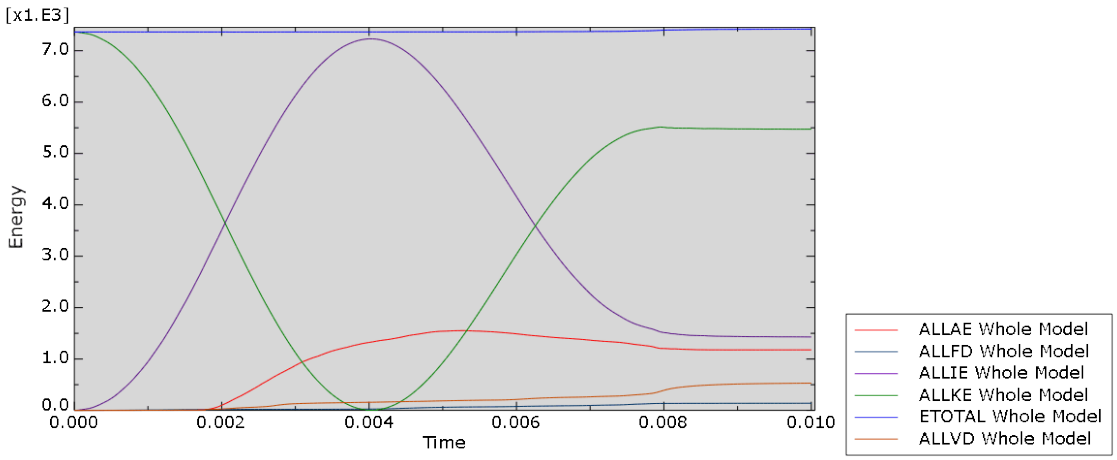

Figure A.3: Global energy evolution for the cross-partition mesh configuration.

A.2 Square-partition mesh

The second configuration introduces a square partition centred on the impact location, motivated by the experimentally observed damage footprint. This strategy allows a more controlled discretization of the impact region while retaining a structured mesh elsewhere.

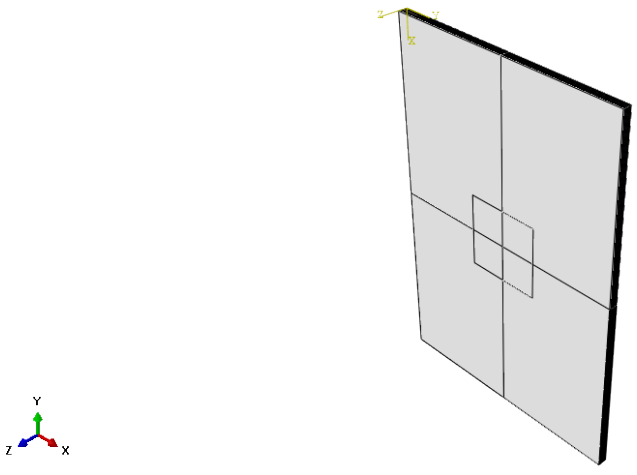

Figure A.4: Part partition of the composite panel using a square-partition strategy.

Compared to the cross-partition mesh, this configuration reduces element sizes in the impact zone and provides improved control of the element aspect ratio at the ply level. In the refined region, the in-plane to through-thickness aspect ratio is approximately AS ≈ 3.24, while elements

located farther from the impact zone exhibit larger aspect ratios, with an average value of approximately AS ≈ 6.31 and a worst-case value of AS ≈ 19.95.

Table A.2: Mesh characteristics for the square-partition configuration.

Number of nodes	171 114
Number of elements	157 165
Element types	SC8R, C3D8R, R3D4

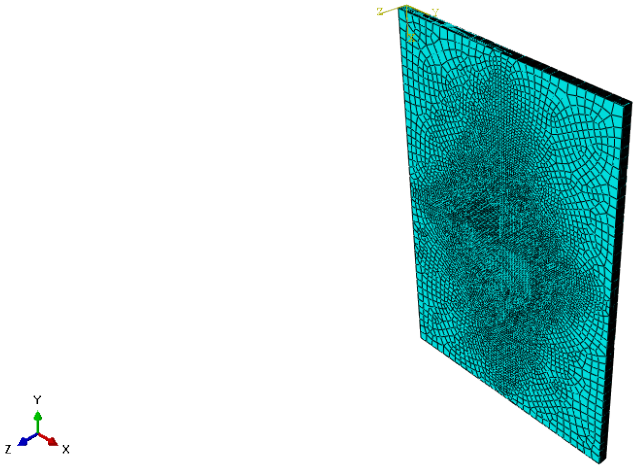

Figure A.5: Finite element mesh corresponding to the square-partition configuration.

The energy evolution for this configuration is shown in Figure A.6. The overall energy balance is well preserved; however, the artificial strain energy contribution remains non-negligible during the contact phase. Despite the improved geometric definition of the impact region, the aspect ratio in the refined zone remains above unity, which contributes to hourglass sensitivity in reduced-integration elements.

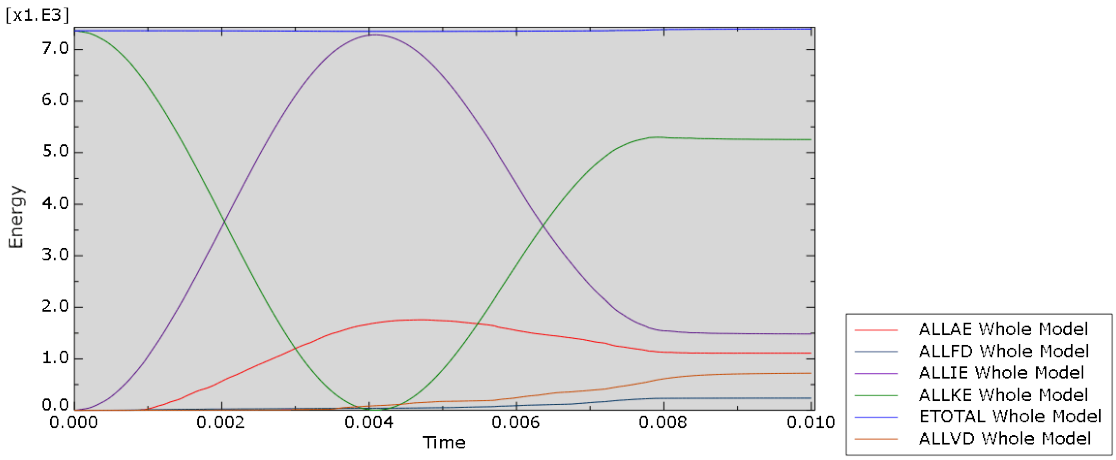

Figure A.6: Global energy evolution for the square-partition mesh configuration.

A.3 Square-partition mesh (alternative seeding configuration)

The third configuration adopts the same square-partition strategy as the previous case, without introducing additional refinement in the impact region. The discretization within the impact zone therefore remains comparable, while differences arise from the meshing and seeding strategy applied to the remainder of the panel.

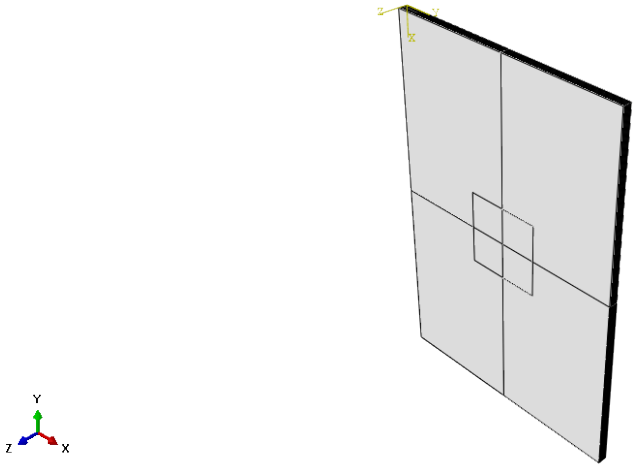

Figure A.7: Part partition of the composite panel using the square-partition strategy (alternative seeding configuration).

Maintaining the same impact-region discretization (AS ≈ 3.24), this configuration presents a modified aspect-ratio distribution over the remainder of the panel. Outside the impact zone, the element aspect ratio exhibits an average value of approximately AS ≈ 5.48, with a worst-case value of AS ≈ 12.49.

Table A.3: Mesh characteristics for the square-partition configuration with alternative seeding.

Number of nodes	223 831
Number of elements	205 917
Element types	SC8R, C3D8R, R3D4

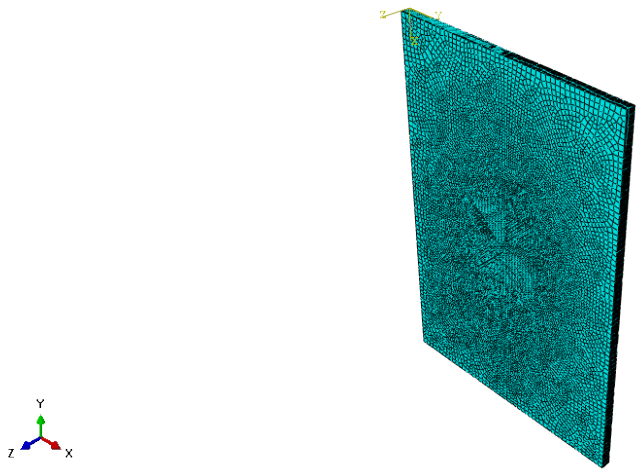

Figure A.8: Finite element mesh corresponding to the square-partition configuration with alternative seeding.

Figure A.9 shows the corresponding global energy evolution. The global energy balance remains stable throughout the simulation, while the artificial strain energy contribution remains non-negligible. Variations relative to the previous square-partition case primarily reflect differences in far-field aspect-ratio distribution rather than changes in the impact-zone discretization.

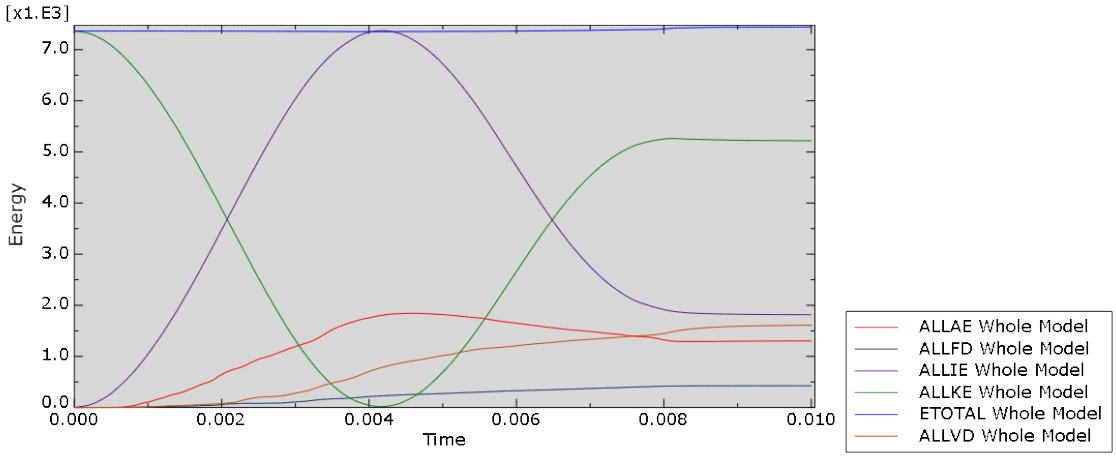

Figure A.9: Global energy evolution for the square-partition mesh configuration with alternative seeding.

A.4 Summary

This mesh sensitivity study indicates that the artificial strain energy contribution (ALLAE) is strongly influenced by element aspect ratio and mesh quality in the region governing the impact response, rather than being determined solely by the adopted partition topology. In the cross-partition baseline mesh, the use of biased seeding increases mesh density towards the panel centre, leading to comparatively favourable element shapes near the impact location and a slightly reduced

hourglass contribution relative to square-partition configurations. Nevertheless, the artificial strain energy remains non-negligible in this case.

Square-partition configurations provide improved geometric control of the impact-region footprint and a more structured discretization in the zones where damage is expected. However, with an impact-region aspect ratio of approximately AS ≈ 3.24, the artificial strain energy contribution remains non-negligible, and variations in the seeding strategy outside the impact zone primarily affect far-field aspect-ratio statistics rather than the contact-dominated response.

The final reference discretization adopted in Chapter 5, enforcing a near-unity aspect ratio in the impact region (AS ≈ 1), exhibits a significantly reduced artificial strain energy contribution. This observation supports the correlation between hourglass sensitivity and aspect-ratio control in the contact-dominated zone, confirming that aspect-ratio optimisation in the impact region is the dominant factor governing numerical stability in the present explicit dynamic simulations.

This appendix supports the mesh design choices discussed in Section 5.1.7 and demonstrates that the numerical results presented in the main body of the thesis are not artefacts of an arbitrary discretization.

It is noted that the AS≈ 1 refinement strategy discussed in the main text corresponds to the final reference discretization adopted for Chapter 5 and is not part of the alternative mesh configurations reported in this appendix.

Appendix B

Python script for intralaminar damage mapping

The following Python script was developed to automatically extract and visualize the intralaminar damage distribution of the composite laminate from Abaqus output databases (ODB). The routine is implemented using the Abaqus Python scripting interface through the `odbAccess` module.

The script performs:

- Extraction of the last-ply element set corresponding to the impact region,
- Combination of the Hashin damage variables into a single damage metric

$$d_{\text{comb}} = \max(\texttt{DAMAGEFT}, \texttt{DAMAGEMT}, \texttt{DAMAGESHR}),$$

- Identification of deleted elements using a damage threshold criterion,
- Projection of element faces onto the laminate mid-plane for two-dimensional visualization,
- Generation of a colour-mapped intralaminar damage contour over the full specimen geometry,
- Automatic measurement of the characteristic cross-shaped damage lengths based on the centroid distribution of deleted elements.

The script was applied consistently to all numerical simulations analysed in this work to generate the intralaminar damage maps presented in Chapter 5.

```
1
2 from abaqus import session
3 from odbAccess import *
4 import numpy as np
5
6 import matplotlib
7 matplotlib.use("Agg")
```

```
import matplotlib.pyplot as plt
from matplotlib.patches import Polygon
from matplotlib.collections import PatchCollection
from matplotlib.colors import BoundaryNorm

# ---------------- USER INPUTS ----------------
step_name  = "Step-1"
frame_idx  = -1

elset_name = "PANEL-1.IMPACT_ZONE-16-1"   # last ply set
plate_instance = "PANEL-1"                  # for full-view bbox & plane basis

f_damage1 = "DAMAGEFT"
f_damage2 = "DAMAGEMT"
f_shear   = "DAMAGESHR"

d_cap   = 0.997
del_eps = 1e-6
del_thr = d_cap - del_eps

levels = [0.05, 0.15, 0.30, 0.50, 0.70, 0.85, 0.95, d_cap]

crop_size = 30.0

plot_deleted_overlay = True
deleted_facecolor = "0.15"
deleted_alpha = 0.95

draw_plate_outline = True

max_plane_fit_points = 4000
max_bbox_points      = 8000

# --- Cross measurement settings ---
cross_band_mm = 1.0
# This is the thickness of the strip used to identify each arm:
# u-arm uses |v - v_c| <= cross_band_mm
# v-arm uses |u - u_c| <= cross_band_mm

dpi = 300
out_dir = r"path/to/user/defined/output_directory"
# ---------------------------------------------

def get_elset_anywhere(odb, name):
    root = odb.rootAssembly
    if name in root.elementSets:
        return root.elementSets[name]
    name_u = name.upper()
    if name_u in root.elementSets:
```

```
57          return root.elementSets[name_u]
58
59      if "." in name:
60          inst_name, set_short = name.split(".", 1)
61          inst_u = inst_name.upper()
62          set_u = set_short.upper()
63          if inst_u not in root.instances:
64              raise RuntimeError("Instance not found: %s" % inst_u)
65          inst = root.instances[inst_u]
66          if set_short in inst.elementSets:
67              return inst.elementSets[set_short]
68          if set_u in inst.elementSets:
69              return inst.elementSets[set_u]
70          raise RuntimeError("Element set '%s' not found in instance '%s'" %
                (set_short, inst_u))
71
72      for inst in root.instances.values():
73          if name in inst.elementSets:
74              return inst.elementSets[name]
75          if name_u in inst.elementSets:
76              return inst.elementSets[name_u]
77      raise RuntimeError("Element set not found: %s" % name)
78
79  def safe_basename(path):
80      p = path.replace("\\", "/")
81      name = p.split("/")[-1]
82      if name.lower().endswith(".odb"):
83          name = name[:-4]
84      return name
85
86  def sanitize_filename(name):
87      bad = ['\\','/',';',':','*','?','"','<','>','|',',',' ']
88      for ch in bad:
89          name = name.replace(ch, '_')
90      return name
91
92  def join_path(folder, filename):
93      f = folder.replace("\\", "/").rstrip("/")
94      return f + "/" + filename
95
96  def must_have(frame, field):
97      if field in frame.fieldOutputs:
98          return field
99      if field.upper() in frame.fieldOutputs:
100         return field.upper()
101     raise RuntimeError("Field not found in frame: %s" % field)
102
103 def scalar_from_value(v):
104     d = v.data
```

```
    try:
        return float(d)
    except:
        try:
            arr = np.array(d, dtype=float).ravel()
            return float(np.max(arr)) if arr.size else 0.0
        except:
            return 0.0

def build_node_coord_map(instance):
    m = {}
    for n in instance.nodes:
        m[n.label] = np.array(n.coordinates, dtype=float)
    return m

def sample_indices(n, nmax):
    if n <= nmax:
        return range(n)
    step = int(max(1, n // nmax))
    return range(0, n, step)

def best_fit_plane_basis(points3d):
    P = np.asarray(points3d, dtype=float)
    if P.size == 0 or P.shape[0] < 3:
        raise RuntimeError("Not enough points to fit a plane (need >=3).")
    mask = np.isfinite(P).all(axis=1)
    P = P[mask]
    if P.shape[0] < 3:
        raise RuntimeError("Not enough finite points to fit a plane.")
    c = P.mean(axis=0)
    Q = P - c
    if np.linalg.norm(Q) < 1e-12:
        raise RuntimeError("Degenerate point cloud: cannot fit plane.")
    u, s, vt = np.linalg.svd(Q, full_matrices=False)
    e1 = vt[0] / np.linalg.norm(vt[0])
    e2 = vt[1] / np.linalg.norm(vt[1])
    return c, e1, e2

def project_points(points3d, origin, e1, e2):
    P = np.asarray(points3d, dtype=float)
    Q = P - origin
    x = np.dot(Q, e1)
    y = np.dot(Q, e2)
    return np.column_stack([x, y])

def plate_bbox_uv(odb, inst_name, origin, e1, e2, max_pts=8000):
    root = odb.rootAssembly
    inst = root.instances[inst_name.upper()]
    nodes = inst.nodes
```

```
    idxs = sample_indices(len(nodes), max_pts)
    pts3d = [np.array(nodes[i].coordinates, dtype=float) for i in idxs]
    pts2d = project_points(pts3d, origin, e1, e2)
    umin = float(np.min(pts2d[:, 0])); umax = float(np.max(pts2d[:, 0]))
    vmin = float(np.min(pts2d[:, 1])); vmax = float(np.max(pts2d[:, 1]))
    return umin, umax, vmin, vmax

def uv_of_point(p, origin, e1, e2):
    q = p - origin
    return float(np.dot(q, e1)), float(np.dot(q, e2))

def apply_full_view(ax, umin, umax, vmin, vmax):
    ax.set_aspect('equal', adjustable='box')
    ax.set_xlim(umin, umax)
    ax.set_ylim(vmin, vmax)
    if draw_plate_outline:
        ax.plot([umin, umax, umax, umin, umin],
                [vmin, vmin, vmax, vmax, vmin],
                linewidth=1.0)

def build_element_max_map(frame, region, field_name):
    sub = frame.fieldOutputs[field_name].getSubset(region=region)
    m = {}
    for v in sub.values:
        lbl = v.elementLabel
        val = scalar_from_value(v)
        if lbl in m:
            if val > m[lbl]:
                m[lbl] = val
        else:
            m[lbl] = val
    return m

def measure_cross_arms(uv_pts, band_mm):
    """
    uv_pts: Nx2 array of deleted element centroids (u,v) in mm
    band_mm: strip half-width around centerline
    Returns: (uc, vc, L_u, L_v, Nu, Nv)
    """
    if uv_pts is None or len(uv_pts) == 0:
        return None, None, 0.0, 0.0, 0, 0

    uv = np.asarray(uv_pts, dtype=float)
    u = uv[:,0]; v = uv[:,1]

    # robust cross center (median)
    uc = float(np.median(u))
    vc = float(np.median(v))

```

```
    # u-arm: points near horizontal centerline
    mask_u = np.abs(v - vc) <= float(band_mm)
    # v-arm: points near vertical centerline
    mask_v = np.abs(u - uc) <= float(band_mm)

    L_u = 0.0; L_v = 0.0
    Nu = int(np.sum(mask_u)); Nv = int(np.sum(mask_v))

    if Nu >= 2:
        L_u = float(np.max(u[mask_u]) - np.min(u[mask_u]))
    if Nv >= 2:
        L_v = float(np.max(v[mask_v]) - np.min(v[mask_v]))

    return uc, vc, L_u, L_v, Nu, Nv

def main():
    if len(session.odbs) == 0:
        raise RuntimeError("No ODB is open in Viewer. Open the ODB first, then run
            the script.")
    odb = session.odbs.values()[0]
    odb_name = sanitize_filename(safe_basename(odb.name))

    if step_name not in odb.steps:
        raise RuntimeError("Step not found: %s" % step_name)
    step = odb.steps[step_name]
    frame = step.frames[frame_idx]

    f1 = must_have(frame, f_damage1)
    f2 = must_have(frame, f_damage2)
    fs = must_have(frame, f_shear)

    root = odb.rootAssembly

    # --- Get last ply element set ---
    elset = get_elset_anywhere(odb, elset_name)
    elements = elset.elements
    if len(elements) == 0:
        raise RuntimeError("Element set is empty: %s" % elset_name)

    # --- Plane basis from plate instance (sample nodes) ---
    inst_plate = root.instances[plate_instance.upper()]
    nodes = inst_plate.nodes
    if len(nodes) < 3:
        raise RuntimeError("Plate instance has too few nodes: %s" % plate_instance)

    idx_plane = sample_indices(len(nodes), max_plane_fit_points)
    pts_plane = [np.array(nodes[i].coordinates, dtype=float) for i in idx_plane]
    origin, e1, e2 = best_fit_plane_basis(pts_plane)

```

```
    # --- Full specimen bbox (for fixed view) ---
    umin, umax, vmin, vmax = plate_bbox_uv(odb, plate_instance, origin, e1, e2,
        max_pts=max_bbox_points)

    # Refined-zone window (centered on full bbox)
    uc0 = 0.5*(umin + umax)
    vc0 = 0.5*(vmin + vmax)
    half = 0.5*float(crop_size)
    u0, u1 = uc0 - half, uc0 + half
    v0, v1 = vc0 - half, vc0 + half

    # --- Per-element maps on last ply region ---
    m_ft = build_element_max_map(frame, elset, f1)
    m_mt = build_element_max_map(frame, elset, f2)
    m_sh = build_element_max_map(frame, elset, fs)

    inst_map = root.instances
    node_maps = {}

    dmin_plot = float(levels[0])

    patches = []
    cvals = []
    del_patches = []
    del_centroids_uv = []  # for cross measurement

    for e in elements:
        inst_name = e.instanceName
        if inst_name not in node_maps:
            node_maps[inst_name] = build_node_coord_map(inst_map[inst_name])
        coord = node_maps[inst_name]

        lbl = e.label
        conn = list(e.connectivity)
        if len(conn) < 4:
            continue

        face = conn[:4]
        try:
            p1 = coord[face[0]]
            p2 = coord[face[1]]
            p3 = coord[face[2]]
            p4 = coord[face[3]]
        except:
            continue

        pc = 0.25*(p1 + p2 + p3 + p4)
        u_c, v_c = uv_of_point(pc, origin, e1, e2)

```

```
        # Only process refined zone (speed)
        if (u_c < u0) or (u_c > u1) or (v_c < v0) or (v_c > v1):
            continue

        # Deletion assumption: if ANY metric reaches cap
        d_ft = m_ft.get(lbl, 0.0)
        d_mt = m_mt.get(lbl, 0.0)
        d_sh = m_sh.get(lbl, 0.0)
        is_del = (d_ft >= del_thr) or (d_mt >= del_thr) or (d_sh >= del_thr)

        poly2d = np.array([uv_of_point(p1, origin, e1, e2),
                           uv_of_point(p2, origin, e1, e2),
                           uv_of_point(p3, origin, e1, e2),
                           uv_of_point(p4, origin, e1, e2)], dtype=float)

        if is_del and plot_deleted_overlay:
            del_patches.append(Polygon(poly2d, closed=True))
            del_centroids_uv.append([u_c, v_c])
            continue

        # Combined damage for contour
        d = d_ft
        if d_mt > d: d = d_mt
        if d_sh > d: d = d_sh
        if d > d_cap: d = d_cap

        if d < dmin_plot:
            continue

        patches.append(Polygon(poly2d, closed=True))
        cvals.append(d)

# ---- Cross arm measurement from deleted centroids (inside refined zone) ----
uc, vc, L_u, L_v, Nu, Nv = measure_cross_arms(del_centroids_uv, cross_band_mm)

print ""
print "--- DELETED CROSS MEASUREMENTS (LAST PLY, refined zone) ---"
print "Deleted criterion: any(DAMAGEFT, DAMAGEMT, DAMAGESHR) >= %.6f" %
    float(del_thr)
print "Deleted elements counted in refined zone:", len(del_centroids_uv)
print "Cross band half-width: %.3f mm" % float(cross_band_mm)
if uc is None:
    print "No deleted elements -> lengths are zero."
else:
    print "Estimated cross center: uc=%.3f mm | vc=%.3f mm" % (float(uc),
        float(vc))
    print "u-arm points used: %d | u-arm length: %.3f mm" % (int(Nu),
        float(L_u))
```

```
        print "v-arm points used: %d | v-arm length: %.3f mm" % (int(Nv),
            float(L_v))
    print "------------------------------------------------------------"
    print ""

    # --- Plot (no title, no info) ---
    fig = plt.figure()
    ax = fig.add_subplot(111)

    cmap = plt.get_cmap("jet")
    bounds = [0.0] + list(levels)
    norm = BoundaryNorm(bounds, cmap.N)

    if len(patches) > 0:
        pcoll = PatchCollection(patches, cmap=cmap, norm=norm, edgecolor="none",
            alpha=0.95)
        pcoll.set_array(np.array(cvals, dtype=float))
        ax.add_collection(pcoll)

        cb = fig.colorbar(pcoll, ax=ax, fraction=0.046, pad=0.04)
        cb.set_label("d_comb = max(DAMAGEFT, DAMAGEMT, DAMAGESHR)")
        cb.set_ticks(levels)

    if plot_deleted_overlay and len(del_patches) > 0:
        ax.add_collection(PatchCollection(del_patches, facecolor=deleted_facecolor,
                                          edgecolor="none", alpha=deleted_alpha))

    ax.set_xlabel("u (mm)")
    ax.set_ylabel("v (mm)")
    apply_full_view(ax, umin, umax, vmin, vmax)

    fig.tight_layout()
    out_path = join_path(out_dir,
        "%s_lastply18_dcomb_full_refined30_delAny_cross.png" % odb_name)
    fig.savefig(out_path, dpi=dpi)
    plt.close(fig)

    print "Saved:", out_path

main()
```

Listing B.1: Python script for post-processing intralaminar Hashin damage maps and deleted-element cross-arm measurements from Abaqus ODB files.

Appendix C

Python Script for Automated Extraction of Delamination Footprints from Cohesive Damage Fields

The following Python script was developed to automatically extract and visualize the delamination footprint predicted by the cohesive interface elements in the finite element simulations. The routine processes Abaqus output databases (ODB) and evaluates the damage state of the interlaminar cohesive elements based on the scalar stiffness degradation variable `SDEG`. The script is implemented using the Abaqus Python scripting interface through the `odbAccess` module.

The script performs:

- Extraction of the cohesive element set corresponding to the laminate interface,
- Identification of delaminated elements based on two damage thresholds

$$\texttt{SDEG} \geq 0.90 \qquad \text{and} \qquad \texttt{SDEG} \geq 0.99,$$

- Computation of the cumulative delaminated area from the projected faces of the cohesive elements,
- Projection of the cohesive element faces onto the laminate mid-plane to generate two-dimensional delamination maps,
- Automatic generation of delamination contour plots for different damage levels,
- Estimation of the delamination footprint area through raster-based envelope detection for comparison with experimentally measured C-scan delamination areas.

The script was applied consistently to all numerical simulations analysed in this work to obtain the predicted delamination areas and maps presented in Chapter 5.

```
1
2  from abaqus import session
3  from odbAccess import *
4  import numpy as np
5
6  import matplotlib
7  matplotlib.use("Agg")
8  import matplotlib.pyplot as plt
9  from matplotlib.patches import Polygon
10 from matplotlib.collections import PatchCollection
11 from matplotlib.path import Path
12
13 # ---------------- USER INPUTS ----------------
14 step_name  = "Step-1"
15 frame_idx  = -1
16
17 # Cohesive set
18 elset_name = "PANEL-1.COHESIVE"
19
20 # Plate instance name
21 plate_instance = "PANEL-1"
22
23 thr_lo = 0.90
24 thr_hi = 0.997
25
26 dpi = 300
27 out_dir = r"path/to/user/defined/output_directory"
28
29 draw_plate_outline = True         # draws a rectangle frame of the plate bbox in
       (u,v)
30
31 # Deterministic axis orientation
32 stable_axes = True
33 stable_axis_preference = "X"
34
35 # Footprint raster resolution (mm)
36 dx = 1.0
37 # ---------------------------------------------
38
39
40 def get_elset_anywhere(odb, name):
41     root = odb.rootAssembly
42
43     # Assembly sets
44     if name in root.elementSets:
45         return root.elementSets[name]
46     name_u = name.upper()
47     if name_u in root.elementSets:
48         return root.elementSets[name_u]
```

```
49
50      # Instance-qualified: "INSTANCE.SET"
51      if "." in name:
52          inst_name, set_short = name.split(".", 1)
53          inst_u = inst_name.upper()
54          set_u = set_short.upper()
55          if inst_u not in root.instances:
56              raise RuntimeError("Instance not found: %s" % inst_u)
57          inst = root.instances[inst_u]
58          if set_short in inst.elementSets:
59              return inst.elementSets[set_short]
60          if set_u in inst.elementSets:
61              return inst.elementSets[set_u]
62          raise RuntimeError("Element set '%s' not found in instance '%s'" %
                (set_short, inst_u))
63
64      # Search all instance sets
65      for inst in root.instances.values():
66          if name in inst.elementSets:
67              return inst.elementSets[name]
68          if name_u in inst.elementSets:
69              return inst.elementSets[name_u]
70
71      raise RuntimeError("Element set not found: %s" % name)
72
73
74  def quad_area_3d(p1, p2, p3, p4):
75      p1 = np.array(p1, dtype=float)
76      p2 = np.array(p2, dtype=float)
77      p3 = np.array(p3, dtype=float)
78      p4 = np.array(p4, dtype=float)
79      a1 = 0.5 * np.linalg.norm(np.cross(p2 - p1, p3 - p1))
80      a2 = 0.5 * np.linalg.norm(np.cross(p4 - p1, p3 - p1))
81      return float(a1 + a2)
82
83
84  def build_node_coord_map(instance):
85      m = {}
86      for n in instance.nodes:
87          m[n.label] = np.array(n.coordinates, dtype=float)
88      return m
89
90
91  def best_fit_plane_basis(points3d):
92      P = np.asarray(points3d, dtype=float)
93
94      if P.size == 0 or P.shape[0] < 3:
95          raise RuntimeError("Not enough points to fit a plane (need at least 3).")
96
```

```
    mask = np.isfinite(P).all(axis=1)
    P = P[mask]
    if P.shape[0] < 3:
        raise RuntimeError("Not enough finite points to fit a plane.")

    c = P.mean(axis=0)
    Q = P - c

    if np.linalg.norm(Q) < 1e-12:
        raise RuntimeError("Degenerate point cloud: cannot fit plane.")

    u, s, vt = np.linalg.svd(Q, full_matrices=False)
    e1 = vt[0] / np.linalg.norm(vt[0])
    e2 = vt[1] / np.linalg.norm(vt[1])
    return c, e1, e2


def stable_plane_basis_from_plate(plate_pts3d, prefer_axis="X"):
    """
    Deterministic in-plane basis:
    1) Fit plane normal from PCA.
    2) Choose e1 aligned with global X (or Y) projected onto plane.
    3) e2 = n x e1.
    """
    origin, e1p, e2p = best_fit_plane_basis(plate_pts3d)

    n = np.cross(e1p, e2p)
    n_norm = np.linalg.norm(n)
    if n_norm < 1e-12:
        return origin, e1p, e2p
    n = n / n_norm

    if prefer_axis.upper() == "Y":
        g1 = np.array([0.0, 1.0, 0.0])
        g2 = np.array([1.0, 0.0, 0.0])
    else:
        g1 = np.array([1.0, 0.0, 0.0])
        g2 = np.array([0.0, 1.0, 0.0])

    e1 = g1 - np.dot(g1, n) * n
    if np.linalg.norm(e1) < 1e-8:
        e1 = g2 - np.dot(g2, n) * n
    if np.linalg.norm(e1) < 1e-8:
        return origin, e1p, e2p

    e1 = e1 / np.linalg.norm(e1)
    e2 = np.cross(n, e1)
    e2 = e2 / np.linalg.norm(e2)

```

```
    return origin, e1, e2


def project_points(points3d, origin, e1, e2):
    P = np.asarray(points3d, dtype=float)
    Q = P - origin
    x = np.dot(Q, e1)
    y = np.dot(Q, e2)
    return np.column_stack([x, y])


def plate_extents_uv(plate_pts3d, origin, e1, e2):
    pts2d = project_points(plate_pts3d, origin, e1, e2)
    umin = float(np.min(pts2d[:, 0])); umax = float(np.max(pts2d[:, 0]))
    vmin = float(np.min(pts2d[:, 1])); vmax = float(np.max(pts2d[:, 1]))
    return umin, umax, vmin, vmax


def enforce_u_is_long_side(plate_pts3d, origin, e1, e2):
    """
    Ensure u corresponds to the LONG side of the plate.
    If v-span > u-span, swap e1 and e2.
    """
    umin, umax, vmin, vmax = plate_extents_uv(plate_pts3d, origin, e1, e2)
    Lu = umax - umin
    Lv = vmax - vmin

    if Lv > Lu:
        e1, e2 = e2, e1
        umin, umax, vmin, vmax = plate_extents_uv(plate_pts3d, origin, e1, e2)

    return origin, e1, e2, umin, umax, vmin, vmax


def sanitize_filename(name):
    bad = ['\\','/',';',':','*','?','"','<','>','|',',',' ']
    for ch in bad:
        name = name.replace(ch, '_')
    return name


def safe_basename(path):
    p = path.replace("\\", "/")
    name = p.split("/")[-1]
    if name.lower().endswith(".odb"):
        name = name[:-4]
    return name


```

```
195 def fmt_thr(t):
196     s = "%.2f" % float(t)
197     return s.replace(".", "p")
198 
199 
200 def join_path(folder, filename):
201     f = folder.replace("\\", "/").rstrip("/")
202     return f + "/" + filename
203 
204 
205 def polys3d_to_patches(polys3d, origin, e1, e2):
206     patches = []
207     for poly3d in polys3d:
208         poly2d = project_points(poly3d, origin, e1, e2)
209         patches.append(Polygon(poly2d, closed=True))
210     return patches
211 
212 
213 def apply_full_plate_view(ax, umin, umax, vmin, vmax):
214     ax.set_aspect('equal', adjustable='box')
215     ax.set_xlim(umin, umax)
216     ax.set_ylim(vmin, vmax)
217     if draw_plate_outline:
218         ax.plot([umin, umax, umax, umin, umin],
219                 [vmin, vmin, vmax, vmax, vmin],
220                 linewidth=1.0)
221 
222 
223 def make_single_plot(patches, title, area_mm2, out_path, umin, umax, vmin, vmax):
224     fig = plt.figure()
225     ax = fig.add_subplot(111)
226 
227     if len(patches) > 0:
228         ax.add_collection(PatchCollection(patches, facecolor="C0",
            edgecolor="none", alpha=0.85))
229 
230     ax.set_xlabel("u (mm)")
231     ax.set_ylabel("v (mm)")
232     ax.set_title(title)
233 
234     apply_full_plate_view(ax, umin, umax, vmin, vmax)
235 
236     ax.text(0.02, 0.98, "Area=%.1f mm^2" % float(area_mm2),
        transform=ax.transAxes, va="top")
237 
238     fig.tight_layout()
239     fig.savefig(out_path, dpi=dpi)
240     plt.close(fig)
241 
```

```
242
243 def make_combined_plot(patches_band, patches_hi, A_lo, A_hi, out_path, umin, umax,
        vmin, vmax):
244     fig = plt.figure()
245     ax = fig.add_subplot(111)
246
247     if len(patches_band) > 0:
248         ax.add_collection(PatchCollection(patches_band, facecolor="C0",
                edgecolor="none", alpha=0.50))
249     if len(patches_hi) > 0:
250         ax.add_collection(PatchCollection(patches_hi, facecolor="C1",
                edgecolor="none", alpha=0.90))
251
252     ax.set_xlabel("u (mm)")
253     ax.set_ylabel("v (mm)")
254     ax.set_title("Delamination map (two-level damage)")
255
256     apply_full_plate_view(ax, umin, umax, vmin, vmax)
257
258     txt = "A(SDEG>=%.2f)=%.1f mm^2\nA(SDEG>=%.2f)=%.1f mm^2" % (
259         float(thr_lo), float(A_lo),
260         float(thr_hi), float(A_hi)
261     )
262     ax.text(0.02, 0.98, txt, transform=ax.transAxes, va="top")
263
264     ax.legend(["%.2f<=SDEG<%.2f" % (float(thr_lo), float(thr_hi)),
265                "SDEG>=%.2f" % float(thr_hi)], loc="best")
266
267     fig.tight_layout()
268     fig.savefig(out_path, dpi=dpi)
269     plt.close(fig)
270
271
272 def footprint_area_raster(patches, umin, umax, vmin, vmax, dx_):
273     if dx_ <= 0.0:
274         raise RuntimeError("dx must be > 0")
275
276     if patches is None or len(patches) == 0:
277         return 0.0, 0.0
278
279     nx = int(np.ceil((umax - umin) / dx_))
280     ny = int(np.ceil((vmax - vmin) / dx_))
281     if nx <= 0 or ny <= 0:
282         return 0.0, 0.0
283
284     xs = umin + (np.arange(nx) + 0.5) * dx_
285     ys = vmin + (np.arange(ny) + 0.5) * dx_
286     XX, YY = np.meshgrid(xs, ys)
287     pts = np.column_stack([XX.ravel(), YY.ravel()])
```

```
288
289     mask = np.zeros((pts.shape[0],), dtype=bool)
290
291     for poly in patches:
292         verts = poly.get_xy()
293         pu_min = float(np.min(verts[:,0])); pu_max = float(np.max(verts[:,0]))
294         pv_min = float(np.min(verts[:,1])); pv_max = float(np.max(verts[:,1]))
295
296         in_bbox = (pts[:,0] >= pu_min) & (pts[:,0] <= pu_max) & (pts[:,1] >=
                pv_min) & (pts[:,1] <= pv_max)
297         if not np.any(in_bbox):
298             continue
299
300         pth = Path(verts)
301         inside = pth.contains_points(pts[in_bbox])
302         if np.any(inside):
303             idx = np.where(in_bbox)[0]
304             mask[idx[inside]] = True
305
306     A_foot = float(np.sum(mask)) * (dx_ * dx_)
307     A_plate = float((umax - umin) * (vmax - vmin))
308     frac = (A_foot / A_plate) if A_plate > 0 else 0.0
309     return A_foot, frac
310
311
312 def main():
313     if len(session.odbs) == 0:
314         raise RuntimeError("No ODB is open in Viewer. Open the ODB first, then run
                the script.")
315     odb = session.odbs.values()[0]
316
317     odb_name = sanitize_filename(safe_basename(odb.name))
318
319     if step_name not in odb.steps:
320         raise RuntimeError("Step not found: %s" % step_name)
321     step = odb.steps[step_name]
322     frame = step.frames[frame_idx]
323
324     if 'SDEG' not in frame.fieldOutputs:
325         raise RuntimeError("SDEG not found in this frame. Add SDEG to Field Output
                Requests and rerun the analysis.")
326
327     elset = get_elset_anywhere(odb, elset_name)
328     elements = elset.elements
329
330     # elementLabel -> SDEG
331     sdeg_field = frame.fieldOutputs['SDEG'].getSubset(region=elset)
332     sdeg_map = {}
333     for v in sdeg_field.values:
```

```
        sdeg_map[v.elementLabel] = float(v.data)

    labels_ge_lo = set([lbl for (lbl, val) in sdeg_map.items() if val >= thr_lo])
    labels_ge_hi = set([lbl for (lbl, val) in sdeg_map.items() if val >= thr_hi])
    labels_band  = set([lbl for (lbl, val) in sdeg_map.items() if (val >= thr_lo
        and val < thr_hi)])

    # Cache node coords per instance
    root = odb.rootAssembly
    inst_map = root.instances
    node_maps = {}

    A_lo = 0.0
    A_hi = 0.0
    A_band = 0.0

    polys_lo_3d = []
    polys_hi_3d = []
    polys_band_3d = []

    # COH3D8 face = first 4 connectivity nodes
    for e in elements:
        lbl = e.label
        if (lbl not in labels_ge_lo) and (lbl not in labels_ge_hi) and (lbl not in
            labels_band):
            continue

        inst_name = e.instanceName
        if inst_name not in node_maps:
            node_maps[inst_name] = build_node_coord_map(inst_map[inst_name])
        coord = node_maps[inst_name]

        conn = list(e.connectivity)
        if len(conn) != 8:
            continue

        face = conn[:4]
        p1 = coord[face[0]]
        p2 = coord[face[1]]
        p3 = coord[face[2]]
        p4 = coord[face[3]]

        A = quad_area_3d(p1, p2, p3, p4)  # mm^2

        if lbl in labels_ge_lo:
            A_lo += A
            polys_lo_3d.append([p1, p2, p3, p4])
        if lbl in labels_ge_hi:
            A_hi += A
```

```
            polys_hi_3d.append([p1, p2, p3, p4])
        if lbl in labels_band:
            A_band += A
            polys_band_3d.append([p1, p2, p3, p4])

    # Print results
    print ""
    print "--- DELAMINATION AREAS FROM SDEG (COH3D8 -> mm^2) ---"
    print "ODB: %s" % odb.name
    print "Step: %s | Frame index: %s" % (step_name, str(frame_idx))
    print "Element set: %s" % elset_name
    print ""
    print "Area(SDEG >= %.2f): %.6f mm^2" % (float(thr_lo), A_lo)
    print "Area(SDEG >= %.2f): %.6f mm^2" % (float(thr_hi), A_hi)
    print "Area(%.2f <= SDEG < %.2f): %.6f mm^2" % (float(thr_lo), float(thr_hi),
        A_band)
    print "-----------------------------------------------------"
    print ""

    # === ALWAYS define projection basis from FULL PLATE instance nodes ===
    if plate_instance.upper() not in root.instances:
        raise RuntimeError("Plate instance not found: %s" % plate_instance)

    inst_plate = root.instances[plate_instance.upper()]
    plate_pts3d = [np.array(n.coordinates, dtype=float) for n in inst_plate.nodes]

    if stable_axes:
        origin, e1, e2 = stable_plane_basis_from_plate(plate_pts3d,
            prefer_axis=stable_axis_preference)
    else:
        origin, e1, e2 = best_fit_plane_basis(plate_pts3d)

    # === Enforce u = long side of the specimen ===
    origin, e1, e2, umin, umax, vmin, vmax = enforce_u_is_long_side(plate_pts3d,
        origin, e1, e2)

    # Convert polygons to patches in the stable plate (u,v)
    patches_lo   = polys3d_to_patches(polys_lo_3d, origin, e1, e2)
    patches_hi   = polys3d_to_patches(polys_hi_3d, origin, e1, e2)
    patches_band = polys3d_to_patches(polys_band_3d, origin, e1, e2)

    # --- FOOTPRINT / ENVELOPE AREA (C-scan comparable) ---
    Afoot_lo, frac_lo = footprint_area_raster(patches_lo, umin, umax, vmin, vmax,
        dx)
    Afoot_hi, frac_hi = footprint_area_raster(patches_hi, umin, umax, vmin, vmax,
        dx)

    print ""
    print "--- FOOTPRINT (UNION) AREAS (C-scan comparable) ---"
```

```
425     print "Raster resolution dx = %.3f mm" % float(dx)
426     print "Footprint Area(SDEG >= %.2f): %.6f mm^2 (%.2f%% of plate bbox)" %
            (float(thr_lo), Afoot_lo, 100.0*frac_lo)
427     print "Footprint Area(SDEG >= %.2f): %.6f mm^2 (%.2f%% of plate bbox)" %
            (float(thr_hi), Afoot_hi, 100.0*frac_hi)
428     print "------------------------------------------------"
429     print ""
430
431     if Afoot_lo > 0:
432         print "Cumulative/Footprint ratio (>=%.2f): %.3f" % (float(thr_lo),
                float(A_lo / Afoot_lo))
433     if Afoot_hi > 0:
434         print "Cumulative/Footprint ratio (>=%.2f): %.3f" % (float(thr_hi),
                float(A_hi / Afoot_hi))
435     print ""
436
437     # Output file paths
438     f_lo = join_path(out_dir, "%s_delam_SDEG_ge_%s.png" % (odb_name,
            fmt_thr(thr_lo)))
439     f_hi = join_path(out_dir, "%s_delam_SDEG_ge_%s.png" % (odb_name,
            fmt_thr(thr_hi)))
440     f_both = join_path(out_dir, "%s_delam_SDEG_%s_to_%s_and_ge_%s.png" %
441                        (odb_name, fmt_thr(thr_lo), fmt_thr(thr_hi),
                               fmt_thr(thr_hi)))
442
443     # 1) SDEG >= 0.90
444     make_single_plot(patches_lo, "Delamination map (SDEG >= %.2f)" %
            float(thr_lo), A_lo,
445                      f_lo, umin, umax, vmin, vmax)
446     print "Saved: %s" % f_lo
447
448     # 2) SDEG >= 0.99
449     make_single_plot(patches_hi, "Delamination map (SDEG >= %.2f)" %
            float(thr_hi), A_hi,
450                      f_hi, umin, umax, vmin, vmax)
451     print "Saved: %s" % f_hi
452
453     # 3) Combined: band + high
454     make_combined_plot(patches_band, patches_hi, A_lo, A_hi, f_both, umin, umax,
            vmin, vmax)
455     print "Saved: %s" % f_both
456
457
458 main()
```

Listing C.1: Python script for automatic delamination area extraction from back-face C-scan images.

Appendix D

Python Script for Ultrasonic C-Scan Delamination Extraction

The following Python script was developed to automatically extract the projected delamination footprint from exported back-face C-scan images.

The script performs:

- Automatic spatial calibration using the embedded scale ruler,
- Brightness-based segmentation of the delamination region,
- Extraction of the largest connected component,
- Pixel counting and conversion to physical area (mm^2),
- Automatic generation of binary masks and overlay visualizations.

The script was applied consistently to all specimens investigated in this work.

```python
import os
import glob
import numpy as np
import matplotlib.pyplot as plt
import matplotlib.image as mpimg

# ---------------- USER SETTINGS ----------------
image_folder = r"path/to/input/images"
out_folder = r"path/to/output/results"
exts = ("*.png", "*.jpg", "*.jpeg", "*.bmp")

SCALE_STRIP_FRAC = 0.18   # bottom fraction where the ruler exists
MAJOR_TICK_MM    = 10.0

# segmentation (tune once, then batch)
BRIGHT_THRESH = 0.90      # threshold on normalized brightness [0..1]
CROP_BOTTOM_FOR_DAMAGE = 0.18  # ignore scale strip when measuring damage
```

```
18
19 # tick detection tuning
20 SMOOTH_WIN = 21           # smoothing window for 1D projection
21 PEAK_REL   = 0.55         # peaks above PEAK_REL * max considered
22 MERGE_PX   = 30           # merge tick peaks closer than this
23 # ------------------------------------------------
24
25 os.makedirs(out_folder, exist_ok=True)
26
27 def to_float_rgb(img):
28     """Ensure float RGB image in [0,1]."""
29     if img.dtype == np.uint8:
30         img = img.astype(np.float32) / 255.0
31     else:
32         img = img.astype(np.float32)
33         if img.max() > 1.5:  # e.g., 0..255 float
34             img /= 255.0
35     if img.ndim == 2:
36         img = np.stack([img, img, img], axis=-1)
37     if img.shape[2] == 4:
38         img = img[..., :3]
39     return np.clip(img, 0, 1)
40
41 def moving_average(x, win):
42     win = int(win)
43     if win < 3:
44         return x
45     if win % 2 == 0:
46         win += 1
47     kernel = np.ones(win, dtype=np.float32) / win
48     return np.convolve(x, kernel, mode="same")
49
50 def detect_ticks_and_mm_per_px(img_rgb):
51     h, w, _ = img_rgb.shape
52     y0 = int(h * (1.0 - SCALE_STRIP_FRAC))
53     strip = img_rgb[y0:h, :, :]
54
55     # Convert to grayscale brightness (simple luminance)
56     gray = 0.2126*strip[...,0] + 0.7152*strip[...,1] + 0.0722*strip[...,2]
57
58     # Emphasize vertical lines: use absolute horizontal gradient
59     gx = np.abs(np.diff(gray, axis=1))
60     proj = gx.sum(axis=0)
61
62     # Smooth it
63     proj_s = moving_average(proj, SMOOTH_WIN)
64
65     m = proj_s.max()
66     if not np.isfinite(m) or m <= 0:
```

```
        raise RuntimeError("Scale detection failed: projection is empty.")

    thr = PEAK_REL * m
    candidates = np.where(proj_s >= thr)[0]
    if len(candidates) < 10:
        raise RuntimeError("Scale detection failed: too few tick candidates.")

    # Group consecutive indices to peaks
    groups = []
    start = candidates[0]
    prev = candidates[0]
    for x in candidates[1:]:
        if x == prev + 1:
            prev = x
        else:
            groups.append((start, prev))
            start = x
            prev = x
    groups.append((start, prev))
    peaks = [int((a+b)/2) for a,b in groups]
    peaks.sort()

    # Merge close peaks (double edges of same tick)
    merged = []
    for p in peaks:
        if not merged or (p - merged[-1]) > MERGE_PX:
            merged.append(p)
        else:
            merged[-1] = int((merged[-1] + p)/2)

    if len(merged) < 4:
        raise RuntimeError("Scale detection failed: too few merged ticks.")

    diffs = np.diff(merged).astype(np.float32)
    med = np.median(diffs)
    good = diffs[(diffs > 0.6*med) & (diffs < 1.6*med)]
    if len(good) < 2:
        raise RuntimeError("Scale detection failed: inconsistent tick spacing.")

    px_per_major = float(np.median(good))
    mm_per_px = MAJOR_TICK_MM / px_per_major
    return mm_per_px, merged, y0

def segment_damage(img_rgb):
    h, w, _ = img_rgb.shape
    y_cut = int(h * (1.0 - CROP_BOTTOM_FOR_DAMAGE))
    roi = img_rgb[:y_cut, :, :]

    # Brightness map
```

```
    bright = roi.max(axis=2)  # "white" tends to have high max channel

    mask = bright >= BRIGHT_THRESH

    return mask, y_cut

def largest_component(mask):
    """Return mask of largest 4-connected component (pure numpy floodfill)."""
    h, w = mask.shape
    visited = np.zeros_like(mask, dtype=bool)

    best_size = 0
    best_coords = None

    # iterate over true pixels
    ys, xs = np.where(mask)
    for y, x in zip(ys, xs):
        if visited[y, x]:
            continue
        # BFS stack
        stack = [(y, x)]
        visited[y, x] = True
        coords = []
        while stack:
            cy, cx = stack.pop()
            coords.append((cy, cx))
            # neighbors
            if cy > 0 and mask[cy-1, cx] and not visited[cy-1, cx]:
                visited[cy-1, cx] = True
                stack.append((cy-1, cx))
            if cy < h-1 and mask[cy+1, cx] and not visited[cy+1, cx]:
                visited[cy+1, cx] = True
                stack.append((cy+1, cx))
            if cx > 0 and mask[cy, cx-1] and not visited[cy, cx-1]:
                visited[cy, cx-1] = True
                stack.append((cy, cx-1))
            if cx < w-1 and mask[cy, cx+1] and not visited[cy, cx+1]:
                visited[cy, cx+1] = True
                stack.append((cy, cx+1))

        if len(coords) > best_size:
            best_size = len(coords)
            best_coords = coords

    out = np.zeros_like(mask, dtype=bool)
    if best_coords is not None:
        for (y, x) in best_coords:
            out[y, x] = True
    return out
```

```
165
166 def save_overlay(img_rgb, mask_roi, y_cut, tick_xs, strip_y0, out_path):
167     overlay = img_rgb.copy()
168
169     # red overlay for damage in ROI
170     alpha = 0.35
171     red = np.zeros_like(overlay)
172     red[...,0] = 1.0
173
174     m = mask_roi
175     roi = overlay[:y_cut]
176     roi[m] = alpha*red[:y_cut][m] + (1-alpha)*roi[m]
177     overlay[:y_cut] = roi
178
179     # green tick lines (in scale strip)
180     for x in tick_xs:
181         x = int(x)
182         if 0 <= x < overlay.shape[1]:
183             overlay[strip_y0:, x:x+1, :] = np.array([0,1,0], dtype=np.float32)
184
185     plt.imsave(out_path, overlay)
186
187 def process_one(path):
188     img = mpimg.imread(path)
189     img_rgb = to_float_rgb(img)
190
191     mm_per_px, tick_xs, strip_y0 = detect_ticks_and_mm_per_px(img_rgb)
192
193     mask, y_cut = segment_damage(img_rgb)
194     mask_big = largest_component(mask) if mask.any() else mask
195
196     white_px = int(mask_big.sum())
197     area_mm2 = white_px * (mm_per_px**2)
198
199     return mm_per_px, white_px, area_mm2, mask_big, y_cut, tick_xs, strip_y0,
        img_rgb
200
201 def main():
202     paths = []
203     for e in exts:
204         paths += glob.glob(os.path.join(image_folder, e))
205     paths = sorted(paths)
206
207     results = []
208     for p in paths:
209         base = os.path.splitext(os.path.basename(p))[0]
210         print(f"\n=== {base} ===")
211         try:
```

```
            mm_per_px, white_px, area_mm2, mask_big, y_cut, tick_xs, strip_y0,
                img_rgb = process_one(p)
            print(f"mm/px = {mm_per_px:.6f} | white_px = {white_px} | area =
                {area_mm2:.2f} mm^2")

            mask_img = np.zeros((y_cut, img_rgb.shape[1]), dtype=np.uint8)
            mask_img[mask_big] = 255
            plt.imsave(os.path.join(out_folder, f"{base}_mask.png"), mask_img,
                cmap="gray")

            save_overlay(img_rgb, mask_big, y_cut, tick_xs, strip_y0,
                         os.path.join(out_folder, f"{base}_overlay.png"))

            results.append((os.path.basename(p), mm_per_px, white_px, area_mm2,
                "OK"))
        except Exception as e:
            print(f"[FAIL] {e}")
            results.append((os.path.basename(p), np.nan, np.nan, np.nan, f"FAIL:
                {e}"))

    csv_path = os.path.join(out_folder, "damage_area_results.csv")
    with open(csv_path, "w", encoding="utf-8") as f:
        f.write("file,mm_per_pixel,white_pixels,damage_area_mm2,status\n")
        for row in results:
            f.write(",".join(str(x) for x in row) + "\n")

    print(f"\nDone. Results saved to: {csv_path}")
    print(f"Masks/overlays saved to: {out_folder}")

if __name__ == "__main__":
    main()
```

Listing D.1: Python script for automatic delamination area extraction from back-face C-scan images.

Appendix E

Computational Cost of the Numerical Models

The computational time required for each numerical model investigated in this work is summarised in Table E.1. The reported values correspond to the total wall-clock simulation time obtained for each impact case using the computational resources indicated in the table footnotes.

Table E.1: Wall-clock simulation time (s) required for the different numerical models and impact energy levels.

Impact case	Elastic	Hashin	Cohesive	Bi-phasic Elastic (FP)	Bi-phasic Elastic (IZ)	Bi-phasic Hashin (FP)	Bi-phasic Hashin (IZ)	Bi-phasic Plastic
4.5 J	9590[a]	12036[a]	76234[b]	29000[a]	27955[a]	33179[a]	31497[a]	–
7.5 J	9887[a]	12291[a]	72839[b]	29191[a]	28175[a]	33595[a]	31271[a]	–
14.2 J	10387[a]	13641[a]	78836[b]	29333[a]	27837[a]	33996[a]	31573[a]	4914[c]
27 J	9924[a]	17075[a]	104641[b]	29405[a]	27474[a]	34392[a]	31372[a]	5147[c]

[a] Simulations performed using 8 CPU cores.

[b] Simulations performed using 15 CPU cores.

[c] Simulations performed using 64 CPU cores.

The results highlight the significant variation in computational cost associated with the different modelling strategies. Compared with the baseline elastic model, the Hashin intralaminar damage formulation increases the simulation time by approximately 1.5×, while the bi-phasic modelling approaches require roughly 3× longer simulation times due to the increased constitutive complexity and additional degrees of freedom introduced by the multi-phase representation. The cohesive-zone formulation represents the most computationally demanding approach, with simulation times approximately 7–10× higher than the elastic baseline. This increase is primarily associated with the large number of cohesive interface elements and the stiffness degradation mechanisms governing interlaminar damage evolution.